PIPING MATERIAL SPECIFICATION

PS	AKM	SJ	124	09-05-2016	3
PS	GJ	GRP	94	22-07-2009	2
PS	GJ	GRP	78	07-08-2008	1
PS	GJ	GRP	78	04-08-2008	0
Prepared By	Reviewed By	Approved By	Number of pages	Date	Revision No.

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15	XG1	0	08.04.08	35	36
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28	A10	0	08.04.08	61	62
29	B10	0	15.01.16	63	64
30	D10	0	15.01.16	65	66
31	E10	0	15.01.16	67	68
32	F10	1	03.05.16	69	70
33	A12	1	15.01.16	71	72
34	A13	1	15.01.16	73	74
35	A1LN	0	22.07.09	75	76
36	A22	0	22.07.09	77	78
37	B22	1	22.07.09	79	80
38	D22	2	15.01.16	81	82
39	E22	0	15.01.16	83	84
40	A23	2	15.01.16	85	86
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43	E23	3	15.01.16	91	92
44	F23	0	15.01.16	93	94
45	G23	0	15.01.16	95	96
46	A15	0	22.07.09	97	98
47	B15	0	22.07.09	99	100
48	D15	1	22.07.09	101	102
49	E15	1	15.01.16	103	104
50	XG15	1	08.04.08	105	106
51	XH15	1	08.04.08	107	108
52	A15A	0	08.04.08	109	110
53	B15A	0	08.04.08	111	112

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53	D15A	0	08.04.08	113	114
54	E15A	0	08.04.08	115	116
55	PE1	1	15.01.16	117	118
56	PB1N	0	15.01.16	119	120
57	PD1N	1	15.01.16	121	122
58	PE1N	1	15.01.16	123	124

Piping Specification Index

SL. No.	Material Specification	Class	Pipe Material	Corro.All. In mm
1	A1	# 150	Carbon steel-(CS)	3
2	B1	# 300	Carbon steel-(CS)	3
3	D1	# 600	Carbon steel-(CS)	3
4	E1	# 900	Carbon steel-(CS)	3
5	A2	# 150	Carbon steel-(CS)	1.5
6	B2	# 300	Carbon steel-(CS)	1.5
7	D2	# 600	Carbon steel-(CS)	1.5
8	E2	#900	Carbon steel-(CS)	1.5
9	A1N	# 150	Carbon steel NACE-(CS-NACE)	6
10	B1N	# 300	Carbon steel NACE-(CS-NACE)	6
11	D1N	# 600	Carbon steel NACE-(CS-NACE)	6
12	E1N	# 900	Carbon steel NACE-(CS-NACE)	6
13	F1N	#1500	Carbon steel NACE-(CS-NACE)	6
14	G1N	#2500	Carbon steel NACE-(CS-NACE)	6
15	XG1	API 5000	Carbon steel-(CS)	3
16	XG1N	API 5000	Carbon steel NACE-(CS-NACE)	6
17	A3	# 150	Carbon steel Galvanised-(CS-Galv.)/SS	1.5/Nil
18	B3	# 300	Carbon steel Galvanised-(CS-Galv.)/SS	1.5/Nil
19	D3	#600	Carbon steel Galvanised-(CS-Galv.)/SS	1.5/Nil
20	A4	# 150	Carbon steel Galvanised-(CS-Galv.)	1.5
21	A5	# 150	Copper Nickel(90/10 Cu-Ni)	Nil
22	A6	# 150	Stainless steel (SS 316L)	Nil
23	A8	# 150	Carbon steel Galvanised-(CS-Galv.)	3
24	A9	# 150	Stainless steel (SS 316)	Nil
25	B9	# 300	Stainless steel (SS 316)	Nil
26	D9	# 600	Stainless steel (SS 316)	Nil
27	E9	# 900	Stainless steel (SS 316)	Nil
28	A10	# 150	Stainless steel NACE(SS 316L-NACE)	1.5
29	B10	# 300	Stainless steel NACE(SS 316L-NACE)	1.5
30	D10	# 600	Stainless steel NACE(SS 316L-NACE)	1.5
31	E10	#900	Stainless steel NACE(SS 316L-NACE)	1.5
32	F10	# 2500	Stainless steel NACE(SS 316L-NACE)	1.5
33	A12	# 150	Titanium	Nil
34	A13	# 150	Copper(Cu)	Nil
35	A1LN	#150	LT CS - NACE	3mm
36	A22	# 150	Stainless steel (SS 316L)	Nil
37	B22	# 300	Stainless steel (SS 316L)	Nil
38	D22	# 600	Stainless steel (SS 316L)	Nil
39	E22	# 900	Stainless steel (SS 316L)	Nil
40	A23	# 150	Duplex Stainless steel-NACE (DSS-NACE)	Nil
41	B23	# 300	Duplex Stainless steel-NACE (DSS-NACE)	Nil
42	D23	# 600	Duplex Stainless steel-NACE (DSS-NACE)	Nil
43	E23	# 900	Duplex Stainless steel-NACE (DSS-NACE)	Nil
44	F23	#1500	Duplex Stainless steel-NACE (DSS-NACE)	Nil
45	G23	#2500	Duplex Stainless steel-NACE (DSS-NACE)	Nil

Note : During detail engineering, if it is observed that certain specification sheets are required as per approved P&ID but not included in this document, EPC contractor shall prepare the same based upon FS 2004A and submit it to company's review and approval.

Piping Specification Index

SL. No.	Material Specification	Class	Pipe Material	Corro.All. In mm
46	A15	#150	Incoloy 825 and Incoloy Cladding - NACE	Nil
47	B15	#300	Incoloy 825 and Incoloy Cladding - NACE	Nil
48	D15	#600	Incoloy 825 and Incoloy Cladding - NACE	Nil
49	E15	#900	Incoloy 825 and Incoloy Cladding - NACE	Nil
50	XG15	API 5000	Incoloy 825 and Incoloy Cladding - NACE	Nil
51	XH15	API 10000	Incoloy 825 and Incoloy Cladding - NACE	Nil
52	A15A	#150	Inconel 625 and Inconel Cladding - NACE	Nil
53	B15A	#300	Inconel 625 and Inconel Cladding - NACE	Nil
54	D15A	#600	Inconel 625 and Inconel Cladding - NACE	Nil
55	E15A	#900	Inconel 625 and Inconel Cladding - NACE	Nil
56	PE1	# 900	Carbon steel-(CS)	3
57	PB1N	# 300	Carbon steel NACE-(CS-NACE)	6
58	PD1N	# 600	Carbon steel NACE-(CS-NACE)	6
59	PE1N	# 900	Carbon steel NACE-(CS-NACE)	6

Note : During detail engineering, if it is observed that certain specification sheets are required as per approved P&ID but not included in this document, EPC contractor shall prepare the same based upon FS 2004A and submit it to company's review and approval.

1 Introduction

The piping material specifcation is prepared based upon Functional Specificastion FS 2004A and Piping Design Criteria. Datasheets attached here is minimum indicative piping thickness, Piping MOC, Fittings, gaskets and bolts etc. and EPC shall not construe it as firm piping material specification. The EPC contractor shall develope piping material specification and valve material specificaion and provide all supporting documents including piping wall thickness reports, material selection basis etc during detail engineering to company. Valve tag nos. indicated in the PMS is for referance only. Contractor shall prepare detail valve material specification based on FS 2004A and other relevent codes. Contractor shall reconfirm the tag nos. indicated in the Piping material Specification.

1.1 Order of precedance

The order of precedance for piping material specification shall be as follows in the order indicated:

i) Piping Design Criteria
ii) P&ID
iii) FS 2004 A (LATEST REV.)
iv) This Specification -PMS
v) Codes and standard

1.2 Genaral

All material shall conform to ASTM or BS standard. Design and fabrication shall be as per ASME B 31.3 for process piping and ASME B 31.4/31.8 liquid and gas transmission pipeline repectively for the piping of top side beyond top of transition bend.The EPC contractor shall also ensure that API Recommended practices (API 14E) Design and installation of offshore production platform piping system is adherred. In case of any conflict between FS 2004A and this document, FS 2004A shall prevail.

1.2.a During detail engineering, if it is observed that certain specification sheets are required as per approved P&ID but not included in this document, EPC contractor shall prepare the same based upon FS 2004A and submit it to company's review and approval. Piping and Valve material specification for non-matellic piping is not included in this document and contractor shall prepare the same based upon vendor information as per manufacturer standard.

1.3 Contractor's Responsibility

The piping and valve material specification prepared by EPC contractor shall necessarily include all the minimum information required by vendor for procurement of piping material. EPC shall prepare Piping material and valve material specification based on FS 2004A,FS 2004B,FS2004D and Piping Design Criteria.

All the piping, piping component and piping speciality shall be painted in accordance with FS 2005 latest revision.

All the piping, piping component and piping speciality shall be insulated in accordance with FS 2006 latest revision.

All the piping, piping component and piping speciality shall be insulated in accordance with FS 2006 latest revision.

All type of welding and NDT in piping, piping component, piping specillaity and supports shall be in accordance with FS 2009 latest revision

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION													Sheet No.		1 of 2	
	Piping Class				A1		# 150		Material		CS		C.A 3mm	Spec.No 2004A,Rev 9		Revision:2 Page No.	7
	Design Code:				**ASME B 31.3**												
	Service:	Vent Gas (VG), Blow Down(BD),Process Hydro Carbon Drain (DH),Nitrogen(N),Chemical (C), Water Injection(WI),Process Drain Water(DW),Diesel Fuel(DF), Hydrocarbon Liquid and Vapour with low CO2 AND H2S.															

Pipe Data	ASME B 36.10																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	XXS	160	160	160	XS	80	80	80	Std	Std	Std	Std	Std	Std	20	20	20
WTmm	7.47	5.56	6.35	7.14	5.54	7.62	8.56	10.97	8.18	9.27	9.52	9.52	9.52	9.52	9.52	9.52	9.52
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5				
TYPE	Seamless													LSAW with 100% Radiography, Note-7			
MOC	ASTM A 106 Gr. B													API 5L Gr. B			
Ends	PE				BE												
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Socket Weld				Butt Weld												
Rating	# 9000	# 6000			XS	80	80	80	Std	Std	Std	Std	Std	Std	20	20	20
MOC	ASTM A 105 U.N.O				ASTM A 234 Gr. WPB, Seamless U.N.O									ASTM A234 Gr.WPB,Welded U.N.O			
Elbow	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Tee	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Red.	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Cap	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Coupl	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Plug	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Union																	
Elbowlet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105									MSS SP 97,ASTM A 105			
Sockolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105									MSS SP 97,ASTM A 105			
Weldolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105									MSS SP 97,ASTM A 105			
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe									MOC Same as pipe			
Swage	MSS SP 95,Same as pipe				MSS SP 95,MOC Same as pipe									MSS SP 95,MOC Same as pipe			
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				WN												
MOC	ASTM A 105				ASTM A 105												
FACE	RF-Serrated Finished				RF, Serrated Finished												
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25												
Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM A 105																
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48					
Mechanical Joints																	
Stud Bolts	ASTM A 193 Gr. B7				ASTM A 193 Gr. B7												
Hex Nuts	ASTM A 194 Gr. 2H				ASTM A 194 Gr. 2H												
Gasket	ASME B 16.21,Flat Ring,1.58mm,CNAF				ASME B 16.21,Flat Ring,1.58mm,CNAF												

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
(-) 4 to 100	285	(-) 20 to 38	20	450 psig	31.6 Kg/cm2
200	260	93	18.3	Limited by Flange considereing flange as the weakest joint in piping system.	
300	230	149	16.2		
400	200	204	14.1		
500	170	260	11.9		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2.All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. All pipe thread shall be as per ASME B 1.20.0
5. PMS to be read in conjuction with FS 2004 A.
6. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and speciifed spectacle blind assemblies.
7. Weld Joint Factor for welded pipe shall be as per ASME B 31.3

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION				Sheet No.	2 of 2
	Piping Class		Material	C.A	Spec.No	Revision:2
	A1	# 150	CS	3mm	2004A,Rev 9	Page No. 8
	Design Code:	**ASME B 31.3**				
	Service:	Vent Gas (VG), Blow Down(BD),Process Hydro Carbon Drain (DH),Nitrogen(N),Chemical (C),				
	Water Injection(WI),Process Drain Water(DW),Diesel Fuel(DF), Hydrocarbon Liquid and Vapour with low CO2 AND H2S.					

Branch Table as per API RP 14E

Run Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T
Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 42,SW				VG 60,FLANGED												
Rating	# 800				# 150,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB 44,SW				VB 61, Lever Operated			VB 63, Gear Operated									
Rating	# 800				# 150, RF												
Globe Valve																	
TAG	VGL 42,SW				VGL 60												
Rating	# 800				# 150, RF												
Check Valve																	
TAG	VC 42,SW,Lift Check				VC 60,Swing Check/ VC 81 (Wafer Check, Note -1)												
Rating	# 800				# 150, RF												
Needle Valve																	
TAG	VN 3,SW																
Rating	# 800																
Butter Fly Valve																	
TAG					VBF 1(WAFFER)												
Rating					#150, RF												

Notes Releted to Valve and Brach Connection

1. Wafer check valve to be avoided as far as possible; unless the avaliable space constraint does not allow normal check valve . The contractor to take approvel for use of wafer check Valve during pipe routing after extablishing the space constraint.

2. Wafer type Butterfly Valve may be used only in fire water(Clean water) service.
3. PMS to be read in conjuction with FS 2004 A.
4. Maximum temperature limit for all Ball Valve shall be 121 Degree C

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision:2	15.01.16	Issued For Bid	PS	AKM	SJ

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**1 of 2**
	Piping Class		Material	C.A	Spec.No	Revision:2
	B1	# 300	CS	3mm	2004A,Rev 9	Page No. 9
	Design Code:	**ASME B 31.3**				
	Service:	Vent Gas (VG), Blow Down(BD),Process Hydro Carbon Drain (DH),Nitrogen(N),Chemical (C), Water Injection(WI),Process Drain Water(DW),Diesel Fuel(DF), Hydrocarbon Liquid and Vapour with low CO2 AND H2S.				

Pipe Data	ASME B 36.10																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	XXS	XXS	160	160	XS	80	80	80	60	60	80	40	40	40	40	40	40
WTmm	7.47	7.82	6.35	7.14	5.54	7.62	8.56	10.97	10.3	12.7	12.70	11.1	12.7	14.27	15.1	15.87	17.47
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5				
TYPE	Seamless													LSAW with 100% Radiography, Note-7			
MOC	ASTM A 106 Gr. B													API 5L Gr. B			
Ends	PE				BE												
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Socket Weld				Butt Weld												
Rating	# 9000		# 6000		XS	80	80	80	60	60	80	40	40	40	40	40	40
MOC	ASTM A 105 U.N.O				ASTM A 234 Gr. WPB, Seamless U.N.O									ASTM A234 Gr.WPB,Welded U.N.O			
Elbow	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Tee	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Red.	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Cap	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Coupl	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Plug	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Union																	
Elbowlet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105									MSS SP 97,ASTM A 105			
Sockolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105									MSS SP 97,ASTM A 105			
Weldolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105									MSS SP 97,ASTM A 105			
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe									MOC Same as pipe			
Swage	MSS SP 95,Same as pipe				MSS SP 95,MOC Same as pipe									MSS SP 95,MOC Same as pipe			
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				WN												
MOC	ASTM A 105				ASTM A 105												
FACE	RF-Smooth Finished				RF, Smooth Finished												
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25												
Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM A 105																
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48					
Mechanical Joints																	
Stud Bolts	ASTM A 193 Gr. B7				ASTM A 193 Gr. B7												
Hex Nuts	ASTM A 194 Gr. 2H				ASTM A 194 Gr. 2H												
Gasket	B 16.20,4.5mm, SS Spiral wound with CNAF filler				ASME B 16.20,4.5mm, SS Spiral wound with CNAF filler												

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
(-) 4 to 100	740	(-) 20 to 38	52	1125 psig	79 Kg/cm2
200	675	93	47.5	Limited by Flange considereing flange as the weakest joint in piping system	
300	655	149	46		
400	635	204	44.6		
500	600	260	44.2		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and speciifed spectacle blind assemblies.
7. Weld Joint Factor for welded pipe shall be as per ASME B 31.3.

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**			Sheet No.	**2 of 2**
	Piping Class	Material	C.A	Spec.No	Revision:2
	B1 # 300	CS	3mm	2004A,Rev 9	Page No. 10
	Design Code:	**ASME B 31.3**			
	Service:	Vent Gas (VG), Blow Down(BD),Process Hydro Carbon Drain (DH),Nitrogen(N),Chemical (C), Water Injection(WI),Process Drain Water(DW),Diesel Fuel(DF), Hydrocarbon Liquid and Vapour with low CO2 AND H2S.			

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 42,SW				VG 62,FLANGED												
Rating	# 800				# 300,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB 44,SW				VB 64, Lever Operated			VB 66, Gear Operated									
Rating	# 800				# 300, RF												
Globe Valve																	
TAG	VGL 42,SW				VGL 62												
Rating	# 800				# 300, RF												
Check Valve																	
TAG	VC 42,SW,Lift Check				VC 62,Swing Check/ VC 83 (Wafer Check, Note -1)												
Rating	# 800				# 300, RF												
Needle Valve																	
TAG	VN 3,SW																
Rating	# 800																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. Wafer check valve to be avoided as far as possible; unless the avaliable space constraint does not allow normal check valve . The contractor to take approvel for use of wafer check Valve during pipe routing after extablishing the space constraint.

2. Wafer type Butter Fly Valve may be used only in fire water(Clean water) service.
3. PMS to be read in conjuction with FS 2004 A.
4. Maximum temperature limit for all Ball Valve shall be 121 Degree C

Revision:2	15.01.16	Issued For Bid	PS	AKM	SJ
Revision	Date	Description	Prepared By	Reviewed By	Approved By

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**													Sheet No.		**1 of 2**	
		Piping Class				Material				C.A		Spec.No			Revision:2		
		D1		# 600		CS				3mm		2004A,Rev 9			Page No.		11
		Design Code:						**ASME B 31.3**									
		Service: Vent Gas (VG), Blow Down(BD),Process Hydro Carbon Drain (DH),Nitrogen(N),Chemical (C), Water Injection(WI),Process Drain Water(DW),Diesel Fuel(DF), Hydrocarbon Liquid and Vapour with low CO2 AND H2S.															

Pipe Data	ASME B 36.10																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	XXS	XXS	XXS	XXS	160	80	80	80	80	80	80	80	80	80	80	80	80
WTmm	7.47	7.82	9.09	10.16	8.74	7.62	8.56	10.97	12.70	15.08	17.47	19.05	21.44	23.82	26.19	28.57	30.96
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5				
TYPE	Seamless													LSAW with 100% Radiography, Note-7			
MOC	ASTM A 106 Gr. B													API 5L Gr. B			
Ends	PE				BE												

Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Socket Weld				Butt Weld												
Rating	# 9000				160	80	80	80	80	80	80	80	80	80	80	80	80
MOC	ASTM A 105 U.N.O				ASTM A 234 Gr. WPB, Seamless U.N.O									ASTM A234 Gr.WPB,Welded U.N.O			
Elbow	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Tee	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Red.	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Cap	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Coupl	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Plug	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Union																	
Elbowlet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105									MSS SP 97,ASTM A 105			
Sockolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105									MSS SP 97,ASTM A 105			
Weldolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105									MSS SP 97,ASTM A 105			
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe									MOC Same as pipe			
Swage	MSS SP 95,Same as pipe				MSS SP 95,MOC Same as pipe									MSS SP 95,MOC Same as pipe			

Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				WN												
MOC	ASTM A 105				ASTM A 105												
FACE	RF-Smooth Finished				RF, Smooth Finished												
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25												

Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM A 105																
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48					

Mechanical Joints		
Stud Bolts	ASTM A 193 Gr. B7	ASTM A 193 Gr. B7
Hex Nuts	ASTM A 194 Gr. 2H	ASTM A 194 Gr. 2H
Gasket	B 16.20,4.5mm, SS Spiral wound with CNAF filler	ASME B 16.20,4.5mm, SS Spiral wound with CNAF filler

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
(-) 4 to 100	1480	(-) 20 to 38	104	2225 psig	157 Kg/cm2
200	1360	93	95.61	Limited by Flange considereing flange as the weakest joint in piping system	
300	1310	149	92.1		
400	1265	204	88.94		
500	1205	260	84.72		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2.All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange speciifed spectacle blind assemblies.
7. Weld Joint Factor for welded pipe shall be as per ASME B 31.3.

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**		Sheet No.	**2 of 2**
		Piping Class	Material	C.A	Spec.No	Revision:2
		D1	# 600	CS	3mm	2004A,Rev 9	Page No.	12
		Design Code:	**ASME B 31.3**					
		Service:	Vent Gas (VG), Blow Down(BD),Process Hydro Carbon Drain (DH),Nitrogen(N),Chemical (C), Water Injection(WI),Process Drain Water(DW),Diesel Fuel(DF), Hydrocarbon Liquid and Vapour with low CO2 AND H2S.					

Branch Table as per API RP 14E

Run Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	T	T	T	T	T	T
Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 42,SW				VG 66,FLANGED												
Rating	# 800				# 600,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB 44,SW				VB 76, Lever Operated			VB 77, Gear Operated									
Rating	# 800				# 600, RF												
Globe Valve																	
TAG	VGL 42,SW				VGL 67												
Rating	# 800				# 600, RF												
Check Valve																	
TAG	VC 42,SW,Lift Check				VC 66,Swing Check/ VC 86 (Wafer Check, Note -1)												
Rating	# 800				# 600, RF												
Needle Valve																	
TAG	VN 3,SW																
Rating	# 800																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. Wafer check valve to be avoided as far as possible; unless the avaliable space constraint does not allow normal check valve . The contractor to take approvel for use of wafer check Valve during pipe routing after extablishing the space constraint.
2. Wafer type Butter Fly Valve may be used only in fire water(Clean water) service.
3. PMS to be read in conjuction with FS 2004 A.
4. Maximum temperature limit for all Ball Valve shall be 121 Degree C

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision:2	15.01.16	Issued For Bid	PS	AKM	SJ

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**													Sheet No.	**1 of 2**		
	Piping Class				Material							C.A	Spec.No		Revision:2		
	E1				# 900				CS			3mm	2004A,Rev 9		Page No.	13	
	Design Code:				**ASME B 31.3**												
	Service:	Vent Gas (VG), Blow Down(BD),Process Hydro Carbon Drain (DH),Nitrogen(N),Chemical (C), Water Injection(WI),Process Drain Water(DW),Diesel Fuel(DF), Hydrocarbon Liquid and Vapour with low CO2 AND H2S.															

Pipe Data	ASME B 36.10																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14.00	16.00	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114	168.3	219	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	XXS	XXS	XXS	XXS	160	160	120	120	120	120	120	120	120	120	120	120	120
WTmm	7.47	7.82	9.09	10.16	8.7	11.12	11.1	14.27	18.3	21.44	25.40	27.79	30.96	34.92	38.10	41.27	46.02
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5				
TYPE	Seamless													LSAW with 100% Radiography, Note-10			
MOC	ASTM A 106 Gr. B													API 5L Gr. B			
Ends	PE				BE												
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Socket Weld				Butt Weld												
Rating	# 9000				160	160	120	120	120	120	120	120	120	120	120	120	120
MOC	ASTM A 105 U.N.O				ASTM A 234 Gr. WPB, Seamless U.N.O									ASTM A234 Gr.WPB,Welded U.N.O			
Elbow	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Tee	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Red.	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Cap	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Coupl	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Plug	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Union																	
Elbowlet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105									MSS SP 97,ASTM A 105			
Sockolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105									MSS SP 97,ASTM A 105			
Weldolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105									MSS SP 97,ASTM A 105			
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe									MOC Same as pipe			
Swage	MSS SP 95,Same as pipe				MSS SP 95,MOC Same as pipe									MSS SP 95,MOC Same as pipe			
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				WN												
MOC	ASTM A 105				ASTM A 105												
FACE	RTJ-Note 7				RTJ, Note 7												
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25												
Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM A 105																
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48					
Mechanical Joints																	
Stud Bolts	ASTM A 193 Gr. B7				ASTM A 193 Gr. B7												
Hex Nuts	ASTM A 194 Gr. 2H				ASTM A 194 Gr. 2H												
Gasket	B 16.20,OCT ring of Soft Iron with Max. 90BHN				ASME B 16.20,OCT ring of Soft Iron with Max. Hardness of 90BHN												

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
(-) 4 to 100	2220	(-) 20 to 38	156	3350 psig	236 Kg/cm2
200	2035	93	143.07	Limited by Flange considereing flange as the weakest joint in piping system	
300	1965	149	138.2		
400	1900	204	133.6		
500	1810	260	127.2		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Gasket Contact Surface shall have maximum roughness of 63AARH
7. Minimum RTJ groove hardness shall be 120BHN
8. Octagonal (OCT) ring shall be Hot Dip Galvanized/electro galvanized.
9. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange, WNRTJ flange 3" and above and specified spectacle blind assemblies.
10. Weld Joint Factor for welded pipe shall be as per ASME B 31.3.

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**			Sheet No.	**2 of 2**
	Piping Class	Material	C.A	Spec.No	Revision:2
	E1 # 900	CS	3mm	2004A,Rev 9	Page No. 14
	Design Code:	**ASME B 31.3**			
	Service:	Vent Gas (VG), Blow Down(BD),Process Hydro Carbon Drain (DH),Nitrogen(N),Chemical (C), Water Injection(WI),Process Drain Water(DW),Diesel Fuel(DF), Hydrocarbon Liquid and Vapour with low CO2 AND H2S.			

Branch Table as per API RP 14E

Run Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

Branch Pipe

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 46,SW				VG 76,FLANGED												
Rating	# 1500				# 900,RTJ												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	Use Gate Valve, VG 46				VB 84, L -			VB 85, Gear Operated									
Rating	# 1500				# 900, RTJ												
Globe Valve																	
TAG	VGL 44,SW				VGL 72												
Rating	# 1500				# 900, RTJ												
Check Valve																	
TAG	VC 44,SW,Lift Check				VC 68,Swing Check												
Rating	# 1500				# 900, RTJ												
Needle Valve																	
TAG	VN 4,SW																
Rating	# 1500																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.
2. Maximum temperature limit for all Ball Valve shall be 121 Degree C

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision:2	15.01.16	Issued For Bid	PS	AKM	SJ

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**1 of 2**
	Piping Class	Material	C.A	Spec.No	Revision:1	
	A2 # 150, Note-6	CS	1.5mm	2004A,Rev 9	Page No.	15
	Design Code:	**ASME B 31.3**				
	Service:	Glycol/ Therminol/Hot Oil				

Pipe Data	ASME B 36.10																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.60
Sch.	XS	XS	XS	XS	80	80	80	80	Std								
WTmm	3.73	3.91	4.55	5.08	5.54	7.62	8.56	10.97	8.18								
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5								
TYPE	Seamless																
MOC	ASTM A 106 Gr. B																
Ends	PE				BE												
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Socket Weld				Butt Weld												
Rating	# 3000				80	80	80	80	Std								
MOC	ASTM A 105 U.N.O				ASTM A 234 Gr. WPB, Seamless U.N.O												
Elbow	ASMEB16.11				ASME B 16.9												
Tee	ASMEB16.11				ASME B 16.9												
Red.	ASMEB16.11				ASME B 16.9												
Cap	ASMEB16.11				ASME B 16.9												
Coupl	ASMEB16.11				ASME B 16.9												
Plug	ASMEB16.11				ASME B 16.9												
Union																	
Elbowlet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105												
Sockolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105												
Weldolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105												
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe												
Swage	MSS SP 95,Same as pipe				MSS SP95,MOC Same as pipe												
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				WN												
MOC	ASTM A 105				ASTM A 105												
FACE	RF-smooth Finished				RF, smooth Finished												
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25												
Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM A 105																
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48					
Mechanical Joints																	
Stud Bolts	ASTM A 193 Gr. B7				ASTM A 193 Gr. B7												
Hex Nuts	ASTM A 194 Gr. 2H				ASTM A 194 Gr. 2H												
Gasket	16.20,4.5mm, SS Spiral wound with CNAF filler				ASME B 16.20,4.5mm, SS Spiral wound with CNAF filler												

Pressure-Temperature Rating

Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2
(-) 4 to 100	285	(-) 20 to 38	20
200	260	93	18.3
300	230	149	16.2
400	200	204	14.1
500	170	260	11.9
600	140	316	9.84
700	110	371	7.7

Maximum Hydrostatic Pressure

Pressure, psig	Pres.,Kg/cm2g
450 psig	31.6 Kg/cm2

Limited by Flange considereing flange as the weakest joint in piping system. For design temperaure higher than test temperature, Test pressure, (Pt) shall be calculated as follows:Pt=1.5xPx(St/S) where P=Internal Design Pressure,St=Allowable Stress at Test Temperature,S=Allowable Stress at Design Temperature.

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2.All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. All pipe thread shall be as per ASME B 1.20.1
6. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange
5. PMS to be read in conjuction with FS 2004 A.
6. To minimize leakage all flanges used under this PMS shall be of #300 instead of #150

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**			Sheet No.	**2 of 2**
	Piping Class	Material	C.A	Spec.No	Revision:1
	A2 # 150, Note-6	CS	1.5mm	2004A,Rev 9	Page No.16
	Design Code:	**ASME B 31.3**			
	Service:	Glycol/ Therminol/Hot Oil			
	0				

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 42,SW				Use Valve Indicated in B2 PMS												
Rating	# 800				# 300,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB 44,SW				Use Valve Indicated in B2 PMS												
Rating	# 800				# 300,RF												
Globe Valve																	
TAG	VGL 42,SW				Use Valve Indicated in B2 PMS												
Rating	# 800				# 300,RF												
Check Valve																	
TAG	VC 42,SW,Lift Check				Use Valve Indicated in B2 PMS												
Rating	# 800				# 300,RF												
Needle Valve																	
TAG	VN 107,SW																
Rating	# 800																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. Wafer check valve to be avoided as far as possible; unless the avaliable space constraint does not allow normal check valve . The contractor to take approvel for use of wafer check Valve during pipe routing after extablishing the space constraint.

3. PMS to be read in conjuction with FS 2004 A.

4. Maximum temperature limit for all Ball Valve shall be 121 Degree C. For high temperature service use special Valves

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision:1	15.01.16	Issued For Bid	PS	AKM	SJ

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**1 of 2**
	Piping Class		Material	C.A	Spec.No	Revision : 1
	B2	# 300	CS	1.5mm	2004A,Rev 9	Page No. 17
	Design Code:		**ASME B 31.3**			
	Service:		Glycol/ Therminol/Hot Oil			

Pipe Data	ASME B 36.10																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273	324	355.6	406.4	457.2	508	558.8	609.6	
Sch.	160	160	XS	XS	80	80	80	80	Std									
WTmm	4.77	5.56	4.55	5.08	5.54	7.62	8.56	10.97	8.18									
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5					
TYPE	Seamless																	
MOC	ASTM A 106 Gr. B																	
Ends	PE				BE													
Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	Socket Weld				Butt Weld													
Rating	# 6000		# 3000		80	80	80	80	Std									
MOC	ASTM A 105 U.N.O				ASTM A 234 Gr. WPB, Seamless U.N.O													
Elbow	ASMEB16.11				ASME B 16.9													
Tee	ASMEB16.11				ASME B 16.9													
Red.	ASMEB16.11				ASME B 16.9													
Cap	ASMEB16.11				ASME B 16.9													
Coupl	ASMEB16.11				ASME B 16.9													
Plug	ASMEB16.11				ASME B 16.9													
Union																		
Elbowlet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105													
Sockolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105													
Weldolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105													
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe													
Swage	MSS SP 95,Same as pipe				MSS SP95,MOC Same as pipe													
Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				WN													
MOC	ASTM A 105				ASTM A 105													
FACE	RF-Serrated Finished				RF, Serrated Finished													
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25													
Spectacle Blind/Spacer Blinds																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
MOC	ASTM A 105																	
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48						
Mechanical Joints																		
Stud Bolts	ASTM A 193 Gr. B7				ASTM A 193 Gr. B7													
Hex Nuts	ASTM A 194 Gr. 2H				ASTM A 194 Gr. 2H													
Gasket	16.20,4.5mm, SS Spiral wound with CNAF filler				ASME B 16.20,4.5mm, SS Spiral wound with CNAF filler													

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
(-) 4 to 100	740	(-) 20 to 38	52	1125 psig	79 Kg/cm2
200	675	93	47.5	Limited by Flange considereing flange as the weakest joint in piping system. For design temperaure higher than test temperature, Test pressure, (Pt) shall be calculated as follows:Pt=1.5xPx(St/S) where P=Internal Design Pressure,St=Allowable Stress at Test Temperature,S=Allowable Stress at Design Temperature.	
300	655	149	46		
400	635	204	44.6		
500	600	260	44.2		
600	550	316	38.7		
700	535	371	37.6		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION				Sheet No.	2 of 2
	Piping Class		Material	C.A	Spec.No	Revision : 1
	B2	# 300	CS	1.5mm	2004A,Rev 9	Page No. 18
	Design Code:		**ASME B 31.3**			
	Service:	Glycol/ Therminol/Hot Oil				
	0					

Branch Table as per API RP 14E

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG		VG 137,SW			VG 77,FLANGED												
Rating		# 800			# 300,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG		VB 137,SW			VB 177 (Note-5)												
Rating		# 800			# 300, RF												
Globe Valve																	
TAG		VGL 137,SW			VGL 177												
Rating		# 800			# 300, RF												
Check Valve																	
TAG		VC 137,SW,Lift Check			VC 177,Swing Check												
Rating		# 800			# 300, RF												
Needle Valve																	
TAG		VN 107,SW															
Rating		# 800															
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. Wafer check valve to be avoided as far as possible; unless the avaliable space constraint does not allow normal check valve . The contractor to take approvel for use of wafer check Valve during pipe routing after extablishing the space constraint.
3. PMS to be read in conjuction with FS 2004 A.
4. Maximum temperature limit for all Ball Valve shall be 371 Degree C
5. Type of Valve metal seated or soft seated shall be firmed up based on service temperature.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision : 1	15.01.16	Issued For Bid	PS	AKM	SJ

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**			Sheet No.	**1 of 2**
	Piping Class	Material	C.A	Spec.No	Revision: 1
	D2 # 600	CS	1.5mm	2004A,Rev 9	Page No. 19
	Design Code:	**ASME B 31.3**			
	Service:	Glycol/ Therminol/Hot Oil			

Pipe Data	ASME B 36.10															
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	160	160	160	160	80	80	80	80	80								
WTmm	4.77	5.56	6.35	7.14	5.54	7.62	8.56	10.97	12.7								
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5				
TYPE	Seamless																
MOC	ASTM A 106 Gr. B																
Ends	PE				BE												

Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Socket Weld				Butt Weld												
Rating	# 6000				80	80	80	80	80								
MOC	ASTM A 105 U.N.O				ASTM A 234 Gr. WPB, Seamless U.N.O												
Elbow	ASMEB16.11				ASME B 16.9												
Tee	ASMEB16.11				ASME B 16.9												
Red.	ASMEB16.11				ASME B 16.9												
Cap	ASMEB16.11				ASME B 16.9												
Coupl	ASMEB16.11				ASME B 16.9												
Plug	ASMEB16.11				ASME B 16.9												
Union																	
Elbowlet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105												
Sockolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105												
Weldolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105												
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe												
Swage	MSS SP 95,Same as pipe				MSS SP 95,MOC Same as pipe												

Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				WN												
MOC	ASTM A 105				ASTM A 105												
FACE	RF-Smooth Finished				RF, Smooth Finished												
STD.	ASME B 16.5				B 16.5, But welding ends as per B 16.25												

Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM A 105																
Spectacle	ASME B 16.48				ASME B 16.48												

Mechanical Joints		
Stud Bolt	ASTM A 193 Gr. B7	ASTM A 193 Gr. B7
Hex Nuts	ASTM A 194 Gr. 2H	ASTM A 194 Gr. 2H
Gasket	B 16.20,4.5mm, SS Spiral wound with CNAF filler	B 16.20,4.5mm, SS Spiral wound CNAF filler

Pressure-Temperature Rating

Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2
(-) 4 to 100	1480	(-) 20 to 38	104
200	1360	93	96
300	1310	149	92
400	1265	204	89
500	1205	260	85
600	1135	316	78
700	1060	371	75

Maximum Hydrostatic Pressure

Pressure, psig	Pres.,Kg/cm2g
2225 psig	157 Kg/cm2

Limited by Flange considereing flange as the weakest joint in piping system. For design temperaure higher than test temperature, Test pressure, (Pt) shall be calculated as follows:Pt=1.5xPx(St/S) where P=Internal Design Pressure,St=Allowable Stress at Test Temperature,S=Allowable Stress at Design Temperature.

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2.All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**			Sheet No.	**2 of 2**
	Piping Class	Material	C.A	Spec.No	Revision: 1
	D2 # 600	CS	1.5mm	2004A,Rev 9	Page No. 20
	Design Code:	**ASME B 31.3**			
	Service:	Glycol/ Therminol/Hot Oil			
	0				

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 137,SW				VG 66,FLANGED												
Rating	# 800				# 600,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB 137,SW				VB 76,FLANGED (Note-4)												
Rating	# 800				# 600,RF												
Globe Valve																	
TAG	VGL 137,SW				VGL 67,FLANGED												
Rating	# 800				# 600,RF												
Check Valve																	
TAG	VC 137,SW,Lift Check				VC 66,Swing Check												
Rating	# 800				# 600, RF												
Needle Valve																	
TAG	VN 107,SW																
Rating	# 800																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. Wafer check valve to be avoided as far as possible; unless the avaliable space constraint does not allow normal check valve . The contractor to take approvel for use of wafer check Valve during pipe routing after extablishing the space constraint.
2. PMS to be read in conjuction with FS 2004 A.
3. Maximum temperature limit for all Ball Valve shall be 371 Degree C
4. Type of Valve metal seated or soft seated shall be firmed up based on service temperature.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 1	15.01.16	Issued For Bid	PS	AKM	SJ

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION			Sheet No.	1 of 2
	Piping Class	Material	C.A	Spec.No	Revision: 0
	E2 # 900	CS	1.5mm	2004A,Rev 9	Page No. 21
	Design Code:	**ASME B 31.3**			
	Service:	Glycol/ Therminol/Hot Oil			

Pipe Data	ASME B 36.10																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273	323.9	356	406	457	508	558.8	609.6	
Sch.	XXS	XXS	XXS	XXS	160	160	120	120	120									
WTmm	7.47	7.82	9.09	10.16	8.74	11.12	11.12	14.27	18.26									
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5					
TYPE	Seamless																	
MOC	ASTM A 106 Gr. B																	
Ends	PE				BE													
Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	Socket Weld				Butt Weld													
Rating	# 9000				160	160	120	120	120									
MOC	ASTM A 105 U.N.O				ASTM A 234 Gr. WPB, Seamless U.N.O													
Elbow	ASMEB16.11				ASME B 16.9													
Tee	ASMEB16.11				ASME B 16.9													
Red.	ASMEB16.11				ASME B 16.9													
Cap	ASMEB16.11				ASME B 16.9													
Coupl	ASMEB16.11				ASME B 16.9													
Plug	ASMEB16.11				ASME B 16.9													
Union																		
Elbowlet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105													
Sockolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105													
Weldolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105													
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe													
Swage	MSS SP 95,Same as pipe				MSS SP 95,MOC Same as pipe													
Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				WN													
MOC	ASTM A 105				ASTM A 105													
FACE	RTJ-Note 7				RTJ-Note 7													
STD.	ASME B 16.5				B 16.5, But welding ends as per B 16.25													
Spectacle Blind/Spacer Blinds																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
MOC	ASTM A 105																	
Spectacle	ASME B 16.48				ASME B 16.48													
Mechanical Joints																		
Stud Bolt	ASTM A 193 Gr. B7				ASTM A 193 Gr. B7													
Hex Nuts	ASTM A 194 Gr. 2H				ASTM A 194 Gr. 2H													
Gasket	B 16.20,OCT ring of Soft Iron, Note-7				B 16.20,OCT ring of Soft Iron, Note-7													

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
(-) 4 to 100	2220	(-) 20 to 38	156	3350 psig	236 Kg/cm2
200	2035	93	143.07	Limited by Flange considereing flange as the weakest joint in piping system. For design temperaure higher than test temperature, Test pressure, (Pt) shall be calculated as follows:Pt=1.5xPx(St/S) where P=Internal Design Pressure,St=Allowable Stress at Test Temperature,S=Allowable Stress at Design Temperature.	
300	1965	149	138.2		
400	1900	204	133.6		
500	1810	260	127.2		
600	1705	316	119.9		
700	1590	371	111.8		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange
7. Minimum RTJ groove hardness shall be more than RTJ ring hardness and Octagonal (OCT) ring shall be Hot Dip Galvanized/electro galvanized.
8. Gasket Contact Surface shall have maximum roughness of 63AARH

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**			Sheet No.	**2 of 2**
	Piping Class	Material	C.A	Spec.No	Revision: 0
	E2 # 900	CS	1.5mm	2004A,Rev 9	Page No. 22
	Design Code:	**ASME B 31.3**			
	Service:	Glycol/ Therminol/Hot Oil			
	0				

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 46,SW				VG 76,FLANGED												
Rating	# 1500				# 900,RTJ												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	Use Gate Valve, VG 46				VB 84, L-Opted		VB 85, Gear Optd (Note-4)										
Rating	# 1500				# 900, RTJ												
Globe Valve																	
TAG	VGL 44,SW				VGL 72												
Rating	# 1500				# 900, RTJ												
Check Valve																	
TAG	VC 44,SW,Lift Check				VC 68,Swing Check												
Rating	# 1500				# 900, RTJ												
Needle Valve																	
TAG	VN 4,SW																
Rating	# 1500																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. Wafer check valve to be avoided as far as possible; unless the avaliable space constraint does not allow normal check valve . The contractor to take approvel for use of wafer check Valve during pipe routing after extablishing the space constraint.
2. PMS to be read in conjuction with FS 2004 A.
3. Maximum temperature limit for all Ball Valve shall be 371 Degree C
4. Type of Valve metal seated or soft seated shall be firmed up based on service temperature.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 0	15.01.16	Issued For Bid	PS	AKM	SJ

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**1 of 2**
		Piping Class		Material	C.A	Spec.No	Revision: 2
		A1N	# 150	CS-NACE	6mm	2004A,Rev 9	Page No. 23
		Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
		Service: Corrosive HydroCarbon Service-Sour Liquid and Vapour.					
		Instrument Gas Header including 1st Bolck Valve					

Pipe Data	ASME B 36.10																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	XXS	XXS	XXS	XXS	160	160	80	80	60	40	40	40	40	40	40	40	40
WTmm	7.47	7.82	9.09	10.16	8.74	11.13	8.56	10.97	10.31	9.27	10.31	11.12	12.70	14.27	15.08	15.87	17.48
M.T (%)	12.5	12.5	11.0	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5				
TYPE	Seamless													LSAW with 100% Radiography			
MOC	ASTM A 106 Gr. B													API 5L Gr. B			
Ends	PE				BE												
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Socket Weld				Butt Weld												
Rating	# 9000				160	160	80	80	60	40	40	40	40	40	40	40	40
MOC	ASTM A 105 U.N.O				ASTM A 234 Gr. WPB, Seamless U.N.O									ASTM A234 Gr.WPB,Welded U.N.O			
Elbow	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Tee	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Red.	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Cap	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Coupl	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Plug	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Union																	
Elbowlet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105									MSS SP 97,ASTM A 105			
Sockolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105									MSS SP 97,ASTM A 105			
Weldolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105									MSS SP 97,ASTM A 105			
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe									MOC Same as pipe			
Swage	MSS SP 95,Same as pipe				MSS SP 95,MOC Same as pipe									MSS SP 95,MOC Same as pipe			
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				WN												
MOC	ASTM A 105				ASTM A 105												
FACE	RF-Serrated Finished				RF, Serrated Finished												
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25												
Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM A 105																
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48					
Mechanical Joints																	
Stud Bolts	ASTM A 193 Gr. B7M				ASTM A 193 Gr. B7M												
Hex Nuts	ASTM A 194 Gr. 2HM				ASTM A 194 Gr. 2HM												
Gasket	ASME B 16.21,Flat Ring,1.58mm,CNAF				ASME B 16.21,Flat Ring,1.58mm,CNAF												

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
(-) 4 to 100	285	(-) 20 to 38	20	450 psig	31.6 Kg/cm2
200	260	93	18.3	Limited by Flange considereing flange as the weakest joint in piping system	
300	230	149	16.2		
400	200	204	14.1		
500	170	260	11.9		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. Weld Joint Factor for welded pipe shall be as per ASME B 31.3
6. PMS to be read in conjuction with FS 2004 A and all the requirement of MR-01-75/ISO-15156-1/2/3 to complied.
7. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and specified spectacle blinds.
8. In instrument gas lines - Use this specification for main Header and SS316L piping for Sub Header. Presssure-temperature rating of SS316L shall compatible to this piping class and both shall comply all the requirement of MR-01-75/ISO-15156-1/2/3.
9. In cae calculated thickness is more than thickness of schedule of XXS for small bore piping (1/2"), reduce MT and use XXS only.

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**							Sheet No.	**2 of 2**								
		Piping Class				Material		C.A	Spec.No							Revision: 2		
		A1N		# 150		CS-NACE		6mm	2004A,Rev 9							Page No.	24	
		Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3																
		Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.															
		Instrument Gas Header including 1st Bolck Valve																

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 45,SW				VG 65,FLANGED												
Rating	# 800				# 150,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB 45,SW and VB 107,SW (Note-5)				VB 65, Lever Operated							VB **, Gear Operated					
Rating	# 800				# 150, RF												
Globe Valve																	
TAG	VGL 45,SW				VGL 65												
Rating	# 800				# 150, RF												
Check Valve																	
TAG	VC 45,SW,Lift Check				VC 65,Swing Check/ VC 82 (Wafer Check, Note -1)												
Rating	# 800				# 150, RF												
Needle Valve																	
TAG	VN 9,SW																
Rating	# 800																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. Wafer check valve to be avoided as far as possible; unless the avaliable space constraint does not allow normal check valve . The contractor to take approvel for use of wafer check Valve during pipe routing after extablishing the space constraint.
2. Wafer type Butter Fly Valve may be used only in fire water(Clean water) service.
3. PMS to be read in conjuction with FS 2004 A.
4. Maximum temperature limit for all Ball Valve shall be 121 Degree C
5. In instrument gas lines - Use VB -45 for main Header and VB-107 for Sub Header. The MOC of VB-107 shall be SS-316L as per VMS and comply all the requirement of MR-01-75/ISO-15156-1/2/3.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 2	15.01.16	Issued For Bid	PS	AKM	SJ

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**1 of 2**
	Piping Class	Material	C.A	Spec.No		Revision: 2
	B1N	# 300	CS-NACE	6mm	2004A,Rev 9	Page No. 25
	Design Code:	**ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3**				
	Service	Corrosive HydroCarbon Service-Sour Liquid and Vapour.				

Pipe Data	ASME B 36.10																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	XXS	XXS	XXS	XXS	160	160	120	XS	XS	60	60	60	60	60	60	60	60
WTmm	7.47	7.82	9.09	10.16	8.74	11.12	11.12	10.97	12.70	12.70	14.27	15.09	16.66	19.05	20.62	22.22	24.61
M.T (%)	12.5	12.5	5.0	5.0	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5				
TYPE	Seamless													LSAW with 100% Radiography			
MOC	ASTM A 106 Gr. B													API 5L Gr. B			
Ends	PE				BE												
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Socket Weld				Butt Weld												
Rating	# 9000				160	160	120	XS	XS	60	60	60	60	60	60	60	60
MOC	ASTM A 105 U.N.O				ASTM A 234 Gr. WPB, Seamless U.N.O									ASTM A234 Gr.WPB,Welded U.N.O			
Elbow	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Tee	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Red.	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Cap	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Coupl	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Plug	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Union																	
Elbowlet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105									MSS SP 97,ASTM A 105			
Sockolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105									MSS SP 97,ASTM A 105			
Weldolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105									MSS SP 97,ASTM A 105			
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe									MOC Same as pipe			
Swage	MSS SP 95,Same as pipe				MSS SP 95,MOC Same as pipe									MSS SP 95 or as applicable,MOC Same as pipe			
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				WN												
MOC	ASTM A 105				ASTM A 105												
FACE	RF-Smooth Finished				RF, Smooth Finished												
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25												
Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM A 105																
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48					
Mechanical Joints																	
Stud Bolts	ASTM A 193 Gr. B7M				ASTM A 193 Gr. B7M												
Hex Nuts	ASTM A 194 Gr. 2HM				ASTM A 194 Gr. 2HM												
Gasket	B 16.20,4.5mm, SS Spiral wound with CNAF filler				ASME B 16.20,4.5mm, SS Spiral wound with CNAF filler												

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
(-) 4 to 100	740	(-) 20 to 38	52	1125 psig	79 Kg/cm2
200	675	93	47.5	Limited by Flange considereing flange as the weakest joint in piping system	
300	655	149	46		
400	635	204	44.6		
500	600	260	44.2		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. Weld Joint Factor for welded pipe shall be as per ASME B 31.3
6. PMS to be read in conjuction with FS 2004 A and all the requirement of MR-01-75/ISO-15156-1/2/3 to complied.
7. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and specified spectacle blinds.
8. In instrument gas lines - Use this specification for main Header and SS316L piping for Sub Header. Presssure-temperature rating of SS316L shall compatible to this piping class and both shall comply all the requirement of MR-01-75/ISO-15156-1/2/3.
9. In cae calculated thickness is more than thickness of schedule of XXS for small bore piping (1/2"), reduce MT and use XXS only.

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**2 of 2**
	Piping Class		Material	C.A	Spec.No	Revision: 2
	B1N	# 300	CS-NACE	6mm	2004A,Rev 9	Page No. 26
	Design Code:	**ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3**				
	Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.				
	0					

Branch Table as per API RP 14E

Run Pipe	Branch Pipe															
	≤ 1	T														
	1-1.5	T	T													
	2	*	*	T												
	3	*	*	T	T											
	4	*	*	T	T	T										
	6	*	*	W	T	T	T									
	8	*	*	W	W	T	T	T								
	10	*	*	W	W	W	T	T	T							
	12	*	*	W	W	W	T	T	T	T						
	14	*	*	W	W	W	W	T	T	T	T					
	16	*	*	W	W	W	W	T	T	T	T	T				
	18	*	*	W	W	W	W	W	T	T	T	T	T			
	20	*	*	W	W	W	W	W	T	T	T	T	T	T		
	24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
	30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T
		≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 45,SW				VG 69,FLANGED												
Rating	# 800				# 300,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB 45,SW and VB 107,SW (Note-5)				VB 69, Lever Operated							VB **, Gear Operated					
Rating	# 800				# 300, RF												
Globe Valve																	
TAG	VGL 45,SW				VGL 69												
Rating	# 800				# 300, RF												
Check Valve																	
TAG	VC 45,SW,Lift Check				VC 69,Swing Check/ VC 89 (Wafer Check, Note -1)												
Rating	# 800				# 300, RF												
Needle Valve																	
TAG	VN 9,SW																
Rating	# 800																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. Wafer check valve to be avoided as far as possible; unless the avaliable space constraint does not allow normal check valve . The contractor to take approvel for use of wafer check Valve during pipe routing after extablishing the space constraint.
2. Wafer type Butter Fly Valve may be used only in fire water(Clean water) service.
3. PMS to be read in conjuction with FS 2004 A.
4. Maximum temperature limit for all Ball Valve shall be 121 Degree C
5. In instrument gas lines - Use VB -45 for main Header and VB-107 for Sub Header. The MOC of VB-107 shall be SS 316L as per VMS and comply all the requirement of MR-01-75/ISO-15156-1/2/3.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 2	15.01.16	Issued For Bid	PS	AKM	SJ

	Offshore Design Section Engineering Services IS0-9001:2000				**PIPING MATERIAL SPECIFICATION**									Sheet No.		**1 of 2**	
					Piping Class			Material			C.A	Spec.No				Revision: 2	
					D1N		# 600	CS-NACE			6mm	2004A,Rev 9			Page No.	27	
					Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3												
					Service: Corrosive HydroCarbon Service-Sour Liquid and Vapour.												
					Instrument Gas Header including 1st Bolck Valve												
Pipe Data	ASME B 36.10																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	XXS	XXS	XXS	XXS	XXS	160	160	120	120	100	100	100	100	100	100	100	100
WTmm	7.47	7.82	9.09	10.16	11.07	11.12	13.49	14.27	18.26	18.26	21.44	23.82	26.19	29.36	32.54	34.92	38.89
M.T (%)	12.5	12.5	5.0	6.0	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5				
TYPE	Seamless													LSAW with 100% Radiography			
MOC	ASTM A 106 Gr. B													API 5L Gr. B			
Ends	PE				BE												
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	XXS	160	160	120	120	100	100	100	100	18	20	22	24
TYPE	Socket Weld				Butt Weld												
Rating	# 9000				XXS	160	160	120	120	120	120	100	100	100	100	100	100
MOC	ASTM A 105 U.N.O				ASTM A 234 Gr. WPB, Seamless U.N.O									ASTM A234 Gr.WPB,Welded U.N.O			
Elbow	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Tee	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Red.	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Cap	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Coupl	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Plug	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Union																	
Elbowlet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105									MSS SP 97,ASTM A 105			
Sockolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105									MSS SP 97,ASTM A 105			
Weldolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105									MSS SP 97,ASTM A 105			
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe									MOC Same as pipe			
Swage	MSS SP 95,Same as pipe				MSS SP 95,MOC Same as pipe									MSS SP 95,MOC Same as pipe			
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				WN												
MOC	ASTM A 105				ASTM A 105												
FACE	RF-Smooth Finished				RF, Smooth Finished												
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25												
Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM A 105																
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48					
Mechanical Joints																	
Stud Bolts	ASTM A 193 Gr. B7M				ASTM A 193 Gr. B7M												
Hex Nuts	ASTM A 194 Gr. 2HM				ASTM A 194 Gr. 2HM												
Gasket	B 16.20,4.5mm, SS Spiral wound with CNAF filler				ASME B 16.20,4.5mm, SS Spiral wound with CNAF filler												

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
(-) 4 to 100	1480	(-) 20 to 38	104	2225 psig	157 Kg/cm2
200	1360	93	95.61	Limited by Flange considereing flange as the weakest joint in piping system	
300	1310	149	92.1		
400	1265	204	88.94		
500	1205	260	84.72		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. Weld Joint Factor for welded pipe shall be as per ASME B 31.3
5. All pipe thread shall be as per ASME B 1.20.1
6. PMS to be read in conjuction with FS 2004 A and all the requirement of MR-01-75/ISO-15156-1/2/3 to complied.
7. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and specified spectacle blinds.
8. In instrument gas lines - Use this specification for main Header and SS316L piping for Sub Header. Presssure-temperature rating of SS316L shall compatible to this piping class and both shall comply all the requirement of MR-01-75/ISO-15156-1/2/3.
9. In cae calculated thickness is more than thickness of schedule of XXS for small bore piping (1/2"), reduce MT and use XXS only.

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**2 of 2**
	Piping Class		Material	C.A	Spec.No	Revision: 2
	D1N	# 600	CS-NACE	6mm	2004A,Rev 9	Page No. 28
	Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
	Service: Corrosive HydroCarbon Service-Sour Liquid and Vapour.					

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 45,SW				VG 71,FLANGED												
Rating	# 800				# 600,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	B 45,SW and VB 107,SW (Note-				VB 71, Lever Operated							VB **, Gear Operated					
Rating	# 800				# 600, RF												
Globe Valve																	
TAG	VGL 45,SW				VGL 71												
Rating	# 800				# 600, RF												
Check Valve																	
TAG	VC 45,SW,Lift Check				VC 71,Swing Check												
Rating	# 800				# 600, RF												
Needle Valve																	
TAG	VN 9,SW																
Rating	# 800																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. Wafer check valve to be avoided as far as possible; unless the avaliable space constraint does not allow normal check valve . The contractor to take approvel for use of wafer check Valve during pipe routing after exstablishing the space constraint.
2. PMS to be read in conjuction with FS 2004 A.
3. Maximum temperature limit for all Ball Valve shall be 121 Degree C
4. In instrument gas lines - Use VB -45 for main Header and VB-107 for Sub Header. The MOC of VB-107 shall be SS 316L as per VMS and comply all the requirement of MR-01-75/ISO-15156-1/2/3.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 2	15.01.16	Issued For Bid	PS	AKM	SJ

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**												Sheet No.		**1 of 2**	
		Piping Class		Material		C.A		Spec.No						Revision: 2			
		E1N	# 900	CS-NACE		6mm		2004A,Rev 9						Page No.	29		
		Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3															
		Service Corrosive HydroCarbon Service-Sour Liquid and Vapour.															

Pipe Data	ASME B 36.10																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6	
Sch.	XXS	XXS	XXS	XXS	XXS	XXS	XXS	160	140	140	140	140	140	120	120	120	120	
WTmm	7.47	7.82	9.09	10.16	11.07	15.24	17.12	18.26	20.63	25.40	28.57	31.75	36.53	34.92	38.10	41.27	46.02	
M.T (%)	12.5	12.5	5.0	6.0	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5					
TYPE	Seamless													LSAW with 100% Radiography				
MOC	ASTM A 106 Gr. B													API 5L Gr. B				
Ends	PE				BE													
Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	Socket Weld				Butt Weld													
Rating	# 9000				XXS	XXS	XXS	160	140	140	140	140	140	120	120	120	120	
MOC	ASTM A 105 U.N.O				ASTM A 234 Gr. WPB, Seamless U.N.O									ASTM A234 Gr.WPB,Welded U.N.O				
Elbow	ASMEB16.11				ASME B 16.9									ASME B 16.9				
Tee	ASMEB16.11				ASME B 16.9									ASME B 16.9				
Red.	ASMEB16.11				ASME B 16.9									ASME B 16.9				
Cap	ASMEB16.11				ASME B 16.9									ASME B 16.9				
Coupl	ASMEB16.11				ASME B 16.9									ASME B 16.9				
Plug	ASMEB16.11				ASME B 16.9									ASME B 16.9				
Union																		
Elbowlet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105									MSS SP 97,ASTM A 105				
Sockolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105									MSS SP 97,ASTM A 105				
Weldolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105									MSS SP 97,ASTM A 105				
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe									MOC Same as pipe				
Swage	MSS SP 95,Same as pipe				MSS SP 95,MOC Same as pipe									MSS SP 95,MOC Same as pipe				
Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				WN													
MOC	ASTM A 105				ASTM A 105													
FACE	RTJ-Note 9				RTJ, Note 9													
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25													
Spectacle Blind/Spacer Blinds																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
MOC	ASTM A 105																	
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48						
Mechanical Joints																		
Stud Bolts	ASTM A 193 Gr. B7M				ASTM A 193 Gr. B7M													
Hex Nuts	ASTM A 194 Gr. 2HM				ASTM A 194 Gr. 2HM													
Gasket	B 16.20,OCT ring of Soft Iron with Max. 90BHN				ASME B 16.20,OCT ring of Soft Iron with Max. Hardness of 90BHN													

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
(-) 4 to 100	2220	(-) 20 to 38	156	3350 psig	236 Kg/cm2
200	2035	93	143.07	Limited by Flange considereing flange as the weakest joint in piping system	
300	1965	149	138.2		
400	1900	204	133.6		
500	1810	260	127.2		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and WNRTJ flanges above 3" size and specified spectacle blind assemblies.
7. Weld Joint Factor for welded pipe shall be as per ASME B 31.3
8. Gasket Contact Surface shall have maximum roughness of 63AARH
9. Minimum RTJ groove hardness shall be 120BHN
10. Octagonal (OCT) ring shall be Hot Dip Galvanized or electrogalvaized.
11. In cae calculated thickness is more than thickness of schedule of XXS for small bore piping (1/2"), reduce MT and use XXS only.

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**											Sheet No.		**2 of 2**		
		Piping Class		Material						C.A		Spec.No			Revision: 2		
		E1N	# 900	CS-NACE						6mm		2004A,Rev 9		Page No.	30		
		Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3															
		Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.														
		0															

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	T	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG		VG 47,SW			VG 73,FLANGED												
Rating		# 1500			# 900,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG		Use Gate Valve, VG 47			VB 73, Lever Operated							VB **, Gear Operated					
Rating		# 1500			# 900, RTJ												
Globe Valve																	
TAG		VGL 47,SW			VGL 73												
Rating		# 1500			# 900, RTJ												
Check Valve																	
TAG		VC 47,SW,Lift Check			VC 73,Swing Check												
Rating		# 1500			# 900, RTJ												
Needle Valve																	
TAG		VN 10,SW															
Rating		# 1500															
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.
2. Maximum temperature limit for all Ball Valve shall be 121 Degree C

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 2	15.01.16	Issued For Bid	PS	AKM	SJ

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**												Sheet No.		**1 of 2**		
		Piping Class			Material			C.A	Spec.No					Revision: 0				
		F1N		# 1500	CS-NACE			6mm	2004A,Rev 9					Page No.	31			
		Design Code:		**ASME B 31.3 and other applicable Code**					**NACE-MR-01-75/ISO-15156-1/2/3**									
		Service		Corrosive HydroCarbon Service-Sour Liquid and Vapour.														

Pipe Data	ASME B 36.10																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114	168	219	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6	
Sch.	XXS	XXS	XXS	XXS	XXS	XXS	XXS	XXS	160	160	160	160	160	160	160	160	160	
WTmm	7.47	7.82	9.09	10.16	11.07	15.24	17.12	21.95	23.01	28.57	33.32	35.71	40.49	45.24	50.01	53.97	59.54	
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	7.0	12.0	12.5	12.5	12.5					
TYPE	Seamless													LSAW with 100% Radiography				
MOC	API 5L Gr, X60 PSL-2													API 5L Gr, X60 PSL-2				
Ends	PE				BE													
Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	Socket Weld				Butt Weld													
Rating	# 9000				XXS	XXS	XXS	XXS	XXS	XXS	160	160	160	160	160	160	160	
MOC	ASTM A 694 F60				ASTM A 860 WPHY 60, Seamless U.N.O									ASTM A860 WPHY 60,Welded U.N.O				
Elbow	ASMEB16.11				ASME B 16.9									ASME B 16.9				
Tee	ASMEB16.11				ASME B 16.9									ASME B 16.9				
Red.	ASMEB16.11				ASME B 16.9									ASME B 16.9				
Cap	ASMEB16.11				ASME B 16.9									ASME B 16.9				
Coupl	ASMEB16.11				ASME B 16.9									ASME B 16.9				
Plug	ASMEB16.11				ASME B 16.9									ASME B 16.9				
Union																		
Elbowlet	MSS SP 97,ASTM A 694 F60				MSS SP 97,ASTM A 694 F60									MSS SP 97,ASTM A 694 F60				
Sockolet	MSS SP 97,ASTM A 694 F60				MSS SP 97,ASTM A 694 F60									MSS SP 97,ASTM A 694 F60				
Weldolet	MSS SP 97,ASTM A 694 F60				MSS SP 97,ASTM A 694 F60									MSS SP 97,ASTM A 694 F60				
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe									MOC Same as pipe				
Swage	MSS SP 95,Same as pipe				MSS SP 95,MOC Same as pipe									MSS SP 95,MOC Same as pipe				
Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				WN													
MOC	ASTM A 694 F60				ASTM A 694 F60													
FACE	RTJ-Note 7				RTJ, Note 7													
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25													
Spectacle Blind/Spacer Blinds																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
MOC	ASTM A 694 F60																	
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48						
Mechanical Joints																		
Stud Bolt	ASTM A 193 Gr. B7M				ASTM A 193 Gr. B7M													
Hex Nuts	ASTM A 194 Gr. 2HM				ASTM A 194 Gr. 2HM													
Gasket	B 16.20,OCT ring of Soft Iron with Max. 90BHN				ASME B 16.20,OCT ring of Soft Iron with Max. Hardness of 90BHN													

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
(-) 4 to 100	3702.81	(-) 20 to 38	260.33	5554.22 psig	390.50 Kg/cm2
122	3634.65	50	255.54	Limited by Flange considereing flange as the weakest joint in piping system	
212	3379.38	100	237.6		
302	3269.15	150	229.85		
392	3176.33	200	223.32		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and WNRTJ flanges above 3" size and specified spectacle
7. Weld Joint Factor for welded pipe shall be as per ASME B 31.3
8. Gasket Contact Surface shall have maximum roughness of 63AARH
9. Minimum RTJ groove hardness shall be 120BHN
10. Octagonal (OCT) ring shall be Hot Dip Galvanized or electrogalvaized.
11. In cae calculated thickness is more than thickness of schedule of XXS for small bore piping (1/2"), reduce MT and use XXS only.

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION				Sheet No.	2 of 2
	Piping Class	Material	C.A	Spec.No	Revision: 0	
	F1N	# 1500	CS-NACE	6mm	2004A,Rev 9	Page No. 32
	Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
	Service	Corrosive HydroCarbon Service-Sour Liquid and Vapour.				
	0					

Branch Table as per API RP 14E

Run Pipe		Branch Pipe														
		≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
	≤ 1	T														
	1-1.5	T	T													
	2	*	*	T												
	3	*	*	T	T											
	4	*	*	T	T	T										
	6	*	*	W	T	T	T									
	8	*	*	W	W	T	T	T								
	10	*	*	W	W	W	T	T	T							
	12	*	*	W	W	W	T	T	T	T						
	14	*	*	W	W	W	W	T	T	T	T					
	16	*	*	W	W	W	W	T	T	T	T	T				
	18	*	*	W	W	W	W	W	T	T	T	T	T			
	20	*	*	W	W	W	W	W	T	T	T	T	T	T		
	24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
	30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 67,SW				VG 273,FLANGED												
Rating	# 2500				# 1500,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	Use Gate Valve, VG 67				VB 273, Lever Operated							VB **, Gear Operated					
Rating	# 2500				# 1500, RTJ												
Globe Valve																	
TAG	VGL 67,SW				VGL 273												
Rating	# 2500				# 1500, RTJ												
Check Valve																	
TAG	VC 67,SW,Lift Check				VC 273,Swing Check												
Rating	# 2500				# 1500, RTJ												
Needle Valve																	
TAG	VN 20,SW																
Rating	# 2500																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.
2. Maximum temperature limit for all Ball Valve shall be 121 Degree C

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 0	15.01.16	Issued For Bid	PS	AKM	SJ

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**1 of 2**
	Piping Class	Material	C.A	Spec.No	Revision: 0	
	G1N # 2500	CS-NACE	6mm	2004A,Rev 9	Page No.	33
	Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
	Service Corrosive HydroCarbon Service-Sour Liquid and Vapour.					

Pipe Data	ASME B 36.10																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114	168	219	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	XXS	XXS	XXS	XXS	XXS	XXS	XXS	XXS	XXS	XXS	160	160	160	160	160	160	160
WTmm	7.47	7.82	9.09	10.16	11.07	15.24	17.12	21.95	22.22	25.40	33.32	35.71	40.40	45.24	50.01	53.97	59.54
M.T (%)	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0									
TYPE	Seamless													LSAW with 100% Radiography			
MOC	API 5L Gr, X60 PSL-2													API 5L Gr, X60 PSL-2			
Ends	PE				BE												
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Socket Weld				Butt Weld												
Rating	# 9000				XXS	XXS	XXS	XXS	XXS	XXS	160	160	160	160	160	160	160
MOC	ASTM A 694 F60				ASTM A 860 WPHY 60, Seamless U.N.O									ASTM A860 WPHY 60,Welded U.N.O			
Elbow	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Tee	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Red.	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Cap	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Coupl	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Plug	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Union																	
Elbowlet	MSS SP 97,ASTM A 694 F60				MSS SP 97,ASTM A 694 F60									MSS SP 97,ASTM A 694 F60			
Sockolet	MSS SP 97,ASTM A 694 F60				MSS SP 97,ASTM A 694 F60									MSS SP 97,ASTM A 694 F60			
Weldolet	MSS SP 97,ASTM A 694 F60				MSS SP 97,ASTM A 694 F60									MSS SP 97,ASTM A 694 F60			
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe									MOC Same as pipe			
Swage	MSS SP 95,Same as pipe				MSS SP 95,MOC Same as pipe									MSS SP 95,MOC Same as pipe			
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				WN												
MOC	ASTM A 694 F60				ASTM A 694 F60												
FACE	RTJ-Note 7				RTJ, Note 7												
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25												
Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM A 694 F60																
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48					
Mechanical Joints																	
Stud Bolt	ASTM A 193 Gr. B7M				ASTM A 193 Gr. B7M												
Hex Nuts	ASTM A 194 Gr. 2HM				ASTM A 194 Gr. 2HM												
Gasket	B 16.20,OCT ring of Soft Iron with Max. 90BHN				ASME B 16.20,OCT ring of Soft Iron with Max. Hardness of 90BHN												

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
(-) 4 to 100	6171.5	(-) 20 to 38	433.9	9257.3 psig	650.85 Kg/cm2
122	6057.7	50	425.9	Limited by Flange considereing flange as the weakest joint in piping system	
212	5632.4	100	396		
302	5447.5	150	383		
392	5293.9	200	372.2		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and WNRTJ flanges above 3" size and specified spectacle
7. Weld Joint Factor for welded pipe shall be as per ASME B 31.3
8. Gasket Contact Surface shall have maximum roughness of 63AARH
9. Minimum RTJ groove hardness shall be 120BHN
10. Octagonal (OCT) ring shall be Hot Dip Galvanized or electrogalvaized.
11. In cae calculated thickness is more than thickness of schedule of XXS for small bore piping (1/2"), reduce MT and use XXS only.

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION				Sheet No.	2 of 2
	Piping Class	Material	C.A	Spec.No	Revision: 0	
	G1N	# 2500	CS-NACE	6mm	2004A,Rev 9	Page No. 34
	Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
	Service	Corrosive HydroCarbon Service-Sour Liquid and Vapour.				
	0					

Branch Table as per API RP 14E

Run Pipe	Branch Pipe																
		≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
	≤ 1	T														
	1-1.5	T	T													
	2	*	*	T												
	3	*	*	T	T											
	4	*	*	T	T	T										
	6	*	*	W	T	T	T									
	8	*	*	W	W	T	T	T								
	10	*	*	W	W	W	T	T	T							
	12	*	*	W	W	W	T	T	T	T						
	14	*	*	W	W	W	W	T	T	T	T					
	16	*	*	W	W	W	W	T	T	T	T	T				
	18	*	*	W	W	W	W	W	T	T	T	T	T			
	20	*	*	W	W	W	W	W	T	T	T	T	T	T		
	24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
	30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 67,SW				VG 273,FLANGED												
Rating	# 2500				# 1500,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	Use Gate Valve, VG 67				VB 273, Lever Operated							VB **, Gear Operated					
Rating	# 2500				# 1500, RTJ												
Globe Valve																	
TAG	VGL 67,SW				VGL 273												
Rating	# 2500				# 1500, RTJ												
Check Valve																	
TAG	VC 67,SW,Lift Check				VC 273,Swing Check												
Rating	# 2500				# 1500, RTJ												
Needle Valve																	
TAG	VN 20,SW																
Rating	# 2500																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.
2. Maximum temperature limit for all Ball Valve shall be 121 Degree C

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 0	15.01.16	Issued For Bid	PS	AKM	SJ

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**												Sheet No.		**1 of 2**	
		Piping Class		Material		C.A		Spec.No						Revision: 0			
		XG1	API 5000	CS		3mm		2004A,Rev 9						Page No.	35		
		Design Code:		**ASME B 31.3**													
		Service		Acidization													

Pipe Data	ASME B 36.10																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	XXS	XXS	XXS	XXS	XXS	XXS											
WTmm	7.47	7.82	9.09	10.16	11.07	15.24											
M.T (%)	8.0	12.5	5.0	12.5	11.0	12.5											
TYPE	Seamless																
MOC	ASTM A 106 Gr. B																
Ends	Threaded				BE												

Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Threaded				Butt Weld												
Rating	# 9000				XXS	XXS											
MOC	ASTM A 105 U.N.O				A 234 Gr. WPB, Seamless												
Elbow	ASMEB16.11				ASME B 16.9												
Tee	ASMEB16.11				ASME B 16.9												
Red.	ASMEB16.11				ASME B 16.9												
Cap	ASMEB16.11				ASME B 16.9												
Coupl	ASMEB16.11				ASME B 16.9												
Plug	ASMEB16.11				ASME B 16.9												
Union	ASMEB16.11				BS 3799												
Elbowlet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105												
Sockolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105												
Weldolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105												
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe												
Swage	MSS SP 95,Same as pipe				MSSSP95,Same as pipe												

Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Threaded				WN,# 5000												
MOC	API 6A Type -2				API 6A Type-4												
FACE	RTJ-Note 7				RTJ, Note 7												
STD.	API 6A Type 6B				API 6A Type 6B												

Miscellenious	
Size(in) Union	Wing union each of forged steel male& female stubs.Forged steel nuts & replaceble resielent ring. Wing Union Shall be suitable to 5000 PSI CWP for sizes less than equal to 3"
Mechanical Joints	
Stud Bolts	ASTM A 193 Gr. B7
Hex Nuts	ASTM A 194 Gr. 2H
Gasket	API 6A Type RX ,OCT ring of Soft Iron with Max. 90BHN

Pressure-Temperature Rating

Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Maximum Hydrostatic Pressure Pressure, psig	 Pres.,Kg/cm2g
60 to 100	5000	16 to 38	351.6	10000 psig	703.3 Kg/cm2
				Limited by Flange considereing flange as the weakest joint in piping system	

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and WNRTJ flanges above 3" size
5. All pipe thread shall be as per ASME B 1.20.1
6. PMS to be read in conjuction with FS 2004 A.
7. Gasket Contact Surface shall have maximum roughness of 63AARH
8. Minimum RTJ groove hardness shall be 120BHN
9. Octagonal (OCT) ring shall be Hot Dip Galvanized or electrogalvaized.
10. Acidization lines to be designed without any pockect
11. All threaded jonts shall be seal welded with full strength fillet weld.
12. Contractor to fulfill the requirement of category M- Fluid of ASME B 31.3
13. In cae calculated thickness is more than thickness of schedule of XXS for small bore piping (1/2"), reduce MT and use XXS only.

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**2 of 2**
		Piping Class	Material	C.A	Spec.No		Revision: 0
		XG1 API 5000	CS	3mm	2004A,Rev 9		Page No. 36
		Design Code:	**ASME B 31.3**				
		Service	Acidization				
		0					

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	0.5	0.75	1	1.5	2	3	4
0.5	T						
0.75	T	T					
1	T	T	T				
1.5	T	T	T	T			
2	*	*	*	T	T		
3	*	*	*	*	T	T	
4	*	*	*	*	*	T	T
6	*	*	*	*	*	*	T
8	*	*	*	*	*	*	*
10	*	*	*	*	*	*	*
12	*	*	*	*	*	*	*
14	*	*	*	*	*	*	*
16	*	*	*	*	*	*	*
18	*	*	*	*	*	*	*
20	*	*	*	*	*	*	*
22	*	*	*	*	*	*	*
24	*	*	*	*	*	*	*

LEGEND	
*	Thredolet/Elbowlet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 57,SCRD																
Rating	# API 5000																
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	Use Gate Valve, VG 57				VB 97, G.O.												
Rating	# API 5000				API 5000, RTJ												
Globe Valve																	
TAG																	
Rating																	
Check Valve																	
TAG					VC 97,Swing Check												
Rating					API 5000, RTJ												
Needle Valve																	
TAG																	
Rating																	
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.
2. Maximum temperature limit for all Ball Valve shall be 121 Degree C

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 0	04-08-2008	Issued For Bid	PS	GJ	GRP

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**												Sheet No.		**1 of 2**	
		Piping Class			Material		C.A	Spec.No						Revision: 0			
		XG1N			API 5000		CS-NACE	6mm	2004A,Rev 9					Page No.	37		
		Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3															
		Service	Corrosive HydroCarbon Service-Sour Liquid and Vapour and Acidization Service														

Pipe Data	ASME B 36.10																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6	
Sch.	XXS	XXS	XXS	XXS	XXS	XXS												
WTmm	7.47	7.82	9.09	7.14	11.07	15.24												
M.T (%)	8.0	12.5	5.0	12.5	11.0	12.5												
TYPE	Seamless																	
MOC	ASTM A 106 Gr. B																	
Ends	Threaded				BE													
Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	Threaded				Butt Weld													
Rating	# 9000				XXS	XXS												
MOC	ASTM A 105 U.N.O				A 234 Gr. WPB, Seamless													
Elbow	ASMEB16.11				ASME B 16.9													
Tee	ASMEB16.11				ASME B 16.9													
Red.	ASMEB16.11				ASME B 16.9													
Cap	ASMEB16.11				ASME B 16.9													
Coupl	ASMEB16.11				ASME B 16.9													
Plug	ASMEB16.11				ASME B 16.9													
Union	ASMEB16.11				BS 3799													
Elbowlet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105													
Sockolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105													
Weldolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105													
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe													
Swage	MSS SP 95,Same as pipe				MSSSP95,MOC Same as pipe													
Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	Threaded				WN,# 5000													
MOC	API 6A Type -2				API 6A Type-4													
FACE	RTJ-Note 7				RTJ, Note 7													
STD.	API 6A Type 6B				API 6A Type 6B													
Miscellenious																		
Size(in)	Wing union each of forged steel male& female stubs.Forged steel nuts & replaceble resielent ring. Wing Union Shall be suitable to 5000 PSI CWP																	
Union	for sizes less than equal to 3"																	
Mechanical Joints																		
Stud Bolts	ASTM A 193 Gr. B7M																	
Hex Nuts	ASTM A 194 Gr. 2HM																	
Gasket	API 6A Type RX ,OCT ring of Soft Iron with Max. 90BHN																	

Pressure-Temperature Rating

Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Maximum Hydrostatic Pressure Pressure, psig	 Pres.,Kg/cm2g
60 to 100	5000	16 to 38	351.6	10000 psig	703.3 Kg/cm2
				Limited by Flange considereing flange as the weakest joint in piping system	

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and WN flanges above 3" size
5. All pipe thread shall be as per ASME B 1.20.1
6. PMS to be read in conjuction with FS 2004 A.
7. Gasket Contact Surface shall have maximum roughness of 63AARH
8. Minimum RTJ groove hardness shall be 120BHN
9. Octagonal (OCT) ring shall be Hot Dip Galvanized or electrogalvaized.
10. Acidization lines to be designed without any pockect
11. All threaded jonts shall be seal welded with full strength fillet weld.
12. Contractor to fulfill the requirement of category M- Fluid of ASME B 31.3
13. In cae calculated thickness is more than thickness of schedule of XXS for small bore piping (1/2"), reduce MT and use XXS only.

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION				Sheet No.	2 of 2
	Piping Class	Material	C.A	Spec.No		Revision: 0
	XG1N	API 5000	CS-NACE	6mm	2004A,Rev 9	Page No. 38
	Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
	Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour and Acidization Service				
	0					

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	0.5	0.75	1	1.5	2	3	4
0.5	T						
0.75	T	T					
1	T	T	T				
1.5	T	T	T	T			
2	*	*	*	T	T		
3	*	*	*	*	T	T	
4							
6							
8							
10							
12							
14							
16							
18							
20							
22							

LEGEND	
*	Thredolet/Elbowlet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 58,SCRD																
Rating	# API 5000																
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	Use Gate Valve, VG 58				VB 98,G.O.												
Rating	# API 5000				API 5000, RTJ												
Globe Valve																	
TAG																	
Rating																	
Check Valve																	
TAG					VC 98,Swing Check												
Rating					API 5000, RTJ												
Needle Valve																	
TAG																	
Rating																	
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.
2. Maximum temperature limit for all Ball Valve shall be 121 Degree C

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 0	04-08-2008	Issued For Bid	PS	GJ	GRP

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**											Sheet No.	**1 of 2**			
		Piping Class		Material								C.A(Nil for SS)	Spec.No		Revision: 3		
		A3	# 150	SS/CS-Galvanized-Note-8								1.5 mm	2004A,Rev 9		Page No.	39	
		Design Code:	**ASME B 31.3**														
		Service	Instrument Air														

Pipe Data	ASME B 36.10																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6	
Sch.	80s	80s	80s	80s	80s	80	80	80	Std	Std	Std	Std	Std					
WTmm	3.73	3.91	4.55	5.08	5.54	7.62	8.56	10.97	8.18	9.27	9.53	9.53	9.53					
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5					
TYPE	Seamless																	
MOC	ASTM A 312 TP 316				ASTM A 106 Gr. B													
Ends	Screwed (SCRD)				BE													
Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	Screwed (SCRD)				Butt Weld													
Rating	#3000				80s	80	80	80	Std	Std	Std	Std	Std					
MOC	ASTM A 182 F 316 U.N.O				ASTM A 234 Gr. WPB, Seamless U.N.O													
Elbow	ASMEB16.11				ASME B 16.9													
Tee	ASMEB16.11				ASME B 16.9													
Red.	ASMEB16.11				ASME B 16.9													
Cap	ASMEB16.11				ASME B 16.9													
Coupl	ASMEB16.11				ASME B 16.9													
Plug	ASMEB16.11				ASME B 16.9													
Union	ASMEB16.11				BS 3799													
Elbowlet	MSS SP 97				MSS SP 97,ASTM A 105,Galvanized													
Sockolet	MSS SP 97				MSS SP 97,ASTM A 105,Galvanized													
Weldolet	MSS SP 97				MSS SP 97,ASTM A 105,Galvanized													
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe													
Swage	MSS SP 95,Same as pipe				MSS SP 95, MOC Same as pipe													
Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	Screwed (SCRD)				WN													
MOC	ASTM A 182 F 316				ASTM A 105													
FACE	RF-Serrated Finished				RF-Serrated Finished													
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25													
Spectacle Blind/Spacer Blinds																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
MOC	ASTM A 182 F 316				ASTM A 105-Galvanized U.N.O													
Spectacle	ASME B 16.48											Spacer and blind as per ASME B 16.48						
Mechanical Joints																		
Stud Bolts	ASTM A 193 Gr. B7																	
Hex Nuts	ASTM A 194 Gr. 2H																	
Gasket	ASME B 16.21,Flat Ring,1.58mm,CNAF																	

Pressure-Temperature Rating is considered same as material group 2.2 of ASME B 16.5

Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2
60/100	275	16/38	19.33
200	235	93	16.52
300	215	149	15.11
400	195	204	13.7

Maximum Hydrostatic Pressure	
Pressure, psig	Pres.,Kg/cm2g
425 psig	29.9 Kg/cm2
Limited by Flange considereing flange as the weakest joint in piping system	

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Flanged galvanized spool to be used to minimized welding at site.
7. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and specified spectacle blind assemblies.
8. All pipe fittings other that SS piping shall be hot dip galvanized with a minimum of 763 gram of galvanizing material per square meter of surface area in accordance with ASTM A 123/ASTM A 153.Pipe spool requireing welding shall be galvanized after welding. Spool shall be of such shape and dimenssion to suite the size and shape that the galvanizing shop can accomodate. Contractor to ensure of the spools such that the can be thouroughly cleaned from inside out before galvanizing.
9. Exposed thread and piping assemblies etc . Shall be coated with zinc primer or epoxy coating after installation.
10. Connection between SS and GALVANIZED piping shall be flanged.

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**												Sheet No.	**2 of 2**			
		Piping Class			Material						C.A(Nil for SS)		Spec.No		Revision: 3		
		A3	# 150		SS/CS-Galvanized-Note-8						1.5 mm	2004A,Rev 9			Page No.	40	
		Design Code:			**ASME B 31.3**													
		Service			Instrument Air													
		0																

Branch Table as per API RP 14E

Run Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	T	T	T	T	T	T

LEGEND	
*	Threadolet/Elbowlet
W	Weldolet
T	TEE

Branch Pipe

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 51-A,SCRD				VG81	VG 60,FLANGED											
Rating	# 800				# 150,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB 51-A,SCRD				VB91	VB 60, Lev. Op											
Rating	# 800				# 150, RF												
Globe Valve																	
TAG	VGL 51-A,SCRD				VGL81	VGL 60											
Rating	# 800				# 150, RF												
Check Valve																	
TAG	VC 51-A,SCRD,Lift Check				VC91	VC 60,Swing Check/ VC 81 (Wafer Check, Note -1)											
Rating	# 800				# 150, RF												
Needle Valve																	
TAG																	
Rating																	
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. Wafer check valve to be avoided as far as possible; unless the avaliable space constraint does not allow normal check valve . The contractor to take approvel for use of wafer check Valve during pipe routing after extablishing the space constraint.
2. PMS to be read in conjuction with FS 2004 A.
3. For instrument main header use CS valves but for instrument sub header and braches use SS 316 Valves.
4. Valve for sizes 1/2" to 1.5" shall be screwed and valve material shall be SS316 same as A9 piping class.
5. Valve for size 2" shall be flanged and valve material shall be SS316 same as A9 Piping Class.
6. Valve for size 3" and above shall be flanged and valve material shall be CS same as A1 Piping Class.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 3	15.01.16	Issued For Bid	PS	AKM	SJ

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION				Sheet No.	1 of 2
	Piping Class	Material	C.A(Nil for SS)	Spec.No	Revision: 3	
	B3 # 300	SS/CS-Galvanized-Note-8	1.5 mm	2004A,Rev 9	Page No	41
	Design Code:	**ASME B 31.3**				
	Service:	Instrument Air				

Pipe Data	ASME B 36.10																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	80s	80s	80s	80s	80s	80	80	80	Std	Std	40	40	XS				
WTmm	3.73	3.91	4.55	5.08	5.54	7.62	8.56	10.97	8.18	9.27	10.31	11.13	12.7				
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5				
TYPE	Seamless																
MOC	ASTM A 312 TP 316					ASTM A 106 Gr. B											
Ends	Screwed (SCRD)					BE											
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Screwed (SCRD)				80s	Butt Weld											
Rating	# 3000					80	80	80	Std	Std	40	40	XS				
MOC	ASTM A 182 F 316 U.N.O					ASTM A 234 Gr. WPB, Seamless U.N.O											
Elbow	ASMEB16.11					ASME B 16.9											
Tee	ASMEB16.11					ASME B 16.9											
Red.	ASMEB16.11					ASME B 16.9											
Cap	ASMEB16.11					ASME B 16.9											
Coupl	ASMEB16.11					ASME B 16.9											
Plug	ASMEB16.11					ASME B 16.9											
Union	ASMEB16.11					BS 3799											
Elbowlet	MSS SP 97					MSS SP 97,ASTM A 105,Galvanized											
Sockolet	MSS SP 97					MSS SP 97,ASTM A 105,Galvanized											
Weldolet	MSS SP 97					MSS SP 97,ASTM A 105,Galvanized											
Nipple	MSS SP 97,Same as pipe					MOC Same as pipe											
Swage	MSS SP 95,Same as pipe					MSS SP 95, MOC Same as pipe											
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Screwed (SCRD)					WN											
MOC	ASTM A 182 F 316					ASTM A 105											
FACE	RF-Smooth Finished					RF-Smooth Finished											
STD.	ASME B 16.5					ASME B 16.5, But welding ends as per ASME B 16.25											
Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM A 182 F 316					ASTM A 105-Galvanized U.N.O											
Spectacle	ASME B 16.48											Spacer and blind as per ASME B 16.48					
Mechanical Joints																	
Stud Bolts	ASTM A 193 Gr. B7																
Hex Nuts	ASTM A 194 Gr. 2H																
Gasket	ASME B 16.21,Flat Ring,1.58mm,CNAF																

Pressure-Temperature Rating is considered same as material group 2.2 of ASME B 16.5				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
60/100	720	16/38	50.6	1100 psig	77 Kg/cm2
200	620	93	43.6	Limited by Flange considereing flange as the weakest joint in piping system	
400	515	204	36.2		
600	450	316	31.6		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Flanged galvanized spool to be used to minimized welding at site.
7. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and specified spectacle blind assemblies.
8. All pipe fittings other that SS piping shall be hot dip galvanized with a minimum of 763 gram of galvanizing material per square meter of surface area in accordance with ASTM A 123/ASTM A 153.Pipe spool requireing welding shall be galvanized after welding. Spool shall be of such shape and dimenssion to suite the size and shape that the galvanizing shop can accomodate. Contractor to ensure of the spools such that the can be thouroughly cleaned from inside out before galvanizing.
9. Exposed thread and piping assemblies etc . Shall be coated with zinc primer or epoxy coating after installation.

	Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION				Sheet No.	2 of 2
		Piping Class	Material	C.A(Nil for SS)	Spec.No	Revision: 3	
		B3	# 300	SS/CS-Galvanized-Note-8	1.5 mm	2004A,Rev 9	Page No 42
		Design Code:	**ASME B 31.3**				
		Service:	Instrument Air				
		0					

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Threadolet/Elbowlet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 51-A,SCRD				VG-82	VG 62,FLANGED											
Rating	# 800				# 300,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB 51-A,SCRD				VB-92	VB 64, Lever Operated											
Rating	# 800				# 300, RF												
Globe Valve																	
TAG	VGL 51-A,SCRD				VGL - 82	VGL 62											
Rating	# 800				# 300, RF												
Check Valve																	
TAG	VC 51-A,SCRD,Lift Check				VC - 92	VC-62,Swing Check/VC 86 (Wafer Check,Note-1)											
Rating	# 800				# 300, RF												
Needle Valve																	
TAG																	
Rating																	
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. Wafer check valve to be avoided as far as possible; unless the avaliable space constraint does not allow normal check valve . The contractor to take approvel for use of wafer check Valve during pipe routing after establishing the space constraint.
2. PMS to be read in conjuction with FS 2004 A.
3. For instrument main header use CS valves but for instrument sub header and braches use SS 316 Valves.
4. Valve for sizes 1/2" to 1.5" shall be screwed and valve material shall be SS316 same as B9 piping class.
5. Valve for size 2" shall be flanged and valve material shall be SS316 same as B9 Piping Class.
6. Valve for size 3" and above shall be flanged and valve material shall be CS same as B1 Piping Class.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 3	15.01.16	Issued For Bid	PS	AKM	SJ

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**													Sheet No.		**1 of 2**	
		Piping Class				Material			C.A(Nil for SS)						Spec.No		Revision: 0	
		D3			# 600			SS/CS-Galvanized-Note-8		1.5 mm				2004A,Rev 9			Page No	43
		Design Code:						**ASME B 31.3**										
		Service:			Instrument Air													

Pipe Data	ASME B 36.10																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168	219.1	273.1	323.9	355.6	406.4	457	508	558.8	609.6	
Sch.	80s	80s	80s	80s	80s	80	80	80	80	80	80	80	80					
WTmm	3.73	3.9	4.6	5.08	5.54	7.62	8.56	10.97	12.7	15.08	17.47	19.05	21.4					
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5					
TYPE	Seamless																	
MOC	ASTM A 312 TP 316					ASTM A 106 Gr. B												
Ends	Screwed (SCRD)					BE												
Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	Screwed (SCRD)				Butt Weld													
Rating	# 3000				80	80	80	80	80	80	80	80						
MOC	ASTM A 182 F 316 U.N.O				ASTM A 234 Gr. WPB, Seamless U.N.O													
Elbow	ASMEB16.11				ASME B 16.9													
Tee	ASMEB16.11				ASME B 16.9													
Red.	ASMEB16.11				ASME B 16.9													
Cap	ASMEB16.11				ASME B 16.9													
Coupl	ASMEB16.11				ASME B 16.9													
Plug	ASMEB16.11				ASME B 16.9													
Union	ASMEB16.11				BS 3799													
Elbowlet	MSS SP 97				MSS SP 97,ASTM A 105,Galvanized													
Sockolet	MSS SP 97				MSS SP 97,ASTM A 105,Galvanized													
Weldolet	MSS SP 97				MSS SP 97,ASTM A 105,Galvanized													
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe													
Swage	MSS SP 95,Same as pipe				MSS SP 95, MOC Same as pipe													
Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	Screwed (SCRD)				WN													
MOC	ASTM A 182 F 316				ASTM A 105													
FACE	RF-Smooth Finished				RF-Smooth Finished													
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25													
Spectacle Blind/Spacer Blinds																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
MOC	ASTM A 182 F 316				ASTM A 105-Galvanized U.N.O													
Spectacle	ASME B 16.48											Spacer and blind as per ASME B 16.48						
Mechanical Joints																		
Stud Bolt	ASTM A 193 Gr. B7																	
Hex Nuts	ASTM A 194 Gr. 2H																	
Gasket	ASME B 16.20 , 4.5mm thick SS Spiral wound gasket with CNAF filler																	

Pressure-Temperature Rating is considered same as material group 2.2 of ASME B 16.5				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
60/100	1440	16/38	101.2	2160 psig	151.9 Kg/cm2
200	1240	93	87.2	Limited by Flange considereing flange as the weakest joint in piping system	

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.

2.All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.

3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.

4. All pipe thread shall be as per ASME B 1.20.1

5. PMS to be read in conjuction with FS 2004 A.

6. Flanged galvanized spool to be used to minimized welding at site.

7. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and specified spectacle blind assemblies.

8. All pipe fittings other that SS piping shall be hot dip galvanized with a minimum of 763 gram of galvanizing material per square meter of surface area in accordance with ASTM A 123/ASTM A 153.Pipe spool requireing welding shall be galvanized after welding. Spool shall be of such shape and dimenssion to suite the size and shape that the galvanizing shop can accomodate. Contractor to ensure of the spools such that the can be thouroughly cleaned from inside out before galvanizing.

9. Exposed thread and piping assemblies etc . Shall be coated with zinc primer or epoxy coating after installation.

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION					Sheet No.	2 of 2
	Piping Class		Material	C.A(Nil for SS)	Spec.No		Revision: 0
	D3	# 600	SS/CS-Galvanized-Note-8	1.5 mm	2004A,Rev 9	Page No	44
	Design Code:		**ASME B 31.3**				
	Service:	Instrument Air					
	0						

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Threadolet/Elbowlet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 51-A,SCRD				VG-92	VG 72,FLANGED											
Rating	# 800				# 600,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB 51-A,SCRD				VB-292	VB 264, Lever Operated											
Rating	# 800				# 600, RF												
Globe Valve																	
TAG	VGL 51-A,SCRD				VGL - 9	VGL 67											
Rating	# 800				# 600, RF												
Check Valve																	
TAG	VC 51-A,SCRD,Lift Check				VC - 29	VC-72,Swing Check/VC 96 (Wafer Check,Note-1)											
Rating	# 800				# 600, RF												
Needle Valve																	
TAG																	
Rating																	
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. Wafer check valve to be avoided as far as possible; unless the avaliable space constraint does not allow normal check valve . The contractor to take approvel for use of wafer check Valve during pipe routing after extablishing the space constraint.
2. PMS to be read in conjuction with FS 2004 A.
3. For instrument main header use CS valves but for instrument sub header and braches use SS 316 Valves.
4. Valve for sizes 1/2" to 1.5" shall be screwed and valve material shall be SS316 same as D9 piping class.
5. Valve for size 2" shall be flanged and valve material shall be SS316 same as D9 Piping Class.
6. Valve for size 3" and above shall be flanged and valve material shall be CS same as D1 Piping Class.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 0	15.01.16	Issued For Bid	PS	AKM	SJ

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**1 of 2**
		Piping Class		Material	C.A	Spec.No	Revision: 1
		A4	# 150	CS-Galvanized-Note-8	1.5mm	2004A,Rev 9	Page No. 45
		Design Code:		***ASME B 31.3***			
		Service:	Potable Water (For non-drinking use)				

Pipe Data	ASME B 36.10																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	80	80	80	80	80	80	80										
WTmm	3.73	3.91	4.55	5.08	5.54	7.62	8.56										
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5										
TYPE	Seamless																
MOC	ASTM A 106 Gr. B-Galvanized					ASTM A106Gr.B											
Ends	Screwed (SCRD)					BE											
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Screwed (SCRD)					Butt Weld											
Rating	# 3000					80	80										
MOC	ASTM A 105-Galvanized U.N.O					ASTM A 234 Gr. WPB, Seamless U.N.O											
Elbow	ASMEB16.11					ASME B 16.9											
Tee	ASMEB16.11					ASME B 16.9											
Red.	ASMEB16.11					ASME B 16.9											
Cap	ASMEB16.11					ASME B 16.9											
Coupl	ASMEB16.11					ASME B 16.9											
Plug	ASMEB16.11					ASME B 16.9											
Union	ASMEB16.11					BS 3799											
Elbowlet	MSS SP 97,ASTM A 105,Galvanized					MSS SP 97,ASTM A 105,Galvanized											
Sockolet	MSS SP 97,ASTM A 105,Galvanized					MSS SP 97,ASTM A 105,Galvanized											
Weldolet	MSS SP 97,ASTM A 105,Galvanized					MSS SP 97,ASTM A 105,Galvanized											
Nipple	MSS SP 97,Same as pipe,Galvanized					MOC Same as pipe											
Swage	MSS SP 95,Same as pipe,Galvanized					MSS SP 95,MOC Same as pipe											
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Screwed (SCRD)					WN											
MOC	ASTM A 105-Galvanized U.N.O					ASTM A 105											
FACE	FF-Serrated Finished					FF-Serrated Finished-When connecte to Valves use FF flange											
STD.	ASME B 16.5					ASME B 16.5, But welding ends as per ASME B 16.25											
Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM A 105-Galvanized U.N.O																
Spectacle	ASME B 16.48											Spacer and blind as per ASME B 16.48					
Mechanical Joints																	
Stud Bolts	ASTM A 307 Gr. B																
Hex Nuts	ASTM A 307 Gr. B																
Gasket	3.175mm Ribber with Maximum 60 Shore Durometer Flat ring as per ASME B 16.21. Use full face gaskets for Valves																

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
(-) 4 to 100	230	(-) 20 to 38	16.3	350 psig	24.5 Kg/cm2
				Limited by Flange considereing flange as the weakest joint in piping system	

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2.All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Flanged galvanized spool to be used to minimized welding at site.
7. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange
8. All pipe fittings galvanized shall be hot dip galvanized with a minimum of 763 gram of galvanizing material per square meter of surface area in accordance with ASTM A 123/ASTM A 153.Pipe spool requireing welding shall be welded after welding. Spool shall be of such shape and dimenssion to suite the size and shape that the galvanizing shop can accomodate. Contractor to ensure of the spools such that the can be thouroughly cleaned from inside out before galvanizing.
9. Exposed thread and piping assemblies etc . Shall be coated with zinc primer or epoxy coating after installation.

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION				Sheet No.	2 of 2
	Piping Class	Material	C.A	Spec.No		Revision: 1
	A4 # 150	CS-Galvanized-Note-8	1.5mm	2004A,Rev 9		Page No. 46
	Design Code:	**ASME B 31.3**				
	Service:	Potable Water (For non-drinking use)				
	0					

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Threadolet/Elbowlet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 50,SCRD				VG 80,FLANGED												
Rating	# 800				# 150,FF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB 50,SCRD				VB 86, Lever Operated												
Rating	# 800				# 150, FF,Flanged												
Globe Valve																	
TAG	VGL 50,SCRD				VGL 80												
Rating	# 800				# 150, FF,Flanged												
Check Valve																	
TAG	VC 50,SCRD,Lift Check				VC 80,Swing Check												
Rating	# 800				# 150, FF,Flanged												
Needle Valve																	
TAG	VN 11,SCRD																
Rating	# 800																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. Wafer check valve to be avoided as far as possible; unless the avaliable space constraint does not allow normal check valve . The contractor to take approvel for use of wafer check Valve during pipe routing after extablishing the space constraint.
2. PMS to be read in conjuction with FS 2004 A.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 1	04-02-2016	Issued For Bid	PS	AKM	SJ

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**1 of 2**
		Piping Class		Material	C.A	Spec.No	Revision: 2
		A5	# 150	90-10 Cu/Ni	Nil	2004A,Rev 9	Page No. 47
		Design Code:	**ASME B 31.3**				
		Service:	Fire Water and Raw Sea Water				

Pipe Data	EEMUA 144 Range																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	16	25	30	44.5	57	88.9	108	159	219.1	267	323.9	368	419	457.2	508	558.8	609.6
Sch.																	
WTmm	2	2	2.5	2.5	2.5	2.5	3	3	4	4.5	5.5	6.5	7				
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5				
TYPE	Seamless													Welded with 100% Radiography			
MOC	Annealed tube as per ASTM B 466 Copper Alloy No. 706 or BS 2871 CN 102																
Ends	PE																
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Brazed					Butt Weld											
Rating						2.5	3	3	4	4.5	5.5	6.5	7				
MOC	Forged 90-10 Cu-Ni					90-10Cu-Ni											
Elbow																	
Tee																	
Red.																	
Cap																	
Coupl																	
Plug																	
Union																	
Elbowlet																	
Sockolet																	
Weldolet																	
Nipple																	
Swage																	
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Slip-On Composite Flange					WN Composite Flange/S.O. Bossed flange (Suitable for welding),											
MOC	Outer flange- A 105(Galv.) to # 150 of B-16.5 upto 24" and B 16.47A for 22" and 28" AND Inner Flange -90-10Ci-Ni																
FACE						FF											
STD.	EEMUA 145																
Blind Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM A 105 with 3mm 90-10 CuNi weld deposite.																
Mechanical Joints																	
Stud Bolts	ASTM A 193 Gr. B7 to be used with insulating gasket/Sleeves																
Hex Nuts	ASTM A 194 Gr. 2H to be used with insulating gasket/Sleeves																
Gasket	3mm thick flat ring of neoprine rubber as ASME B 16.21																

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
up to 167	230	Up to 75	16	345 psig	24 Kg/cm2
				Limited by Flange considereing flange as the weakest joint in piping system	

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. For flanged valves full face gaskets to be used and S.O. Bossed flage.
4. PMS to be read in conjuction with FS 2004 A.
5. For joints with material other than Cu-Ni Sacrificial spool and insulating gaskets are to used as indicated in design criteria.
6. Threading on Cu-Ni piping is not allowed.However, screwed fittings may be used for vent and drain connections only with company's approval.

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**2 of 2**
	Piping Class		Material	C.A	Spec.No	Revision: 2
	A5	# 150	90-10 Cu/Ni	Nil	2004A,Rev 9	Page No. 48
	Design Code:		**ASME B 31.3**			
	Service:	Fire Water and Raw Sea Water				
	0					

Branch Table as per API RP 14E

Run Pipe	Branch Pipe																
		≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30	
	≤ 1	T															
	1-1.5	T	T														
	2	*	*	T													
	3	*	*	T	T												
	4	*	*	T	T	T											
	6	*	*	W	T	T	T										
	8	*	*	W	W	T	T	T									
	10	*	*	W	W	W	T	T	T								
	12	*	*	W	W	W	T	T	T	T							
	14	*	*	W	W	W	W	T	T	T	T						
	16	*	*	W	W	W	W	T	T	T	T	T					
	18	*	*	W	W	W	W	W	T	T	T	T	T				
	20	*	*	W	W	W	W	W	T	T	T	T	T	T			
	24	*	*	W	W	W	W	W	W	T	T	T	T	T	T		
	30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T	

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 54,Brazing union end as per BS 1952 Series B Cl. 200					VG 87,FLANGED											
Rating	# 150					# 150, FF											
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB 54,Brazing union end as per BS 1952 Series B Cl. 200					VB 87,FLANGED											
Rating	# 150					# 150, FF											
Globe Valve																	
TAG	VGL 54,Brazing union end as per BS 1952 Series B Cl. 200					VGL 87,FLANGED											
Rating	# 150					# 150, FF											
Check Valve																	
TAG	VC 54,Brazing union end as per BS 1952 Series B Cl. 200					VC 87,FLANGED											
Rating	# 150					# 150, FF											
Needle Valve																	
TAG	VN 5,Brazing union end as per BS 1952 Series B Cl. 200																
Rating	# 150					# 150, FF											
Butter Fly Valve																	
TAG						VBF-22											
Rating						# 150, FF											

Notes Releted to Valve and Brach Connection

1. Wafer type Butterfly Valve may be used only in fire water(Clean water) service.
2. PMS to be read in conjuction with FS 2004 A.
3. Maximum temperature limit for all Ball Valve shall be 121 Degree C

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 2	15.01.16	Issued For Bid	PS	AKM	SJ

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**1 of 2**
		Piping Class	Material	C.A	Spec.No		Revision: 1
		A6	# 150	SS 316L	Nil	2004A,Rev 9	Page No. 49
		Design Code:	**ASME B 31.3**				
		Service	Gas Turbine and Engine exhaust				

Pipe Data	ASME B 36.19																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	40s	40s	40s	40s	10s	10s	10s	10s	10s								
WTmm	2.77	2.87	3.38	3.68	2.77	3.05	3.05	3.4	3.76	4mm	4mm	4mm	4mm	4mm	4mm	4.8mm	4.8mm
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5
TYPE	Seamless									EFW with 100% Radiography							
MOC	ASTM A 312 TP 316L									ASTM A 358 Gr. 316L Cl.1							
Ends	SW				BE												
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				Butt Weld												
Rating	# 3000				10s	10s	10s	10s	10s	4mm	4mm	4mm	4mm	4mm	4mm	4.8mm	4.8mm
MOC	ASTM A 182 F 316L U.N.O				ASTM A 403 Gr. WP 316 L, Seamless U.N.O					ASTM A 403 Gr. WP 316 L, Welded with Joint Factor -1, U.N.O							
Elbow	ASMEB16.11				ASME B 16.9					ASME B 16.9							
Tee	ASMEB16.11				ASME B 16.9					ASME B 16.9							
Red.	ASMEB16.11				ASME B 16.9					ASME B 16.9							
Cap	ASMEB16.11				ASME B 16.9					ASME B 16.9							
Coupl	ASMEB16.11				ASME B 16.9					ASME B 16.9							
Plug	ASMEB16.11				ASME B 16.9					ASME B 16.9							
Union																	
Elbowlet	MSS SP 97				MSS SP 97,ASTM A 182 F316L					MSS SP 97,ASTM A 182 F316L							
Sockolet	MSS SP 97				MSS SP 97,ASTM A 182 F316L					MSS SP 97,ASTM A 182 F316L							
Weldolet	MSS SP 97				MSS SP 97,ASTM A 182 F316L					MSS SP 97,ASTM A 182 F316L							
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe					MOC Same as pipe							
Swage	MSS SP 95,Same as pipe				MSS SP 95,MOC Same as pipe					MSS SP 95,MOC Same as pipe							
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				FF Lap Joint (LJ)												
MOC	ASTM A 182 F 316L				Stub end SS 316L + ASTM A 105 Backing flange (While connecting to FF flange use Slip -on(SO),FF, SS 316L flange												
FACE	RF-Serrated Finished-125 AARH				FF, Serrated Finished,Stub end and FF shall have 125AARH												
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25												
Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14.00	16.00	18	20	22	24
MOC	ASTM A 182 F 316L																
Spectacle	ASME B 16.48											Spacer and blind as per ASME B 16.48					
Mechanical Joints																	
Stud Bolts	ASTM A 193 Gr. B8M																
Hex Nuts	ASTM A 194 Gr. 8M																
Gasket	ASME B 16.21,Flat Ring,1.58mm,CNAF																

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
446	99.6	230	7	25 psig (Air)	Kg/cm2
850	65	454	4.5	Limited by Flange considereing flange as the weakest joint in piping system	

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Flanged galvanized spool to be used to minimize welding at site.
7. Weld Joint Factor for welded pipe shall be as per ASME B 31.3

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**			Sheet No.	**2 of 2**
	Piping Class	Material	C.A	Spec.No	Revision: 1
	A6 # 150	SS 316L	Nil	2004A,Rev 9	Page No. 50
	Design Code:	**ASME B 31.3**			
	Service	Gas Turbine and Engine exhaust			
	0				

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	T	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG																	
Rating																	
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG																	
Rating																	
Globe Valve																	
TAG																	
Rating																	
Check Valve																	
TAG																	
Rating																	
Needle Valve																	
TAG																	
Rating																	
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 1	15.01.16	Issued For Bid	PS	AKM	SJ

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**1 of 2**
	Piping Class	Material	C.A	Spec.No	Revision:2	
	A8	# 150	CS- Galvanized	3mm	2004A,Rev 9	Page No. 51
	Design Code:	**ASME B 31.3**				
	Service	Over Board Lines				

Pipe Data	ASME B 36.10																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	XXS	160	160	160	160	80	80	80	Std	Std	Std	Std	Std	Std	20	20	20
WTmm	7.47	5.56	6.35	7.14	8.74	7.62	8.56	10.97	8.18	9.27	9.52	9.52	9.52	9.52	9.52	9.52	9.52
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5				
TYPE	Seamless													LSAW with 100% Radiography			
MOC	ASTM A 106 Gr. B-Galvanized					ASTM A 106 Gr. B								API 5L Gr. B			
Ends	Screwed (SCRD)					BE								BE			
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Screwed (SCRD)					Butt Weld											
Rating	# 9000	# 6000			160	80	80	80	Std	Std	Std	Std	Std	Std	20	20	20
MOC	ASTM A 105-Galvanized U.N.O					ASTM A 234 Gr. WPB, Seamless U.N.O								ASTM A234 Gr.WPB,Welded U.N.O			
Elbow	ASMEB16.11					ASME B 16.9								ASME B 16.9			
Tee	ASMEB16.11					ASME B 16.9								ASME B 16.9			
Red.	ASMEB16.11					ASME B 16.9								ASME B 16.9			
Cap	ASMEB16.11					ASME B 16.9								ASME B 16.9			
Coupl	ASMEB16.11					ASME B 16.9								ASME B 16.9			
Plug	ASMEB16.11					ASME B 16.9								ASME B 16.9			
Union	ASMEB16.11					BS 3799								BS 3799			
Elbowlet	MSS SP 97,ASTM A 105,Galvanized					MSS SP 97,ASTM A 105,Galvanized								MSS SP 97,ASTM A 105,Galvanized			
Sockolet	MSS SP 97,ASTM A 105,Galvanized					MSS SP 97,ASTM A 105,Galvanized								MSS SP 97,ASTM A 105,Galvanized			
Weldolet	MSS SP 97,ASTM A 105,Galvanized					MSS SP 97,ASTM A 105,Galvanized								MSS SP 97,ASTM A 105,Galvanized			
Nipple	MSS SP 97,Same as pipe,Galvanized					MOC Same as pipe								MOC Same as pipe			
Swage	MSS SP 95,Same as pipe,Galvanized					MSS SP95,MOC Same as pipe								MOC Same as pipe			
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Screwed (SCRD)					WN											
MOC	ASTM A 105					ASTM A 105											
FACE	RF-Serrated Finished					RF, Serrated Finished											
STD.	ASME B 16.5					ASME B 16.5, But welding ends as per ASME B 16.25											
Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM A 105																
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48					
Mechanical Joints																	
Stud Bolts	ASTM A 193 Gr. B7				ASTM A 193 Gr. B7												
Hex Nuts	ASTM A 194 Gr. 2H				ASTM A 194 Gr. 2H												
Gasket	ASME B 16.21,Flat Ring,1.58mm,CNAF				ASME B 16.21,Flat Ring,1.58mm,CNAF												

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
(-) 4 to 100	285	(-) 20 to 38	20	450 psig	31.6 Kg/cm2
200	260	93	18.3	Limited by Flange considereing flange as the weakest joint in piping system	
300	230	149	16.2		
400	200	204	14.1		
500	170	260	11.9		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange
7. Weld Joint Factor for welded pipe shall be as per ASME B 31.3

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**											Sheet No.		**2 of 2**		
		Piping Class				Material				C.A		Spec.No			Revision:2		
		A8		# 150		CS- Galvanized				3mm		2004A,Rev 9			Page No.	52	
		Design Code:				**ASME B 31.3**											
		Service				Over Board Lines											
		0															

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14.00	16.00	18	20	22	24	
Gate Valve																		
TAG		VG 41, SCRD			VG 61,FLANGED													
Rating		# 800			# 150,RF													
Plug Valve																		
TAG																		
Rating																		
Ball Valve																		
TAG		VB 42,SCRD			VB 61, Lever Operated			VB 63, Gear Operated										
Rating		# 800			# 150, RF													
Globe Valve																		
TAG		VGL 42,SW			VGL 60													
Rating		# 800			# 150, RF													
Check Valve																		
TAG		VC 41,SCRD,Lift Check			VC 85 (Wafer Check, Note -1)													
Rating		# 800			# 150, RF													
Needle Valve																		
TAG		VN 8,SCRD																
Rating		# 800																
Butter Fly Valve																		
TAG					VBF 1(WAFFER)													
Rating					#150, RF													

Notes Releted to Valve and Brach Connection

1. Wafer check valve to be avoided as far as possible; unless the available space constraint does not allow normal check valve. The contractor to take approvel for use of wafer check Valve during pipe routing after extablishing the space constraint.
2. Wafer type Butterfly Valve may be used only in fire water(Clean water) service.
3. PMS to be read in conjuction with FS 2004 A.
4. Maximum temperature limit for all Ball Valve shall be 121 Degree C
5. Exposed thread and piping assemblies etc . Shall be coated with zinc primer or epoxy coating after installation.
6. Flanged galvanized spool to be used to minimized welding at site.
7. All pipe fittings galvanized shall be hot dip galvanized with a minimum of 763 gram of galvanizing material per square meter of surface area in accordance with ASTM A 123/ASTM A 153.Pipe spool requireing welding shall be welded after welding. Spool shall be of such shape and dimenssion to suite the size and shape that the galvanizing shop can accomodate. Contractor to ensure of the spools such that the can be thouroughly cleaned from inside out before galvanizing.

Revision:2	15.01.16	Issued For Bid	PS	AKM	SJ
Revision	Date	Description	Prepared By	Reviewed By	Approved By

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**													Sheet No.		**1 of 2**	
	Piping Class				Material				C.A	Spec.No						Revision: 1	
	A9				# 150				SS 316	Nil	2004A,Rev 9					Page No.	53
	Design Code:				**ASME B 31.3**												
	Service				Lub Oil, Seal Oil and Chemical												

Pipe Data	ASME B 36.19																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	40s	40s	40s	40s	10s	10s	10s	10s									
WTmm	2.77	2.87	3.38	3.68	2.77	3.05	3.05	3.4									
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5									
TYPE	Seamless																
MOC	ASTM A 312 TP 316																
Ends	PE				BE												
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				Butt Weld												
Rating	# 3000				10s	10s	10s	10s									
MOC	ASTM A 182 F 316 U.N.O				ASTM A 403 Gr. WP 316 , Seamless												
Elbow	ASMEB16.11				ASME B 16.9												
Tee	ASMEB16.11				ASME B 16.9												
Red.	ASMEB16.11				ASME B 16.9												
Cap	ASMEB16.11				ASME B 16.9												
Coupl	ASMEB16.11				ASME B 16.9												
Plug	ASMEB16.11				ASME B 16.9												
Union																	
Elbowlet	MSS SP 97				MSS SP 97,ASTM A 182 F316												
Sockolet	MSS SP 97				MSS SP 97,ASTM A 182 F316												
Weldolet	MSS SP 97				MSS SP 97,ASTM A 182 F316												
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe												
Swage	MSS SP 95,Same as pipe				MSS SP 95MOC Same as pipe												
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				FF Lap Joint (LJ)												
MOC	ASTM A 182 F 316				Stub end A 182 F316 + ASTM A 105 Backing flange (While connecting to FF flange use Slip -on(SO),FF, 316 flange												
FACE	RF-Serrated Finished				FF, Serrated Finished												
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25												
Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM A 182 F 316																
Spectacle	ASME B 16.48											Spacer and blind as per ASME B 16.48					
Mechanical Joints																	
Stud Bolts	ASTM A 193 Gr. B7																
Hex Nuts	ASTM A 194 Gr. 2H																
Gasket	ASME B 16.20,5mm, SS Spiral wound with CNAF fille																

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
60/100	275	16/38	19.3	425 psig	29.9 Kg/cm2
200	240	93	16.87	Limited by Flange considereing flange as the weakest joint in piping system	
400	195	204	13.7		
600	140	454	4.6		
850	65				

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Contractor use Schedule 80 for sizes less than 1/1/2" for threaded pipe and fittings.
7. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange.

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**			Sheet No.	**2 of 2**
	Piping Class	Material	C.A	Spec.No	Revision: 1
	A9 # 150	SS 316	Nil	2004A,Rev 9	Page No. 54
	Design Code:	**ASME B 31.3**			
	Service	Lub Oil, Seal Oil and Chemical			
	0				

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG		VG-51			VG-81												
Rating		# 800, SW			# 150, RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG		VB-51			VB-91												
Rating		# 800, SW			# 150, RF												
Globe Valve																	
TAG		VGL - 51			VGL - 81												
Rating		# 800, SW			# 150, RF												
Check Valve																	
TAG		VC - 51, Lift Check			VC - 91, Swing Check												
Rating		# 800, SW			# 150, RF												
Needle Valve																	
TAG																	
Rating																	
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 1	22/7/2009	Issued For Bid	PS	GJ	GRP

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**1 of 2**
	Piping Class	Material	C.A	Spec.No	Revision: 2	
	B9	# 300	SS 316	Nil	2004A,Rev 9	Page No. 55
	Design Code:	**ASME B 31.3**				
	Service	Lub Oil, Seal Oil and Chemical				

Pipe Data	ASME B 36.19																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	40s	40s	40s	40s	10s	10s	10s	40s									
WTmm	2.77	2.87	3.38	3.68	2.77	3.05	3.05	7.11									
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5									
TYPE	Seamless																
MOC	ASTM A 312 TP 316									ASTM A 358 TP 316 with 100% Radiography.							
Ends	PE				BE												
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				Butt Weld												
Rating	# 3000				10s	10s	10s	40s									
MOC	ASTM A 182 F 316 U.N.O				ASTM A 403 Gr. WP 316 , Seamless												
Elbow	ASMEB16.11				ASME B 16.9												
Tee	ASMEB16.11				ASME B 16.9												
Red.	ASMEB16.11				ASME B 16.9												
Cap	ASMEB16.11				ASME B 16.9												
Coupl	ASMEB16.11				ASME B 16.9												
Plug	ASMEB16.11				ASME B 16.9												
Union																	
Elbowlet	MSS SP 97				MSS SP 97,ASTM A 182 F316												
Sockolet	MSS SP 97				MSS SP 97,ASTM A 182 F316												
Weldolet	MSS SP 97				MSS SP 97,ASTM A 182 F316												
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe												
Swage	MSS SP 95,Same as pipe				MSS SP 95MOC Same as pipe												
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				FF Lap Joint (LJ)												
MOC	ASTM A 182 F 316				Stub end A 182 F316 + ASTM A 105 Backing flange (While connecting to FF flange use Slip -on(SO),FF, 316 flange												
FACE	RF-Smooth Finished				FF, Smooth Finished												
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25												
Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM A 182 F 316																
Spectacle	ASME B 16.48											Spacer and blind as per ASME B 16.48					
Mechanical Joints																	
Stud Bolts	ASTM A 193 Gr. B7																
Hex Nuts	ASTM A 194 Gr. 2H																
Gasket	ASME B 16.20,5mm, SS Spiral wound with CNAF filler																

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
60/100	720	16/38	50.6	1100 psig	77 Kg/cm2
200	620	93	43.6	Limited by Flange considereing flange as the weakest joint in piping system	
400	515	204	36.2		
600	450	316	31.6		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2.All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Contractor use Schedule 80s for sizes less than 1/1/2" for threaded pipe and fittings.
7. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange.

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION				Sheet No.	2 of 2
	Piping Class		Material	C.A	Spec.No	Revision: 2
	B9	# 300	SS 316	Nil	2004A,Rev 9	Page No. 56
	Design Code:		**ASME B 31.3**			
	Service	Lub Oil, Seal Oil and Chemical				
	0					

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG		VG-51			VG-82												
Rating		# 800, SW			# 300, RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG		VB-51			VB-92												
Rating		# 800, SW			# 300, RF												
Globe Valve																	
TAG		VGL - 51			VGL - 82												
Rating		# 800, SW			# 300, RF												
Check Valve																	
TAG		VC - 51, Lift Check			VC - 92, Swing Check												
Rating		# 800, SW			# 300, RF												
Needle Valve																	
TAG																	
Rating																	
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 2	15.01.16	Issued For Bid	PS	AKM	SJ

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**												Sheet No.		**1 of 2**	
		Piping Class				Material		C.A		Spec.No				Revision: 0			
		D9		# 600		SS 316		Nil		2004A,Rev 9				Page No		57	
		Design Code:				**ASME B 31.3**											
		Service		Lub Oil, Seal Oil and Chemical													

Pipe Data	ASME B 36.19																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168	219	273	324	355.6	406	457	508	558.8	609.6	
Sch.	80s	80s	80s	80s	80s	80s	80s	80s										
WTmm	3.73	3.9	4.6	5.08	5.54	7.62	8.56	11										
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5										
TYPE	Seamless																	
MOC	ASTM A 312 TP 316								ASTM A 358 TP 316 with 100% Radiography.									
Ends	PE				BE													
Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				Butt Weld													
Rating	# 3000				80s	80s	80s	80s										
MOC	ASTM A 182 F 316 U.N.O				STM A 403 Gr. WP 316 , Seamless													
Elbow	ASMEB16.11				ASME B 16.9													
Tee	ASMEB16.11				ASME B 16.9													
Red.	ASMEB16.11				ASME B 16.9													
Cap	ASMEB16.11				ASME B 16.9													
Coupl	ASMEB16.11				ASME B 16.9													
Plug	ASMEB16.11				ASME B 16.9													
Union																		
Elbowlet	MSS SP 97				SS SP 97,ASTM A 182 F316													
Sockolet	MSS SP 97				SS SP 97,ASTM A 182 F316													
Weldolet	MSS SP 97				SS SP 97,ASTM A 182 F316													
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe													
Swage	MSS SP 95,Same as pipe				SS SP 95MOC Same as pipe													
Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				WN													
MOC	ASTM A 182 F 316				ASTM A 182 F316													
FACE	RF-Smooth Finished				RF-Smooth Finish													
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25													
Spectacle Blind/Spacer Blinds																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
MOC	ASTM A 182 F 316																	
Spectacle	ASME B 16.48											Spacer and blind as per ASME B 16.48						
Mechanical Joints																		
Stud Bolt	ASTM A 193 Gr. B7																	
Hex Nuts	ASTM A 194 Gr. 2H																	
Gasket	ASME B 16.20,5mm, SS 316 Spiral wound with CNAF filler																	

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
60/100	1440	16/38	101.2	2160 psig	151.9 Kg/cm2
200	1240	93	87.2	Limited by Flange considereing flange as the weakest joint in piping system	
400	1025	204	72.06		
600	900	316	63,28		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be not dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Contractor use Schedule 80s for sizes less than 1/1/2" for threaded pipe and fittings.
7. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange.

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION				Sheet No.	2 of 2
	Piping Class		Material	C.A	Spec.No	Revision: 0
	D9	# 600	SS 316	Nil	2004A,Rev 9	Page No 58
	Design Code:		**ASME B 31.3**			
	Service	Lub Oil, Seal Oil and Chemical				
	0					

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG-51				VG-92												
Rating	# 800, SW				# 600, RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB-51				VB-92												
Rating	# 800, SW				# 600, RF												
Globe Valve																	
TAG	VGL - 51				VGL - 92												
Rating	# 800, SW				# 600, RF												
Check Valve																	
TAG	VC - 51, Lift Check				VC - 92, Swing Check												
Rating	# 800, SW				# 600, RF												
Needle Valve																	
TAG																	
Rating																	
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 0	15.01.16	Issued For Bid	PS	AKM	GJ

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**											Sheet No.		**1 of 2**			
		Piping Class				Material					C.A	Spec.No			Revision: 0			
		E9		# 900		SS 316					Nil	2004A,Rev 9			Page No	59		
		Design Code:				**ASME B 31.3**												
		Service		Lub Oil, Seal Oil and Chemical														

Pipe Data	ASME B 36.19																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168	219	273	324	355.6	406	457	508	558.8	609.6	
Sch.	80S	80s	80s	160	80s	80s	80s	80s										
WTmm	3.73	3.9	4.6	7.14	5.54	7.62	8.56	11										
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5										
TYPE	Seamless																	
MOC	ASTM A 312 TP 316																	
Ends	PE				BE													
Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				Butt Weld													
Rating	# 3000			#6000	80s	80s	80s	80s										
MOC	ASTM A 182 F 316 U.N.O				STM A 403 Gr. WP 316 , Seamless													
Elbow	ASMEB16.11				ASME B 16.9													
Tee	ASMEB16.11				ASME B 16.9													
Red.	ASMEB16.11				ASME B 16.9													
Cap	ASMEB16.11				ASME B 16.9													
Coupl	ASMEB16.11				ASME B 16.9													
Plug	ASMEB16.11				ASME B 16.9													
Union																		
Elbowlet	MSS SP 97				SS SP 97,ASTM A 182 F316													
Sockolet	MSS SP 97				SS SP 97,ASTM A 182 F316													
Weldolet	MSS SP 97				SS SP 97,ASTM A 182 F316													
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe													
Swage	MSS SP 95,Same as pipe				SS SP 95MOC Same as pipe													
Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				WN													
MOC	ASTM A 182 F 316				A 182 F316													
FACE	RTJ-Note 8				RTJ-Note 8													
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25													
Spectacle Blind/Spacer Blinds																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
MOC	ASTM A 182 F 316																	
Spectacle	ASME B 16.48											Spacer and blind as per ASME B 16.48						
Mechanical Joints																		
Stud Bolt	ASTM A 193 Gr. B7																	
Hex Nuts	ASTM A 194 Gr. 2H																	
Gasket	ASME B 16.20,OCT ring of SS316 with Max. Hardness of 120 BHN																	

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
60/100	2160	16/38	151.9	3240 psig	227.8 Kg/cm2
200	1860	93	130.8	Limited by Flange considereing flange as the weakest joint in piping system	
400	1540	204	108.3		
600	1355	316	95.3		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be not dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Contractor use Schedule 80s for sizes less than 1/1/2" for threaded pipe and fittings.
7. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange.
8. RTJ Groove hardness shall be 140 BHN

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION				Sheet No.	2 of 2
	Piping Class		Material	C.A	Spec.No	Revision: 0
	E9	# 900	SS 316	Nil	2004A,Rev 9	Page No 60
	Design Code:		**ASME B 31.3**			
	Service	Lub Oil, Seal Oil and Chemical				
	0					

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG-61				VG-102												
Rating	# 1500, SW				# 900, RTJ												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB-61				VB-102												
Rating	# 1500, SW				# 900, RTJ												
Globe Valve																	
TAG	VGL - 61				VGL - 102												
Rating	# 1500, SW				# 900, RTJ												
Check Valve																	
TAG	VC - 51, Lift Check				VC - 102, Swing Check												
Rating	# 1500, SW				# 900, RTJ												
Needle Valve																	
TAG																	
Rating																	
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 0	15.01.16	Issued For Bid	PS	AKM	SJ

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**												Sheet No.		**1 of 2**	
		Piping Class			Material			C.A		Spec.No				Revision: 0			
		A 10		# 150		SS 316 L		1.5mm		2004A,Rev 9				Page No.	61		
		Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3															
		Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.														
			Instrument Gas Header including 1st Bolck Valve														

Pipe Data	ASME B 36.19																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6	
Sch.	80s	80s	80s	80s	40s	40s	40s	40s	40s									
WTmm	3.73	3.91	4.55	5.04	3.91	5.49	6.02	7.11	8.18									
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5									
TYPE	Seamless																	
MOC	ASTM A 312 TP 316L									ASTM A 358 TP 316L with 100% Radiography.								
Ends	PE				BE													
Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14.00	16.00	18	20	22	24	
TYPE	SW				Butt Weld													
Rating	# 3000				40s	40s	40s	40s	40s									
MOC	ASTM A 182 F 316L U.N.O				ASTM A 403 Gr. WP 316L , Seamless													
Elbow	ASMEB16.11				ASME B 16.9													
Tee	ASMEB16.11				ASME B 16.9													
Red.	ASMEB16.11				ASME B 16.9													
Cap	ASMEB16.11				ASME B 16.9													
Coupl	ASMEB16.11				ASME B 16.9													
Plug	ASMEB16.11				ASME B 16.9													
Union																		
Elbowlet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L													
Sockolet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L													
Weldolet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L													
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe													
Swage	MSS SP 95,Same as pipe				MSS SP 95,Same as pipe													
Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				FF Lap Joint (LJ)													
MOC	ASTM A 182 F 316 L				Stub end A 182 F316L + ASTM A 105 Backing flange (While connecting to FF flange use Slip -on(SO),FF, 316L flange													
FACE	RF-Serrated Finished				FF, Serrated Finished													
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25													
Spectacle Blind/Spacer Blinds																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
MOC	ASTM A 182 F 316 L																	
Spectacle	ASME B 16.48											Spacer and blind as per ASME B 16.48						
Mechanical Joints																		
Stud Bolt	ASTM A 193 Gr. B7M																	
Hex Nuts	ASTM A 194 Gr. 2HM																	
Gasket	ASME B 16.20,5mm, SS Spiral wound with CNAF filler																	

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
60/100	230	16/38	16.17	350 psig	25 Kg/cm2
200	195	93	13.7	Limited by Flange considereing flange as the weakest joint in piping system	
300	160	149	12.3		
400	145	204	11.25		
500	65	260	10.2		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2.All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Contractor use Schedule 80 for sizes less than 1/1/2" for threaded pipe and fittings.
7. Minimum sizes shall be 0.75 inch for piping other than air,instrument water and manufacturer's standard piping.

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION				Sheet No.	2 of 2
	Piping Class	Material	C.A	Spec.No	Revision: 0	
	A 10 # 150	SS 316 L	1.5mm	2004A,Rev 9	Page No.	62
	Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
	Service: Corrosive HydroCarbon Service-Sour Liquid and Vapour.					
	Instrument Gas Header including 1st Bolck Valve					

Branch Table as per API RP 14E

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Run Pipe	Branch Pipe																
	≤ 1	T															
	1-1.5	T	T														
	2	*	*	T													
	3	*	*	T	T												
	4	*	*	T	T	T											
	6	*	*	W	T	T	T										
	8	*	*	W	W	T	T	T									
	10	*	*	W	W	W	T	T	T								
	12	*	*	W	W	W	T	T	T	T							
	14	*	*	W	W	W	W	T	T	T	T						
	16	*	*	W	W	W	W	T	T	T	T	T					
	18	*	*	W	W	W	W	W	T	T	T	T	T				
	20	*	*	W	W	W	W	W	T	T	T	T	T	T			
	24	*	*	W	W	W	W	W	W	T	T	T	T	T	T		
	30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T	
		≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30	

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG																	
Rating	# 800, SW				# 150, RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB-107				VB-116												
Rating	# 800, SW				# 150, RF												
Globe Valve																	
TAG	VGL - 107				VGL - 116												
Rating	# 800, SW				# 150, RF												
Check Valve																	
TAG	VC - 107, Lift Check				VC - 116, Swing Check												
Rating	# 800, SW				# 150, RF												
Needle Valve																	
TAG																	
Rating																	
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 0	04-08-2008	Issued For Bid	PS	GJ	GRP

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**			Sheet No.	**1 of 2**
	Piping Class	Material	C.A	Spec.No	Revision: 0
	B 10 # 300	SS 316 L	1.5mm	2004A,Rev 9	Page No. 63
	Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3				
	Service	Corrosive HydroCarbon Service-Sour Liquid and Vapour.			

Pipe Data	ASME B 36.19																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273	323.9	355.6	406.4	457.2	508	558.8	609.6	
Sch.	160	160	160	160	40s	40s	40s	40s	40s									
WTmm	4.77	5.6	6.4	7.14	3.91	5.49	6.02	7.11	8.18									
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5									
TYPE	Seamless																	
MOC	ASTM A 312 TP 316L									ASTM A 358 TP 316L with 100% Radiography.								
Ends	PE				BE													
Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14.00	16.00	18	20	22	24	
TYPE	SW				Butt Weld													
Rating	# 6000				40s	40s	40s	40s	40s									
MOC	ASTM A 182 F 316L U.N.O				ASTM A 403 Gr. WP 316L , Seamless													
Elbow	ASMEB16.11				ASME B 16.9													
Tee	ASMEB16.11				ASME B 16.9													
Red.	ASMEB16.11				ASME B 16.9													
Cap	ASMEB16.11				ASME B 16.9													
Coupl	ASMEB16.11				ASME B 16.9													
Plug	ASMEB16.11				ASME B 16.9													
Union																		
Elbowlet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L													
Sockolet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L													
Weldolet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L													
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe													
Swage	MSS SP 95,Same as pipe				MSS SP 95,Same as pipe													
Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				WN													
MOC	ASTM A 182 F 316 L				A 182 F316L													
FACE	RF-Smooth Finished				RF, Smooth Finished													
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25													
Spectacle Blind/Spacer Blinds																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
MOC	ASTM A 182 F 316 L																	
Spectacle	ASME B 16.48											Spacer and blind as per ASME B 16.48						
Mechanical Joints																		
Stud Bolt	ASTM A 193 Gr. B7M																	
Hex Nuts	ASTM A 194 Gr. 2HM																	
Gasket	ASME B 16.20,5mm, SS 316L Spiral wound with CNAF filler																	

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
60/100	600	16/38	42.19	900 psig	63.3 Kg/cm2
200	505	93	35.51	Limited by Flange considereing flange as the weakest joint in piping system	

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Contractor use Schedule 80 for sizes less than 1/1/2" for threaded pipe and fittings.
7. Minimum sizes shall be 0.75 inch for piping other than air,instrument water and manufacturer's standard piping.

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION				Sheet No.	2 of 2
	Piping Class	Material	C.A	Spec.No	Revision: 0	
	B 10 # 300	SS 316 L	1.5mm	2004A,Rev 9	Page No.	64
	Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
	Service	Corrosive HydroCarbon Service-Sour Liquid and Vapour.				
	0					

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG					VB-151												
Rating	# 800, SW				# 300, RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB-107				VB-152												
Rating	# 800, SW				# 300, RF												
Globe Valve																	
TAG	VGL - 107				VGL - 154												
Rating	# 800, SW				# 300, RF												
Check Valve																	
TAG	VC - 107, Lift Check				VC - 154, Swing Check												
Rating	# 800, SW				# 300, RF												
Needle Valve																	
TAG																	
Rating																	
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 0	15.01.16	Issued For Bid	PS	AKM	SJ

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**			Sheet No.	**1 of 2**
	Piping Class	Material	C.A	Spec.No	Revision:0
	D 10 # 600	SS 316 L	1.5mm	2004A,Rev 9	Page No. 65
	Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3				
	Service: Corrosive HydroCarbon Service-Sour Liquid and Vapour.				

Pipe Data	ASME B 36.19																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6	
Sch.	XXS	160	160	160	80s	80s	80s	80s	80s									
WTmm	7.47	5.6	6.4	7.14	5.54	7.62	8.56	10.97	12.7									
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5									
TYPE	Seamless																	
MOC	ASTM A 312 TP 316L																	
Ends	PE				BE													
Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				Butt Weld													
Rating	# 9000	# 6000			80s	80s	80s	80s	80s									
MOC	ASTM A 182 F 316L U.N.O				ASTM A 403 Gr. WP 316L , Seamless													
Elbow	ASMEB16.11				ASME B 16.9													
Tee	ASMEB16.11				ASME B 16.9													
Red.	ASMEB16.11				ASME B 16.9													
Cap	ASMEB16.11				ASME B 16.9													
Coupl	ASMEB16.11				ASME B 16.9													
Plug	ASMEB16.11				ASME B 16.9													
Union																		
Elbowlet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L													
Sockolet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L													
Weldolet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L													
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe													
Swage	MSS SP 95,Same as pipe				MSS SP 95, MOC Same as pipe													
Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				WN													
MOC	ASTM A 182 F 316 L				ASTM A 182 F 316 L													
FACE	RF-Smooth Finished				RF, Smooth Finished													
STD.	ASME B 16.5				B 16.5, But welding ends as per B 16.25													
Spectacle Blind/Spacer Blinds																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
MOC	ASTM A 182 F 316 L																	
Spectacle	ASME B 16.48																	
Mechanical Joints																		
Stud Bolt	ASTM A 193 Gr. B7M																	
Hex Nuts	ASTM A 194 Gr. 2HM																	
Gasket	ASME B 16.20,5mm, SS 361L Spiral wound with CNAF filler																	

Pressure-Temperature Rating

Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2
60/100	1200	16/38	84.4
200	1015	93	71.38

Maximum Hydrostatic Pressure

Pressure, psig	Pres.,Kg/cm2g
1800 psig	126.6 Kg/cm2
Limited by Flange considereing flange as the weakest joint in piping system	

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Contractor use Schedule 80 for sizes less than 1/1/2" for threaded pipe and fittings.
7. Minimum sizes shall be 0.75 inch for piping other than air,instrument water and manufacturer's standard piping.

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION				Sheet No.	2 of 2
	Piping Class	Material	C.A	Spec.No	Revision:0	
	D 10 # 600	SS 316 L	1.5mm	2004A,Rev 9	Page No.	66
	Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
	Service: Corrosive HydroCarbon Service-Sour Liquid and Vapour.					
	0					

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG-132				VB-255												
Rating	# 1500, SW				# 900, RTJ FLGD												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB-131				VB-258												
Rating	# 800, SW				# 600, RF												
Globe Valve																	
TAG	VGL - 131				VGL - 255												
Rating	# 800, SW				# 600, RF												
Check Valve																	
TAG	VC - 131, Lift Check				VC - 255, Swing Check												
Rating	# 800, SW				# 600, RF												
Needle Valve																	
TAG	VN-111																
Rating	# 800, SW																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision:0	15.01.16	Issued For Bid	PS	AKM	SJ

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**			Sheet No.	**1 of 2**
	Piping Class	Material	C.A	Spec.No	Revision:0
	E 10 # 900	SS 316 L	1.5mm	2004A,Rev 9	Page No. 67
	Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3				
	Service: Corrosive HydroCarbon Service-Sour Liquid and Vapour.				

Pipe Data	ASME B 36.19																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6	
Sch.	XXS	160	160	160	160	80s	120	120	100									
WTmm	7.47	5.6	6.4	7.14	8.74	7.62	11.12	14.27	15.1									
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5									
TYPE	Seamless																	
MOC	ASTM A 312 TP 316L																	
Ends	PE				BE													
Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				Butt Weld													
Rating	# 9000	# 6000			160	80s	120	120	100									
MOC	ASTM A 182 F 316L U.N.O				ASTM A 403 Gr. WP 316L , Seamless													
Elbow	ASMEB16.11				ASME B 16.9													
Tee	ASMEB16.11				ASME B 16.9													
Red.	ASMEB16.11				ASME B 16.9													
Cap	ASMEB16.11				ASME B 16.9													
Coupl	ASMEB16.11				ASME B 16.9													
Plug	ASMEB16.11				ASME B 16.9													
Union																		
Elbowlet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L													
Sockolet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L													
Weldolet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L													
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe													
Swage	MSS SP 95,Same as pipe				MSS SP 95, MOC Same as pipe													
Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				WN													
MOC	ASTM A 182 F 316 L				ASTM A 182 F 316 L													
FACE	RTJ-Note 8				RTJ-Note 8													
STD.	ASME B 16.5				B 16.5, But welding ends as per B 16.25													
Spectacle Blind/Spacer Blinds																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
MOC	ASTM A 182 F 316 L																	
Spectacle	ASME B 16.48																	
Mechanical Joints																		
Stud Bolt	ASTM A 193 Gr. B7M																	
Hex Nuts	ASTM A 194 Gr. 2HM																	
Gasket	ASME B 16.20,OCT ring of SS316 with Max. Hardness of 120 BHN																	

Pressure-Temperature Rating

Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Maximum Hydrostatic Pressure	
				Pressure, psig	Pres.,Kg/cm2g
- 20/100	1799.96	-28.8/38	126.55	2700 psig	189.83
200	1534.98	93	107.92	Limited by Flange considereing flange as the weakest joint in piping system	

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Contractor use Schedule 80 for sizes less than 1/1/2" for threaded pipe and fittings.
7. Minimum sizes shall be 0.75 inch for piping other than air,instrument water and manufacturer's standard piping.
8. RTJ groove hardness shall be 140 BHN

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION				Sheet No.	2 of 2
	Piping Class	Material	C.A	Spec.No	Revision:0	
	E 10 # 900	SS 316 L	1.5mm	2004A,Rev 9	Page No.	68
	Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
	Service: Corrosive HydroCarbon Service-Sour Liquid and Vapour.					
	0					

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VB-131				VB-251												
Rating	# 1500, SW				# 900, RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	Use Gate Valve				VB-258												
Rating	# 1500, SW				# 900, RTJ FLGD												
Globe Valve																	
TAG	VGL - 132				VGL - 254												
Rating	# 1500, SW				# 900, RTJ, FLGD												
Check Valve																	
TAG	VC - 132, Lift Check				VC - 255, Swing Check												
Rating	# 1500, SW				# 900, RTJ, FLGD												
Needle Valve																	
TAG	VN-112																
Rating	# 1500, SW																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision:0	15.01.16	Issued For Bid	PS	AKM	SJ

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION				Sheet No.	1 of 2
	Piping Class	Material	C.A	Spec.No	Revision:0	
	F 10 # 1500	SS 316 L	1.5mm	2004A,Rev 9	Page No.	69
	Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
	Service: Corrosive HydroCarbon Service-Sour Liquid and Vapour.					

Pipe Data	ASME B 36.19																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	XXS	XXS	XXS	XXS	160	160	160	160	160								
WTmm	7.47	7.8	9.1	10.16	8.74	11.1	13.49	18.26	23								
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5								
TYPE	Seamless																
MOC	ASTM A 312 TP 316L																
Ends	PE				BE												
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				Butt Weld												
Rating	# 9000				160	160	160	160	160								
MOC	ASTM A 182 F 316L U.N.O				ASTM A 403 Gr. WP 316L , Seamless												
Elbow	ASMEB16.11				ASME B 16.9												
Tee	ASMEB16.11				ASME B 16.9												
Red.	ASMEB16.11				ASME B 16.9												
Cap	ASMEB16.11				ASME B 16.9												
Coupl	ASMEB16.11				ASME B 16.9												
Plug	ASMEB16.11				ASME B 16.9												
Union																	
Elbowlet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L												
Sockolet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L												
Weldolet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L												
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe												
Swage	MSS SP 95,Same as pipe				MSS SP 95, MOC Same as pipe												
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				WN												
MOC	ASTM A 182 F 316 L				ASTM A 182 F 316 L												
FACE	RTJ-Note 8				RTJ-Note 8												
STD.	ASME B 16.5				B 16.5, But welding ends as per B 16.25												
Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM A 182 F 316 L																
Spectacle	ASME B 16.48																
Mechanical Joints																	
Stud Bolt	ASTM A 193 Gr. B7M																
Hex Nuts	ASTM A 194 Gr. 2HM																
Gasket	ASME B 16.20,OCT ring of SS316 with Max. Hardness of 120 BHN																

Pressure-Temperature Rating

Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Maximum Hydrostatic Pressure	
				Pressure, psig	Pres.,Kg/cm2g
- 20/100	2999.7	-28.8/38	210.9	4499.5 psig	316.35
122	2901.6	50	204	Limited by Flange considereing flange as the weakest joint in piping system	

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Contractor use Schedule 80 for sizes less than 1/1/2" for threaded pipe and fittings.
7. Minimum sizes shall be 0.75 inch for piping other than air,instrument water and manufacturer's standard piping.
8. RTJ groove hardness shall be 140 BHN

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION				Sheet No.	2 of 2
	Piping Class	Material	C.A	Spec.No	Revision:0	
	F 10 # 1500	SS 316 L	1.5mm	2004A,Rev 9	Page No.	70
	Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
	Service: Corrosive HydroCarbon Service-Sour Liquid and Vapour.					
	0					

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VB-231				VB-351												
Rating	# 2500, SW				# 2500, RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	Use Gate Valve				VB-358												
Rating	# 2500, SW				# 2500, RTJ FLGD												
Globe Valve																	
TAG	VGL - 232				VGL - 354												
Rating	# 2500, SW				# 2500, RTJ, FLGD												
Check Valve																	
TAG	VC - 232, Lift Check				VC - 355, Swing Check												
Rating	# 2500, SW				# 2500, RTJ, FLGD												
Needle Valve																	
TAG	VN-212																
Rating	# 2500, SW																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision:0	15.01.16	Issued For Bid	PS	AKM	SJ

	Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION				Sheet No.	1 of 2
		Piping Class		Material	C.A	Spec.No	Revision : 1
		A12	# 150	Titanium	Nil	2004A,Rev 9	Page No. 71
		Design Code:		**ASME B 31.3**			
		Service:	Chemical (Ferric Chloride),Coagulant				

Pipe Data	ASME B 36.19 (BS-1600)																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3									
Sch.	40	40	40	40	40	40	10	10									
WTmm	2.77	2.87	3.88	3.68	3.91	5.49	3.05	3.4									
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5									
TYPE	Seamless																
MOC	ASTM B 861 Gr. 2																
Ends	BE																
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Butt Welded																
Rating	2.77	2.87	3.88	3.68	3.91	5.49	3.05	3.4									
MOC	ASTM B 363 Gr. WPT 2																
Elbow	ASME B 16.9 and ASME B 16.28 for short radius elbow and returns																
Tee	ASME B 16.9																
Red.	ASME B 16.9																
Cap	ASME B 16.9																
Coupl	ASME B 16.9																
Plug	ASME B 16.9																
Union																	
Elbowlet	ASME B 16.9																
Sockolet																	
Weldolet	MSS SP 97																
Nipple	MOC Same as pipe																
Swage	MSS SP95, MOC Same as pipe																
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Lap Joint Flange / WN Flange																
MOC	LJ=Innner Flange-B 363 WPT2, Outer Flange A 105 and WN= B 381 Gr. F2																
FACE	RF																
STD.	ASME B 16.5																
Blind Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM B 381 Gr. F 2 as per ASME B 16.5																
Mechanical Joints																	
Stud Bolts	ASTM A 193 Gr. B7 to be used with insulating gasket																
Hex Nuts	ASTM A 194 Gr. 2H to be used with insulating gasket																
Gasket	ASME B 16.21,Flat Ring,1.58mm,CNAF																

Pressure-Temperature Rating

Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2
347	199	175	14

Maximum Hydrostatic Pressure

Pressure, psig	Pres.,Kg/cm2g
300 psig	21 Kg/cm2
Limited by Flange considereing flange as the weakest joint in piping system	

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Pulled cold worked bend shall not be used.
2.All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. PMS to be read in conjuction with FS 2004 A.
4. All flanges are to be epoxy coated for corrosion protection.

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**2 of 2**
		Piping Class	Material	C.A	Spec.No		Revision : 1
		A12 # 150	Titanium	Nil	2004A,Rev 9	Page No.	72
		Design Code:	**ASME B 31.3**				
		Service:	Chemical (Ferric Chloride),Coagulant				
		0					

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6
≤ 1	T					
1-1.5	T	T				
2	T	T	T			
3	W	W	T	T		
4	W	W	T	T	T	
6	W	W	W	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG																	
Rating																	
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB-99																
Rating	# 150, RF																
Globe Valve																	
TAG	VGL-99																
Rating	# 150, RF																
Check Valve																	
TAG	VC-99																
Rating	# 150, RF																
Needle Valve																	
TAG	VN-111																
Rating																	
Butter Fly Valve																	
TAG	VN-111																
Rating																	

Notes Releted to Valve and Brach Connection

1. Valve body and seat test pressure shall be as per BS-5146.
2. PMS to be read in conjuction with FS 2004 A.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision : 1	15.01.16	Issued For Bid	PS	AKM	SJ

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**								Sheet No.		**1 of 2**						
		Piping Class				Material		C.A		Spec.No				Revision: 1				
		A 13			# 150		Copper		Nil		2004A,Rev 9			Page No.	73			
		Design Code:					**ASME B 31.3**											
		Service:					Potable Water - Hot and Cold											

Pipe Data	ASME B 36.19 (BS-1600)																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3											
Sch.																		
WTmm	2.72	2.9	3.2	3.81	3.96	5.56	6.35											
M.T (%)																		
TYPE	Seamles Hard Drawn H 80(Regular)				Seamless light Drawn H55(Regular)													
MOC	ASTM B 42 UNS C1 2200				ASTM B 42 UNS C1 2200													
Ends	BE																	
Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	Brazed Fittings					Butt Welded												
Rating	2.72	2.9	3.2	3.81	3.96	5.56	6.35											
MOC	ASTM B 124 UNS C1 1000					ASTM B 42 UNS C1 2200												
Elbow																		
Tee																		
Red.																		
Cap																		
Coupl																		
Plug																		
Union																		
Elbowlet																		
Sockolet																		
Weldolet																		
Nipple																		
Swage																		
Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	Slip on composite					WN Composite												
MOC	Innner Flange B 61, Outer Flange A 105(Galv.Epoxy coated)																	
FACE	FF																	
STD.	ISO-7005-3																	
Blind Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
MOC	ASTM A 105 With 3mm copper over lay																	
Mechanical Joints																		
Stud Bolts	ASTM A 193 Gr. B7 to be used with insulating gasket																	
Hex Nuts	ASTM A 194 Gr. 2H to be used with insulating gasket																	
Gasket	ASME B 16.21,Flat Ring,2mm,CNAF																	

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
194	227	90	16	340	24 Kg/cm2
				Limited by Flange considereing flange as the weakest joint in piping system	

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Threading on piping is not allowed. Only screwed fittings shall be used for vents and drains.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. PMS to be read in conjuction with FS 2004 A.
4. Whereever there is connection between CS and copper,scrificial spool and electrical isolation kit i.e. insulating gaskets shall be used.
5. For flanged valves FF valves are to be used for sizes 2" and above. FF solid slip on # 150 flanges conforming to ASTM B 124 shall be used.

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**			Sheet No.	**2 of 2**	
		Piping Class	Material	C.A	Spec.No	Revision: 1	
		A 13 # 150	Copper	Nil	2004A,Rev 9	Page No.	74
		Design Code:	**ASME B 31.3**				
		Service:	Potable Water - Hot and Cold				
		0					

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4
≤ 1	T				
1-1.5	T	T			
2	*	T	T		
3	*	*	T	T	
4	*	*	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG - 50				VG 80												
Rating	230 PSI CWP				# 150												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB 50				VB 86												
Rating	230 PSI CWP				# 150												
Globe Valve																	
TAG	VGL 59				VGL 80												
Rating	230 PSI CWP				# 150												
Check Valve																	
TAG	VC 50				VC 80												
Rating	230 PSI CWP				# 150												
Needle Valve																	
TAG	VN 11																
Rating	230 PSI CWP																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.

Revision: 1	15.01.16	Issued For Bid	PS	AKM	SJ
Revision	Date	Description	Prepared By	Reviewed By	Approved By

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**			Sheet No.	**1 of 2**
	Piping Class	Material	C.A	Spec.No	Revision : 0
	A1LN	# 150	LTCS-NACE	3mm	2004A,Rev 9	Page No. 75
	Design Code:	**ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3**			
	Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour (Low temperature).			

Pipe Data	ASME B 36.10																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	XXS	160	160	160	80	80	80	80	Std.	Std.	Std.	Std.	Std.				
WTmm	7.47	5.56	6.35	7.14	5.54	7.62	8.56	10.97	8.18	9.27	9.52	9.52	9.52				
M.T (%)	12.5	12.5	11.0	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5				
TYPE	Seamless																
MOC	ASTM A 333 Gr.6																
Ends	PE				BE												
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Socket Weld				Butt Weld												
Rating	# 9000		# 6000		80	80	80	80	Std.	Std.	Std.	Std.	Std.				
MOC	ASTM A 350 Gr. LF2 U.N.O				ASTM A 420 Gr. WPL6, Seamless U.N.O												
Elbow	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Tee	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Red.	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Cap	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Coupl	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Plug	ASMEB16.11				ASME B 16.9									ASME B 16.9			
Union																	
Elbowlet	MSS SP 97,ASTM A 350 Gr. LF2				MSS SP 97,ASTM A 350 Gr. LF2									MSS SP 97,ASTM A 350 Gr. LF2			
Sockolet	MSS SP 97,ASTM A 350 Gr. LF2				MSS SP 97,ASTM A 350 Gr. LF2									MSS SP 97,ASTM A 350 Gr. LF2			
Weldolet	MSS SP 97,ASTM A 350 Gr. LF2				MSS SP 97,ASTM A 350 Gr. LF2									MSS SP 97,ASTM A 350 Gr. LF2			
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe									MOC Same as pipe			
Swage	MSS SP 95,Same as pipe				MSS SP 95,MOC Same as pipe									MSS SP 95,MOC Same as pipe			
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				WN												
MOC	ASTM A 350 Gr. LF2				ASTM A 350 Gr. LF2												
FACE	RF-Serrated Finished				RF, Serrated Finished												
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25												
Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM A 350 Gr. LF2																
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48					
Mechanical Joints																	
Stud Bolts	ASTM A 320 Gr. L7				ASTM A 320 Gr. L7												
Hex Nuts	ASTM A 194 Gr. 4				ASTM A 194 Gr.4												
Gasket	ASME B 16.21,Flat Ring,1.58mm,CNAF				ASME B 16.21,Flat Ring,1.58mm,CNAF												

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
(-) 50 to 100	285	(-) 45 to 38	20	450 psig	31.6 Kg/cm2
200	260	93	18.3	Limited by Flange considereing flange as the weakest joint in piping system	
300	230	149	16.2		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A and all the requirement of MR-01-75/ISO-15156-1/2/3 to complied.
6. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and specified spectacle blinds.
7. In instrument gas lines - Use this specification for main Header and SS316L piping for Sub Header. Presssure-temperature rating of SS316L shall compatible to this piping class and both shall comply all the requirement of MR-01-75/ISO-15156-1/2/3

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION				Sheet No.	2 of 2
	Piping Class	Material	C.A	Spec.No	Revision : 0	
	A1LN	# 150 LTCS-NACE	3mm	2004A,Rev 9	Page No.	76
	Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
	Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour (Low temperature).				
	0					

Branch Table as per API RP 14E

Run Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Branch Pipe

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 45N,SW				VG 65N,FLANGED												
Rating	# 800				# 150,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB 45N,SW and VB 107N,SW (Note-5)				VB 65N, Lever Operated							VB **, Gear Operated					
Rating	# 800				# 150, RF												
Globe Valve																	
TAG	VGL 45N,SW				VGL 65N												
Rating	# 800				# 150, RF												
Check Valve																	
TAG	VC 45N,SW,Lift Check				VC 65N,Swing Check/ VC 82N (Wafer Check, Note -1)												
Rating	# 800				# 150, RF												
Needle Valve																	
TAG	VN 9N,SW																
Rating	# 800																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. Wafer check valve to be avoided as far as possible; unless the avaliable space constraint does not allow normal check valve . The contractor to take approvel for use of wafer check Valve during pipe routing after extablishing the space constraint.
2. Wafer type Butter Fly Valve may be used only in fire water(Clean water) service.
3. PMS to be read in conjuction with FS 2004 A.
4. Maximum temperature limit for all Ball Valve shall be 121 Degree C
5. In instrument gas lines - Use VB -45 for main Header and VB-107 for Sub Header. The MOC of VB-107 shall be SS-316L as per VMS and comply all the requirement of MR-01-75/ISO-15156-1/2/3

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision : 0	22/7/2009	Issued For Bid	PS	GJ	GRP

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**1 of 2**
		Piping Class		Material	C.A	Spec.No	Revision: 0
		A 22	# 150	SS 316 L	Nil	2004A,Rev 9	Page No. 77
		Design Code:		**ASME B 31.3**			
		Service:		Chemical			

Pipe Data	ASME B 36.19																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	40s	40s	40s	40s	10s	10s	10s	10s									
WTmm	2.77	2.87	3.38	3.68	2.77	3.05	3.05	3.4									
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5									
TYPE	Seamless																
MOC	ASTM A 312 TP 316L																
Ends	PE				BE												
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	PE				Butt Weld												
Rating	# 3000				10s	10s	10s	10s									
MOC	ASTM A 182 F 316L U.N.O				ASTM A 403 Gr. WP 316L , Seamless												
Elbow	ASMEB16.11				ASME B 16.9												
Tee	ASMEB16.11				ASME B 16.9												
Red.	ASMEB16.11				ASME B 16.9												
Cap	ASMEB16.11				ASME B 16.9												
Coupl	ASMEB16.11				ASME B 16.9												
Plug	ASMEB16.11				ASME B 16.9												
Union																	
Elbowlet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L												
Sockolet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L												
Weldolet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L												
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe												
Swage	MSS SP 95,Same as pipe				MSS SP 95,MOC Same as pipe												
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				Lap Joint (LJ)												
MOC	ASTM A 182 F 316 L				Stub end A 182 F316L + ASTM A 105 Backing flange (While connecting to FF flange use Slip -on(SO),FF, F 316L flange												
FACE	RF-Serrated Finished				RF, Serrated Finished												
STD.	ASME B 16.5				B 16.5, But welding ends as per B 16.25												
Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM A 182 F 316 L																
Spectacle	ASME B 16.48																
Mechanical Joints																	
Stud Bolts	ASTM A 193 Gr. B7																
Hex Nuts	ASTM A 194 Gr. 2H																
Gasket	ASME B 16.20,5mm, SS Spiral wound with CNAF filler																

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
60/100	230	16/38	16.17	350 psig	25 Kg/cm2
200	195	93	13.7	Limited by Flange considereing flange as the weakest joint in piping system	
300	160	149	12.3		
400	145	204	11.25		
500	65	260	10.2		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2.All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Contractor use Schedule 80 for sizes less than 1/1/2" for threaded pipe and fittings.
7. Minimum sizes shall be 0.75 inch for piping other than air,instrument water and manufacturer's standard piping.

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**2 of 2**
	Piping Class		Material	C.A	Spec.No	Revision: 0
	A 22	# 150	SS 316 L	Nil	2004A,Rev 9	Page No. 78
	Design Code:		**ASME B 31.3**			
	Service:	Chemical				
	0					

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VB-131				VB-151												
Rating	# 800, SW				# 150, RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB-131				VB-151												
Rating	# 800, SW				# 150, RF												
Globe Valve																	
TAG	VGL - 131				VGL - 151												
Rating	# 800, SW				# 150, RF												
Check Valve																	
TAG	VC - 131, Lift Check				VC - 151, Swing Check												
Rating	# 800, SW				# 150, RF												
Needle Valve																	
TAG	VN-111																
Rating	# 800, SW																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.

Revision: 0	04-08-2008	Issued For Bid	PS	GJ	GRP
Revision	Date	Description	Prepared By	Reviewed By	Approved By

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**1 of 2**
		Piping Class		Material	C.A	Spec.No	Revision: 1
		B 22	# 300	SS 316 L	Nil	2004A,Rev 9	Page No. 79
		Design Code:	**ASME B 31.3**				
		Service:	Chemical				

Pipe Data	ASME B 36.19																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	40s	40s	40s	40s	10s	10s	10s	40s									
WTmm	2.77	2.87	3.38	3.68	2.77	3.05	3.05	7.11									
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5									
TYPE	Seamless																
MOC	ASTM A 312 TP 316L																
Ends	PE				BE												
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				Butt Weld												
Rating	# 3000				10s	10s	10s	40s									
MOC	ASTM A 182 F 316L U.N.O				ASTM A 403 Gr. WP 316L , Seamless												
Elbow	ASMEB16.11				ASME B 16.9												
Tee	ASMEB16.11				ASME B 16.9												
Red.	ASMEB16.11				ASME B 16.9												
Cap	ASMEB16.11				ASME B 16.9												
Coupl	ASMEB16.11				ASME B 16.9												
Plug	ASMEB16.11				ASME B 16.9												
Union																	
Elbowlet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L												
Sockolet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L												
Weldolet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L												
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe												
Swage	MSS SP 95,Same as pipe				MSS SP 95,MOC Same as pipe												
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				Lap Joint (LJ)												
MOC	ASTM A 182 F 316 L				Stub end A182 F 316L + ASTM A 105 Backing flange (While connecting to FF flange use Slip -on(SO),FF, F 316L flange												
FACE	RF-Smooth Finished				RF, Smooth Finished												
STD.	ASME B 16.5				B 16.5, But welding ends as per B 16.25												
Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM A 182 F 316 L																
Spectacle	ASME B 16.48																
Mechanical Joints																	
Stud Bolts	ASTM A 193 Gr. B7																
Hex Nuts	ASTM A 194 Gr. 2H																
Gasket	ASME B 16.20,5mm, SS Spiral wound with CNAF filler																

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
60/100	600	16/38	42.19	900 psig	63.3 Kg/cm2
200	505	93	35.51	Limited by Flange considereing flange as the weakest joint in piping system	

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2.All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Contractor use Schedule 80 for sizes less than 1/1/2" for threaded pipe and fittings.
7. Minimum sizes shall be 0.75 inch for piping other than air,instrument water and manufacturer's standard piping.

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION				Sheet No.	2 of 2
	Piping Class	Material	C.A	Spec.No		Revision: 1
	B 22 # 300	SS 316 L	Nil	2004A,Rev 9	Page No.	80
	Design Code:	**ASME B 31.3**				
	Service:	Chemical				
	0					

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG		VB-131			VB-151												
Rating		# 800, SW			# 300, RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG		VB-131			VB-152												
Rating		# 800, SW			# 300, RF												
Globe Valve																	
TAG		VGL - 131			VGL - 154												
Rating		# 800, SW			# 300, RF												
Check Valve																	
TAG		VC - 131, Lift Check			VC - 154, Swing Check												
Rating		# 800, SW			# 300, RF												
Needle Valve																	
TAG		VN-111															
Rating		# 800, SW															
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 1	22/7/2009	Issued For Bid	PS	GJ	GRP

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**			Sheet No.	**1 of 2**
	Piping Class	Material	C.A	Spec.No	Revision:2
	D 22 # 600	SS 316 L	Nil	2004A,Rev 9	Page No. 81
	Design Code:	**ASME B 31.3**			
	Service:	Chemical			

Pipe Data	ASME B 36.19																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	40s	40s	40s	80s	40s	40s	40s	80s									
WTmm	2.77	2.87	3.38	5.08	3.91	5.49	6.02	10.97									
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5									
TYPE	Seamless																
MOC	ASTM A 312 TP 316L																
Ends	PE				BE												
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				Butt Weld												
Rating	# 3000			# 6000	40s	40s	40s	80s									
MOC	ASTM A 182 F 316L U.N.O				ASTM A 403 Gr. WP 316L., Seamless												
Elbow	ASMEB16.11				ASME B 16.9												
Tee	ASMEB16.11				ASME B 16.9												
Red.	ASMEB16.11				ASME B 16.9												
Cap	ASMEB16.11				ASME B 16.9												
Coupl	ASMEB16.11				ASME B 16.9												
Plug	ASMEB16.11				ASME B 16.9												
Union																	
Elbowlet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L												
Sockolet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L												
Weldolet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L												
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe												
Swage	MSS SP 95,Same as pipe				MSS SP 95, MOC Same as pipe												
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				WN												
MOC	ASTM A 182 F 316 L				ASTM A 182 F 316 L												
FACE	RF-Smooth Finished				RF, Smooth Finished												
STD.	ASME B 16.5				B 16.5, But welding ends as per B 16.25												
Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM A 182 F 316 L																
Spectacle	ASME B 16.48																
Mechanical Joints																	
Stud Bolts	ASTM A 193 Gr. B7																
Hex Nuts	ASTM A 194 Gr. 2H																
Gasket	ASME B 16.20,5mm, SS Spiral wound with CNAF filler																

Pressure-Temperature Rating

Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2
60/100	1200	16/38	84.4
200	1015	93	71.38

Maximum Hydrostatic Pressure

Pressure, psig	Pres.,Kg/cm2g
1800 psig	126.6 Kg/cm2
Limited by Flange considereing flange as the weakest joint in piping system	

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Contractor use Schedule 80 for sizes less than 1/1/2" for threaded pipe and fittings.
7. Minimum sizes shall be 0.75 inch for piping other than air,instrument water and manufacturer's standard piping.

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**													Sheet No.	**2 of 2**	
		Piping Class			Material		C.A		Spec.No							Revision:2	
		D 22	# 600		SS 316 L		Nil		2004A,Rev 9							Page No.	82
		Design Code:			**ASME B 31.3**												
		Service:	Chemical														
		0															

Branch Table as per API RP 14E

Run Pipe	Branch Pipe															
	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30	
≤ 1	T															
1-1.5	T	T														
2	*	*	T													
3	*	*	T	T												
4	*	*	T	T	T											
6	*	*	W	T	T	T										
8	*	*	W	W	T	T	T									
10	*	*	W	W	W	T	T	T								
12	*	*	W	W	W	T	T	T	T							
14	*	*	W	W	W	W	T	T	T	T						
16	*	*	W	W	W	W	T	T	T	T	T					
18	*	*	W	W	W	W	W	T	T	T	T	T				
20	*	*	W	W	W	W	W	T	T	T	T	T	T			
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T		
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T	

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG		VB-131			VB-251												
Rating		# 800, SW			# 600, RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG		VB-131			VB-258												
Rating		# 800, SW			# 600, RF												
Globe Valve																	
TAG		VGL - 131			VGL - 255												
Rating		# 800, SW			# 600, RF												
Check Valve																	
TAG		VC - 131, Lift Check			VC - 255, Swing Check												
Rating		# 800, SW			# 600, RF												
Needle Valve																	
TAG		VN-111															
Rating		# 800, SW															
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.

Revision:2	15.01.16	Issued For Bid	PS	AKM	SJ
Revision	Date	Description	Prepared By	Reviewed By	Approved By

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**												Sheet No.		**1 of 2**	
		Piping Class		Material		C.A	Spec.No									Revision:0	
		E 22	#900	SS 316 L		Nil	2004A,Rev 9									Page No.	83
		Design Code:		**ASME B 31.3**													
		Service:		Chemical													

Pipe Data	ASME B 36.19																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	80s	80s	80s	160	80s	80s	80s	80s									
WTmm	3.73	3.9	4.6	7.14	5.54	7.62	8.56	10.97									
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5									
TYPE	Seamless																
MOC	ASTM A 312 TP 316L																
Ends	PE				BE												

Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				Butt Weld												
Rating	# 3000			# 6000	80s	80s	80s	80s									
MOC	ASTM A 182 F 316L U.N.O				ASTM A 403 Gr. WP 316L , Seamless												
Elbow	ASMEB16.11				ASME B 16.9												
Tee	ASMEB16.11				ASME B 16.9												
Red.	ASMEB16.11				ASME B 16.9												
Cap	ASMEB16.11				ASME B 16.9												
Coupl	ASMEB16.11				ASME B 16.9												
Plug	ASMEB16.11				ASME B 16.9												
Union																	
Elbowlet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L												
Sockolet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L												
Weldolet	MSS SP 97				MSS SP 97,ASTM A 182 F 316L												
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe												
Swage	MSS SP 95,Same as pipe				MSS SP 95, MOC Same as pipe												

Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				WN												
MOC	ASTM A 182 F 316 L				ASTM A 182 F 316 L												
FACE	RTJ-Note 8				RTJ-Note 8												
STD.	ASME B 16.5				6.5, But welding ends as per B 16.25												

Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM A 182 F 316 L																
Spectacle	ASME B 16.48																

Mechanical Joints	
Stud Bolt	ASTM A 193 Gr. B7
Hex Nuts	ASTM A 194 Gr. 2H
Gasket	ME B 16.20,OCT ring of SS316L with Max. Hardness of 120 BHN

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
- 20/100	1799.96	-28.8/38	126.55	2700 psig	189.83
200	1534.98	93	107.92	Limited by Flange considereing flange as the weakest joint in piping system	

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Contractor use Schedule 80 for sizes less than 1/1/2" for threaded pipe and fittings.
7. Minimum sizes shall be 0.75 inch for piping other than air,instrument water and manufacturer's standard piping.
8. RTJ groove hardness shall be 140 BHN

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION				Sheet No.	2 of 2
	Piping Class	Material	C.A	Spec.No	Revision:0	
	E 22 #900	SS 316 L	Nil	2004A,Rev 9	Page No.	84
	Design Code:	**ASME B 31.3**				
	Service:	Chemical				
	0					

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG-132				VB-255												
Rating	# 1500, SW				# 900, RTJ FLGD												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	Use Gate Valve				VB-258												
Rating	# 1500, SW				# 900, RTJ FLGD												
Globe Valve																	
TAG	VGL - 132				VGL - 254												
Rating	# 1500, SW				# 900, RTJ, FLGD												
Check Valve																	
TAG	VC - 132, Lift Check				VC - 255, Swing Check												
Rating	# 1500, SW				# 900, RTJ, FLGD												
Needle Valve																	
TAG	VN-112																
Rating	# 1500, SW																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision:0	15.01.16	Issued For Bid	PS	AKM	SJ

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**													Sheet No.	**1 of 2**			
		Piping Class				Material				C.A	Spec.No					Revision: 2			
		A23				# 150		DSS-NACE		Nil	2004A,Rev 9					Page No.	85		
		Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3																	
		Service: Corrosive HydroCarbon Service-Sour Liquid and Vapour.																	

Pipe Data	ASME B 36.10																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6	
Sch.	40s	40s	40s	40s	10s	10s	10s	10s	10s	10s	10s	10s	10s	10s	10s	10s	10s	
WTmm	2.77	2.87	3.38	3.68	2.77	3.05	3.05	3.4	3.76	4.19	4.57	4.77	4.77	4.77	5.54	5.54	6.35	
M.T (%)	12.5	12.5	5.0	6.0	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	
TYPE	Seamless									Welded (Longitudinally) with 100% radiography								
MOC	ASTM A 790 Gr. S 32205									ASTM A 790 Gr. 32205								
Ends	PE				BE													
Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	Socket Weld				Butt Weld													
Rating	# 3000				10s	10s	10s	10s	10s	10s	10s	10s	10s	10s	10s	10s	10s	
MOC	ASTM A 182 Gr. F60				ASTM A 815 Gr.WP-S U.N.S 32205					ASTM A 815 Gr.WP-WX U.N.S 32205								
Elbow	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Tee	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Red.	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Cap	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Coupl	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Plug	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Union																		
Elbowlet	MSS SP 97,ASTM A 182 Gr. F60				MSS SP 97,ASTM A 182 Gr. F60					MSS SP 97,ASTM A 182 Gr. F60								
Sockolet	MSS SP 97,ASTM A 182 Gr. F60				MSS SP 97,ASTM A 182 Gr. F60					MSS SP 97,ASTM A 182 Gr. F60								
Weldolet	MSS SP 97,ASTM A 182 Gr. F60				MSS SP 97,ASTM A 182 Gr. F60					MSS SP 97,ASTM A 182 Gr. F60								
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe					MSS SP 97,ASTM A 182 Gr. F60								
Swage	MSS SP 95,Same as pipe				MSS SP 95, MOC Same as pipe					MSS SP 97,ASTM A 182 Gr. F60								
Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				WN													
MOC	ASTM A 182 Gr. F60				ASTM A 182 Gr. F60													
FACE	RF,Serrated Finnished				RF,Serrated Finnished													
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25													
Spectacle Blind/Spacer Blinds																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
MOC	ASTM A 182 Gr. F60																	
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48						
Mechanical Joints																		
Stud Bolts	ASTM A 453 Gr. 660 CL A																	
Hex Nuts	ASTM A 453 Gr. 660 CL A																	
Gasket	ASME B 16.20 , 5mm thick Duplex SS Spiral wound gasket with CNAF filler																	

Pressure-Temperature Rating

Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2
60 to 100	290	16 to 38	20.4
200	260	93	18.3
250	245	121	17.2

Maximum Hydrostatic Pressure

Pressure, psig	Pres.,Kg/cm2g
435 psig	30.6 Kg/cm2
Limited by Flange considereing flange as the weakest joint in piping system	

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval
2. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange, specified spectacle blinds and
3. All pipe thread shall be as per ASME B 1.20.1
4. PMS to be read in conjuction with FS 2004 A.
5. Weld Joint Factor for welded pipe shall be as per ASME B 31.3
6. Contractor to use Schedule 80 for sizes less than 1/1/2" for threaded pipe and fittings.
7. Material shall comply heat treatment requirement of NACE MR 01 75/ISO-15156-3 inadditon to the requirement of this material standard.
8. Allowable stress considered for this material shall be based on testing and actual evaluation of allowable stress of the material within the temperature range for which this material is to be used..
9. Thickness indicated is for referance only. To be finalized based on Note- 7,8 above.

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**												Sheet No.	**2 of 2**			
		Piping Class						Material				C.A		Spec.No			Revision: 2	
		A23					# 150			DSS-NACE		Nil		2004A,Rev 9		Page No.	86	
		Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3																
		Service:																
		0																

Branch Table as per API RP 14E

Run Pipe																	
≤ 1	T																
1-1.5	T	T															
2	*	*	T														
3	*	*	T	T													
4	*	*	T	T	T												
6	*	*	W	T	T	T											
8	*	*	W	W	T	T	T										
10	*	*	W	W	W	T	T	T									
12	*	*	W	W	W	T	T	T	T								
14	*	*	W	W	W	W	T	T	T	T							
16	*	*	W	W	W	W	T	T	T	T	T						
18	*	*	W	W	W	W	W	T	T	T	T	T					
20	*	*	W	W	W	W	W	T	T	T	T	T	T				
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T			
30	*	*	W	W	W	W	W	W	W	T	T	T	T	T	T		
	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30		
	Branch Pipe																

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 133,SW				VG 158,FLANGED												
Rating	# 800				# 150,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB 133,SW				VB 165, Lever Operated												
Rating	# 800				# 150,RF												
Globe Valve																	
TAG	VGL 133,SW				VGL 158												
Rating	# 800				# 150,RF												
Check Valve																	
TAG	VC 133,SW,Lift Check				VC 159,Swing Check												
Rating	# 800				# 150,RF												
Needle Valve																	
TAG	VN 103,SW																
Rating	# 800																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.
2. Maximum temperature limit for all Ball Valve shall be 121 Degree C

Revision: 2	15.01.16	Issued For Bid	PS	AKM	SJ
Revision	Date	Description	Prepared By	Reviewed By	Approved By

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**												Sheet No.		**1 of 2**	
		Piping Class				Material			C.A	Spec.No				Revision: 2			
		B23		# 300		DSS-NACE			Nil	2004A,Rev 9				Page No.	87		
		Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3															
		Service: Corrosive HydroCarbon Service-Sour Liquid and Vapour.															

Pipe Data	ASME B 36.19																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6	
Sch.	40s	40s	40s	40s	40s	10s	10s	10s	10s	10s	20	10	10	20	20	20	20	
WTmm	2.77	2.87	3.38	3.68	3.91	3.05	3.05	3.4	3.76	4.19	6.35	6.35	6.35	7.92	9.52	9.52	9.53	
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	
TYPE	Seamless									Welded (Longitudinally) with 100% radiography								
MOC	ASTM A 790 Gr. S 32205									ASTM A 790 Gr. 32205				ASTM A 928 Class 1 ,Gr. 32205(Note-6,7,8)				
Ends	PE				BE													
Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	Socket Weld				Butt Weld													
Rating	# 3000				10s	10s	10s	10s	10s	10s	20	10	10	20	20	20	20	
MOC	ASTM A 182 Gr. F60				ASTM A 815 Gr.WP-S U.N.S 32205					ASTM A 815 Gr.WP-WX U.N.S 32205								
Elbow	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Tee	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Red.	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Cap	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Coupl	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Plug	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Union																		
Elbowlet	MSS SP 97,ASTM A 182 Gr. F60				MSS SP 97,ASTM A 182 Gr. F60					MSS SP 97,ASTM A 182 Gr. F60								
Sockolet	MSS SP 97,ASTM A 182 Gr. F60				MSS SP 97,ASTM A 182 Gr. F60					MSS SP 97,ASTM A 182 Gr. F60								
Weldolet	MSS SP 97,ASTM A 182 Gr. F60				MSS SP 97,ASTM A 182 Gr. F60					MSS SP 97,ASTM A 182 Gr. F60								
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe					MSS SP 97,ASTM A 182 Gr. F60								
Swage	MSS SP 95,Same as pipe				MSS SP 95, MOC Same as pipe					MSS SP 97,ASTM A 182 Gr. F60								
Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				WN													
MOC	ASTM A 182 Gr. F60				ASTM A 182 Gr. F60													
FACE	RF,Smooth Finnished				RF,Smooth Finnished													
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25													
Spectacle Blind/Spacer Blinds																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
MOC	ASTM A 182 Gr. F60																	
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48						
Mechanical Joints																		
Stud Bolts	ASTM A 453 Gr. 660 CL A																	
Hex Nuts	ASTM A 453 Gr. 660 CL A																	
Gasket	ASME B 16.20 , 5mm thick Duplex SS Spiral wound gasket with CNAF filler																	

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
60 to 100	750	16 to 38	52.74	1125 psig	79.1 Kg/cm2
200	720	93	50.63	Limited by Flange considereing flange as the weakest joint in piping system	
250	692.5	121	48.69		
302	664	150	46.69		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
2. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange, specified spectacle blinds and flanges.
3. All pipe thread shall be as per ASME B 1.20.1
4. PMS to be read in conjuction with FS 2004 A.
5. Weld Joint Factor for welded pipe shall be as per ASME B 31.3
6. Contractor to use Schedule 80 for sizes less than 1/1/2" for threaded pipe and fittings.
7. Material shall comply heat treatment requirement of NACE MR 01 75/ISO-15156-3 inadditon to the requirement of this material standard.
8. Allowable stress considered for this material shall be based on testing and actual evaluation of allowable stress of the material within the temperature range for which this material is to be used..
9. Thickness indicated is for referance only. To be finalized based on Note- 7,8 above.

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**		Sheet No.	**2 of 2**
		Piping Class	Material	C.A	Spec.No	Revision: 2
		B23	# 300	DSS-NACE	Nil	2004A,Rev 9	Page No. 88
		Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3				
		Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.			
		0				

Branch Table as per API RP 14E

Run Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Branch Pipe

Valve Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 133,SW				VG 159,FLANGED												
Rating	# 800				# 300,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB 133,SW				VB 166												
Rating	# 800				# 300,RF												
Globe Valve																	
TAG	VGL 133,SW				VGL 159												
Rating	# 800				# 300,RF												
Check Valve																	
TAG	VC 133,SW,Lift Check				VC 159,Swing Check												
Rating	# 800				# 300,RF												
Needle Valve																	
TAG	VN 103,SW																
Rating	# 800																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.
2. Maximum temperature limit for all Ball Valve shall be 121 Degree C

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 2	15.01.16	Issued For Bid	PS	AKM	SJ

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**													Sheet No.	**1 of 2**			
		Piping Class				Material				C.A		Spec.No				Revision: 2			
		D23				# 600				DSS-NACE		Nil		2004A,Rev 9		Page No	89		
		Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3																	
		Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.																

Pipe Data	ASME B 36.19																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	40s	40s	40s	40s	10s	10s	40s	40s	40s	40s	40s	40	40	40	40	60	60
WTmm	2.77	2.87	3.38	3.68	2.77	3.05	6.02	7.11	8.18	9.27	9.52	11.12	12.7	14.27	15.08	22.22	24.61
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5
TYPE	Seamless									Welded (Longitudinally) with 100% radiography							
MOC	ASTM A 790 Gr. S 32205									ASTM A 928 Class 1,Gr. 32205 (Note-6,7,8)							
Ends	PE				BE												
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Socket Weld				Butt Weld												
Rating	# 3000				10s	10s	40s	40s	40s	40s	40s	40	40	40	40	60	60
MOC	ASTM A 182 Gr. F60				ASTM A 815 Gr.WP-S U.N.S 32205					ASTM A 815 Gr.WP-WX U.N.S 32205							
Elbow	ASMEB16.11				ASME B 16.9					ASME B 16.9							
Tee	ASMEB16.11				ASME B 16.9					ASME B 16.9							
Red.	ASMEB16.11				ASME B 16.9					ASME B 16.9							
Cap	ASMEB16.11				ASME B 16.9					ASME B 16.9							
Coupl	ASMEB16.11				ASME B 16.9					ASME B 16.9							
Plug	ASMEB16.11				ASME B 16.9					ASME B 16.9							
Union																	
Elbowlet	MSS SP 97,ASTM A 182 Gr. F60				MSS SP 97,ASTM A 182 Gr. F60					MSS SP 97,ASTM A 182 Gr. F60							
Sockolet	MSS SP 97,ASTM A 182 Gr. F60				MSS SP 97,ASTM A 182 Gr. F60					MSS SP 97,ASTM A 182 Gr. F60							
Weldolet	MSS SP 97,ASTM A 182 Gr. F60				MSS SP 97,ASTM A 182 Gr. F60					MSS SP 97,ASTM A 182 Gr. F60							
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe					MSS SP 97,ASTM A 182 Gr. F60							
Swage	MSS SP 95,Same as pipe				MSS SP 95, MOC Same as pipe					MSS SP 97,ASTM A 182 Gr. F60							
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				WN												
MOC	ASTM A 182 Gr. F60				ASTM A 182 Gr. F60												
FACE	RF,Smooth Finnished				RF,Smooth Finnished												
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25												
Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM A 182 Gr. F60																
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48					
Mechanical Joints																	
Stud Bolts	ASTM A 453 Gr. 660 CL A																
Hex Nuts	ASTM A 453 Gr. 660 CL A																
Gasket	ASME B 16.20 , 5mm thick Duplex SS Spiral wound gasket with CNAF filler																

Pressure-Temperature Rating

Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2
60 to 100	1500	16 to 38	105.5
200	1440	93	101.24
250	1385	121	97.38

Maximum Hydrostatic Pressure

Pressure, psig	Pres.,Kg/cm2g
2250 psig	158.2 Kg/cm2
Limited by Flange considereing flange as the weakest joint in piping system	

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
2. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange, specified spectacle blinds and flanges.
3. All pipe thread shall be as per ASME B 1.20.1
4. PMS to be read in conjuction with FS 2004 A.
5. Weld Joint Factor for welded pipe shall be as per ASME B 31.3
5. Contractor to use Schedule 80 for sizes less than 1/1/2" for threaded pipe and fittings.
6. Material shall comply heat treatment requirement of NACE MR 01 75/ISO-15156-3 inadditon to the requirement of this material standard.
7. Allowable stress considered for this material shall be based on testing and actual evaluation of allowable stress of the material within the temperature range for which this material is to be used..
8. Thickness indicated is for referance only. To be finalized based on Note- 7 above.

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**											Sheet No.	**2 of 2**			
		Piping Class						Material		C.A	Spec.No					Revision: 2	
		D23					# 600	DSS-NACE		Nil	2004A,Rev 9					Page No	90
		Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3															
		Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.														
		0															

Branch Table as per API RP 14E

Run Pipe																	
≤ 1	T																
1-1.5	T	T															
2	*	*	T														
3	*	*	T	T													
4	*	*	T	T	T												
6	*	*	W	T	T	T											
8	*	*	W	W	T	T	T										
10	*	*	W	W	W	T	T	T									
12	*	*	W	W	W	T	T	T	T								
14	*	*	W	W	W	W	T	T	T	T							
16	*	*	W	W	W	W	T	T	T	T	T						
18	*	*	W	W	W	W	W	T	T	T	T	T					
20	*	*	W	W	W	W	W	T	T	T	T	T	T				
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T			
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T		
	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30		
	Branch Pipe																

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 133,SW				VG 160,FLANGED												
Rating	# 800				# 600,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB 133,SW				VB 167												
Rating	# 800				# 600,RF												
Globe Valve																	
TAG	VGL 133,SW				VGL 160												
Rating	# 800				# 600,RF												
Check Valve																	
TAG	VC 133,SW,Lift Check				VC 160,Swing Check												
Rating	# 800				# 600,RF												
Needle Valve																	
TAG	VN 103,SW																
Rating	# 800																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.
2. Maximum temperature limit for all Ball Valve shall be 121 Degree C

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 2	15.01.16	Issued For Bid	PS	AKM	SJ

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**1 of 2**
	Piping Class	Material	C.A	Spec.No	Revision:3	
	E23 # 900	DSS-NACE	Nil	2004A,Rev 9	Page No.	91
	Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
	Service: Corrosive HydroCarbon Service-Sour Liquid and Vapour.					

Pipe Data	ASME B 36.19																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6	
Sch.	40s	40s	80s	80s	80s	40s	40s	40s	60	80s	60	60	80	60	80	80	80	
WTmm	2.77	2.87	4.55	5.08	5.54	5.49	6.02	7.11	10.31	12.7	14.27	15.09	21.44	19.05	26.19	28.57	30.96	
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	
TYPE	Seamless									Welded (Longitudinally) with 100% radiography								
MOC	ASTM A 790 Gr. S 32205									ASTM A 928 Class 1,Gr. 32205 (Note-9,10,11)								
Ends	PE				BE													
Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	Socket Weld				Butt Weld													
Rating	# 3000		# 6000		80s	40s	40s	40s	60	80s	60	60	80	60	80	80	80	
MOC	ASTM A 182 Gr. F60				ASTM A 815 Gr.WP-S U.N.S 32205					ASTM A 815 Gr.WP-WX U.N.S 32205								
Elbow	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Tee	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Red.	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Cap	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Coupl	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Plug	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Union																		
Elbowlet	MSS SP 97,ASTM A 182 Gr. F60				MSS SP 97,ASTM A 182 Gr. F60					MSS SP 97,ASTM A 182 Gr. F60								
Sockolet	MSS SP 97,ASTM A 182 Gr. F60				MSS SP 97,ASTM A 182 Gr. F60					MSS SP 97,ASTM A 182 Gr. F60								
Weldolet	MSS SP 97,ASTM A 182 Gr. F60				MSS SP 97,ASTM A 182 Gr. F60					MSS SP 97,ASTM A 182 Gr. F60								
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe					MSS SP 97,ASTM A 182 Gr. F60								
Swage	MSS SP 95,Same as pipe				MSS SP 95, MOC Same as pipe					MSS SP 97,ASTM A 182 Gr. F60								
Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				WN													
MOC	ASTM A 182 Gr. F60				ASTM A 182 Gr. F60													
FACE	RTJ-Note 7				RTJ, Note 7													
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25													
Spectacle Blind/Spacer Blinds																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
MOC	ASTM A 182 Gr. F60																	
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48						
Mechanical Joints																		
Stud Bolts	ASTM A 453 Gr. 660 CL A																	
Hex Nuts	ASTM A 453 Gr. 660 CL A																	
Gasket	ASME B 16.20,OCT ring of UNS S 32205 with Max. Hardness of 22 HRC																	

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
60 to 100	2250	16 to 38	158.2	3375 psig	237.3 Kg/cm2
200	2160	93	151.9	Limited by Flange considereing flange as the weakest joint in piping system	
250	2077.5	121	146.1		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
3. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and WNRTJ flanges above 3" size and specified snectacle blind.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Gasket Contact Surface shall have maximum roughness of 63AARH
7. RTJ groove hardness shall be 24 -25 HRC
8. Octagonal (OCT) ring shall be of UNS S 32205.
9. Material shall comply heat treatment requirement of NACE MR 01 75/ISO-15156-3 inadditon to the requirement of this material standard.
10. Allowable stress considered for this material shall be based on testing and actual evaluation of allowable stress of the material within the temperature range for which this material is to be used..
11. Thickness indicated is for referance only. To be finalized based on Note- 10 above.
12. To use Schedule 80 for sizes less than 1/1/2" for threaded pipe and fittings.
13. Weld Joint Factor for welded pipe shall be as per ASME B 31.3

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**2 of 2**
	Piping Class		Material	C.A	Spec.No	Revision:3
	E23	# 900	DSS-NACE	Nil	2004A,Rev 9	Page No. 92
	Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
	Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.				
	0					

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 134,SW				VG 161,FLANGED												
Rating	# 1500				# 900,RTJ												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	Use VG 134,SW				VB 168												
Rating	# 1500				# 900,RTJ												
Globe Valve																	
TAG	VGL 134,SW				VGL 161												
Rating	# 1500				# 900,RTJ												
Check Valve																	
TAG	VC 134,SW,Lift Check				VC 161,Swing Check												
Rating	# 1500				# 900,RTJ												
Needle Valve																	
TAG	VN 104,SW																
Rating	# 1500																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.
2. Maximum temperature limit for all Ball Valve shall be 121 Degree C

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision:3	15.01.16	Issued For Bid	PS	AKM	SJ

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**1 of 2**
	Piping Class	Material	C.A	Spec.No	Revision:0	
	F23 # 1500	DSS-NACE	Nil	2004A,Rev 9	Page No.	93
	Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
	Service: Corrosive HydroCarbon Service-Sour Liquid and Vapour.					

Pipe Data	ASME B 36.19																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6	
Sch.	80s	80s	80s	160	80s	80s	80s	120	120	120	120	120	120	120	120	120	120	
WTmm	3.73	3.91	4.55	7.14	5.54	7.62	8.56	14.27	18.26	21.44	25.4	27.7	30.96	34.92	38.1	41.27	46.02	
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	
TYPE	Seamless									Welded (Longitudinally) with 100% radiography								
MOC	ASTM A 790 Gr. S 32205									ASTM A 928 Class 1,Gr. 32205 (Note-9,10,11)								
Ends	PE				BE													
Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	Socket Weld				Butt Weld													
Rating	# 3000			# 6000	80s	80s	80s	120	120	120	120	120	120	120	120	120	120	
MOC	ASTM A 182 Gr. F60				ASTM A 815 Gr.WP-S U.N.S 32205					ASTM A 815 Gr.WP-WX U.N.S 32205								
Elbow	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Tee	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Red.	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Cap	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Coupl	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Plug	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Union																		
Elbowlet	MSS SP 97,ASTM A 182 Gr. F60				MSS SP 97,ASTM A 182 Gr. F60					MSS SP 97,ASTM A 182 Gr. F60								
Sockolet	MSS SP 97,ASTM A 182 Gr. F60				MSS SP 97,ASTM A 182 Gr. F60					MSS SP 97,ASTM A 182 Gr. F60								
Weldolet	MSS SP 97,ASTM A 182 Gr. F60				MSS SP 97,ASTM A 182 Gr. F60					MSS SP 97,ASTM A 182 Gr. F60								
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe					MSS SP 97,ASTM A 182 Gr. F60								
Swage	MSS SP 95,Same as pipe				MSS SP 95, MOC Same as pipe					MSS SP 97,ASTM A 182 Gr. F60								
Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				WN													
MOC	ASTM A 182 Gr. F60				ASTM A 182 Gr. F60													
FACE	RTJ-Note 7				RTJ, Note 7													
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25													
Spectacle Blind/Spacer Blinds																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
MOC	ASTM A 182 Gr. F60																	
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48						
Mechanical Joints																		
Stud Bolts	ASTM A 453 Gr. 660 CL A																	
Hex Nuts	ASTM A 453 Gr. 660 CL A																	
Gasket	ASME B 16.20,OCT ring of UNS S 32205 with Max. Hardness of 22 HRC																	

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
60 to 100	3750.7	16 to 38	263.7	5626.8 psig	395.6 Kg/cm2
122	3750.7	50	263.7	Limited by Flange considereing flange as the weakest joint in piping system	
212	3673.9	100	258.3		
302	3329.7	150	234.1		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
3. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and WNRTJ flanges above 3" size and specified snectacle blind.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Gasket Contact Surface shall have maximum roughness of 63AARH
7. RTJ groove hardness shall be 24 -25 HRC
8. Octagonal (OCT) ring shall be of UNS S 32205.
9. Material shall comply heat treatment requirement of NACE MR 01 75/ISO-15156-3 inadditon to the requirement of this material standard.
10. Allowable stress considered for this material shall be based on testing and actual evaluation of allowable stress of the material within the temperature range for which this material is to be used..
11. Thickness indicated is for referance only. To be finalized based on Note- 10 above.
12. To use Schedule 80 for sizes less than 1/1/2" for threaded pipe and fittings.
13. Weld Joint Factor for welded pipe shall be as per ASME B 31.3

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**2 of 2**
	Piping Class		Material	C.A	Spec.No	Revision:0
	F23	# 1500	DSS-NACE	Nil	2004A,Rev 9	Page No. 94
	Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
	Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.				
	0					

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 144,SW				VG 171,FLANGED												
Rating	# 2500				#1500,RTJ												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	Use VG 144,SW				VB 178												
Rating	# 2500				# 1500,RTJ												
Globe Valve																	
TAG	VGL 144,SW				VGL 171												
Rating	# 2500				# 1500,RTJ												
Check Valve																	
TAG	VC 144,SW,Lift Check				VC 171,Swing Check												
Rating	# 2500				# 1500,RTJ												
Needle Valve																	
TAG	VN 204,SW																
Rating	# 2500																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.
2. Maximum temperature limit for all Ball Valve shall be 121 Degree C

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision:0	15.01.16	Issued For Bid	PS	AKM	SJ

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**1 of 2**
	Piping Class	Material	C.A	Spec.No	Revision:0	
	G23 # 2500	DSS-NACE	Nil	2004A,Rev 9	Page No.	95
	Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
	Service: Corrosive HydroCarbon Service-Sour Liquid and Vapour.					

Pipe Data	ASME B 36.19																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6	
Sch.	160	160	160	XXS	160	160	160	XXS	160	160	160	160	160	160	160	160	160	
WTmm	4.77	5.56	6.35	10.16	8.74	11.12	13.49	21.95	23.01	28.57	33.32	35.71	40.49	45.24	50.01	53.97	59.54	
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	5.0	4.0	3.0	2.0	2.0	2.0	2.0	2.0	2.0	
TYPE	Seamless									Welded (Longitudinally) with 100% radiography								
MOC	ASTM A 790 Gr. S 32205									ASTM A 928 Class 1,Gr. 32205 (Note-9,10,11)								
Ends	PE				BE													
Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	Socket Weld				Butt Weld													
Rating	# 6000			# 9000	160	160	160	XXS	160	160	160	160	160	160	160	160	160	
MOC	ASTM A 182 Gr. F60				ASTM A 815 Gr.WP-S U.N.S 32205					ASTM A 815 Gr.WP-WX U.N.S 32205								
Elbow	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Tee	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Red.	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Cap	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Coupl	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Plug	ASMEB16.11				ASME B 16.9					ASME B 16.9								
Union																		
Elbowlet	MSS SP 97,ASTM A 182 Gr. F60				MSS SP 97,ASTM A 182 Gr. F60					MSS SP 97,ASTM A 182 Gr. F60								
Sockolet	MSS SP 97,ASTM A 182 Gr. F60				MSS SP 97,ASTM A 182 Gr. F60					MSS SP 97,ASTM A 182 Gr. F60								
Weldolet	MSS SP 97,ASTM A 182 Gr. F60				MSS SP 97,ASTM A 182 Gr. F60					MSS SP 97,ASTM A 182 Gr. F60								
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe					MSS SP 97,ASTM A 182 Gr. F60								
Swage	MSS SP 95,Same as pipe				MSS SP 95, MOC Same as pipe					MSS SP 97,ASTM A 182 Gr. F60								
Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				WN													
MOC	ASTM A 182 Gr. F60				ASTM A 182 Gr. F60													
FACE	RTJ-Note 7				RTJ, Note 7													
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25													
Spectacle Blind/Spacer Blinds																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
MOC	ASTM A 182 Gr. F60																	
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48						
Mechanical Joints																		
Stud Bolts	ASTM A 453 Gr. 660 CL A																	
Hex Nuts	ASTM A 453 Gr. 660 CL A																	
Gasket	ASME B 16.20,OCT ring of UNS S 32205 with Max. Hardness of 22 HRC																	

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
60 to 100	6249.7	16 to 38	439.4	9374.6 psig	659.1 Kg/cm2
122	6249.7	50	439.4	Limited by Flange considereing flange as the weakest joint in piping system	
212	6123.1	100	430.5		
302	5549.9	150	390.2		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
3. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and WNRTJ flanges above 3" size and specified snectacle blind.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A.
6. Gasket Contact Surface shall have maximum roughness of 63AARH
7. RTJ groove hardness shall be 24 -25 HRC
8. Octagonal (OCT) ring shall be of UNS S 32205.
9. Material shall comply heat treatment requirement of NACE MR 01 75/ISO-15156-3 inadditon to the requirement of this material standard.
10. Allowable stress considered for this material shall be based on testing and actual evaluation of allowable stress of the material within the temperature range for which this material is to be used..
11. Thickness indicated is for referance only. To be finalized based on Note- 10 above.
12. To use Schedule 80 for sizes less than 1/1/2" for threaded pipe and fittings.
13. Weld Joint Factor for welded pipe shall be as per ASME B 31.3

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION				Sheet No.	2 of 2
	Piping Class		Material	C.A	Spec.No	Revision:0
	G23	# 2500	DSS-NACE	Nil	2004A,Rev 9	Page No. 96
	Design Code: ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
	Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.				
	0					

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 144,SW				VG 381,FLANGED												
Rating	# 2500				#2500,RTJ												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	Use VG 144,SW				VB 388												
Rating	# 2500				# 2500,RTJ												
Globe Valve																	
TAG	VGL 144,SW				VGL 381												
Rating	# 2500				# 2500,RTJ												
Check Valve																	
TAG	VC 144,SW,Lift Check				VC 381,Swing Check												
Rating	# 2500				# 2500,RTJ												
Needle Valve																	
TAG	VN 204,SW																
Rating	# 2500																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.
2. Maximum temperature limit for all Ball Valve shall be 121 Degree C

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision:0	15.01.16	Issued For Bid	PS	AKM	SJ

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**1 of 2**
	Piping Class		Material	C.A	Spec.No	Revision: 0
	A15	# 150	Incoloy 825	Nil	2004A,Rev 9	Page No. 97
	Code:ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
	Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.				

Pipe Data	ASME B 36.10																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	40s	40s	40s	40s	10s	10s	10s	10s									
WTmm	2.77	2.87	3.38	3.68	2.77	3.05	3.05	3.40									
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	5.0	5.0	5.0	5.0	5.0	5.0
TYPE	Seamless											Welded pipe from clad plate. Note -5					
MOC	ASTM B 423 UNSN08825 Cold finnished Annealed								ASTM B 423 UNSN8825			API 5L Gr.B Pipe with Incoly cladding, Note 7					
Ends	PE				BE												
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Socket Weld				Butt Weld												
Rating	# 3000				10s	10s	10s	10s									
MOC	ASTM B 564 UNS N08825				ASTM B 366 WP NIC MC S UNS N08825							ASTM A 234 WPB with cladding min. 3mm. Note 8					
Elbow	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Tee	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Red.	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Cap	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Coupl	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Plug	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Union																	
Elbowlet	MSS SP 97,B 564 UNS N08825				MSS SP 97,ASTM B 564 UNS N08825							MSSSP 97,ASTMA105 ,cladding min.3mm.Note8					
Sockolet	MSS SP 97,B 564 UNS N08825				MSS SP 97,ASTM B 564 UNS N08825							MSSSP 97,ASTMA105 ,cladding min.3mm.Note8					
Weldolet	MSS SP 97,B 564 UNS N08825				MSS SP 97,ASTM B 564 UNS N08825							MSSSP 97,ASTMA105 ,cladding min.3mm.Note8					
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe							MOC Same as pipe					
Swage	MSS SP 95,Same as pipe				MSS SP 95, MOC Same as pipe							MSS SP 95, MOC Same as pipe					
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				WN							WN					
MOC	ASTM B 564 UNS N08825				ASTM B 564 UNS N08825							MSSSP 97,ASTMA105 ,cladding min.3mm.Note8					
FACE	RF,Serrated Finnished				RF,Serrated Finnished												
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25												
Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM B 564 UNS N08825																
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48					
Mechanical Joints																	
Stud Bolt	ASTM A 453 Gr. 660 CL A																
Hex Nuts	ASTM A 453 Gr. 660 CL A																
Gasket	ASME B 16.20 , 5mm thick Duplex SS Spiral wound gasket with CNAF filler																

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
(-) 20 to 100	290	(-) 29 to 38	20.3	435psig	31 Kg/cm2
200	260	93	18.3	Limited by Flange considereing flange as the weakest joint in piping system	
300	230	149	16.2		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
2. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and specified flange and spectacle blind.
3. All pipe thread shall be as per ASME B 1.20.1
4. PMS to be read in conjuction with FS 2004 A.
5. Only the base material thickness shall be considered.
6. Pipe sizes 8" to 12" shall be hotfinnished annealed (Solution annealed), pickled and end shall be bevelled. Allwable stress to be established for the temperature range based on actual evaluation after testing .
7. For sizes 14" and above welded pipe of clad plate with 100% radiography to be used. Cladding of minimum 3mm shall be done with Incoloy 825 or Inconell 625 on API 5L Gr. B pipe.
8. Cladding shall be weld over lay type with Incoly 825 or Inconel 625 of min. 3mm
9. To use Schedule 80 for sizes less than 1/1/2" for threaded pipe and fittings.

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**			Sheet No.	**2 of 2**
	Piping Class	Material	C.A	Spec.No	Revision: 0
	A15 # 150	Incoloy 825	Nil	2004A,Rev 9	Page No. 98
	Code:ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3				
	Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.			
	0				

Branch Table as per API RP 14E

Run Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T
Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 105,SW				VG 111,FLANGED												
Rating	# 800				# 150,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB 105,SW				VB 111, Flanged												
Rating	# 800				# 150,RF												
Globe Valve																	
TAG	VGL 104,SW				VGL 111,flanged												
Rating	# 800				# 150,RF												
Check Valve																	
TAG	VC 104,SW,Lift Check				VC 111,Swing Check												
Rating	# 800				# 150,RF												
Needle Valve																	
TAG	VN 105,SW																
Rating	# 800																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.
2. Maximum temperature limit for all Ball Valve shall be suitable to maximum temperature of the project.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 0	22/7/2009	Issued For Bid	PS	GJ	GRP

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**											Sheet No.		**1 of 2**			
		Piping Class				Material			C.A	Spec.No				Revision: 0			
		B15		# 300		Incoloy 825			Nil	2004A,Rev 9				Page No.	99		
		Code:ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3															
		Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.														

Pipe Data	ASME B 36.10																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	40s	40s	40s	40s	10s	10s	10s	10s									
WTmm	2.77	2.87	3.38	3.68	2.77	3.05	3.05	3.40									
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	5.0	5.0	5.0	5.0	5.0	5.0
TYPE	Seamless											Welded pipe from clad plate. Note -5					
MOC	ASTM B 423 UNSN08825 Cold finnished Annealed								ASTM B 423 UNSN8825			API 5L Gr.B Pipe with Incoly cladding, Note 7					
Ends	PE				BE												
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Socket Weld				Butt Weld												
Rating	# 3000				10s	10s	10s	10s									
MOC	ASTM B 564 UNS N08825				ASTM B 366 WP NIC MC S UNS N08825							ASTM A 234 WPB with cladding min. 3mm. Note 8					
Elbow	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Tee	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Red.	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Cap	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Coupl	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Plug	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Union																	
Elbowlet	MSS SP 97,B 564 UNS N08825				MSS SP 97,ASTM B 564 UNS N08825							MSSSP 97,ASTMA105 ,cladding min.3mm.Note8					
Sockolet	MSS SP 97,B 564 UNS N08825				MSS SP 97,ASTM B 564 UNS N08825							MSSSP 97,ASTMA105 ,cladding min.3mm.Note8					
Weldolet	MSS SP 97,B 564 UNS N08825				MSS SP 97,ASTM B 564 UNS N08825							MSSSP 97,ASTMA105 ,cladding min.3mm.Note8					
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe							MOC Same as pipe					
Swage	MSS SP 95,Same as pipe				MSS SP 95, MOC Same as pipe							MSS SP 95, MOC Same as pipe					
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				WN							WN					
MOC	ASTM B 564 UNS N08825				ASTM B 564 UNS N08825							MSSSP 97,ASTMA105 ,cladding min.3mm.Note8					
FACE	RF,Smooth Finnished				RF,Smooth Finnished												
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25												
Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM B 564 UNS N08825																
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48					
Mechanical Joints																	
Stud Bolt	ASTM A 453 Gr. 660 CL A																
Hex Nuts	ASTM A 453 Gr. 660 CL A																
Gasket	ASME B 16.20 , 5mm thick Duplex SS Spiral wound gasket with CNAF filler																

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
(-) 20 to 100	750	(-) 29 to 38	52.8	1125psig	80 Kg/cm2
200	750	93	52.8	Limited by Flange considereing flange as the weakest joint in piping system	
300	730	149	51.3		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
2. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and specified flange and spectacle blind.
3. All pipe thread shall be as per ASME B 1.20.1
4. PMS to be read in conjuction with FS 2004 A.
5. Only the base material thickness shall be considered.
6. Pipe sizes 8" to 12" shall be hotfinnished annealed (Solution annealed), pickled and end shall be bevelled. Allwable stress to be established for the temperature range based on actual evaluation after testing .
7. For sizes 14" and above welded pipe of clad plate with 100% radio graphy to be used. Cladding of minimum 3mm shall be done with Incoloy 825 or Inconell 625 on API 5L Gr. B pipe.
8. Cladding shall be weld over lay type with Incoly 825 or Inconel 625 of min. 3mm
9. To use Schedule 80 for sizes less than 1/1/2" for threaded pipe and fittings.

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**											Sheet No.	**2 of 2**			
		Piping Class					Material		C.A	Spec.No				Revision: 0			
		B15	# 300				Incoloy 825		Nil	2004A,Rev 9				Page No.	100		
		Code:ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3															
		Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.														
		0															

Branch Table as per API RP 14E

Run Pipe																	
≤ 1	T																
1-1.5	T	T															
2	*	*	T														
3	*	*	T	T													
4	*	*	T	T	T												
6	*	*	W	T	T	T											
8	*	*	W	W	T	T	T										
10	*	*	W	W	W	T	T	T									
12	*	*	W	W	W	T	T	T	T								
14	*	*	W	W	W	W	T	T	T	T							
16	*	*	W	W	W	W	T	T	T	T	T						
18	*	*	W	W	W	W	W	T	T	T	T	T					
20	*	*	W	W	W	W	W	T	T	T	T	T	T				
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T			
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T		
	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30		
	Branch Pipe																

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 105,SW				VG 113,FLANGED												
Rating	# 800				# 300,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB 105,SW				VB 113,												
Rating	# 800				# 300,RF												
Globe Valve																	
TAG	VGL 104,SW				VGL 113												
Rating	# 800				# 300,RF												
Check Valve																	
TAG	VC 104,SW,Lift Check				VC 113,Swing Check												
Rating	# 800				# 300,RF												
Needle Valve																	
TAG	VN 105,SW																
Rating	# 800																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.
2. Maximum temperature limit for all Ball Valve shall be suitable to maximum temperature of the project.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 0	22/7/2009	Issued For Bid	PS	GJ	GRP

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**												Sheet No.		**1 of 2**	
		Piping Class				Material				C.A		Spec.No				Revision:1	
		D15				# 600				Incoloy 825		Nil		2004A,Rev 9		Page No.	101
		Code:ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3															
		Service:				Corrosive HydroCarbon Service-Sour Liquid and Vapour.											

Pipe Data	ASME B 36.10																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6	
Sch.	40s	40s	40s	80s	10s	30	30	40s										
WTmm	2.77	2.87	3.38	5.08	2.77	4.78	4.78	7.11										
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	5.0	5.0	5.0	5.0	5.0	5.0	
TYPE	Seamless											Welded pipe from clad plate. Note -5						
MOC	ASTM B 423 UNSN08825 Cold finnished Annealed								ASTM B 423 UNSN8825			API 5L Gr.B Pipe with Incoly cladding, Note 7						
Ends	PE				BE													
Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	Socket Weld				Butt Weld													
Rating	# 3000				10s	30	30	40s										
MOC	ASTM B 564 UNS N08825				ASTM B 366 WP NIC MC S UNS N08825							ASTM A 234 WPB with cladding min. 3mm. Note 8						
Elbow	ASMEB16.11				ASME B 16.9							ASME B 16.9						
Tee	ASMEB16.11				ASME B 16.9							ASME B 16.9						
Red.	ASMEB16.11				ASME B 16.9							ASME B 16.9						
Cap	ASMEB16.11				ASME B 16.9							ASME B 16.9						
Coupl	ASMEB16.11				ASME B 16.9							ASME B 16.9						
Plug	ASMEB16.11				ASME B 16.9							ASME B 16.9						
Union																		
Elbowlet	MSS SP 97,B 564 UNS N08825				MSS SP 97,ASTM B 564 UNS N08825							MSSSP 97,ASTMA105 ,cladding min.3mm.Note8						
Sockolet	MSS SP 97,B 564 UNS N08825				MSS SP 97,ASTM B 564 UNS N08825							MSSSP 97,ASTMA105 ,cladding min.3mm.Note8						
Weldolet	MSS SP 97,B 564 UNS N08825				MSS SP 97,ASTM B 564 UNS N08825							MSSSP 97,ASTMA105 ,cladding min.3mm.Note8						
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe							MOC Same as pipe						
Swage	MSS SP 95,Same as pipe				MSS SP 95, MOC Same as pipe							MSS SP 95, MOC Same as pipe						
Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				WN							WN						
MOC	ASTM B 564 UNS N08825				ASTM B 564 UNS N08825							MSSSP 97,ASTMA105 ,cladding min.3mm.Note8						
FACE	RF,Smooth Finnished				RF,Smooth Finnished													
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25													
Spectacle Blind/Spacer Blinds																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
MOC	ASTM B 564 UNS N08825																	
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48						
Mechanical Joints																		
Stud Bolts	ASTM A 453 Gr. 660 CL A																	
Hex Nuts	ASTM A 453 Gr. 660 CL A																	
Gasket	ASME B 16.20 , 5mm thick Duplex SS Spiral wound gasket with CNAF filler																	

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
(-) 20 to 100	1500	(-) 29 to 38	105.4	2250psig	158.2 Kg/cm2
200	1500	93	105.4	Limited by Flange considereing flange as the weakest joint in piping system	
300	1455	149	102.3		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
2. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and specified flange and spectacle blind.
3. All pipe thread shall be as per ASME B 1.20.1
4. PMS to be read in conjuction with FS 2004 A.
5. Only the base material thickness shall be considered.
6. Pipe sizes 8" to 12" shall be hotfinnished annealed (Solution annealed), pickled and end shall be bevelled. Allwable stress to be established for the temperature range based on actual evaluation after testing .
7. For sizes 14" and above welded pipe of clad plate with 100% radiography to be used. Cladding of minimum 3mm shall be done with Incoloy 825 or Inconell 625 on API 5L Gr. B pipe.
8. Cladding shall be weld over lay type with Incoly 825 or Inconel 625 of min. 3mm
9. To use Schedule 80 for sizes less than 1/1/2" for threaded pipe and fittings.

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**			Sheet No.	**2 of 2**
	Piping Class	Material	C.A	Spec.No	Revision:1
	D15	# 600	Incoloy 825	Nil	2004A,Rev 9	Page No.	102
	Code:ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3				
	Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.			
	0				

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 105,SW				VG 115,FLANGED												
Rating	# 800				# 600,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB 105,SW				VB 115,												
Rating	# 800				# 600,RF												
Globe Valve																	
TAG	VGL 104,SW				VGL 115												
Rating	# 800				# 600,RF												
Check Valve																	
TAG	VC 104,SW,Lift Check				VC 115,Swing Check												
Rating	# 800				# 600,RF												
Needle Valve																	
TAG	VN 105,SW																
Rating	# 800																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.
2. Maximum temperature limit for all Ball Valve shall be suitable to maximum temperature of the project.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision:1	22/7/2009	Issued For Bid	PS	GJ	GRP

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**1 of 2**
	Piping Class		Material	C.A	Spec.No	Revision: 1
	E15	# 900	Incoloy 825	Nil	2004A,Rev 9	Page No. 103
	Code:ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
	Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.				

Pipe Data	ASME B 36.10																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	80s	80s	80s	80s	40s	40s	Std.	80s									
WTmm	3.73	3.91	4.55	5.08	3.91	5.49	6.02	10.97									
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	5.0	5.0	5.0	5.0	5.0	5.0
TYPE	Seamless											Welded pipe from clad plate. Note -5					
MOC	ASTM B 423 UNSN08825 Cold finnished Annealed								ASTM B 423 UNSN8825			API 5L Gr.B Pipe with Incoly cladding, Note 7					
Ends	PE				BE												
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Socket Weld				Butt Weld												
Rating	# 9000				40s	40s	Std.	80s									
MOC	ASTM B 564 UNS N08825				ASTM B 366 WP NIC MC S UNS N08825							ASTM A 234 WPB with cladding min. 3mm. Note 8					
Elbow	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Tee	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Red.	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Cap	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Coupl	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Plug	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Union																	
Elbowlet	MSS SP 97,B 564 UNS N08825				MSS SP 97,ASTM B 564 UNS N08825							MSSSP 97,ASTMA105 ,cladding min.3mm.Note8					
Sockolet	MSS SP 97,B 564 UNS N08825				MSS SP 97,ASTM B 564 UNS N08825							MSSSP 97,ASTMA105 ,cladding min.3mm.Note8					
Weldolet	MSS SP 97,B 564 UNS N08825				MSS SP 97,ASTM B 564 UNS N08825							MSSSP 97,ASTMA105 ,cladding min.3mm.Note8					
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe							MOC Same as pipe					
Swage	MSS SP 95,Same as pipe				MSS SP 95, MOC Same as pipe							MSS SP 95, MOC Same as pipe					
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				WN							WN					
MOC	ASTM B 564 UNS N08825				ASTM B 564 UNS N08825							MSSSP 97,ASTMA105 ,cladding min.3mm.Note8					
FACE	RTJ-Note 8				RTJ-Note 8												
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25												
Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM B 564 UNS N08825																
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48					
Mechanical Joints																	
Stud Bolts	ASTM A 453 Gr. 660 CL A																
Hex Nuts	ASTM A 453 Gr. 660 CL A																
Gasket	ASME B 16.20,OCT ring of UNS S 32205 with Max. Hardness of 22 HRC. Gasket contact surface roughness max. 63 AARH																

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
(-) 20 to 100	2250	(-) 29 to 38	158.1	3375psig	237.2 Kg/cm2
200	2250	93	158.1	Limited by Flange considereing flange as the weakest joint in piping system	
300	2185	149	153.6		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and
2. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and specified flange and spectacle blind.
3. All pipe thread shall be as per ASME B 1.20.1
4. PMS to be read in conjuction with FS 2004 A.
5. Only the base material thickness shall be considered.
6. Pipe sizes 8" to 12" shall be hotfinnished annealed (Solution annealed), pickled and end shall be bevelled. Allwable stress to be established for the temperature range based on actual evaluation after testing .
7. For sizes 14" and above welded pipe of clad plate with 100% radiography shall be used. Cladding of minimum 3mm shall be done with Incoloy 825 or Inconell 625 on API 5L Gr. B pipe.
8. RTJ groove hardness shall be 35 HRC.
9. Cladding shall be weld over lay type with Incoly 825 or Inconel 625 of min. 3mm

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION				Sheet No.	2 of 2
	Piping Class		Material	C.A	Spec.No	Revision: 1
	E15	# 900	Incoloy 825	Nil	2004A,Rev 9	Page No. 104
	Code:ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
	Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.				
	0					

Branch Table as per API RP 14E

Run Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Branch Pipe

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 107,SW				VG 117,FLANGED												
Rating	# 1500				# 900,RTJ												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB 107,SW				VB 117,												
Rating	# 1500				# 900,RTJ												
Globe Valve																	
TAG	VGL 106,SW				VGL 117												
Rating	# 1500				# 900,RTJ												
Check Valve																	
TAG	VC 106,SW,Lift Check				VC 117,Swing Check												
Rating	# 1500				# 900,RTJ												
Needle Valve																	
TAG	VN 107,SW																
Rating	# 1500																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.
2. Maximum temperature limit for all Ball Valve shall be suitable to maximum temperature of the project.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 1	15.01.16	Issued For Bid	PS	AKM	SJ

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**											Sheet No.		**1 of 2**			
		Piping Class				Material				C.A		Spec.No				Revision: 1		
		XG15				# 5000 API				Incoloy 825		Nil		2004A,Rev 9		Page No.		105
		Code:ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3,API 6A																
		Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.															

Pipe Data	ASME B 36.10																	
Size(in)	0.5	0.75	1	1.5	2 1/16	3 1/8	4 1/16	7 1/16	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6	
Sch.	80s	80s	80s	160	160	160	160	XXS										
WTmm	3.73	3.91	4.55	7.14	8.74	11.13	13.49	10.97										
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	5.0	5.0	5.0	5.0	5.0	5.0	
TYPE	Seamless											Welded pipe from clad plate. Note -5						
MOC	ASTM B 423 UNSN08825 Cold finnished Annealed								ASTM B 423 UNSN8825			API 5L Gr.B Pipe with Incoly cladding, Note 7						
Ends	PE				BE													
Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	Socket Weld				Butt Weld													
Rating	# 9000																	
MOC	ASTM B 564 UNS N08825				ASTM B 366 WP NIC MC S UNS N08825							ASTM A 234 WPB with cladding min. 3mm. Note 9						
Elbow	ASMEB16.11				ASME B 16.9							ASME B 16.9						
Tee	ASMEB16.11				ASME B 16.9							ASME B 16.9						
Red.	ASMEB16.11				ASME B 16.9							ASME B 16.9						
Cap	ASMEB16.11				ASME B 16.9							ASME B 16.9						
Coupl	ASMEB16.11				ASME B 16.9							ASME B 16.9						
Plug	ASMEB16.11				ASME B 16.9							ASME B 16.9						
Union																		
Elbowlet	MSS SP 97,B 564 UNS N08825				MSS SP 97,ASTM B 564 UNS N08825							MSSSP 97,ASTMA105 ,cladding min.3mm.Note9						
Sockolet	MSS SP 97,B 564 UNS N08825				MSS SP 97,ASTM B 564 UNS N08825							MSSSP 97,ASTMA105 ,cladding min.3mm.Note9						
Weldolet	MSS SP 97,B 564 UNS N08825				MSS SP 97,ASTM B 564 UNS N08825							MSSSP 97,ASTMA105 ,cladding min.3mm.Note9						
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe							MOC Same as pipe						
Swage	MSS SP 95,Same as pipe				MSS SP 95, MOC Same as pipe							MSS SP 95, MOC Same as pipe						
Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				WN							WN						
MOC	ASTM B 564 UNS N08825				ASTM B 564 UNS N08825							MSSSP 97,ASTMA105 ,cladding min.3mm.Note9						
FACE	RTJ-Note 8				RTJ-Note 8													
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25													
Spectacle Blind/Spacer Blinds																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
MOC	ASTM B 564 UNS N08825																	
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48						
Mechanical Joints																		
Stud Bolts	ASTM A 453 Gr. 660 CL A																	
Hex Nuts	ASTM A 453 Gr. 660 CL A																	
Gasket	ASME B 16.20,OCT ring of UNS S 32205 with Max. Hardness of 22 HRC. Gasket contact surface roughness max. 63 AARH.																	

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
60 to 100	5000	16 to 38	352	10000psig	704 Kg/cm2
				Limited by Flange considereing flange as the weakest joint in piping system	

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.

2. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and WNRTJ flanges above 3" size and specified spectacle blind.

3. All pipe thread shall be as per ASME B 1.20.1

4. PMS to be read in conjuction with FS 2004 A.

5. Only the base material thickness shall be considered.

6. Pipe sizes 8" to 12" shall be hotfinnished annealed (Solution annealed), pickled and end shall be bevelled. Allwable stress to be established for the temperature range based on actual evaluation after testing .

7. For sizes 14" and above welded pipe of clad plate with 100% radiography shall be used. Cladding of minimum 3mm shall be done with Incoloy 825 or Inconell 625 on API 5L Gr. B pipe.

8. RTJ groove hardness shall be 35 HRC.

9. Cladding shall be weld over lay type with Incoly 825 or Inconel 625 of min. 3mm

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**			Sheet No.	**2 of 2**
	Piping Class	Material	C.A	Spec.No	Revision: 1
	XG15 # 5000 API	Incoloy 825	Nil	2004A,Rev 9	Page No. 106
	Code:ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3,API 6A				
	Service: Corrosive HydroCarbon Service-Sour Liquid and Vapour.				
	0				

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 108,SW				VG 118,FLANGED												
Rating	# 5000				# 5000,RTJ												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB 108,SW				VB 118,												
Rating	# 5000				# 5000,RTJ												
Globe Valve																	
TAG	VGL 108,SW				VGL 118												
Rating	# 5000				# 5000,RTJ												
Check Valve																	
TAG	VC 108,SW,Lift Check				VC 118,Swing Check												
Rating	# 5000				# 5000,RTJ												
Needle Valve																	
TAG	VN 108,SW																
Rating	# 5000																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.
2. Maximum temperature limit for all Ball Valve shall be suitable to maximum temperature of the project.
3. Groove hardness for DSS flange shall be 25HRC and 35 HRC for Incoloy flange.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 1	22/7/2009	Issued For Bid	PS	GJ	GRP

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**											Sheet No.		**1 of 2**	
		Piping Class				Material			C.A	Spec.No				Revision: 1	
		XH15		# 10000 API		Incoloy 825			Nil	2004A,Rev 9				Page No.	107
		Code:ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3,API 6A													
		Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.												

Pipe Data	ASME B 36.10																	
Size(in)	0.5	0.75	1	1.5	2 1/16	3 1/8	4 1/16	7 1/16	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6	
Sch.	160	XXS	XXS	XXS														
WTmm	4.78	7.82	9.09	10.15														
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	5.0	5.0	5.0	5.0	5.0	5.0	
TYPE	Seamless											Welded pipe from clad plate. Note -5						
MOC	ASTM B 423 UNSN08825 Cold finnished Annealed								ASTM B 423 UNSN8825			API 5L Gr.B Pipe with Incoly cladding, Note 7						
Ends	PE				BE													
Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	Socket Weld				Butt Weld													
Rating	# 9000																	
MOC	ASTM B 564 UNS N08825				ASTM B 366 WP NIC MC S UNS N08825							ASTM A 234 WPB with cladding min. 3mm. Note 9						
Elbow	ASMEB16.11				ASME B 16.9							ASME B 16.9						
Tee	ASMEB16.11				ASME B 16.9							ASME B 16.9						
Red.	ASMEB16.11				ASME B 16.9							ASME B 16.9						
Cap	ASMEB16.11				ASME B 16.9							ASME B 16.9						
Coupl	ASMEB16.11				ASME B 16.9							ASME B 16.9						
Plug	ASMEB16.11				ASME B 16.9							ASME B 16.9						
Union																		
Elbowlet	MSS SP 97,B 564 UNS N08825				MSS SP 97,ASTM B 564 UNS N08825							MSSSP 97,ASTMA105 ,cladding min.3mm.Note9						
Sockolet	MSS SP 97,B 564 UNS N08825				MSS SP 97,ASTM B 564 UNS N08825							MSSSP 97,ASTMA105 ,cladding min.3mm.Note9						
Weldolet	MSS SP 97,B 564 UNS N08825				MSS SP 97,ASTM B 564 UNS N08825							MSSSP 97,ASTMA105 ,cladding min.3mm.Note9						
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe							MOC Same as pipe						
Swage	MSS SP 95,Same as pipe				MSS SP 95, MOC Same as pipe							MSS SP 95, MOC Same as pipe						
Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				WN							WN						
MOC	ASTM B 564 UNS N08825				ASTM B 564 UNS N08825							MSSSP 97,ASTMA105 ,cladding min.3mm.Note9						
FACE	RTJ-Note 8				RTJ-Note 8													
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25													
Spectacle Blind/Spacer Blinds																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
MOC	ASTM B 564 UNS N08825																	
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48						
Mechanical Joints																		
Stud Bolts	ASTM A 453 Gr. 660 CL A																	
Hex Nuts	ASTM A 453 Gr. 660 CL A																	
Gasket	ASME B 16.20,OCT ring of UNS S 32205 with Max. Hardness of 22 HRC. Gasket contact surface roughness max. 63 AARH.																	

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
60 to 100	10000	16 to 38	704	15000psig	1055 Kg/cm2
				Limited by Flange considereing flange as the weakest joint in piping system	

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
2. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and WNRTJ flanges above 3" size and specified spectacle blind.
3. All pipe thread shall be as per ASME B 1.20.1
4. PMS to be read in conjuction with FS 2004 A.
5. Only the base material thickness shall be considered.
6. Pipe sizes 8" to 12" shall be hotfinnished annealed (Solution annealed), pickled and end shall be bevelled. Allwable stress to be established for the temperature range based on actual evaluation after testing .
7. For sizes 14" and above welded pipe of clad plate 100% radiography shall be used. Cladding of minimum 3mm shall be done with Incoloy 825 or Inconell 625 on API 5L Gr. B pipe.
8. RTJ groove hardness shall be 35 HRC.
9. Cladding shall be weld over lay type with Incoly 825 or Inconel 625 of min. 3mm

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**			Sheet No.	**2 of 2**
	Piping Class	Material	C.A	Spec.No	Revision: 1
	XH15 / # 10000 API	Incoloy 825	Nil	2004A,Rev 9	Page No. 108
	Code:ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3,API 6A				
	Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.			
	0				

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 106,SW				VG 116,FLANGED												
Rating	# 10000				# 10000,RTJ												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB 106,SW				VB 116,												
Rating	# 10000				# 10000,RTJ												
Globe Valve																	
TAG	VGL 106,SW				VGL 116												
Rating	# 10000				# 10000,RTJ												
Check Valve																	
TAG	VC 106,SW,Lift Check				VC 116,Swing Check												
Rating	# 10000				# 10000,RTJ												
Needle Valve																	
TAG	VN 106,SW																
Rating	# 10000																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.
2. Maximum temperature limit for all Ball Valve shall be suitable to maximum temperature of the project.
3. Groove hardness for DSS flange shall be 25HRC and 35 HRC for Incoloy flange.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 1	22/7/2009	Issued For Bid	PS	GJ	GRP

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**											Sheet No.		**1 of 2**			
		Piping Class				Material				C.A		Spec.No				Revision: 0	
		A15A		# 150		Inconel 625				Nil		2004A,Rev 9				Page No.	109
		Code:ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3															
		Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.														

Pipe Data	ASME B 36.10																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6	
Sch.	40s	40s	40s	40s	10s	10s	10s	10s	10s	10s	10s							
WTmm	2.77	2.87	3.38	3.68	2.77	3.05	3.05	3.40	3.76	4.19	4.57							
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	5.0	5.0	5.0	5.0	5.0	5.0	
TYPE	Seamless											Welded pipe from clad plate. Note -5						
MOC	ASTM B 444 UNSN06625 Gr.1											API 5L Gr.B Pipe with Inconel cladding, Note 6						
Ends	PE				BE													
Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	Socket Weld				Butt Weld													
Rating	# 3000				10s	10s	10s	10s	10s	10s	10s							
MOC	ASTM B 564 UNS N06625				ASTM B 366 WP NC MC S UNS N06625							ASTM A 234 WPB with cladding min. 3mm. Note 7						
Elbow	ASMEB16.11				ASME B 16.9							ASME B 16.9						
Tee	ASMEB16.11				ASME B 16.9							ASME B 16.9						
Red.	ASMEB16.11				ASME B 16.9							ASME B 16.9						
Cap	ASMEB16.11				ASME B 16.9							ASME B 16.9						
Coupl	ASMEB16.11				ASME B 16.9							ASME B 16.9						
Plug	ASMEB16.11				ASME B 16.9							ASME B 16.9						
Union																		
Elbowlet	MSS SP 97,B 564 UNS N06625				MSS SP 97,ASTM B 564 UNS N06625							MSSSP 97,ASTMA105 ,cladding min.3mm.Note7						
Sockolet	MSS SP 97,B 564 UNS N06625				MSS SP 97,ASTM B 564 UNS N06625							MSSSP 97,ASTMA105 ,cladding min.3mm.Note7						
Weldolet	MSS SP 97,B 564 UNS N06625				MSS SP 97,ASTM B 564 UNS N06625							MSSSP 97,ASTMA105 ,cladding min.3mm.Note7						
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe							MOC Same as pipe						
Swage	MSS SP 95,Same as pipe				MSS SP 95, MOC Same as pipe							MSS SP 95, MOC Same as pipe						
Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				WN							WN						
MOC	ASTM B 564 UNS N06625				ASTM B 564 UNS N06625							MSSSP 97,ASTMA105 ,cladding min.3mm.Note7						
FACE	RF,Serrated Finnished				RF,Serrated Finnished													
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25													
Spectacle Blind/Spacer Blinds																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
MOC	ASTM B 564 UNS N06625																	
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48						
Mechanical Joints																		
Stud Bolt	ASTM A 453 Gr. 660 CL A																	
Hex Nuts	ASTM A 453 Gr. 660 CL A																	
Gasket	ASME B 16.20 , 5mm thick Duplex SS Spiral wound gasket with CNAF filler																	

Pressure-Temperature Rating

Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2
(-) 20 to 100	290	(-) 29 to 38	20.3
200	260	93	18.3
300	230	149	16.2

Maximum Hydrostatic Pressure

Pressure, psig	Pres.,Kg/cm2g
435psig	31 Kg/cm2
Limited by Flange considereing flange as the weakest joint in piping system	

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
2. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and specified flange and spectacle blind.
3. All pipe thread shall be as per ASME B 1.20.1
4. PMS to be read in conjuction with FS 2004 A.
5. Only the base material thickness shall be considered.
6. For sizes 14" and above welded pipe of clad plate with 100% radiography shall be used. Cladding of minimum 3mm shall be done with Inconel 625 on API 5L Gr. B pipe.
7. Cladding shall be weld over lay type with Inconel 625 of min. 3mm
8. To use Schedule 80 for sizes less than 1/1/2" for threaded pipe and fittings.

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**			Sheet No.	**2 of 2**	
	Piping Class	Material	C.A	Spec.No	Revision: 0	
	A15A # 150	Inconel 625	Nil	2004A,Rev 9	Page No.	110
	Code:ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
	Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.				
	0					

Branch Table as per API RP 14E

Run Pipe																	
	≤ 1	T															
	1-1.5	T	T														
	2	*	*	T													
	3	*	*	T	T												
	4	*	*	T	T	T											
	6	*	*	W	T	T	T										
	8	*	*	W	W	T	T	T									
	10	*	*	W	W	W	T	T	T								
	12	*	*	W	W	W	T	T	T	T							
	14	*	*	W	W	W	W	T	T	T	T						
	16	*	*	W	W	W	W	T	T	T	T	T					
	18	*	*	W	W	W	W	W	T	T	T	T	T				
	20	*	*	W	W	W	W	W	T	T	T	T	T	T			
	24	*	*	W	W	W	W	W	W	T	T	T	T	T	T		
	30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T	
		≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30	
		Branch Pipe															

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG		VG 105-1,SW			VG 111-1,FLANGED												
Rating		# 800			# 150,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG		VB 105-1,SW			VB 111-1, Flanged												
Rating		# 800			# 150,RF												
Globe Valve																	
TAG		VGL 104-1,SW			VGL 111-1,flanged												
Rating		# 800			# 150,RF												
Check Valve																	
TAG		VC 104-1,SW,Lift Check			VC 111-1,Swing Check												
Rating		# 800			# 150,RF												
Needle Valve																	
TAG		VN 105-1,SW															
Rating		# 800															
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.
2. Maximum temperature limit for all Ball Valve shall be suitable to maximum temperature of the project.

Revision: 0	22/7/2009	Issued For Bid	PS	GJ	GRP
Revision	Date	Description	Prepared By	Reviewed By	Approved By

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**											Sheet No.		**1 of 2**		
		Piping Class			Material			C.A	Spec.No				Revision: 0				
		B15A		# 300	Inconel 625			Nil	2004A,Rev 9				Page No.	111			
		Code:ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3															
		Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.														

Pipe Data	ASME B 36.10																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	80s	80s	80s	80s	10s	10s	10s	10s	10s	10s	10s						
WTmm	3.73	3.91	4.55	5.08	2.77	3.05	3.05	3.40	3.76	4.19	4.57						
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	5.0	5.0	5.0	5.0	5.0	5.0
TYPE	Seamless											Welded pipe from clad plate. Note -5					
MOC	ASTM B 444 UNSN06625 Gr.1											API 5L Gr.B Pipe with Inconel cladding, Note 6					
Ends	PE				BE												
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Socket Weld				Butt Weld												
Rating	# 3000				10s	10s	10s	10s	10s	10s	10s						
MOC	ASTM B 564 UNS N06625				ASTM B 366 WP NC MC S UNS N06625							ASTM A 234 WPB with cladding min. 3mm. Note 7					
Elbow	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Tee	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Red.	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Cap	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Coupl	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Plug	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Union																	
Elbowlet	MSS SP 97,B 564 UNS N06625				MSS SP 97,ASTM B 564 UNS N06625							MSSSP 97,ASTMA105 ,cladding min.3mm.Note7					
Sockolet	MSS SP 97,B 564 UNS N06625				MSS SP 97,ASTM B 564 UNS N06625							MSSSP 97,ASTMA105 ,cladding min.3mm.Note7					
Weldolet	MSS SP 97,B 564 UNS N06625				MSS SP 97,ASTM B 564 UNS N06625							MSSSP 97,ASTMA105 ,cladding min.3mm.Note7					
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe							MOC Same as pipe					
Swage	MSS SP 95,Same as pipe				MSS SP 95, MOC Same as pipe							MSS SP 95, MOC Same as pipe					
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				WN							WN					
MOC	ASTM B 564 UNS N06625				ASTM B 564 UNS N06625							MSSSP 97,ASTMA105 ,cladding min.3mm.Note7					
FACE	RF,Smooth Finnished				RF,Smooth Finnished												
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25												
Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM B 564 UNS N06625																
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48					
Mechanical Joints																	
Stud Bolt	ASTM A 453 Gr. 660 CL A																
Hex Nuts	ASTM A 453 Gr. 660 CL A																
Gasket	ASME B 16.20 , 5mm thick Duplex SS Spiral wound gasket with CNAF filler																

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
(-) 20 to 100	750	(-) 29 to 38	52.8	1125psig	80 Kg/cm2
200	750	93	52.8	Limited by Flange considereing flange as the weakest joint in piping system	
300	730	149	51.3		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
2. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and specified flange and spectacle blind.
3. All pipe thread shall be as per ASME B 1.20.1
4. PMS to be read in conjuction with FS 2004 A.
5. Only the base material thickness shall be considered.
6. For sizes 14" and above welded pipe of clad plate with 100% radiography shall be used. Cladding of minimum 3mm shall be done with Inconel 625 on API 5L Gr. B pipe.
7. Cladding shall be weld over lay type with Inconel 625 of min. 3mm

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION			Sheet No.	2 of 2
	Piping Class	Material	C.A	Spec.No	Revision: 0
	B15A	# 300	Inconel 625	Nil	2004A,Rev 9	Page No.	112
	Code:ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3				
	Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.			
	0				

Branch Table as per API RP 14E

Run Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

Branch Pipe

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 105-1,SW				VG 113-1,FLANGED												
Rating	# 800				# 300,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB 105-1,SW				VB 113-1,												
Rating	# 800				# 300,RF												
Globe Valve																	
TAG	VGL 104-1,SW				VGL 113-1												
Rating	# 800				# 300,RF												
Check Valve																	
TAG	VC 104-1,SW,Lift Check				VC 113-1,Swing Check												
Rating	# 800				# 300,RF												
Needle Valve																	
TAG	VN 105-1,SW																
Rating	# 800																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.
2. Maximum temperature limit for all Ball Valve shall be suitable to maximum temperature of the project.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 0	22/7/2009	Issued For Bid	PS	GJ	GRP

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**											Sheet No.			**1 of 2**	
		Piping Class				Material				C.A	Spec.No				Revision: 0	
		D15A		# 600		Inconel 625				Nil	2004A,Rev 9				Page No.	113
		Code:ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3														
		Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.													

Pipe Data	ASME B 36.10																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	80s	80s	80s	80s	10s	40s	40s	40s	40s	40s	40s						
WTmm	3.73	3.91	4.55	5.08	2.77	5.49	6.02	7.11	8.18	9.27	9.52						
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	5.0	5.0	5.0	5.0	5.0	5.0
TYPE	Seamless											Welded pipe from clad plate. Note -5,					
MOC	ASTM B 444 UNSN06625 Gr.1											API 5L Gr.B Pipe with Inconel cladding, Note 6					
Ends	PE				BE												
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Socket Weld				Butt Weld												
Rating	# 3000				10s	40s	40s	40s	40s	40s	40s						
MOC	ASTM B 564 UNS N06625				ASTM B 366 WP NC MC S UNS N06625							ASTM A 234 WPB with cladding min. 3mm. Note 7					
Elbow	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Tee	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Red.	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Cap	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Coupl	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Plug	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Union																	
Elbowlet	MSS SP 97,B 564 UNS N06625				MSS SP 97,ASTM B 564 UNS N06625							MSSSP 97,ASTMA105 ,cladding min.3mm.Note7					
Sockolet	MSS SP 97,B 564 UNS N06625				MSS SP 97,ASTM B 564 UNS N06625							MSSSP 97,ASTMA105 ,cladding min.3mm.Note7					
Weldolet	MSS SP 97,B 564 UNS N06625				MSS SP 97,ASTM B 564 UNS N06625							MSSSP 97,ASTMA105 ,cladding min.3mm.Note					
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe							MOC Same as pipe					
Swage	MSS SP 95,Same as pipe				MSS SP 95, MOC Same as pipe							MSS SP 95, MOC Same as pipe					
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				WN							WN					
MOC	ASTM B 564 UNS N06625				ASTM B 564 UNS N06625							MSSSP 97,ASTMA105 ,cladding min.3mm.Note7					
FACE	RF,Smooth Finnished				RF,Smooth Finnished												
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25												
Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM B 564 UNS N06625																
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48					
Mechanical Joints																	
Stud Bolt	ASTM A 453 Gr. 660 CL A																
Hex Nuts	ASTM A 453 Gr. 660 CL A																
Gasket	ASME B 16.20 , 5mm thick Duplex SS Spiral wound gasket with CNAF filler																

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
(-) 20 to 100	1500	(-) 29 to 38	105.4	2250psig	158.2 Kg/cm2
200	1500	93	105.4	Limited by Flange considereing flange as the weakest joint in piping system	
300	1455	149	102.3		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
2. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and specified flange and spectacle blind.
3. All pipe thread shall be as per ASME B 1.20.1
4. PMS to be read in conjuction with FS 2004 A.
5. Only the base material thickness shall be considered.
6. For sizes 14" and above welded pipe of clad plate with100% radiography shall be used. Cladding of minimum 3mm shall be done with Inconel 625 on API 5L Gr. B pipe.
7. Cladding shall be weld over lay type with Inconel 625 of min. 3mm

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**			Sheet No.	**2 of 2**
	Piping Class	Material	C.A	Spec.No	Revision: 0
	D15A # 600	Inconel 625	Nil	2004A,Rev 9	Page No. 114
	Code:ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3				
	Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.			
	0				

Branch Table as per API RP 14E

Run Pipe

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 105-1,SW				VG 115-1,FLANGED												
Rating	# 800				# 600,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB 105-1,SW				VB 115-1,												
Rating	# 800				# 600,RF												
Globe Valve																	
TAG	VGL 104-1,SW				VGL 115-1												
Rating	# 800				# 600,RF												
Check Valve																	
TAG	VC 104-1,SW,Lift Check				VC 115-1,Swing Check												
Rating	# 800				# 600,RF												
Needle Valve																	
TAG	VN 105-1,SW																
Rating	# 800																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.
2. Maximum temperature limit for all Ball Valve shall be suitable to maximum temperature of the project.

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 0	22/7/2009	Issued For Bid	PS	GJ	GRP

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**1 of 2**
	Piping Class	Material	C.A	Spec.No	Revision: 0	
	E15A	# 900	Inconel 625	Nil	2004A,Rev 9	Page No. 115
	Code:ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
	Service: Corrosive HydroCarbon Service-Sour Liquid and Vapour.					

Pipe Data	ASME B 36.10																
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6
Sch.	80s	80s	80s	80s	40s	40s	40s	80s	80s	80	80						
WTmm	3.73	3.91	4.55	5.08	3.91	5.49	6.02	10.97	12.7	15.1	17.47						
M.T (%)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	5.0	5.0	5.0	5.0	5.0	5.0
TYPE	Seamless											Welded pipe from clad plate. Note -5,					
MOC	ASTM B 444 UNSN06625 Gr.1											API 5L Gr.B Pipe with Inconel cladding, Note 6					
Ends	PE				BE												
Fittings Data																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	Socket Weld				Butt Weld												
Rating	# 3000				40s	40s	40s	80s	80s	80	80						
MOC	ASTM B 564 UNS N06625				ASTM B 366 WP NC MC S UNS N06625							ASTM A 234 WPB with cladding min. 3mm. Note 7					
Elbow	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Tee	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Red.	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Cap	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Coupl	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Plug	ASMEB16.11				ASME B 16.9							ASME B 16.9					
Union																	
Elbowlet	MSS SP 97,B 564 UNS N06625				MSS SP 97,ASTM B 564 UNS N06625							MSSSP 97,ASTMA105 ,cladding min.3mm.Note7					
Sockolet	MSS SP 97,B 564 UNS N06625				MSS SP 97,ASTM B 564 UNS N06625							MSSSP 97,ASTMA105 ,cladding min.3mm.Note7					
Weldolet	MSS SP 97,B 564 UNS N06625				MSS SP 97,ASTM B 564 UNS N06625							MSSSP 97,ASTMA105 ,cladding min.3mm.Note					
Nipple	MSS SP 97,Same as pipe				MOC Same as pipe							MOC Same as pipe					
Swage	MSS SP 95,Same as pipe				MSS SP 95, MOC Same as pipe							MSS SP 95, MOC Same as pipe					
Flange																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
TYPE	SW				WN							WN					
MOC	ASTM B 564 UNS N06625				ASTM B 564 UNS N06625							MSSSP 97,ASTMA105 ,cladding min.3mm.Note7					
FACE	RTJ-Note 8				RTJ-Note 8												
STD.	ASME B 16.5				ASME B 16.5, But welding ends as per ASME B 16.25												
Spectacle Blind/Spacer Blinds																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
MOC	ASTM B 564 UNS N06625																
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48					
Mechanical Joints																	
Stud Bolt	ASTM A 453 Gr. 660 CL A																
Hex Nuts	ASTM A 453 Gr. 660 CL A																
Gasket	ASME B 16.20,OCT ring of UNS S 32205 with Max. Hardness of 22 HRC. Gasket contact surface roughness max. 63 AARH.																

Pressure-Temperature Rating

Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Maximum Hydrostatic Pressure	
				Pressure, psig	Pres.,Kg/cm2g
(-) 20 to 100	2250	(-) 29 to 38	158.1	3375psig	237.2 Kg/cm2
200	2250	93	158.1	Limited by Flange considereing flange as the weakest joint in piping system	
300	2185	149	153.6		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
2. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and specified flange and spectacle blind.
3. All pipe thread shall be as per ASME B 1.20.1
4. PMS to be read in conjuction with FS 2004 A.
5. Only the base material thickness shall be considered.
6. For sizes 14" and above welded pipe of clad plate with 100% radiography shall be used. Cladding of minimum 3mm shall be done with Inconel 625 on API 5L Gr. B pipe.
7. For sizes 14" and above welded pipe of clad plate with joint efficiency 1 shall be used. Cladding of minimum 3mm shall be done with Inconell 625 on API 5L Gr. B pipe.
8. RTJ groove hardness shall be 35 HRC.
9. Cladding shall be weld over lay type with Inconel 625 of min. 3mm

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION			Sheet No.	2 of 2
	Piping Class	Material	C.A	Spec.No	Revision: 0
	E15A # 900	Inconel 625	Nil	2004A,Rev 9	Page No. 116
	Code:ASME B 31.3 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3				
	Service:	Corrosive HydroCarbon Service-Sour Liquid and Vapour.			
	0				

Branch Table as per API RP 14E

Run Pipe	Branch Pipe																
	≤ 1	T															
	1-1.5	T	T														
	2	*	*	T													
	3	*	*	T	T												
	4	*	*	T	T	T											
	6	*	*	W	T	T	T										
	8	*	*	W	W	T	T	T									
	10	*	*	W	W	W	T	T	T								
	12	*	*	W	W	W	T	T	T	T							
	14	*	*	W	W	W	W	T	T	T	T						
	16	*	*	W	W	W	W	T	T	T	T	T					
	18	*	*	W	W	W	W	W	T	T	T	T	T				
	20	*	*	W	W	W	W	W	T	T	T	T	T	T			
	24	*	*	W	W	W	W	W	W	T	T	T	T	T	T		
	30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T	
		≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30	

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG		VG 107-1,SW			VG 117-1,FLANGED												
Rating		# 1500			# 900,RTJ												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG		VB 107-1,SW			VB 117-1,												
Rating		# 1500			# 900,RTJ												
Globe Valve																	
TAG		VGL 106-1,SW			VGL 117-1												
Rating		# 1500			# 900,RTJ												
Check Valve																	
TAG		VC 106-1,SW,Lift Check			VC 117-1,Swing Check												
Rating		# 1500			# 900,RTJ												
Needle Valve																	
TAG		VN 107-1,SW															
Rating		# 1500															
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.
2. Maximum temperature limit for all Ball Valve shall be suitable to maximum temperature of the project.

Revision: 0	22/7/2009	Issued For Bid	PS	GJ	GRP
Revision	Date	Description	Prepared By	Reviewed By	Approved By

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**												Sheet No.		**1 of 2**	
		Piping Class			Material			C.A	Spec.No							Revision: 1	
		PE1		# 900	CS			3mm	2004A,Rev 9							Page No.	117
		Design Code:		**ASME B 31.4/31.8**													
		Service:	Hydro Carbon Liquid and Vapour, Injection Water (Piping from transition bend to Hinged closers Only)														

Pipe Data	ASME B 36.10																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6	
Sch.	XXS	XXS	XXS	XXS	160	160	120	Thickness shall be same as that of Pipe line for smooth Pigging										
WTmm	7.47	7.82	9.09	10.16	8.74	11.12	11.12	Thickness shall be same as that of Pipe line for smooth Pigging										
M.T (%)																		
TYPE	Seamless													LSAW with 100% Radiography				
MOC	ASTM A 106 Gr. B							API 5L Gr. X 60 or Same as Pipeline						API 5L Gr. 60 or same as pipeline				
Ends	PE				BE													
Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	Socket Weld				Butt Weld, Seamless			Butt Welded, Seamless						Welded with 100% Radiography				
Rating	# 9000				160	160	120											
MOC	ASTM A 105 U.N.O				A 234 Gr. WPB, Seamless			MSS SP -75 Gr. WPHY rest portion of grade designation shall be compatible to pipe. U.N.O										
Elbow	ASMEB16.11				ASME B 16.9			MSS SP -75 and for piggable portion only 5D bends shall be used UNO.										
Tee	ASMEB16.11				ASME B 16.9			MSS SP -75										
Red.	ASMEB16.11				ASME B 16.9			MSS SP-75										
Cap	ASMEB16.11				ASME B 16.9			MSS SP-75										
Coupl	ASMEB16.11				ASME B 16.9			MSS SP-75										
Plug	ASMEB16.11				ASME B 16.9			MSS SP-75										
Elbowlet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105			ASTM A 694 Gr. rest portion of grade designation shall be compatible to pipe. U.N.O, MSS SP 97										
Sockolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105			ASTM A 694 Gr. rest portion of grade designation shall be compatible to pipe. U.N.O, MSS SP 97										
Weldolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105			ASTM A 694 Gr. rest portion of grade designation shall be compatible to pipe. U.N.O, MSS SP 97										
Nipple	MSS SP 95,Same as pipe				MSS SP 95MOC Same as pipe			MSS SP 95 MOC Same as pipe										
Swage	MSS SP 95,Same as pipe				MSS SP 95MOC Same as pipe			MSS SP 95 MOC Same as pipe										
Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				WN			WN										
MOC	ASTM A 105				ASTM A 105			ASTM A 694 Grade designation shall be compatible to pipe. U.N.O										
FACE	RTJ-Note 7				RTJ, Note 7			RTJ, Note 7										
STD.	ASME B 16.5				B 16.5, But welding ends B 16.25			ASME B 16.5 with Butt Welding end as per ASME B 16.25										
Spectacle Blind/Spacer Blinds																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
MOC	ASTM A 105																	
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48						
Mechanical Joints																		
Stud Bolts	ASTM A 193 Gr. B7				ASTM A 193 Gr. B7													
Hex Nuts	ASTM A 194 Gr. 2H				ASTM A 194 Gr. 2H													
Gasket	B 16.20,OCT ring of Soft Iron with Max. 90BHN				ASME B 16.20,OCT ring of Soft Iron with Max. Hardness of 90BHN													

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
(-) 4 to 100	2220	(-) 20 to 38	156	3350 psig	236 Kg/cm2
200	2035	93	143.07	Limited by Flange considereing flange as the weakest joint in piping system	
300	1965	149	138.2		
400	1900	204	133.6		
500	1810	260	127.2		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2.All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and specified flange and spectacle blind.
5. All pipe thread shall be as per ASME B 1.20.1
6. PMS to be read in conjuction with FS 2004 A.
7. Gasket Contact Surface shall have maximum roughness of 63AARH
8. Minimum RTJ groove hardness shall be 120BHN
9. Octagonal (OCT) ring shall be Hot Dip Galvanized or electro galvanized.
10.Piggable portion of piping shall be critically checked for passage of pigs with respect to internal diameter of each segment of pipeline.
11. Weld Joint Factor for welded pipe shall be as per ASME B 31.4/31.8

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**2 of 2**
	Piping Class	Material	C.A	Spec.No		Revision: 1
	PE1 # 900	CS	3mm	2004A,Rev 9	Page No.	118
	Design Code:	**ASME B 31.4/31.8**				
	Service:	Hydro Carbon Liquid and Vapour, Injection Water (Piping from transition bend to Hinged closers Only)				

Branch Table as per API RP 14E

Run Pipe	Branch Pipe															
	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30	
≤ 1	T															
1-1.5	T	T														
2	*	*	T													
3	*	*	T	T												
4	*	*	T	T	T											
6	*	*	W	T	T	T										
8	*	*	W	W	T	T	T									
10	*	*	W	W	W	T	T	T								
12	*	*	W	W	W	T	T	T	T							
14	*	*	W	W	W	W	T	T	T	T						
16	*	*	W	W	W	W	T	T	T	T	T					
18	*	*	W	W	W	W	W	T	T	T	T	T				
20	*	*	W	W	W	W	W	T	T	T	T	T	T			
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T		
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T	

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 46,SW				VG 78,FLANGED												
Rating	# 1500				# 900,RTJ												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	Use Gate Valve, VG 46				VB 84,LO	VB 85, Gear Operated											
Rating	# 1500				# 900, RTJ												
Globe Valve																	
TAG	VGL 44,SW				VGL 72												
Rating	# 1500				# 900, RTJ												
Check Valve																	
TAG	VC 44,SW,Lift Check				VC 68,Swing Check			VCS - 4 Full Port Swing Check									
Rating	# 1500				# 900, RTJ												
Needle Valve																	
TAG	VN 4,SW																
Rating	# 1500																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.
2. Maximum temperature limit for all Ball Valve shall be 121 Degree C

Revision: 1	15.01.16	Issued For Bid	PS	AKM	SJ
Revision	Date	Description	Prepared By	Reviewed By	Approved By

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**			Sheet No.	**1 of 2**
	Piping Class	Material	C.A	Spec.No	Revision: 0
	PB1N / # 300	CS-NACE	6mm	2004A,Rev 9	Page No. 119
	Design Code: ASME B 31.4/31.8 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3				
	Service: Corrosive Hydro Carbon Liquid and Vapour, Injection Water (Piping from from transition bend to Hinged closers Only)				

Pipe Data	ASME B 36.10																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6	
Sch.	XXS	XXS	XXS	XXS	160	160	120	Thickness shall be same as that of Pipe line for smooth Pigging										
WTmm	7.47	7.8	9.09	10.16	8.74	9.52	11.12	Thickness shall be same as that of Pipe line for smooth Pigging										
M.T (%)																		
TYPE	Seamless													LSAW with 100% Radiography				
MOC	ASTM A 106 Gr. B							API 5L Gr. X 60 or Same as Pipeline						API 5L Gr. 60 or same as pipeline				
Ends	PE				BE													

Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	Socket Weld				Butt Weld, Seamless			Butt Welded, Seamless						Welded with 100% Radiography				
Rating	# 9000				160	160	120											
MOC	ASTM A 105 U.N.O				A 234 Gr. WPB, Seamless			MSS SP -75 Gr. WPHY rest portion of grade designation shall be compatible to pipe. U.N.O										
Elbow	ASMEB16.11				ASME B 16.9			MSS SP -75 and for piggable portion only 5D bends shall be used UNO.										
Tee	ASMEB16.11				ASME B 16.9			MSS SP -75										
Red.	ASMEB16.11				ASME B 16.9			MSS SP-75										
Cap	ASMEB16.11				ASME B 16.9			MSS SP-75										
Coupl	ASMEB16.11				ASME B 16.9			MSS SP-75										
Plug	ASMEB16.11				ASME B 16.9			MSS SP-75										
Elbowlet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105			ASTM A 694 Gr. rest portion of grade designation shall be compatible to pipe. U.N.O, MSS SP 97										
Sockolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105			ASTM A 694 Gr. rest portion of grade designation shall be compatible to pipe. U.N.O, MSS SP 97										
Weldolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105			ASTM A 694 Gr. rest portion of grade designation shall be compatible to pipe. U.N.O, MSS SP 97										
Nipple	MSS SP 95,Same as pipe				MSS SP 95MOC Same as pipe			MSS SP 95 MOC Same as pipe										
Swage	MSS SP 95,Same as pipe				MSS SP 95MOC Same as pipe			MSS SP 95 MOC Same as pipe										

Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				WN			WN										
MOC	ASTM A 105				ASTM A 105			ASTM A 694 Grade designation shall be compatible to pipe. U.N.O										
FACE	RF, Smooth Finnished				RF, Smooth Finnished			RF, Smooth Finnished										
STD.	ASME B 16.5				B 16.5, But welding ends B 16.25			ASME B 16.5 with Butt Welding end as per ASME B 16.25										

Spectacle Blind/Spacer Blinds																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
MOC	ASTM A 105																	
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48						

Mechanical Joints		
Stud Bolt	ASTM A 193 Gr. B7M	ASTM A 193 Gr. B7M
Hex Nuts	ASTM A 194 Gr. 2HM	ASTM A 194 Gr. 2HM
Gasket	B 16.20,4.5mm, SS Spiral wound with CNAF filler	ASME B 16.20,4.5mm, SS Spiral wound with CNAF filler

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
(-) 4 to 100	740	(-) 20 to 38	52	1125 psig	79 Kg/cm2
200	675	93	47.5	Limited by Flange considereing flange as the weakest joint in piping system	
300	655	149	46		
400	635	204	44.6		
500	600	260	44.2		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A and all the requirement of MR-01-75/ISO-15156-1/2/3 to complied.
6. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and specified flange and spectacle blind.
7. Piggable portion of piping shall be critically checked for passage of pigs with respect to internal diameter of each segment of pipeline.
8. Weld Joint Factor for welded pipe shall be as per ASME B 31.4/31.8

Offshore Design Section Engineering Services IS0-9001:2000	PIPING MATERIAL SPECIFICATION				Sheet No.	2 of 2
	Piping Class	Material	C.A	Spec.No	Revision: 0	
	PB1N	# 300	CS-NACE	6mm	2004A,Rev 9	Page No. 120
	Design Code: ASME B 31.4/31.8 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
	Service	Corrosive Hydro Carbon Liquid and Vapour, Injection Water (Piping from from transition bend to Hinged closers Only)				

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 45,SW				VG 69,FLANGED												
Rating	# 800				# 300,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	45,SW and VB 107,SW (Note-5				VB 69, Lever Operated							VB **, Gear Operated					
Rating	# 800				# 300, RF												
Globe Valve																	
TAG	VGL 45,SW				VGL 69												
Rating	# 800				# 300, RF												
Check Valve																	
TAG	VC 45,SW,Lift Check				VC 69,Swing Check/ VC 89 (Wafer Check, Note -1)												
Rating	# 800				# 300, RF												
Needle Valve																	
TAG	VN 9,SW																
Rating	# 800																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.
2. Maximum temperature limit for all Ball Valve shall be 121 Degree C

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 0	15.01.16	Issued For Bid	PS	AKM	SJ

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**											Sheet No.		**1 of 2**	
		Piping Class		Material		C.A	Spec.No							Revision: 1	
		PD1N	# 600	CS-NACE		6mm	2004A,Rev 9							Page No.	121
		Design Code: ASME B 31.4/31.8 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3													
		Service:	Corrosive Hydro Carbon Liquid and Vapour, Injection Water (Piping from from transition bend to Hinged closers Only)												

Pipe Data	ASME B 36.10																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6	
Sch.	XXS	XXS	XXS	XXS	160	160	120	Thickness shall be same as that of Pipe line for smooth Pigging										
WTmm	7.47	7.82	9.09	10.16	8.74	9.52	11.12	Thickness shall be same as that of Pipe line for smooth Pigging										
M.T (%)																		
TYPE	Seamless													LSAW with 100% Radiography				
MOC	ASTM A 106 Gr. B							API 5L Gr. X 60 or Same as Pipeline						API 5L Gr. 60 or same as pipeline				
Ends	PE				BE													
Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	Socket Weld				Butt Weld, Seamless			Butt Welded, Seamless						Welded with 100% Radiography				
Rating	# 9000				160	160	120											
MOC	ASTM A 105 U.N.O				A 234 Gr. WPB, Seamless			MSS SP -75 Gr. WPHY rest portion of grade designation shall be compatible to pipe. U.N.O										
Elbow	ASMEB16.11				ASME B 16.9			MSS SP -75 and for piggable portion only 5D bends shall be used UNO.										
Tee	ASMEB16.11				ASME B 16.9			MSS SP -75										
Red.	ASMEB16.11				ASME B 16.9			MSS SP-75										
Cap	ASMEB16.11				ASME B 16.9			MSS SP-75										
Coupl	ASMEB16.11				ASME B 16.9			MSS SP-75										
Plug	ASMEB16.11				ASME B 16.9			MSS SP-75										
Elbowlet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105			ASTM A 694 Gr. rest portion of grade designation shall be compatible to pipe. U.N.O, MSS SP 97										
Sockolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105			ASTM A 694 Gr. rest portion of grade designation shall be compatible to pipe. U.N.O, MSS SP 97										
Weldolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105			ASTM A 694 Gr. rest portion of grade designation shall be compatible to pipe. U.N.O, MSS SP 97										
Nipple	MSS SP 95,Same as pipe				MSS SP 95MOC Same as pipe			MSS SP 95 MOC Same as pipe										
Swage	MSS SP 95,Same as pipe				MSS SP 95MOC Same as pipe			MSS SP 95 MOC Same as pipe										
Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				WN			WN										
MOC	ASTM A 105				ASTM A 105			ASTM A 694 Grade designation shall be compatible to pipe. U.N.O										
FACE	RF, Smooth Finnished				RF, Smooth Finnished			RF, Smooth Finnished										
STD.	ASME B 16.5				B 16.5, But welding ends B 16.25			ASME B 16.5 with Butt Welding end as per ASME B 16.25										
Spectacle Blind/Spacer Blinds																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
MOC	ASTM A 105																	
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48						
Mechanical Joints																		
Stud Bolt	ASTM A 193 Gr. B7M				ASTM A 193 Gr. B7M													
Hex Nuts	ASTM A 194 Gr. 2HM				ASTM A 194 Gr. 2HM													
Gasket	B 16.20,4.5mm, SS Spiral wound with CNAF filler				ASME B 16.20,4.5mm, SS Spiral wound with CNAF filler													

Pressure-Temperature Rating				Maximum Hydrostatic Pressure	
Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Pressure, psig	Pres.,Kg/cm2g
(-) 4 to 100	1480	(-) 20 to 38	104	2225 psig	157 Kg/cm2
200	1360	93	95.61	Limited by Flange considereing flange as the weakest joint in piping system	
300	1310	149	92.1		
400	1265	204	88.94		
500	1205	260	84.72		

Notes Releted to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. All pipe thread shall be as per ASME B 1.20.1
5. PMS to be read in conjuction with FS 2004 A and all the requirement of MR-01-75/ISO-15156-1/2/3 to complied.
6. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and specified flange and spectacle blind.
7. Piggable portion of piping shall be critically checked for passage of pigs with respect to internal diameter of each segment of pipeline.
8. Weld Joint Factor for welded pipe shall be as per ASME B 31.4/31.8

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**2 of 2**
	Piping Class	Material	C.A	Spec.No		Revision: 1
	PD1N	# 600	CS-NACE	6mm	2004A,Rev 9	Page No. 122
	Design Code: ASME B 31.4/31.8 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
	Service: Corrosive Hydro Carbon Liquid and Vapour, Injection Water (Piping from from transition bend to Hinged closers Only)					

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG	VG 45,SW				VG 71,FLANGED												
Rating	# 800				# 600,RF												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG	VB 45,SW and VB 107,SW (Note-5				VB 71, Lever Operated							VB **, Gear Operated					
Rating	# 800				# 600, RF												
Globe Valve																	
TAG	VGL 45,SW				VGL 71												
Rating	# 800				# 600, RF												
Check Valve																	
TAG	VC 45,SW,Lift Check				VC 71,Swing Check							VCS - 8, Swing Check Full Port					
Rating	# 800				# 600, RF												
Needle Valve																	
TAG	VN 9,SW																
Rating	# 800																
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.
2. Maximum temperature limit for all Ball Valve shall be 121 Degree C

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision: 1	15.01.16	Issued For Bid	PS	AKM	SJ

	Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**													Sheet No.		**1 of 2**	
		Piping Class				Material			C.A	Spec.No						Revision:1	
		PE1N	# 900			CS-NACE			6mm	2004A,Rev 9						Page No.	123
		Design Code:ASME B 31.4/31.8 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3															
		Service: Corrosive Hydro Carbon Liquid and Vapour, Injection Water (Piping from transition bend to Hinged closers Only)															

Pipe Data	ASME B 36.10																	
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
O.D.mm	21.3	26.7	33.4	48.3	60.3	88.9	114.3	168.3	219.1	273.1	323.9	355.6	406.4	457.2	508	558.8	609.6	
Sch.	XXS	XXS	XXS	XXS	XXS	XXS	160	Thickness shall be same as that of Pipe line for smooth Pigging										
WTmm	7.47	7.82	9.09	10.16	11.07	15.24	13.49	Thickness shall be same as that of Pipe line for smooth Pigging										
M.T (%)																		
TYPE	Seamless													LSAW with 100% Radiography				
MOC	ASTM A 106 Gr. B							API 5L Gr. X 60 or Same as Pipeline						API 5L Gr. 60 or same as pipeline				
Ends	PE				BE													

Fittings Data																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	Socket Weld				Butt Weld, Seamless			Butt Welded, Seamless						Welded with 100% Radiography				
Rating	# 9000				XXS	XXS	160											
MOC	ASTM A 105 U.N.O				A 234 Gr. WPB, Seamless			MSS SP -75 Gr. WPHY rest portion of grade designation shall be compatible to pipe. U.N.O										
Elbow	ASMEB16.11				ASME B 16.9			MSS SP -75 and for piggable portion only 5D bends shall be used UNO.										
Tee	ASMEB16.11				ASME B 16.9			MSS SP -75										
Red.	ASMEB16.11				ASME B 16.9			MSS SP-75										
Cap	ASMEB16.11				ASME B 16.9			MSS SP-75										
Coupl	ASMEB16.11				ASME B 16.9			MSS SP-75										
Plug	ASMEB16.11				ASME B 16.9			MSS SP-75										
Elbowlet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105			ASTM A 694 Gr. rest portion of grade designation shall be compatible to pipe. U.N.O, MSS SP 97										
Sockolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105			ASTM A 694 Gr. rest portion of grade designation shall be compatible to pipe. U.N.O, MSS SP 97										
Weldolet	MSS SP 97,ASTM A 105				MSS SP 97,ASTM A 105			ASTM A 694 Gr. rest portion of grade designation shall be compatible to pipe. U.N.O, MSS SP 97										
Nipple	MSS SP 95,Same as pipe				MSS SP 95MOC Same as pipe			MSS SP 95 MOC Same as pipe										
Swage	MSS SP 95,Same as pipe				MSS SP 95MOC Same as pipe			MSS SP 95 MOC Same as pipe										

Flange																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
TYPE	SW				WN			WN										
MOC	ASTM A 105				ASTM A 105			ASTM A 694 Grade designation shall be compatible to pipe. U.N.O										
FACE	RTJ-Note 7				RTJ, Note 7			RTJ, Note 7										
STD.	ASME B 16.5				B 16.5, But welding ends B 16.25			ASME B 16.5 with Butt Welding end as per ASME B 16.25										

Spectacle Blind/Spacer Blinds																		
Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	
MOC	ASTM A 105																	
Spectacle	ASME B 16.48				ASME B 16.48							Spacer and blind as per ASME B 16.48						

Mechanical Joints		
Stud Bolt	ASTM A 193 Gr. B7M	ASTM A 193 Gr. B7M
Hex Nuts	ASTM A 194 Gr. 2HM	ASTM A 194 Gr. 2HM
Gasket	0,OCT ring of Soft Iron with Max. 9	ASME B 16.20,OCT ring of Soft Iron with Max. Hardness of 90BHN

Pressure-Temperature Rating

Temp.,Deg F	Pressure, PSI	Temp,Deg C	Pressure, Kg/cm2	Maximum Hydrostatic Pressure	
				Pressure, psig	Pres.,Kg/cm2g
(-) 4 to 100	2220	(-) 20 to 38	156	3350 psig	236 Kg/cm2
200	2035	93	143.07	Limited by Flange considereing flange as the weakest joint in piping system	
300	1965	149	138.2		
400	1900	204	133.6		
500	1810	260	127.2		

Notes Related to Pipe,Fittings,Flanges,Gasket and Stud Bolts -Nuts

1. Welded fittings shall be 100% radiographed.
2. All alloy steel stud bolts and nuts shall be hot dip galvanized or electrogalvanized with minimum coating as per spec.
3. Spectacle blinds and spacer sizes and rating that are not available in ASME B 16.48 shall be as per manufacturer standard. Design shall be submitted to company for review and approval.
4. Two jackscrew, 180 degree apart shall be provided in one of the flanges for all orifice flange and WNRTJ flanges above 3" size and specified spectacle blind.
5. All pipe thread shall be as per ASME B 1.20.1
6. PMS to be read in conjuction with FS 2004 A.
7. Gasket Contact Surface shall have maximum roughness of 63AARH
8. Minimum RTJ groove hardness shall be 120BHN
9. Octagonal (OCT) ring shall be Hot Dip Galvanized or electrogalvanized
10. Piggable portion of piping shall be critically checked for passage of pigs with respect to internal diameter of each segment of pipeline.
11. Weld Joint Factor for welded pipe shall be as per ASME B 31.3

Offshore Design Section Engineering Services IS0-9001:2000	**PIPING MATERIAL SPECIFICATION**				Sheet No.	**2 of 2**
	Piping Class	Material	C.A	Spec.No	Revision:1	
	PE1N	# 900	CS-NACE	6mm	2004A,Rev 9	Page No. 124
	Design Code:ASME B 31.4/31.8 and other applicable Code NACE-MR-01-75/ISO-15156-1/2/3					
	Service:	Corrosive Hydro Carbon Liquid and Vapour, Injection Water (Piping from transition bend to Hinged closers Only)				

Branch Table as per API RP 14E

Run Pipe \ Branch Pipe	≤ 1	1-1.5	2	3	4	6	8	10	12	14	16	18	20	24	30
≤ 1	T														
1-1.5	T	T													
2	*	*	T												
3	*	*	T	T											
4	*	*	T	T	T										
6	*	*	W	T	T	T									
8	*	*	W	W	T	T	T								
10	*	*	W	W	W	T	T	T							
12	*	*	W	W	W	T	T	T	T						
14	*	*	W	W	W	W	T	T	T	T					
16	*	*	W	W	W	W	T	T	T	T	T				
18	*	*	W	W	W	W	W	T	T	T	T	T			
20	*	*	W	W	W	W	W	T	T	T	T	T	T		
24	*	*	W	W	W	W	W	W	T	T	T	T	T	T	
30	*	*	W	W	W	W	W	W	W	W	T	T	T	T	T

LEGEND	
*	Sockolet
W	Weldolet
T	TEE

Valve Data

Size(in)	0.5	0.75	1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24
Gate Valve																	
TAG		VG 47,SW			VG 73,FLANGED												
Rating		# 1500			# 900,RTJ												
Plug Valve																	
TAG																	
Rating																	
Ball Valve																	
TAG		Use Gate Valve, VG 47			VB 73, Lever Operated							VB **, Gear Operated					
Rating		# 1500			# 900, RTJ												
Globe Valve																	
TAG		VGL 44,SW			VGL 72												
Rating		# 1500			# 900, RTJ												
Check Valve																	
TAG		VC 47,SW,Lift Check			VC 73,Swing Check			VCS - 9, Full Port Swing Check									
Rating		# 1500			# 900, RTJ												
Needle Valve																	
TAG		VN 10,SW															
Rating		# 1500															
Butter Fly Valve																	
TAG																	
Rating																	

Notes Releted to Valve and Brach Connection

1. PMS to be read in conjuction with FS 2004 A.
2. Maximum temperature limit for all Ball Valve shall be 121 Degree C

Revision	Date	Description	Prepared By	Reviewed By	Approved By
Revision:1	15.01.16	Issued For Bid	PS	AKM	SJ

FUNCTIONAL SPECIFICATION

FOR

"Welding & NDT"

Prepared / Revised By	Reviewed By	Approved By	Total No. of Pages	Date	Rev. No.
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Goutam Jana	M.R.Shete	S.C. Jain			

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1.0. SCOPE

This specification covers the minimum technical requirements for welding & inspection of Piping, Pressure Vessels, Structural & Subsea Pipeline work. For piping, this specification covers CS, CS NACE, SS and Cu-Ni. For piping DSS material welding & NDT, Annexure I shall be referred. This specification modifies and/or supplements the referenced codes that apply to all piping, pressure vessels, structural and pipeline welding. These requirements cover welding done at Contractor's fabrication plant, Contractor's Yard, Vendor / Supplier Premises or field installation, either onshore or offshore. This specification also covers the acceptance standards to be applied to production welds inspected by various NDT Techniques like RT, UT, AUT, MPI & LPT.

1.1. Regulatory Considerations

In addition to this specification, Contractor shall also comply with all applicable regulatory and national code requirements.

1.2. Abbreviations

AISC	American Institute of Steel Construction	NDE	Non Destructive Examination
API	American Petroleum Institute	NDT	Non Destructive Testing
ASME	American Society of Mechanical Engineers	ONGC	Oil and Natural Gas Corporation Ltd.
ASNT	American Society for Nondestructive Testing	PAUT	Phased Array Ultrasonic Testing
ASTM	American Society for Testing and Materials	PCN	Personnel Certification in NDT
AUT	Automated Ultrasonic Testing	PQR	Procedure Qualification Record
AWS	American Welding Society	PWHT	Post Weld Heat Treatment
BPVC	Boiler and Pressure Vessel Code	pWPS	Preliminary / Proposed Welding Procedure Specification
CRA	Corrosion Resistant Alloy	QA	Quality Assurance
CS	Carbon Steel	RT	Radiographic Testing
CTOD	Crack Tip Opening Displacement	SMYS	Specified Minimum Yield Strength
Cu-Ni	Cupro Nickel	SS	Stainless Steel
DCN	Design Change Note	TMCP	Thermo Mechanically Controlled Process
DCR	Design Change Report	TOFD	Time Of Flight Diffraction
DNV	Det Norske Veritas	TPI	Third Party Inspection
DSS	Duplex Stainless Steel	UT	Ultrasonic Testing
ECA	Engineering Critical Assessment	WPQR	Welding Procedure Qualification Record
FCAW	Flux Cored Arc Welding	WPQT	Welding Procedure Qualification Testing
GMAW	Gas Metal Arc Welding	WPS	Welding Procedure Specification
HAZ	Heat Affected Zone	Company	Shall mean ONGC
HRC	Rockwell Hardness		
HV 10	Vicker's Hardness (10 Kg Load)		
ITP	Inspection & Test Plan		
LPT	Liquid Penetrant Testing		
MPI	Magnetic Particle Inspection		
NACE	National Association of Corrosion Engineers		
NCR	Non Conformance Report		

1.3. Quality Assurance

QA 1.3.1 Procedures/ITP:

Before commencing fabrication / installation, the Contractor shall submit a quality plan and a set of fabrication procedures to Company for review and approval. These documents shall include the following minimum requirement:

1. Welding & Consumable Control Procedures,
2. Welder Qualification Procedures,
3. PWHT Procedures and illustration of their proposed areas of application.
4. List of proposed all type of Welding Procedures and their areas of application.
5. List of Qualified all type of Welding Procedures and their areas of application.
6. Material identification & control procedure.
7. Detailed procedures for Non Destructive Testing, Technique sheet & Inspection.
8. Procedures for control of tolerances during fabrication.
9. DCN / DCR Control Procedure
10. Material handling procedure at Fabrication shop, Yard, Offshore installation separately
11. Painting/coating procedure
12. Name of fabrication/NDT agencies involved at Fabrication shop, Yard, Offshore installation with complete address of works & respective activities/job of the project and valid ISO 9001: 2008 certificate.

The quality plan shall include brief details and the sequence of all examinations that will be performed by the Contractor. The names of the individuals responsible for the implementation of all quality assurance and quality control functions shall also be included.

The contractor shall prepare a comprehensive ITP showing the role & scope of various agencies involved in inspection and same shall be submitted for Company's review and approval before commencement of fabrication / installation activity.

Contractor shall be required to do of his own QA internal audit at least 2 times at each yard/offshore activities.

QA 1.3.2 Manpower:

All personnel involved in welding related tasks shall have adequate qualifications and understanding of welding technology. The qualification level shall reflect the tasks and responsibilities of each person in order to obtain the specified quality level. The organization responsible for welding shall nominate at least one authorized welding coordinator in accordance with ISO 14731 – Welding Coordination – Tasks and Responsibilities (or equivalent) to be present on the location where welding is performed. The coordinator shall have comprehensive technical knowledge according to ISO 14731, paragraph 6.2a or equivalent.

The Contractor shall employ sufficient number of qualified inspectors. When an item is ready for inspection, the contractor QA personnel shall inspect it first and should be satisfied before offering the same to the Company inspector.

The contractor shall give tentative inspection schedules in advance so that the Company inspectors can effectively plan for the inspection.

QA 1.3.3 NDT:

In addition to visual inspection, NDT Techniques like RT, UT (Manual UT, PAUT, AUT & TOFD), MPI & LPT are required to be used for inspection. Radiography shall preferably be with X-ray.

γ (Gamma) ray can be used as explained below. However, use of Cobalt 60 Isotope is not allowed.

RT/PAUT as applicable for Piping & Structural welding(CS & CS NACE) is as follows:

1.The X-ray must be used till 10mm thickness joint of Piping & Structural welding.

2. RT-Gamma ray can be used without separate permission of ONGC, from 11 mm thickness till 25mm for Piping & Structural welding joint. However the "Project specific procedure qualification" is must ..

3. Beyond 10mm in case of Piping joints, beyond 19 mm in case of Structural joints , the PAUT can be done without separate permission of ONGC . However the "Project specific procedure qualification" is must.

Only personnel certified in accordance with PCN or ASNT recommended practice SNT-TC-1A shall be allowed to carry out Non Destructive Examination. PCN is the preferred qualification.

1.The technique sheet for the referred NDT shall be prepared by the LSTK contractor. The technique sheet preferably should be concise & precise in single page separately for each type of job & each type of NDT method used.

2.The technique sheet along with NDT procedure shall have approval of PCN or ASNT -Level III qualified (respective NDT method) personal of CA /ONGC.

3.The same technique sheet shall be displayed in the laboratory/workplace by the NDT operator.

4.The NDT operator of LSTK contractor must have to demonstrate/ go for the "Project specific procedure qualification" to be witnessed by ONGC/CA/TPI for the 1st time during start of NDT techniques".

This entire point referred above shall be applicable for all type of NDT method as applicable in piping, pipeline & structural application.

QA 1.3.4 Calibration of NDT instruments:

For standardisation of calibration of instruments used for referred NDT method ,it is essential to have valid calibration certificates for all NDT instruments, Portable Gas detectors, tong tester etc. from NABL accelerated laboratories (India) or from labs which are authorised by local government bodies or from the original manufacturer of the instrument.

Further for RT regarding NDT instruments (RT machine, isotope) shall require valid permission from the local government-regulatory body at the location of use"

QA 1.3.5 *Mechanical integrity* :

Mechanical integrity of all flange joints has to be ensured in proper record.

A standard checklist for the activity along with post box-up checkup should be developed & practiced. The checklist shall include the following:

a) i) Alignment of flanges

ii) Inspection of gasket surface for possible defects such as bends or greases.

iii)Inspection of mating flanges for dirt, mechanical damage and corrosion. Use suitable solvent to clean the surface. The contact area of flanges shall be free from excess pitting and radial scratches.

iv) Ensuring the gasket is properly seated

v) Ensuring that fasteners are tightened as per tightening procedures and using specific values with well maintained tools and torque wrenches.

b) Each joint shall be boxed up and tightened by only skilled technician.

c) Ensuring right type of gaskets & fasteners etc. confirming to the specifications.

d) Ensuring that the stud & nuts are free of dirt and corrosion. The studs should be straight and threads free from nicks, burr and chips etc.

e) Checking of individual joint for any leak during commissioning/ pressure testing by special tapes, soap solution or as per relevant code."

Digital radiography in lieu of film radiography can be used with the permission from ONGC on submission &approval of following documents.

1.Past track record of the technology/vendor in offshore application on similar material (Parent & welding consumable). PO copy & performance report of the user is required.

2.Complete procedure of using the technology with specification, protection of 1st image, calibration, validation of software, Technique etc. with respect to all the relevant standard.

3. The specific training & "procedure qualification "on the subject is required to be provided to ONGC/CA/TPI free of cost.

4. For storage of data a dedicated PC of sufficient memory for entire project with viewing software etc. must be provided without extra cost to ONGC..

5. Optimum nos. of re-use of film is restricted to 150 nos. or damaged whichever is earlier..

QA 1.3.6 ISO QMS certification :

ISO QMS 9001: 2008 *– valid certification of all the agencies/ subcontractors engaged (by LSTK contractors) for NDT and fabrication/welding activities shall be required."*

QA 1.3.7 Inspection of material::

It shall be the responsibility of the contractor to inspect all materials upon receipt and to ensure that the correct grade of material has been supplied and that identification, dimensions, material quality and end preparation are in accordance with the requisite standards and specifications. Different materials shall be kept in discrete sections of the storage area and all material shall be marked in a manner that allows it to be related to the original manufacturer's certification.

Procedures for the transfer of material identification marks shall be agreed with the Company prior to the commencement of fabrication. Any material which is not readily identifiable shall be removed from the worksite and quarantined until its material grade and source can be confirmed.

QA 1.3.8 Welding machines :

Welding shall not be performed when the ambient temperature is lower than zero degree C, when surfaces are wet or exposed to rain, snow or high wind velocities, when welders are exposed to inclement conditions, or when conditions prevent required inspections.

Only welding equipment that is in good working condition and that is properly grounded shall be used. All welding machines shall be calibrated prior to commencement of fabrication and calibration certificates shall be available for ONGC QA audits. Certification shall be valid for 6 month periods, or as per manufacturer's recommendation.

All the welding machines/transformers shall be based of ***Inverter technology*** *or better one instead of conventional transformer/rectifiers based one.*

All welding machines shall be calibrated *prior to commencement of fabrication from NABL accelerated laboratories (India) or from labs which are authorised by local government bodies of from the original manufacturer of the machines.*

All the welding cables, Grinder cables shall be free from any intermittent joints and shall have proper industrial explosion proof electrical plug-top. All used pipes for Gas- cutter shall be in good conditions with armoured & quality stamped.

QA 1.3.9 Grounding of Equipment & air strikes:

- Each welding machine and panels shall be individually grounded to the platform or portion of the platform being welded.
- No machine or panels shall be grounded to floating equipment during welding on the structure.

Arc strikes should be made in weld groove. Arc strikes on the surface of base metal shall be removed by grinding, including any hardened zone beneath the strike. Any such repair shall be visually and magnetic particle inspected.

QA 1.3.10 Welding consumables :

Company shall approve all consumables including brands. Welding Consumables shall have physical, chemical, fatigue and corrosion resistance properties comparable to base metal. Manufacturer's batch test certificate shall be submitted for each batch of consumables used. Test results shall include chemical, physical (including impact), corrosion resistance (if welding material for sour service application or with NACE requirements) & fracture toughness (if welding material or WPS with CTOD Requirements). If these batch test results are not readily available (like corrosion test results, fracture toughness etc.), contractor shall carry out additional batch testing to confirm the same.

The welding consumables which shall not be in no case of a date manufacture older than one year with effect from date of starting the fabrication.

Apart from above, prior to Production welding, the contractor has to do HIC test separately for welding consumables as guided by CA or Company."

The control of Welding Consumables shall be in accordance with a well documented procedure. This procedure shall include techniques for storage, handling, recycling and re-baking of welding consumables to ensure that the diffusible Hydrogen content of weld metal is maintained at less than 5ml per 100 g of weld metal. The procedure shall, as a minimum, be in accordance with the manufacturer's recommendations. Since a variety of welding consumables may be in use at any manufacturing location, strict control shall be exercised to prevent loss of identification or the use of incorrect weld metal.

QA 1.3.11 ***Miscellaneous requirement*** **:**

The welder and weld area shall be provided with protection during periods of inclement weather and/or excessive wind conditions. The procedures shall include means of protecting electrodes, wires, fluxes, etc. Good House Keeping is essential in the job-area

Gas shielded welding processes should generally be used only for shop fabrication or in enclosed conditions. When used outdoor, suitable windshields must be provided to allow adequate gas coverage of molten puddle.

Heated quivers or rod boxes shall be placed at each welding location when low hydrogen welding electrodes are used.

The implementation of the procedures will be audited by ONGC QA team during the phase of Design, Fabrication & Installation activity at least two times par phase. Apart from audit the surprise checking also may be carried out by ONGC QA team at any time. ONGC shall be informed about the timely corrective actions taken by the fabricator as a result of any CAR/NCRs issued during such audit.

Copies of various QA procedures shall be readily available and shall be referenced in the quality plan and shall be made available during QA audit by Company.

Welders and welding operators shall be qualified as per the applicable specification. When using welding processes which have high potential for non-fusing type defects, in addition to the NDT requirements specified in the respective Codes, bend test shall also be carried out. GMAW and FCAW are regarded as welding processes with high potential for non-fusing type defects.

Welder Qualification Tests shall be witnessed by Company representative and/or by a reputed TPI agency. The Company representative shall be advised in advance that the Contractor is conducting welder qualification to enable auditing of test facilities.

Qualification test welds shall be made on test coupons prepared in accordance with the relevant standard. Certified as per QA1.3.4, the "Amp/volt tong tester" shall be available at the Contractor's establishment at all times.

The Contractor shall be responsible for all costs, including labor and laboratory testing, associated with welder qualification tests and retests.

All welders working on the project shall wear identification cards made at contractor's cost which shall contains photograph, welders name, welder no. Procedure qualified and Company's/Engineer's representative's signature. If the fabricator has a different system for welder's identity card, same can be followed after agreement.

The period of validity of welder qualification shall be in accordance with the Standard used for qualification. A qualification can be cancelled if the welder / welding operator show inadequate skill, knowledge and performance.

No wooden planks shall be used in any stage of fabrication job as platform of scaffolding.

During fabrication at yard there must have two independent escape route for workers..

QA 1.3.12 Production Records :

At all stages of fabrication, the Contractor shall maintain all relevant production records using a recording system agreed by the Company. The records shall include:

1. Material & Welding consumable certificates.
2. PWHT Records.
3. Fit-up reports.
4. Weld visual reports.
5. Painting/top coat report with surface finish report.
6. Dimensional inspection together with NDE & pressure test records.
7. Welder & welding operator qualification.
8. Approved Isometric drawings marked with unique weld numbers / NDT Drawings.
9. Authenticated copies of NDT operator's certificates.
10. Records of any agreed deviations to the fabrication standards.

Welding Procedures shall be tested to demonstrate that acceptable welds can be made by the procedure. During WPS Qualification, the quality of welds shall be determined by both non-destructive & destructive testing as per the applicable code. Additional testing requirements specified in this specification shall also be carried out. The Welding Procedure Qualification Testing (WPQT) shall be witnessed by Company representative and also by a reputed TPI agency. Only qualified and approved WPSs shall be used for production welding.

QA 1.3.13 CTOD Testing:

1. Material subjected for CTOD Testing shall be tested and shall meet or exceed CTOD value of 0.35mm.
2. Welding consumable subjected for CTOD Testing shall be tested and shall meet or exceed CTOD value of 0.25mm.
3. WPS subjected for CTOD testing shall be tested and shall meet or exceed CTOD value of 0.20 mm.
4. CTOD testing shall be done from Govt. labs or Govt. accelerated labs with prior information to ONGC.

PP2.0. PIPING & PRESSURE VESSELS

PP2.1. Codes & Standards

The following Codes shall be the minimum acceptable standards for welding and inspection during piping & pressure vessel fabrication. The latest edition, addenda, code cases and supplement available at the time of bidding will be used. Any part of any other non-listed code referred to in these listed codes as augmentation is to be considered applicable.

ASME B31.3	Process Piping
ASME B31.4	Pipeline Transportation System for Liquid Hydrocarbons and other Liquids
ASME B31.8	Gas Transmission and Distribution Piping System
ASME BPVC Sec.VIII, Div.1	Rules for Construction of Pressure Vessels
AWS A2.4	Standard Symbols for Welding, Brazing and Nondestructive Examination.
ASME BPVC Sec.IX	Welding & Brazing Qualification
NACE MR0175	Standard Material Requirements - Metals for Sulfide Stress Cracking and Stress Corrosion Cracking Resistance in Sour Oilfield Environments
NACE TM0284	Standard Test Method - Evaluation of Pipeline and Pressure Vessel Steels for Resistance to Hydrogen-Induced Cracking
NACE TM0177	Standard Test Method - Laboratory Testing of Metals for Resistance to Sulfide Stress Cracking and Stress Corrosion Cracking in H2S Environments
ASME BPVC Sec. V	Nondestructive Examination
ASME Sec. II, Part C	Specification for Welding Rods, Electrodes and Filler Metals
ASTM E164	Standard Practice for Contact Ultrasonic Testing of Weldments
ASTM E165	Practice for Liquid Penetrant Inspection Methods
ASTM E709	Practice for Magnetic Particle Examination
ASTM E94	Standard Guide for Radiographic Examination
ASTM E384	Standard Test Method for Knoop and Vickers Hardness of Materials
ISO 14731	Welding Coordination – Tasks and Responsibilities
ASNT SNT-TC-1A	Personnel Qualification and Certification in Nondestructive Testing

PP2.2. Weld Procedure Qualification & Documentation

The Contractor shall not commence production welding until appropriately qualified welding procedures have been accepted by the Company.

A. Piping

Weld procedures including brazing procedures shall be qualified as per ASME Code Section IX & ASME B31.3

The applicable WPQT (Welding Procedure Qualification Testing) was done in last 2 years for ONGC Job dated with effect from date of starting fabrication. If it is not of ONGC project, it shall be fresh WPQT for project specific."

Such previously qualified welding procedures may be considered for use for CS & CS NACE Piping, where they comply with the present document and are appropriate to the scope of work. These WPSs should have been qualified by the same fabricator in the presence of reputed TPI agencies. New, project specific WPS shall be prepared using the old WPQT as explained above of earlier qualified WPSs as supporting documents. In this exercise, good engineering practice and guidelines given in ASME Sec. IX shall be followed and using PQRs in an arbitrary manner to support a new WPS will not be acceptable.

However, for critical applications involving the welding of CRA materials (SS, DSS, Cu-Ni, Cu etc.), previously qualified procedures are not acceptable and fresh project specific WPQT & WPS is required.

Mechanical testing of WPQT coupons for CS piping shall confirm to the requirements of ASME Sec.IX / ASME B31.3. In addition to the above requirement, hardness testing shall be performed on each test coupon. Maximum hardness value shall be 325 HV10 for normal service piping. For Sour Service piping, a maximum permitted hardness of 248 HV10 (22 HRC) shall apply. CVN (Charpy V Notch) Impact Testing of Carbon Steel Pipe work shall be performed in accordance with ASME B31.3 Table 323.2.2.

For each welding process, the welding procedures shall specify all equipment settings. The Contractor's Welding Supervisor shall check daily and record machine settings for each weld procedure used during welding activities. This record shall be available for audit by the Company representative.

B. Pressure Vessels

Welding Procedures shall be qualified as per ASME Sec. IX & ASME BPVC Sec. VIII Div.I

PP2.3. Welder Qualification

Welders shall be qualified as per ASME Boiler and Pressure Vessel Code, Section IX.

Qualification on production welds is not permitted.

A welder shall not be permitted to weld on pipe work or attachments to pipe work unless they are qualified to the procedure in use and the documentary evidence has been sighted by the Company representative.

Tack welds shall be made by a qualified welder using the same type of electrode as is used for the root pass.

The Company may request a retest of any welder at any time and from time to time during the work. If a welder fails to qualify, then at the discretion of the company representative, all non- installed welds completed by that welder shall be examined by additional radiographic tests, over and above that which would normally be required or specified in the drawings for that pipe work and the same shall be charged to the Contractor's account.

Welds not identified and recorded, or welded by unqualified welders, shall automatically be rejected. It will be the Contractor's responsibility to prove that the welds conform to the applicable Specification. This may require 100% radiography.

PP2.4. Production Welding

1. The quality of piping welding shall be such that a weld efficiency factor as defined in ASME B 31.3 of 1.0 is achieved.

2. The deposited filler metal shall match the chemistry, corrosion resistance (if required) and minimum physical properties of the parent metal when similar metals are welded.

3. Circumferential welds on pipes shall be staggered at least four times the pipe wall thickness or 25 mm, whichever is greater, apart.

4. Branch connections shall be joined to their headers with full penetration welds.

5. Backing rings, back-up rings or chill rings shall not be used.

6. Back or seal welding of screwed fittings is prohibited unless specified in the drawing or is part of the approved welding procedure.

7. Oxy-acetylene torches shall not be used for pre-heating.

8. Pipes of dissimilar materials shall not be welded to each other but shall be flanged unless otherwise approved.

9. Compressor and turbine piping, including lube and seal oil piping, and other piping requiring special cleaning as shown in the drawings provided by the supplier shall have the root pass deposited by the gas tungsten arc process. The backside of the root pass shall be purged with inert gas.

10. If the bore of the pipe is different from the bore of fittings or flanges, to which it is welded, by more than 3mm (total bore diameter), the thicker member shall be bored, taper bored or ground smooth to match the specified bore. Figure 328.4.3 of ASME B 31.3 will governs the geometry of all taper boring and beveling.

11. Coupling or other weld-on type branch connection shall be located at least 80 mm away from any weld joint.

12. When socket weld fittings or valves are used, pipe shall be spaced approximately 1/16 inch to avoid "bottoming" which could result in excessive welds stress.

13. The inside of the brazed fitting and outside of the tube shall be cleaned with sand Paper. Flux paste to be evenly applied to the joint.

14. Reinforcing pads shall be added only after external and internal visual inspection of the attachment. Reinforcing pads shall be provided with ¼" tapped weep hole. Weep holes should be plugged after welding of pads is over.

15. Exposed machined and threaded surfaces shall be protected from oxidation during heat treatment.

16. Flange bolt holes shall straddle the established horizontal and vertical centerlines of the pipe except where connection to equipment dictates otherwise.

17. Cleaning of the piping after fabrication and heat treatment shall be performed externally and internally to remove all loose scale, weld spatter, sand and other foreign materials.

18. Welding shall not be done when quality of completed weld would be impaired by prevailing weather conditions, air borne moisture, blowing sand or high winds. Windshields may be used when practical.

19. Temporary attachments to the outside of surface of the pipe shall not be made without the approval of Company. Any such attachments which are permitted shall be removed by grinding or by using Oxy – Acetylene torch to 3 mm above permanent material surface, removal of the remaining 3mm shall be by grinding, followed by MPI / DPI. Company may require an ultrasonic check to establish that the remaining wall thickness is not less than the design minimum. Arc air gouging, hammering or other mechanical means shall not be used for the removal of temporary attachments.

20. Welder shall be supplied with temple stick – thermal crayons or pended gauges so that welders and inspector can check and control the temperature of element.

21. No weld shall be coated, painted, hydro tested before it has been inspected and accepted.

22. No weld is to be cooled by quenching or by any means other than natural cooling.

23. Welding for Severe Cyclic Condition: The weld reinforcement shall be smooth and shall blend smoothly with the external surface of the pipe to minimize possible stress concentration effects.

24. Field Welding

- Where field welds are designated, the prefabricated pipe shall be cut off 150 mm longer than the dimensions shown on the pipe fabrication drawings to allow for modification then precise fit-up in place.
- Additional field fit welds may be included in a spool by the Contractor for those spots which may have restricted site access or which may be cumbersome for transportation.
- The location of field welds, where not indicated on the drawings, will be the responsibility of the Contractor.

PP2.5. NDT Techniques

NDT Techniques like RT, UT, MPI & LPT shall be used for the examination of welds as per the requirements given in this specification.

Radiography:

1. Radiography will be performed for every pipe thickness and material grouping and for each welding process and procedure, progressively throughout entire job. At least one of each type and position of weld made by each welder will be examined. A record shall be kept by the Contractor of the quality and extent of each welder's work.

2. Radiography is not required for the welds on slip-on flanges or socket welds or seal welds.

3. Fluorescent intensifying screens such as calcium tungsten shall not be used.

4. For each weld found to be defective two additional welds made by the same welder who produced the defective weld will be subjected to radiographic examination. These additional examinations will be made immediately after the defective welds is found and are in addition to the minimum examination requirements for the line class as specified in the drawings or specifications.

5. Weld repairs shall be re-examined by the same method used to detect the original defect.

PP2.6. NDT Procedure

Contractor shall prepare NDT Procedures covering all aspects of the work. NDT Procedures shall be submitted to the Company representative for review and approval. Written NDT Procedures shall be qualified as per the requirements of ASME Sec. V, in the presence of Company representative. Qualification of the NDT procedures shall be required before NDT can be performed on production welds.

A. Piping

NDT Procedures shall be prepared as per the requirements of ASME B31.3, ASME Section V and the applicable ASME Code Cases.

B. Pressure Vessels

NDT Procedures shall be prepared as per the requirements of ASME BPVC Sec. VIII Div.1 and ASME Sec. V

PP2.7. Extent of NDT Coverage

A. Piping

I. Random radiography (10%) shall be performed covering on one weld in each 10 welds for each welder on the following:

1. Water (in case of carbon steel)
2. Air
3. Chemical
4. Diesel
5. Closed & Open drain
6. Relief
7. Hydraulic Oil
8. Oily water
9. Glycol & Thermoil
10. Vent gas
11. Lube oil & Seal oil

II. 100% Radiography shall be performed on size 2" and above and 10% on below 2" for the following:

1. Crude oil
2. Jet fuel
3. Fuel gas / instrument gas
4. Process gas
5. Production flow lines (well fluid)
6. Injection water
7. Hydrocarbon (process)
8. Water (in case of 90/10 Cu-Ni)
9. H.P. and L.P. Flare lines.
10. Piping system not covered under any of the above services

III. For fillet welds and brazed joints where carrying out radiography is not possible, magnetic particle test or dye penetrant test shall be carried out. The extent of inspection shall remain same as for radiography.

All the lines which are stress relieved or have design pressure more than 50 Kg/ cm2 shall be radiographed for 100% of weld joints even if not required as per I to III

Field welded joints, which cannot be leak tested due to unavoidable circumstances & reasons shall be 100 % examined by RT and MPI/LPT.

In addition to the requirements mentioned above, for welded joints of NACE material, 100 % hardness test shall also be carried out.

Welds, which cannot be radiographed because of their location, inaccessibility, excessive & impractical radiographic exposure time required or due to safety reasons, can be examined by ultrasonic and / or liquid penetrant / magnetic particle method as applicable with permission from Company. The extent of inspection shall be the same as for radiography

B. Pressure Vessels

NDT Coverage shall be as per the requirements given in ASME BPVC Sec.VIII Div.I and the design documents.

PP2.8. NDT Operator

NDT Operator Qualification & experience records shall be submitted for Company's review. NDT Operators shall be minimum Level II qualified, PCN qualification is preferred. Skill levels of NDT Operators shall be further examined by Company through practical mock tests & written examination. Contractor shall make necessary arrangements for the same with no additional cost to Company.

PP2.9. Advanced UT Techniques

PAUT (Phased Array Ultrasonic Testing) can be used for the examination of CS & CS NACE piping welded joints with thickness beyond 10 mm for which separate permission from Company is not required. However "*Project specific procedure qualification" to be witnessed by ONGC/CA/TPI for the 1st time during start of NDT techniques..*

The permission to use PAUT will be subject to satisfying the following conditions:

1. Past track record of the agency performing PAUT shall be submitted for Company's review and approval.
2. Written PAUT Procedure shall be prepared as per the applicable code and shall be submitted for Company's review and approval.
3. PAUT Procedure shall be demonstrated and qualified as per the Code requirements and to the satisfaction of the Company in the presence of Company Representative. During the demonstration, it shall be established that the scanning technique employed covers the entire weld volume and HAZ.
4. NDT Operator Qualification records (minimum level II) including experience in PAUT shall be submitted for Company's review and approval.
5. Skill levels of NDT Operators will be further examined through practical mock tests and written tests. Contractor shall make the necessary arrangements for the same with no additional cost to Company.
6. Familiarisation and interpretation Training on PAUT shall be provided to Company's inspectors as and when required with no cost to ONGC.

7. For any ambiguous indication observed in PAUT, especially in the case of volumetric discontinuities, if ONGC desires, RT of that joint shall be carried out without any additional cost to ONGC.
8. Soft copy of scan data (A-Scan, C-Scan, Sector Scan etc.) of each joint along with the software for viewing the same shall be submitted to ONGC.

PP2.10. NDT Acceptance Criteria

A. Piping

Technique	Material	Acceptance Criteria
RT	CS , SS	ASME B31.3
	CS NACE, Cu-Ni	ASME BPVC Sec.VIII Div. I, UW 51
UT	CS, CS NACE	ASME B31.3
PAUT	CS, CS NACE	ASME B31.3 & Code Case 181
MPI	CS, CS NACE	ASME B31.3
LPT	CS, CS NACE, SS, Cu-Ni	ASME B31.3

a. Internal weld protrusion in pigged lines shall not exceed 1.6 mm.
b. Orifice flange butt weld shall be internally ground smooth and flush.

B. Pressure Vessels:

ASME BPVC Sec. VIII Div.1

PP2.11. Heat Treatment

1. Preheat and post-weld heat treatment (PWHT) shall be in accordance with ASME B 31.3. ASME B31.4,ASME B31.8 or ASME BPVC Sec.VIII, Div.I, as applicable.

2. The method and equipment used in heat treatment shall be acceptable to the Company.

3. All threaded connections shall be protected from oxidation during heat treatment and be chased or gauge checked after heat treatment.

4. Exposed machined and threaded surfaces shall be protected from oxidation during heat treatment.

5. Cleaning of pipe work after fabrication and heat treatment shall be performed, externally and internally to remove all loose scale, weld spatter, sand and other foreign materials.

6. For all welds requiring PWHT, the specified inspection and NDT shall be performed after completion of PWHT process but not before the welds have cooled to ambient temperature. NDT prior to PWHT, performed in addition to mandatory NDT following PWHT, shall be at the discretion of the Contractor. However, where defects requiring repair are located by this inspection, it shall be brought to the notice of Company.

7. Machined surfaces shall be protected by a suitable paint or compound to prevent damage from scaling during PWHT.

8. After final heat treatment, the Contractor shall identify the piping as having received PWHT. The method of identification shall be recorded on the as-built isometric and P&ID together with the other recorded information.

9. A legible heat treatment chart shall be provided to the Company. This chart must show the rate of increase of temperature, the holding temperature and time and the rate of decrease of temperature.

PP2.12. Repair Welding

If the company representative considers a weld to be grossly defective, it shall be cut out and the joint re-welded and all costs associated therewith shall be the Contractor's responsibility.

Mechanical defects such as scratches and gouges may be ground smooth provided the depth of the defect does not exceed 10% of the nominal wall thickness of the pipe.

Repair welding shall only be carried out in accordance with an approved repair procedure. Minimum Pre-heat for repair welding shall be 50^0C more than that used for production welding.

After gouging to remove the defects, the area shall be ground smooth and shall be examined with MPI to ensure that the entire defect has been removed. LPT shall be used for non- ferromagnetic materials. As a minimum, completed repairs shall be examined by the same NDT methods used for the original weld.

Only two times repairs are allowed on any welded joint.

Dents or grooves who's depth is greater than 10% of the nominal pipe wall thickness shall be removed by cutting out and replacing that length of pipe in which the defect occurs. The minimum length of a cutout shall be four times the nominal diameter or 150 mm, whichever is the lesser.

Internal weld metal projecting into the pipe on weld neck orifice flanges shall be removed and ground smooth with the pipe.

ST3.0. STRUCTURAL

ST3.1. Codes & Standards

The following Codes shall be the minimum acceptable standards for welding and inspection during Structural Fabrication. The latest edition, addenda, and supplement available at the time of bidding will be used. Any part of any other non-listed code referred to in these listed codes as augmentation is to be considered applicable.

AWS D1.1	Structural Welding Code - Steel
AWS D1.3	Structural Welding Code – Sheet Steel
AWS A2.4	Standard Symbols for Welding, Brazing and Nondestructive Examination.
AISC	Manual of Steel Construction – Allowable Stress Design. Specification for Structural Steel Buildings
API RP 2A	Recommended Practice for Planning, Designing and Constructing fixed Offshore Platforms – Working Stress Design
API Spec. 2B	Specification for Fabricated Structural Steel Pipe
API RP 2X	Recommended Practice for Ultrasonic & Magnetic Particle Examination of Offshore Structural Fabrication and Guidelines for Qualification of Ultrasonic Technicians
ASME BPVC Sec. V	Nondestructive Examination
ASME Sec. II, Part C	Specification for Welding Rods, Electrodes and Filler Metals
ASTM E164	Standard Practice for Contact Ultrasonic Testing of Weldments
ASTM E165	Practice for Liquid Penetrant Inspection Methods
ASTM E709	Practice for Magnetic Particle Examination
ASTM E94	Standard Guide for Radiographic Examination
ASTM E384	Standard Test Method for Knoop and Vickers Hardness of Materials
BS7448	Fracture Mechanics Toughness Tests
EEMUA158	Construction Specification for Fixed Offshore Structures in the North Se
ASNT SNT-TC-1A	Personnel Qualification and Certification in Nondestructive Testing
ISO 14731	Welding Coordination – Tasks and Responsibilities

ST3.2. Welding Processes

Only Gas Metal Arc (GMAW), Gas Tungsten Arc (GTAW), Submerged Arc (SAW), Shielded Metal Arc (SMAW), or Flux Cored Arc (FCAW) welding process shall be used for production welding. Other welding processes shall not be used. Automatic stud welding machines can be used for welding of studs to steel (for fixing insulation etc.).

Use of GMAW process shall be limited to the following:

a) GMAW welding shall not be exposed to air drafts greater than 8 Km/h.

b) GMAW-S can be used for groove welds in materials 10 mm or less using a procedure qualified by the contractor.

c) GMAW-S can be used for fillet welds of 20 mm or less

d) GMAW-S may be used for the root pass only in combination with SAW welding, and for the root pass in non-critical joint applications.

e) Spray transfer mode of metal deposition for GMAW is limited to the flat position. f)

Weld areas shall be blasted or ground to white metal prior to welding when GMAW or GMAW-S is used.

Use of FCAW process shall be limited to the following:

a) FCAW welding with external gas shielding shall not be exposed to air drafts greater than 8 Km/h.

b) Electrode diameters greater than 3 mm are not acceptable.

c) Self – shielded flux core shall not be used as root or fill pass beneath other processes.

d) FCAW welding shall not be used at any case for piping & structural welding during installation or otherwise in offshore environment."

ST3.3. Weld Procedure Qualification & Documentation

The Contractor shall not commence production welding until appropriately qualified welding procedures have been accepted by the Company. All weld procedures to be utilized on this project shall be qualified.

Welding procedures shall be qualified in accordance with AWS D1.1 Structural Welding Code – Steel, except as modified by this specification: The essential variable list of all welding processes shall be increased to include any change in the manufacturing process for the base metal, i..e. separate procedures are required for As Rolled, Controlled Rolled, Normalized, Quench & Tempered or TMCP steel.

Welding of thin sheet shall be as per AWS D1.3 – Structural Welding Code – Sheet Steel.

Unless otherwise specified, all welds shall be full penetration multi-pass welds.

Draft / Proposed WPS shall be agreed with the Company prior to the commencement of procedure qualification.

The applicable WPQT (Welding Procedure Qualification Testing) was done in last 2 years for ONGC Job dated with effect from date of starting fabrication. If it is not of ONGC project, it shall be fresh WPQT for project specific."

Such previously qualified welding procedures may be considered for Structural Welding where they comply with the present document and are appropriate to the scope of work. These WPSs should have been qualified by the same Contractor in the presence of reputed TPI agencies. New, project specific WPS shall be prepared using the old WPQTs as explained above of earlier qualified WPSs as supporting documents. In this exercise, good engineering practice & AWS D1.1 guidelines shall be followed and using PQRs in an arbitrary manner to support a new WPS will not be acceptable.

In addition to the testing requirements specified in AWS D1.1 following tests as applicable shall also be carried out during WPS qualification.

A. Hardness Testing

Welding procedure qualification shall include a hardness traverse of the HAZ, Weld Metal and base metal.

1. Hardness testing shall be performed along the transverse shown in Fig. 1.

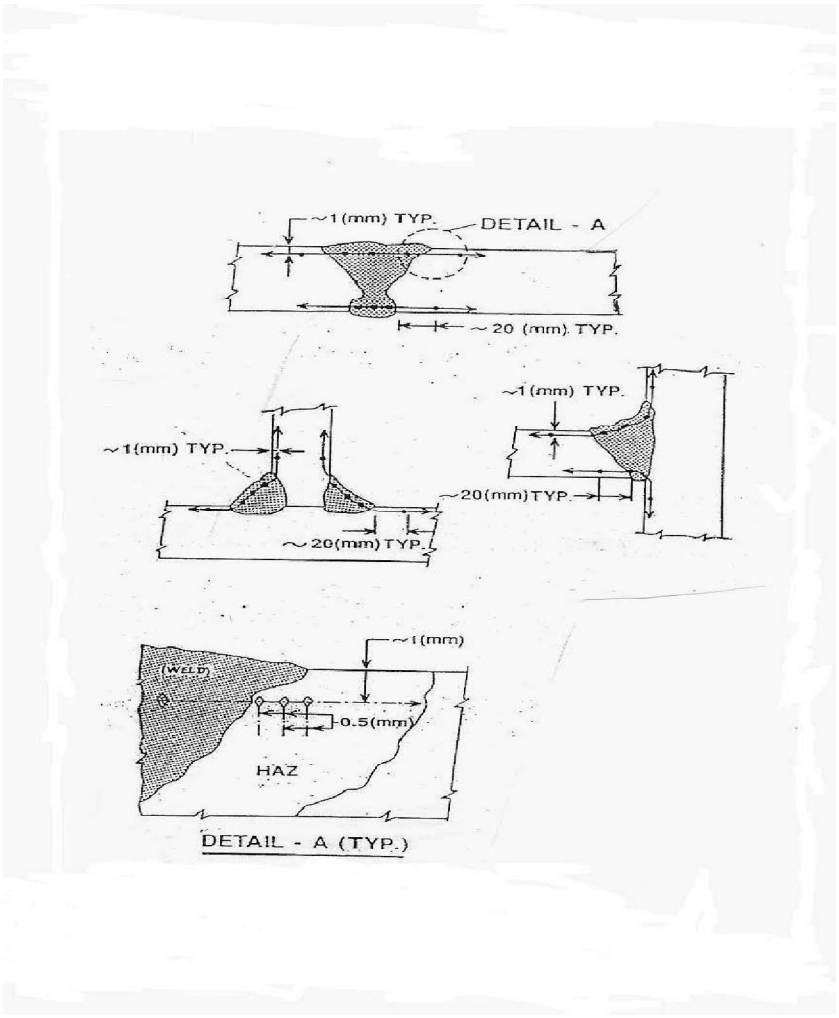
2. The required hardness survey shall be tested on a suitable macro section machined from the same welded test assembly made to determine other weld joint properties.
3. The hardness surveys shall be prepared and tested in accordance with ASTM E 384, Standard Test Method for Knoop and Vickers Hardness (Hv) of Materials, using an applied load of 10 kgf.

4. Indentations shall be made along all of the transverses shown for each type of weld tested, approximately 1 mm below the surface of the base metal. In the weld metal, a minimum of three equally spaced indentations along the traverse shall be made. In the heat-affected zone, the indentations shall start as close to the fusion line as practicable. A minimum of three readings shall be taken at

each weld toe. One additional reading shall be taken at least 20 mm from the fusion line at each traverse to represent unaffected base metal.

5. The maximum hardness value obtained from any indentation shall not exceed 350 Hv. If any single value exceeds 350 Hv and a retest adjacent to the failed test also produces one or more values exceeding 350 Hv, the procedure qualification test has failed these requirements and a new test weld shall be made with some planned modifications of essential variables or techniques (change heat input, increase preheat, later bead sequence, etc.) and retest.

6. A photomacrograph of the hardness test section shall be included in the PQR clearly showing the hardness impressions.



B. Impact Testing

All groove welding procedure qualifications for Class A ad Class B steels shall include Charpy V-notch impact testing of the deposited weld metal and the heat affected zone (HAZ). Locations for the weld and HAZ specimens shall be generally as shown in AWS D1.1, except that three sets of HAZ specimens shall be located at: (1) the fusion line (FL) (2); a position 2.0 mm into the HAZ from the fusion line (FL+2mm); and (3) a position 5.0 mm into the HAZ from the fusion line (FL+5mm) . An additional set of specimens shall also be taken from the location centred within the root pass of the weld. For acceptance the average impact energy shall meet the minimum requirement of the relevant Specification with no single value more than 6.5J lower than the average requirement. When different classes of material are to be joined the required energy level is equal to that of the lower class joined. Testing temperature shall be -10ºC.

C. CTOD Testing

CTOD testing may be used to justify an increase in the maximum thickness of material that can be welded without the application of PWHT as defined elsewhere in this Spec.

CTOD testing shall be carried out as per BS 7448/ISO15653. Each set of CTOD tests shall comprise a minimum of three valid specimens.

CTOD testing is required on full penetration test welds representing the maximum thickness of materials that are to be welded by any single process or combination of processes.

Separate tests are required to qualify single and double sided preparation.

The steel selected for performance of the test programme shall have minimum Carbon Equivalent not less than 0.02 of the maximum CE value to be used on the project. All test plates shall be welded using heat inputs representative of the highest to be used during production welding.

For each test the least values of δc, δu or δ

0.20mm when tested at 0^0C unless specified otherwise in the relevant drawings.

Heat Input

Charpy qualification of weld deposit and HAZ is valid only for welding procedures with heat inputs maintained with in ±25% of the tested weld. This may require more stringent tolerance on travel, speed, voltage and amperage.

In similar fashion the inter pass temperature is limited to that actually achieved and sustained during the procedure test.

ST3.4. Welder & Welding Operator Qualification

Welders shall be qualified as per AWS D1.1 Structural Welding Code – Steel. All welder qualification must have been performed on the entire weld thickness. 'Tack Welders' or 'Fillet Welders' partial qualification is not allowed. Qualification on production welds shall not be allowed.

ST3.5. Production Welding

1. **Chill rings and backing strips** can provide improved root beads on single sides welds. Their use, however is prohibited as they may cause fatigue crack initiation.

2. Unless otherwise specified, all welds shall be **full penetration multi-pass welds.** Full Penetration Welds shall be used for built-up-girders.

3. **All faying surfaces shall be seal welded.**

4. Welding shall proceed in a manner that **minimizes residual stresses or distortions.** The Contractor shall be responsible for preventing or correcting distortions. If excessive distortion becomes evident, corrective methods shall be applied as agreed with the Company Representative.

5. Welds shall **not be quenched or cooled other than by contact with the natural atmosphere. Welds shall be protected from rapid cooling caused by rain, wind, or other inclement weather.**

6. **Gas shielded welding processes (eg. FCAW, GTAW and GMAW), when used outdoor shall be provided with proper wind protection.**

7. **Pre-heat requirements**

   Welding pre-heat shall be as specified in the Qualified Welding Procedure, but shall not be less than as specified in AWS D1.1 (Table 3.2). The guidelines given in Appendix XI of AWS D1.1 shall also be taken into account and adopted as appropriate.

   If any metal is damp or less than $10^{o}C$, a minimum preheat of $38^{o}C$ shall be required regardless of thickness.

Preheating shall be carried out by using Oxy-fuel bulbous type torches, induction or electric resistance heating. Oxy-Acetylene cutting torches shall under no circumstances be used for preheat application.

8. **Inter – pass temperature**

The temperature of previously deposited weld metal and surrounding base metal within 75 mm of each side of the weld and throughout the wall thickness shall meet the following requirements prior to depositing subsequent beads

Minimum – Not less than the specified pre heat.
Maximum - 200ºC or the maximum sustained and documented during procedure qualification, not to exceed 260ºC.

9. **Arc Strikes and Temporary Attachments**

Arc strikes shall be ground out and repaired. After repair, MPI shall be carried out. Temporary attachments shall be removed by grinding or by using Oxy – Acetylene torch to 3 mm above permanent material surface, removal of the remaining 3mm shall be by grinding, followed by MPI / DPI. Company may require an ultrasonic check to establish that the remaining wall thickness is not less than the design minimum. Arc air gouging, hammering or other mechanical means shall not be used for the removal of temporary attachments.

10. **Post Weld Heat Treatment**

PWHT shall be required in the following circumstances or when shown on the design drawings:

a) When fabricating tubulars with d/t ratios less than 20.
b) For nodal joints where the minimum throat thickness exceeds 40mm. c) For other areas where the minimum throat thickness exceeds 50mm.

PWHT shall be carried out for all above three cases individually for Onshore-Yard fabrication in compliance to ST 3.5-point no.11. PWHT shall be carried out for all type of non-TMCP steel. For any special steel as specified by steel manufacturer and approved by ONGC for obviating PWHT with CTOD may be considered.

However PWHT is waved off for welding during installation phase at offshore and CTOD shall be carried out with qualified PQR (fresh) per every new project & subsequent WPS (fresh) is required for structural welding. In this case (offshore) the Fracture Mechanics Assessment (FMA) , its(FMA) separate report, Crack Tip Opening Displacement (CTOD)method and its (CTOD)test report , as per BS 7448/ISO15653 shall be used to obviate the requirement for PWHT (as for all applicable cases separately for b & c only).

Fracture Mechanics Assessment Report and CTOD, as per BS 7448/ISO15653 may be used to obviate the requirement for PWHT (as per b c), if approval in writing is given by the Company. In this case CTOD test shall be carried out during the WPS qualification.

This entire clause of ST .3.5.10 shall also be applicable for welding jobs with piping.

11. **Criteria for PWHT**

All PWHT shall be performed in accordance with AWS D1.1 and the applicable welding procedure qualification.

a) The CONTRACTOR shall inform the COMPANY prior to any PWHT operation.

b) No welding shall be performed on fabrications subsequent to PWHT.

c) PWHT requirements shall be stipulated on the relevant WPS and shall comply with AWS D1.1, Section 5.8. The temperature for PWHT shall be $600^{o}C \pm 20^{o}C$ unless recommended otherwise by the steel manufacturer.

d) For all welds requiring PWHT, the specified inspection and NDT shall be performed after completion of PWHT process but not before the welds have cooled to ambient temperature. NDT prior to PWHT, performed in addition to mandatory NDT following PWHT, shall be at the discretion of the Contractor. However, where defects requiring repair are located by this inspection, it shall be brought to the notice of Company.

e) The contractor shall submit a heat treatment procedure for approval by the Company. This procedure shall define heat and cooling rate, holding time at temperature and details of temperature measurement and control methods.

f) PWHT temperature/time charts shall be clearly identified and retained for inclusion in the project records.

12. **Weld Through Primers**

Weld through primers to be used during fabrication are subject to the approval by the Company and shall be included in the weld procedure qualification. During Weld Procedure Qualification, the thickness of weld through primer shall be measured. Any increase in thickness of primer in production shall necessitate re-qualification of the affected WPS.

ST3.6. NDT Techniques

NDT Techniques which are to be used for inspection are Visual Inspection, RT, UT and LPT / MPI.

ST3.7. NDT Procedures

Contractor shall prepare detailed written NDT Procedures and shall submit the same for Company's review and approval.

Structural	Technique	Specification
Structural General	RT, MPI & LPT	ASME Sec.V & AWS D1.1
Butt & T, K, Y Joints	UT	API RP 2X
UT of Tubular – (Factory Manufactured) Long Seam	UT	ASTM E-273 or ASTM E-164 and AWS D1.1
Butt Joints	Advanced UT System	ASME Sec. V, AWS D1.1 and applicable Code Cases

NDT Procedures including equipments shall be qualified as per the requirements of AWD D1.1, API RP 2X and ASME Sec. V as applicable in the presence of Company representative.

ST3.8. Extent of NDT Coverage

NDT coverage shall be as per API RP 2A with the following modifications.

A. Structural Tubulars

Radiographic Examination shall be performed for the following welds

1. 10 % RT of Longitudinal Weld Seams (L)
2. 100 % RT of all girth welds in tubular - Circumferential Weld Seams (C)
3. 100 % RT of Intersection of long seams (L) and Circumferential Seams (C)

Inspection requirement for mill fabricated structural tubular, who are authorized to put API monogram, shall be as per the applicable code (like API 2B).

B. Tubular Joints:

1. 100 % UT & MPI of Major brace to chord welding.
2. 100% UT & MPI of Major brace to brace welding.
3. Before fit-up, UT Lamination check shall be carried out on the area of the chord where bracing will be welded to ensure that the area is free from laminations.

C. Miscellaneous Bracing

1. 10% UT & MPI of Conductor Guide Bracing
2. 10% UT & MPI of secondary bracing and subassemblies, i.e., splash zone and / or mudline secondary bracing, boat landing etc.
3. 100% UT & MPI of attachment weld connecting secondary bracing / sub assemblies to main members

D. Deck Members

1. 100 % RT of all primary full penetration welds like splices in main skid beams, main truss beams and deck beams.
2. 10 % RT of all secondary full penetration welds including secondary deck beams.
3. 100 % Visual & MPI of all partial penetration welds including fillet welds.

Chevron / Transverse crack check using UT is required for primary steel with thickness > 25 mm welded using SAW process. Scanning shall be carried out along the weld cap.

100% UT and / MPI shall be carried out on all welds in built-up girders.

Notes:

1. Partial inspection should be conducted as 10 percent of each piece, not 100 percent of 10 percent of the number of pieces. Partial inspection should include a minimum of three segments randomly selected unless specific problems are known or suspected to exist. All suspect areas (e.g., areas of tack welds) shall be included in the areas to be inspected. If rejectable flaws are found from such 10% inspection, additional inspection should be performed until the extent of rejects has been determined and the cause corrected.
2. Welds, which cannot be radiographed because of their location, geometry, inaccessibility, excessive & impractical radiographic exposure time required or due to safety reasons, can be examined by ultrasonic and / or liquid penetrant / magnetic particle method as applicable with permission from Company. The extent of inspection shall be the same as for radiography

ST3.9. NDT Acceptance Criteria

Technique	Acceptance Criteria
RT	AWS D1.1
UT – Butt & TKY Joints	API RP 2X Level C
UT – Long Seam of Mill Fabricated Tubular	API RP 2X Level A

ST3.10. NDT Operator

NDT Operator Qualification & experience records shall be submitted for Company's review. NDT Operators shall be minimum Level II qualified, PCN qualification is preferred. NDT Operators doing examination of TKY joints should have previous verifiable experience in similar job. Skill levels of NDT Operators will be further examined through practical mock tests & written examination. Contractor shall make necessary arrangements for the same with no additional cost to Company.

ST3.11. Advanced Ultrasonic Systems

Phased Array UT (PAUT)/TOFD shall be used where the thickness is beyond 19 mm, for which permission is not required from Company. However *The PAUT/TOFD used shall satisfy the following conditions which has to be established by the NDT agency during the " Project specific procedure qualification" to be witnessed by ONGC/CA/TPI for the 1st time during start of NDT techniques.*

The **PAUT/TOFD** will be used subject to satisfying the following conditions:

1. Past track record of the agency performing PAUT / TOFD shall be submitted for Company's review and approval.

2. Written UT Procedure shall be prepared as per the applicable code and shall be submitted for Company's review and approval.
3. PAUT / TOFD Procedure shall be demonstrated and qualified as per the Code requirements and to the satisfaction of the Company in the presence of Company Representative. During the demonstration, it shall be established that the scanning technique employed covers the entire weld volume and HAZ.
4. NDT Operator Qualification records (minimum level II) including experience in PAUT / TOFD shall be submitted for Company's review and approval.
5. Skill levels of NDT Operators will be further examined through practical mock tests and written tests. Contractor shall make the necessary arrangements for the same with no additional cost to Company.
6. Familiarisation & interpretation Training on PAUT / TOFD shall be provided to Company's inspectors as and when required with no cost to ONGC.
7. For any ambiguous indication observed in PAUT / TOFD, especially in the case of volumetric discontinuities, if ONGC desires, RT of that joint shall be carried out without any additional cost to ONGC.
8. Soft copy of scan data (A-Scan, C-Scan, Sector Scan etc.) of each joint along with the software for viewing the same shall be submitted to ONGC.

ST3.12. NDT Waiting Period

NDT of primary structural welded members should generally be carried out 24 hrs after completion of welding.

ST3.13. Inspection of Low Fatigue Joints

All low fatigue joints shall be inspected for acceptance to required profile and toe grinding shall be performed, if required.

Disk test shall be performed as mentioned in AWS D1.1.

ST3.14. Repair Welding

Repair welding shall be performed strictly in accordance with the qualified repair welding procedure.

Removal of Defects

Removal of defects for repair may be by any of the following methods: machining, grinding, chipping, or air carbon arc gouging. The method shall produce a clean uncontaminated surface for installation of the repair weld.

Gouging

Oxygen gouging shall not be used in quenched and tempered, normalized or TMCP steels. All gouges shall be ground, power wire brushed, or grit blasted to remove all traces of residual carbon and oxidation.

After gouging to remove the defects, the area shall be ground smooth and shall be examined with LPT to ensure that the entire defect has been removed.

Completed repairs shall be examined by the same NDT methods used for the original weld

Only two times repairs are allowed on any joint.

PL4.0. SUBSEA PIPELINE SYSTEM

PL4.1. Codes & Standards

The following Codes shall be the minimum acceptable standards for subsea pipeline welding and inspection. The latest edition, addenda, code cases and supplement available at the time of bidding will be used. Any part of any other non-listed code referred to in these listed codes as augmentation is to be considered applicable.

DNV OS F101	Submarine Pipeline System
BS 7910	Guide to methods for assessing the acceptability of flaws in metallic structures
ASTM E1961	Standard Practice for Mechanized Ultrasonic Testing of Girth Welds Using Zonal Discrimination with Focused Search Units
AWS A2.4	Standard Symbols for Welding, Brazing and Nondestructive Examination.
NACE MR0175	Standard Material Requirements - Metals for Sulfide Stress Cracking and Stress Corrosion Cracking Resistance in Sour Oilfield Environments
NACE TM0284	Standard Test Method - Evaluation of Pipeline and Pressure Vessel Steels for Resistance to Hydrogen-Induced Cracking
NACE TM0177	Standard Test Method - Laboratory Testing of Metals for Resistance to Sulfide Stress Cracking and Stress Corrosion Cracking in H2S Environments
ASME BPVC Sec. V	Nondestructive Examination
ASME Sec. II, Part C	Specification for Welding Rods, Electrodes and Filler Metals
ASTM E164	Standard Practice for Contact Ultrasonic Testing of Weldments
ASTM E165	Practice for Liquid Penetrant Inspection Methods
ASTM E709	Practice for Magnetic Particle Examination
ASTM E94	Standard Guide for Radiographic Examination
ASTM E384	Standard Test Method for Knoop and Vickers Hardness of Materials
ISO 14731	Welding Coordination – Tasks and Responsibilities
ASNT SNT-TC-1A	Personnel Qualification and Certification in Nondestructive Testing

PL4.2. Welding Processes

Various welding processes as given in DNV-OS-F101 can be used for production welding. If the contractor propose to use mechanized welding process, a complete description of the process including examples of previously qualified welding procedures and subsea pipeline

projects completed using proposed process shall be submitted for Company's review and approval. Mechanised and Automatic welding systems shall be subject to a detailed pre- qualification programme or documentation before they may be used. The extent and the contents of a pre-qualification programme for such mechanised welding systems shall be agreed before start up. The Contractor shall prove and document that the welding systems are reliable and that the process can be continuously monitored and controlled.

PL4.3. Welding Consumables

Welding consumables and welding processes shall give a diffusible hydrogen content of maximum 5ml / 100g of weld metal.

Cellulose coated electrodes may be used only subject to agreement for welding of Pipeline Girth welds in CS Line pipe with SMYS $\leq$ 450 MPa. If used, the delay between completion of root pass and the deposition of hot pass shall be simulated during welding procedure qualification. If the use of cellulosic electrode has been agreed, the following additional requirements shall apply:

1. Pre-heat shall be minimum 100^0C.
2. Delay between the completion of root pass and the start of depositing the hot pass shall be minimum 6 minutes.
3. Immediately upon completion of welding during welding procedure qualification, the test pieces shall be water quenched as soon as the temperature of the test piece is below 300^0C.
4. Non destructive testing of the test pieces shall be by Automated Ultrasonic Testing (AUT) or by Radiographic Testing and Manual Ultrasonic Testing.

Low Hydrogen Electrodes with uphill techniques shall be used for all repairs, when external lineup clamp is used, tie-in, special crossings and **fillet welds.**

PL4.4 Welding Procedure Qualification

Welding Procedure shall be qualified as per DNV-OS-F101(Submarine Pipeline Systems) except as modified by this specification. Previously qualified welding procedures are not acceptable. However, for welding of CS & CS NACE Line Pipes (Girth welds) of SMYS < 450 MPa, WPS for new production job may be based on a previously qualified WPQR (Welding Procedure Qualification Record) provided

a. The applicable WPQT (Welding Procedure Qualification Testing) was done in last 2 years for ONGC Job *dated with effect from date of starting fabrication. If it is not of ONGC project, it shall be fresh WPQT for project specific.*
b. WPQT was witnessed by a reputed TPI agency.
c. Meet all the requirements of this specification and DNV-OS-F101. d.

Match all the essential parameters as given in DNV-OS-F101.

e. Written approval shall be obtained from Company.

For new WPS, the contractor shall first submit the Preliminary WPS (pWPS) for Company's review and approval. The pWPS shall contain all the relevant information required for the applicable welding process, as given in Clause D700 of DNV-OS-F101. After getting the approval of pWPS, the contractor can go ahead with the WPS qualification as per the requirements given in

DNV-OS-F101. Qualification welding shall be performed using the type of welding equipment to be used for production welding and under conditions that are representative of the actual working environment for the work (for WPSs to be used on pipelay barges, qualification welding has to be carried out on floating vessels / barges).
WPS qualification shall be witnessed by Company Representative and a reputed TPI agency.

Repair welding procedure shall be qualified separately. Pre-heat for repair welding shall be minimum 50^0C above the minimum specified pre-heat for production welding. Re-repair welding procedure shall also be qualified separately.

In case of riser line, as it is considered part of pipeline, this WPQT & WPS is applicable till the flange joint of shutdown valve or launcher whichever is earlier. In case, this riser line is prefabricated in yard , the same WPQT & WPS is applicable. However internal/external line clamp may not be required as mentioned in clause PL4.6."

PL4.5. Welder and Welding Operator Qualification

Welders and Welding Operators shall be qualified as per the requirements of DNV-OS-F101 in the presence of Company Representative and a reputed TPI agncy. Welders shall be qualified under conditions that are representative of the actual working environment. (Pipeline welders shall be qualified on floating vessel / barge)

PL4.6. Production Welding

The type of welding Equipment and Welding Procedure shall be qualified prior to the start of installation welding.
Internal line-up clamp shall be used for pipeline of size 6" and above. Internal lineup clamp shall remain in place at least until the root pass is completed around the full circumference.

External lineup clamp shall not be removed until a minimum of 50% of root pass, uniformly spaced around the circumference, has been completed.

Arc strikes – arc shall be struck only on fusion faces and not on other areas of the pipe.

The pipelay barge shall not be moved before the first hot pass is completed.

PL4.7. Non Destructive Testing

The extent of NDT for installation girth welds shall be 100% Automated Ultrasonic Testing (AUT) or Radiographic Testing. Radiography shall be carried out using X-ray with panoramic exposure. Gama ray may be used in case of inaccessible joints with approval of ONGC (with submission of specific joint details & with AFC).
For wall thickness > 25 mm, AUT should be used. For >25mm, if RT is used, it shall be supplemented with Ultrasonic Testing.
While using GMAW (like mechainsed or semi-automatic welding) or FCAW welding process, AUT is the preferred NDT technique.

100% Ultrasonic Testing of first 10 welds shall be carried out while using GMAW or FCAW processes, when starting installation or when resuming production after suspension of welding and when radiographic testing is the primary NDT technique. If Ultrasonic Testing reveals defects not discovered by radiography, the extent of Ultrasonic Testing shall be 100% for the next 10 welds. If the results of this extended testing are unsatisfactory, the welding shall be suspended until the causes of the defects have been established and rectified.

UT lamination check shall be carried out on 50 mm wide band at the flame cut pipe ends.

For "Golden Welds" (critical welds e.g. tie-in welds that will not be subject to pressure testing, etc.) 100% ultrasonic testing, 100% radiographic testing, and 100% magnetic particle testing or 100% liquid penetrant testing of non- ferromagnetic materials shall be performed. If the ultrasonic testing is performed as automated ultrasonic testing, the radiographic and magnetic particle/liquid penetrant testing may be omitted subject to agreement.

Prior to commencement of WPS qualification, contractor shall prepare and submit NDT Procedures (RT, MPI, LPT, UT/AUT) for Company's review and approval.

NDT Procedures, Equipment qualification and Acceptance Criteria shall be as per the requirements given in Appendix D of DNV-OS-F101.

AUT Procedures, Equipment and qualification for AUT shall be as per the requirements given in Appendix E of DNV-OS-F101. Defect Acceptance Criteria shall be arrived at using ECA. AUT

shall be performed in accordance with DNV-OS-F101 Appendix E and ASTM E1961.

If the Contractor proposes to use AUT, the contractor shall arrange familiarization and interpretation training on AUT for Company inspectors as and when required without any additional cost to Company

If using AUT, Contractor shall carry out ECA to arrive at weld acceptability criteria which is subject to approval by Company. As a minimum, requirements of Appendix A of DNV-OS-F101 shall be met. Details of laboratory / agency performing ECA shall be submitted to Company for review and acceptance. The agency should have skilled personnel and previous experience to carry out ECA as per BS 7910 requirements.

The effectiveness of the AUT Procedure shall be demonstrated by a qualification programme in accordance with DNV-OS-F-101, Appendix E, Section H. The AUT system must demonstrate the ability to detect and accurately size length and vertical height of indications with a resolution compatible with the applicable acceptance criteria.

NDT Operators shall be qualified and certified in accordance with the requirements of Level II of the latest edition of SNT-TC-1A or equivalent PCN or CSWIP Level II.

Additionally, AUT operators shall have received a sufficient level of training of Automated Ultrasonic Testing, including practical and theoretical aspects applicable to the equipment and configuration to be examined. This training shall be documented and will be verified by the Company.

PL4.8. Repair Welding

Repair welding shall be performed in accordance with a qualified repair welding procedure. After grinding / gouging, Magnetic particle testing or liquid penetrant testing of non-ferromagnetic materials shall be performed to verify complete removal of defects before commencing weld repairs.
Pre-heat for repair welding shall be 50^0C more than that used for production welding.
Only two times repairs are allowed on any joint.
If the repair welding station is after the pipe tensioner, a weld repair analysis shall be carried out.
The analysis shall determine the maximum allowable excavation length and depth combinations taking in to account all stresses acting at the area of the repair. The weld repair analysis shall be subject to agreement.

PL4.9. Other Materials

For materials other than CS & CS NACE, contractor shall prepare various procedures like WPS, NDT etc. and shall submit for Company's review and approval.

ANNEXURE I

WELDING OF DUPLEX STAINLESS STEEL

			7	10/01/2005	1
PDD / PNN	SKS / MP	JSS			
PREPARED / REVISED BY	REVIEWED BY	APPROVED BY	TOTAL No. OF PAGES	DATE	REV. No.

WELDING SPECIFICATION FOR DUPLEX STAINLESS STEELS

1.0 WELDING PROCESS

Shielded Metal Arc Welding (SHAW), Gas Tungsten Arc Welding (GTAW), Submerged Arc Welding (SAW) and Plasma Arc Welding (PAW) processes are suitable for Duplex stainless steel. These Welding methods along with the related operational precautions such as cleanliness and protective gas shielding should be used for making duplex stainless steel welds. Shielding / purging gases shall be selected ensuring freedom from hydrogen.

2.0 MICROSTRUCTURE

The ferrite content of the weld metal estimated through metallographic examination, should be between 30% to 55% to obtain weld joints with the best ductility and corrosion resistance. The filler metal composition and the welding heat input should be so selected as to give the ferrite content of the weld metal between 30% to 55%.

3.0 FILLER METALS (Welding Consumables)

Table 1 enclosed herewith tentatively lists the different filler metals used for duplex stainless steels conforming to UNS S 31803. Welding consumables shall be used only after conducting consumable qualification tests. Consumables shall be degreased or baked and stored in accordance with the manufacturer's recommendations.

4.0 MECHANICAL PROPERTY REQUIREMENTS

The weld metal should have the following mechanical properties :

4.1	Tensile Strength	-	680 – 880 N/mm^2
4.2	Yield Strength (0.2% Offset)	-	450 N/mm^2 (Min.)
4.3	Elongation	-	25% (Min.)
4.4	Impact Energy (Charpy V-notch Valves)		
	At + 20°C (68°F)	-	100 Joules Average (72 ft. lb) 75 Joules Average (54 ft. lb)

At –30°C (-22°F) - 60 Joules Average (43.2 ft. lb)
40 Joules Average (28.8 ft. lb)

5.0 QUALIFICATION TESTS

5.1 Welding Procedure Qualification Tests

5.1.1 General

Each particular welding procedure shall be specially qualified for the job; no previous test reports shall be recognized as a qualification of the work to be performed. The welding filler metal to be used shall be of a grade, which will yield a weld metal equivalent to that of the base metal. Welding consumables shall be subjected to initial qualification testing and the consumable shall require company's approval. During Procedure Qualification, the heat input for 22% Cr Duplex steels shall be limited to 0.8 – 2.5 KJ/mm. In production welding, the heat input shall not deviate from that qualified by more than +/- 10%. P numbers shall not apply to duplex stainless steels. Each alloy designation shall be separately qualified in accordance with this specification.

5.1.2 Welding consumable qualification tests and welding procedure qualification tests shall be conducted before the start of procurement by employing an approved welding procedure in presence of the Company's Representative. The welding procedure shall be prepared by the Contractor based on the information provided by the Supplier of raw materials and/or electrodes Supplier and approved by the Company. Welding procedure shall be such that it gives maximum hardness on the outer surface of pipe.

All the mills supplying the raw material/pipes etc. shall be required to pass the tests required for procedure qualification testing. Approved procedure shall be common to all the mills.

Laboratory tests shall be performed at an official Laboratory approved by the Company. Sufficient advance notice shall allow the Company to witness all or part of the tests.

5.1.3 Test Samples

Tests mentioned below shall be performed on welded test pieces and shall include the parent metal, heat affected zone and the weld metal.

5.1.4 Procedure Qualification Tests shall be carried out as given below :

i) Chemical Composition and Microstructure Examination

The ferrite content measurement of test pieces shall be carried out by both the following methods:

a) Metallographic determination and

b) Magnegage method (According to Kotecki)

The procedure of ferrite measurement and the calibration of measuring instrument shall bear the approval of the Company before being employed.

ii) Mechanical Properties

These tests shall include tensile testing, impact strength and bend test. Bend tests shall be performed in accordance with ASTN A 370 with a plunger dia equal to 4 times the thickness. No defect shall be visible on bending.

iii) Hardness Tests

Each qualification test shall include a cross sectional hardness survey.

A complete survey of the weld, heat affected zone and the base metal shall be performed according to the following procedure :

- the test method shall be as per ASTM E 18.
- the distance between indentations shall be a maximum of 1 mm.
- The results shall not exceed RC 28.

iv) Corrosion Tests

a) Intergranular Corrosion Tests (HUEY)

The welds shall be subjected to ASTM A 162 practice C (HUEY Test). Weight loss shall not exceed 3 to 4 mils/month.

b) Chloride Stress Corrosion Cracking Test.

This type of test shall be conducted on the material at 155° C with aeration in 45% $MgCl_2$ solution as per ASTM G 36. Stress to

cause repute in 500 hours shall exceed 0.35 times the ultimate tensile strength. This test is required to be done per heat for each mill before start of procurement.

c) SSCC Test

Resistance of the material to SSCC shall be tested using :

a) NACE TM 01-77 test solution (Test temp. 24 Deg. C).Minimum stress for cracking in 720 Hrs shall be 350 N/mm2.

b) NACE TM 01-77 test solution (test temp. 90 Deg C and total pressure of H2S = 16 Bars).

Min. stress for cracking in 720 Hrs = 325 N/mm2.

v) Radiography

The welds shall be 100% radio graphed shall be as per ASME, Section – VIII, Division – I, UW – 51.

vi) Dye Penetrant / MP Testing

All fillet and groove welds other than these radio graphed shall be tested by dye penetrant testing or by magnetic particle testing method. No defect shall be accepted.

vii) Heat Treatment (if required)

Suitable heat treatment procedure shall be established for retaining the Duplex S.S. properties of welded joints which shall be approved by the Company before implementing.

5.2 Welder Qualification Tests

Welders shall be qualified at the fabrication yard / barge employing the qualified welding procedure. The test pieces of each welder shall be subjected to mechanical tests, radiographic examination, DPT/MP Test, microstructure examination and cross-sectional hardness survey with the same test procedure and acceptance criteria as mentioned in 5.1 above.

6.0 FABRICATION

6.1 No fabrication shall start until the Company has approved the chemistry of the materials, the results of welding procedure qualification tests and welders are qualified as per the approved procedure.

6.2 Working practices shall be designed to minimize contamination and before welding, internal and external surfaces shall be cleaned for a distance of at least 50mm from the fusion face.

6.2 In production welding the heat input shall be monitored and the same shall not deviate from the qualified WPS by more than +/- 10%.

6.2 Production/Fabrication Tests

Shop/Site Fabrication Tests for each weld shall include :

- Non – destructive hardness tests by portable hardness tester
- Radiography
- DPI/MP Examination
- Ferrite Measurement by either

  a) Magnegage method (extended range, according to Kotecki)

  OR

  b) Forster Probe Method.

Procedure	Supplier Commercial Name	Typical Chemical Composition in %	Comments
Plasma auto or Manual	Smitweld SW 4462 (Revised Composition)	C:0.010 Mn:1.6 Sl:0.25 P:0.20 S:0.008 Cr:23.0 Ni:8.8 Mo:3.1 N:0.012	Filler metal recommended when as welded
	Thyssem	C:0.016 Mn:1.57 Si:0.17 Cr:22.8 Ni:9.18 N:0.04	Very low nitrogen (traces) ferrite level practically identical to 22-6-31.
	Thermanit 22-09		Excellent results when welded with nitrogen in welding gas.
SAW	Thyssen Thermanit	C:0.016 Mn:1.57 Si:0.17 Cr:22.8 Ni:9.18 N:0.04	With flux SAP CN 100 – as welded condition gives ferrite range 35-65%
SMAW	Thermanit 22-09	C $\leq$ 0.030 Cr:22.0 to 23.0	
	Sandvik 22-9-3LR Smitweld Arosta 4462 Philips Rs 22-9-3LC Avesta 223 FAL-PW	Ni:9 to 10 Mo $\approx$ 3, N:0.10 to 0.15	These electrodes even with low heat input, give welds with specified ferrite range.

SPECIFICATION FOR PROTECTIVE COATING

Prepared/ Revised by	Reviewed by	Approved by	Pages	Date	Rev. No.
MRS	RMK	HMG	--	15.02.2012	3
KCD /MRS	RMK	GRP	67	27.03.2012	4
MRS	SKJ	RMK	65	12.12.2014	5
MRS	SKJ	RMK	64	05.03.2015	5.1
RCS	MRS	RKJ	65	25.08.2016	6
KKD	MRS	RKJ	71	01.08.2018	7
MK	GM	KKD	71	26.11.2019	8
GOURANG GARG	GUNAJIT MEDHI	KANAK CHANDRA DEKA Digitally signed by KANAK CHANDRA DEKA Date: 2022.12.15 11:42:41 +05'30'	78	12.12.2022	9
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2.2.2 American Society for Testing and Materials (ASTM)

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1.0 INTRODUCTION

1.1 Scope

This specification covers the minimum requirements governing surface preparation, selection and application of the protective coating system to be used on the interior and exterior surfaces of all types of production facilities including structural steel, vessels, piping and equipment on offshore platforms.

Individual equipment specifications and /or drawings, when furnished, are to be used with these specifications. If conflict exists, the individual specifications and/or drawings shall govern. This functional specification is applicable for both green field projects/job as well as brown field projects/job.

It includes the requirements with respect to protection against corrosion of both Ferrous and non-ferrous metals of all types of production facilities including structural Steel, vessels, piping and equipment on offshore platforms, SBMs.

This specification also covers the performance-based coating standard and is Applicable to painting and coating, for new construction, modifications and Maintenance of offshore facilities. All paint and coating systems to be used shall meet the qualification requirements and the minimum coating system requirements as specified in this specification.

The coating systems to be used in accordance with this specification shall be Suitable for a Marine environment C5-M, High Durability (H) as specified in ISO 12944, NORSOK M-501 & NACE SP 0108.

1.2 Definitions

The following definitions shall apply:

COMPANY : Shall mean ONGC or the designated representative.

CONTRACTOR : Shall mean the party contracted to perform the work in accordance with the drawings, specifications and work scope.

2.0 CODES AND STANDARDS

2.1 Mandatory Statutory Requirements

This document has been prepared to the International Standards detailed within. The CONTRACTOR shall ensure that the Work is executed in accordance with international standards, Statutory & Regulatory requirements.

2.2 Codes and Standards & Regulations

The requirements of the latest published versions of the following listed Codes, Recommended Practices. Specifications and standards shall be met.

2.2.1 Steel Structure Painting Council (SSPC) (Latest Revision)

SSPC-PA1	Shop, Field and Maintenance Painting of Steel.
SSPC-PA2	Measurement of Dry Coating Thickness with Magnetic Gauges
SSPC-SP1	Solvent Cleaning
SSSC-SP2	Hand Tool Cleaning
SSPC-SP3	Power Tool Cleaning
SSPC-SP5	White Metal Blast Cleaning
SSPC-SP6	Commercial Blast Cleaning
SSPC-SP7	Brush –Off Blast Cleaning
SSPC-SP10	Near White Blast Cleaning
SSPC-SP11	Power Tool Cleaning to bare metal
SSPC-SP12	Surface Preparation & cleaning of Steel and Other Hard material by High and Ultra High Pressure Water Jetting prior to recoating
SSPC-AB1	Mineral and Slag Abrasive
SSPC-AB3	Ferrous Metallic Abrasive
SSPC-SP20	zinc rich coating Type-I(Inorganic) & Type-II (Organic)
SSPC-SP COM	Surface Preparation and Abrasives Commentary, SSPC Painting Manual, Volume 2, “Systems and Specifications”
SSPC VIS-1	Guide and Reference Photographs for Steel Surfaces Prepared by Dry Abrasive Blast Cleaning
	SSPC Painting Manual
SSPC Vol.2	

2.2.2 American Society for Testing and Materials (ASTM) (Latest Revision)

ASTM A123	Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM A153	Standard Specification for Zinc-Coating (Hot-Dip) on iron and Steel Hardware
ASTM D4228	Standard practice for qualification of coating applicators for application of coating on steel surfaces.
ASTM B117	Salt Spray Test
ASTM G8	Cathodic Disbonding Test
ASTM G50	Standard practice for conducting atmospheric corrosion test
ASTM G53	Weathering Test (Part B)
ASTM D520	Standard Specification for Zinc Dust Pigment (Metallic Zinc Powder)
ASTM D523	Specification for Gloss
ASTM D610	Standard Practice for evaluation degree of rusting of painted

steel surfaces

ASTM C 633	Test Method for Adhesive/Cohesive Strength of Flame Sprayed Coating
ASTM D1200	Viscosity
ASTM D1640	Drying time
ASTM D1653	Standard test method for evaluation of painted or quoted specimens subject to corrosive environment
ASTM D2247	relative Humidity Test
ASTM D2697	Volume of Solids
ASTM D3359	Standard test method for measuring adhesion by tape test.
ASTM D4060	Abrasion Resistance of Coating
ASTM D4285	Standard Test Method for Indicating Oil or Water in compressed air
ASTM D4417	Test Method for field measurement of surface profile of Blasted steel
ASTM D4541	Test Method for Pull-off Strength of Coating Using Portable Adhesion Testers
ASTM D4752	MEK Test for Testing of Zinc Silicate Paint
ASTM D4940	Standard Test Method for conductometric Analysis of water soluble ionic contamination of Blasting Abrasive
ASTM D5894	Standard test method for evaluating drying or curing during film. Corrosion resistance under cyclic condensation/UV
ASTM D6386	Standard Practice for Preparation of Zinc (Hot-Dip Galvanized) Coated Iron and Steel Product and Hardware Surfaces for Painting
ASTM E119	Standard Test Methods for Fire Test of Building Construction and Materials.
ASTM E1529	Standard Test Methods for Effect of Large Hydrocarbon Pool Fires on Structural Members and Assemblies

2.2.3 Indian Standards (Latest Revision)

IS 5	Colours for Ready Mixed Paints and Enamels
IS 2379	Pipelines Identification Color Code

2.2.4 British Standard (Latest Revision)

BS 476: Part 20/21 Intumescent Coating
BS 2569: Specification for Sprayed Metal Coating
BS 3900 Part G7 Heat Resistance Test

2.2.5 International Standards Organization (Latest Revision)

ISO –898	International Standard for Mechanical Properties of fasteners made of carbon steel and alloy steel
ISO –8501	International Standard for Preparation of Surface
ISO-8502-6	Preparation of steel substrates before application of paints & related products Tests for the assessment for surface cleanliness-:Part-6:Extraction of soluble contaminants Analysis: The Bressle Method
ISO –8504	Preparation of steel substrates before application of paints and related products
ISO 1461	Hot dip galvanized coating on fabricated iron and steel articles-Specification and test methods
ISO 14713	Protection against corrosion of iron and steel structure-zinc and aluminum coating.
ISO 4624	Adhesion test of paint
ISO 12944	A Global Corrosion standard
ISO 4628	Evaluation of degradation of paint coating
ISO 834	Hydrocarbon Resistance Design (PFP)
ISO 20340	Paints and varnishes – Performance requirements for protective paint systems for offshore and related structures.
ISO 4628-6	Paints and varnishes – Evaluation of degradation of paint coatings – Designation of intensity, quantity and size of common types of defect – Part 6: Rating of degree of chalking by tape method

2.2.6 Occupational Safety and Health Act

OSHA	Occupational Safety and Health Act

2.2.7 NACE Standards (Latest Revision)

NACE SP0188	Discontinuity (Holiday) Testing of New Protective Coatings on Conductive Substrates
NACE Standard RP0287	Field Measurement of Surface profile Measurement of abrasive blast cleaned steel surfaces using a replica tape.
NACE RP0198	The control of Corrosion Under Thermal Insulation & Fire Proofing Materials-A System Approach
NACE Standard RP 0176-2003	Corrosion control of Steel Fixed Offshore Platforms Associated with Petroleum Production
NACE SP 0108	Corrosion control of Offshore Structure by protective coating
NACE 012	Specification for Application of Thermally Sprayed Coatings (Metallizing) of Aluminum, Zinc, and Their Alloys and Composites for the Corrosion Protection of

	Steel

2.2.8 Underwriters Laboratories

UL1709	Hydrocarbon Fire Resistant Design (PFP)

2.2.9 NORSOK Standard

M-501	Surface Preparation and Protective coating
M-001	Material Selection

2.2.10 American Welding Society (AWS)

AWS C.2.17	Recommended Practice for Electric Arc Spray
AWS C.2.18	Guide for Protection of steel with Thermal Spray Coating of Aluminium ,Zinc and Their alloys and composites
AWS C.2.23	Specification for Application of Thermally Sprayed Coatings (metalizing) of aluminium ,Zinc and Their alloys and composites for corrosion protection of steel

2.2.11 RAL 840 HR ACQPA: COLOUR CHART

2.2.12 Abbreviations and Definitions

APAS	Australian Paint Approval Scheme
ASTM	The American Society for Testing and Materials
C	Coating Thickness of metallic Zn (HDG coating)
COT	College of Occupational Therapists, (UK)
CP	Cathodic Protection DFT Dry Film Thickness
CPS	coating procedure specification
CPT	coating procedure test
CSDS	coating system data sheet
DFT	Dry Film Thickness
GRP	Glass Reinforced Plastic (fiber glass)
HB	High Build Epoxy
HDG	Hot-Dip Galvanized
HP	High Pressure (as in Water Washing at pressures above 300 Bar)
ITP	Inspection and Test Plan µm Micron
IMO	International Maritime Organization
ISO	International Organization for Standardization
MSDS	Material Safety Data Sheet
MEK	methyl ethyl ketone
MSC	Maritime Safety Committee
NACE	National Association of Corrosion Engineers

NSF	National Science Foundation
OH&S	Occupational Health and Safety
PFP	Passive Fire Protection
PIG	Paint Inspection Gauge
PPE	Personal Protection Equipment
PSPC	Performance Standard for protective Coatings
QA	Quality Assurance as defined by ISO 9001 and 9002 Quality Systems
QC	Quality Control of production processes by activity
RH	Relative Humidity
S/S	Stainless steel Supplier
UHP WJ	Ultra High Pressure Water Jetting (> 2000 Bar)
UL	Underwriters Laboratories
VOC	volatile organic compound
WB-IZS	Water Borne - Inorganic Zinc Silicate
WFT	Wet Film Thickness

3.0 GENERAL

3.1 Selection of Coating systems and application procedure shall be made with due consideration to conditions during fabrication, installation, and service of Installation.

All painting activities shall be incorporated in the fabrication plan.

Details of Management, Inspectors, operators, facilities, equipments and qualified procedures shall be established and document before commencing work.

Following items shall not be coated unless otherwise specified.

- Fibre-glass
- plastic or other non-metallic finish
- Equipments, valves, etc. having factory coated finish
- Indicators
- Sprinklers, fusible plugs and fire detectors
- Control Valve Stems
- Stainless Steel Control Panels
- Stainless Steel Tubing and pipe – work

- Cupro-Nickel (CuNi) pipe work
- Glass Reinforced Epoxy (GRE) pipe work

The following specific items shall not be coated and shall be protected from blasting and coating being applied to adjacent equipment

- Bearings and seals
- Flange mating faces, Raised Face (RF) and Ring Type Joint (RTJ) and flat faced
- Instrument dials and/or cases
- Cable trays and cables
- Level gauge glasses
- Nameplates Shafts and similar polished or machined surfaces Instrument Tags and Valve Position.

3.2 Standard Coating

The manufacturer's standard coating shall be used for the following equipment

- Indoor electrical equipment
- Instrument and control panels
- Insulated rotating equipment

3.3 Equipment Cleaning

The following equipment shall be cleaned with biodegradable, water soluble cleaner and an epoxy primer shall be applied (tie-coat) to the manufacturer's standard coatings prior to the specified intermediate coat.

- Fan and blower housing
- Outdoor electrical equipment
- Engines and electric motors
- Pumps, compressors and other non-insulated rotating equipment

- Control and relief valves

Equipment shall be recoated as specified if the coating system applied by equipment manufacturer/packager does not comply with this specification or if coating repairs is necessary.

3.4 Flanges

Flanges on piping and valves (including control and relief valves) shall have a primer coat applied to bolt holes and the non-contact area of the face prior or being made-up. After make-up of these connections intermediate and finish coats shall be applied. Flanged ends shall have a finish coat as required in this Specification. Flange faces must not be coated.

3.5 Piping Spools

Primer and intermediate coatings and finish coat as per clause 13.2.2 shall be applied to spools pre-fabricated for offshore installation. If these spools have an end prepared for field welding, the coatings shall stop 150 mm from the prepared end.

3.6 Seal Welding

Where enclosed or inaccessible areas cannot be adequately painted, the areas shall be completely enclosed and seal welded. Small areas inaccessible to blasting and painting that cannot be boxed by welding shall be filled with epoxy mastic, caulking compound or other suitable filler to prevent retention of dirt and moisture.

3.7 Overspray

Adjacent structures, equipment and all other items shall be protected from blasting, over-spray and drips with tarpaulin, plastic, tapes etc.

3.8 Nozzles Through Insulation

Flanged nozzles, man ways, platform clips, saddles and other attachments that extend through insulation shall be coated in accordance with the coating schedule for equipment and piping.

3.9 Coated Bolts

Galvanized bolts, studs and nuts, where used to bolt up piping, valves etc. or otherwise used to connect painted items, shall receive a top coat of paint over exposed areas after the connections are made. And shall be top coated with anti rust grease.

4.0 SAFETY

Safety conditions shall be met, as required by the Occupational Safety and Health Act (OSHA) or other governing bodies, as well as those that are the deemed necessary.

Particular care must be exercised when working in close or confined spaces, especially when spraying. The maximum allowable concentration of solvent in the air shall not be exceeded. Refer to the manufacturer's recommendations to determine the maximum allowable concentration value. When volatile solvents are flammable, the concentration in air shall be kept below 25 percent of the lower explosive limit by use of adequate exhaust or ventilation facilities.

5.0 ENVIRONMENTAL REQUIREMENTS

The contractor will recognize COMPANY's commitment to preserving the environment and shall comply with local codes and standards for transporting, storing, and disposing of hazardous materials and hazardous wastes.

Upon completion of the job, the contractor shall notify the COMPANY of the volume and type of hazardous waste generated.

Upon completion of the job, all non-hazardous wastes, such as empty paint cans, clothes blasting abrasives and equipment, shall be removed by the contractor from the job sites and properly disposed.

6.0 SURFACE PREPARATION

6.1 General

The surface preparation procedures and requirements except for galvanizing and cadmium plating shall be in accordance with Steel Structural Painting Council (SSPC) - SP5, SP6, SP7 & SP10 and ISO–8501-1.

All fabrication and assembly shall be completed before surface preparation is taken up. Blast and application of coating to structural and piping items prior to assembly will be permitted provided surface preparation for splice ends are taken up by portable blasting tools before application of prime coat. All field splice surface preparation for Structure & Piping Items shall be carried out using portable blasting tools at field. All welding slag, weld spatters and burrs shall be removed prior to blasting. All bolt holes shall be drilled and their edges smoothed prior to blasting.

First step of surface preparation: Solvent cleaning is to be taken up for any surface preparation.

Second step of surface preparation: Steel surface shall be blast cleaned to develop specific anchor pattern/profile to develop efficient bond between paint & steel surface using blasting tool or portable blasting tools for all Structural, Piping & Pressure Vessel items.

Surface preparation for Equipment, Equipment Manufacturer's procedure shall be submitted for approval of Company.

In case of modification work on Old Platform or Maintenance work, portable blasting tool shall be used for surface preparation. However, use of Power Tool / Hand Tool cleaning methodology can be used with approval of Company establishing constraint in use of portable blast tool and shall be applicable for specific location (not for as whole of the work/project) only.

6.2 Requirements of Blasting

Only dry blasting procedures are allowed. Definitions of and requirements for, the various methods of surface cleaning are given below:

A. White Metal Blast: As per SSPC SP5 & visual reference Sa3 as per ISO 8501-1.

B. Near-White Blast: As per SSPC SP10 & visual reference Sa2.5 as per ISO 8501-1.

C. Commercial Blast: As per SSPC SP6 & visual reference Sa2 as per ISO 8501-1.

D. Brush-off Blast: As per SSPC SP7 & visual reference Sa1 as per ISO 8501-1

6.3 Pre Blasting preparation

6.3.1 Rough Edges

Sharp edges, fillets, corners and welds shall be rounded or smoothened by grinding (minimum radius 2 mm).Hard surface layers (e.g. resulting from flame cutting) shall be removed by grinding prior to blast cleaning according to ISO 8501-3, Grade-3.

All surfaces should be washed with clean fresh water prior to blast cleaning.

Any major surface defects, particularly surface laminations or scabs detrimental to the protective coating system shall be removed by suitable dressing. Where such defects have been revealed during blast cleaning, and dressing has been performed, the dressed area shall be re-blasted to the specified standard. Surface pores, cavities etc. shall be removed by suitable dressing or weld repair.

6.3.2 Weld Flux and Spatter

Weld flux, slag spatter, slivers etc. shall be ground smooth before blasting. Welding surface imperfections shall be removed and surface profile shall be prepared as per ISO 12944-3.

Any surface on which grinding is done shall be spot blast cleaned or power tool cleaned to obtain required anchor pattern.

All welds shall be inspected and if necessary repaired prior to final blast cleaning of the area.

6.3.3 Surface Cleaning

Prior to blasting, all deposits or grease or oil shall be removed from the surface in accordance with SSPC-SP1 Solvent Cleaning using biodegradable water soluble cleaner.

6.3.4 Chemical Contamination

All chemical contamination shall be neutralized and/or flushed off prior to any other surface preparation.

6.3.5 Equipment Protection

Items such as motors, machined surfaces, gauges, electrical and instrumentation items tags and nameplates, stainless steel galvanized steel, aluminum, brass, plated surfaces etc. shall be protected to prevent damage or contamination during blasting or painting.

Prior to blasting, openings on engines, pump, vessels, piping etc. shall be effectively sealed to prevent abrasive entering and damaging internal components. All packaged equipment shall be covered and special care taken to cover and seal all instrumentation.

6.4 Blasting Operations

6.4.1 Weather conditions

Blast cleaning shall not be done on any surface that is moist, or that may become moist, before the application of a primer.

No blasting is permitted when the steel temperature is less than 3°C above the dew point, as measured by a sling hydrometer, or when the relative humidity of the air is more than 85 %.

6.4.2 Preliminary Blasting

If blasting is performed at night, the surfaces shall be re-blasted the following day to provide the specified surface preparation standard and the anchor profile required for the specified coating system.

6.4.3 Blasting and Painting

Blasting shall not be done adjacent to painting operations or coated surfaces that are not fully dry. Blasting shall overlap previously coated surfaces by at least 150 mm.

6.4.4 Post – Blasting Procedure

The surface to be coated shall be clean, dry, free from oil/grease, and have the **specified anchor pattern/ roughness** and cleanliness until the priming coat is applied. Blast cleaned steel surfaces shall not be touched by bare hands.

Chloride contamination shall be checked **on the blasted steel surface prior to coating application** using Salt Contamination Meter - SCM 400 / **Bresle** patches / Quantab strips. Maximum permissible limit shall be 20 mg/m^2 for external surface & 50 mg/m^2 for internal of vessels.

Testing of soluble chloride ion content shall be carried out at least on each component, once per 200m² and a minimum of three times per shift during the progress of work. Special attention shall be given to areas where water has been trapped and dried out.

The blast cleaned surface shall be rendered dust free and coated with the specified primer a soon as possible to avoid formation of oxidation on the surface, but **in any case within four hours from the time of blasting**, and at least one hour prior to sunset on the same day. Any steel surface not primed within these limits or that is wet shall be re-blasted.

No acid washes or other cleaning solutions or solvents shall be used on metal surfaces after they are blasted. This includes washes intended to prevent rusting.

All areas around the intended paint surface shall be cleaned of sand prior to coating. Drains shall be purged of sand and flushed.

Biodegradable water – soluble cleaning solution used to clean previously painted surfaces shall not lift softens or otherwise damages the existing coating.

6.5 Blasting Equipment

6.5.1 Compressed Air

The air compressor shall be capable of maintaining a minimum of 700 kpa (7 kg/cm^2 or 100 psi) air pressure at each blasting nozzle.

The compressed air supply shall be free of water and oil. Adequate separators and traps shall be provided on the equipment, which shall be regularly purged of water and oil to maintain efficiency.

6.5.2 Nozzle

The nozzle shall be a 10 mm (maximum) internal diameter venture style nozzle.

6.5.3 Power Tools (Use can be permitted with specific approval for specific location)

Power tools may be used to obtain a metal surface finish as per SSPC SP11 where blasting is not possible, or on items which might be damaged by blasting.

6.5.4 Shot Blasting Equipment

Shot blasting equipment may be used for specific applications. Shot shall be changed as required to maintain the angular profile requirement.

6.6 Blasting Abrasive

6.6.1 Abrasive

The abrasive shall be as per SSPC-AB-1. The abrasives shall be copper slag, steel balls, garnet or coal slag and shall be free of contamination of dust and chlorides to produce the required anchor profile and graded as to be free from clay, silt or other matter likely to become embedded in the steel surface. Abrasives which have a tendency to shatter and adhere or embed in the steel surface shall not be acceptable. Recycled abrasive shall not be used. Use of silica sand is not permitted.

6.6.2 Shot Blasting Material

Shot blasting material shall pass through a 16 mesh screen. At least 15% steel grit shall be mixed with the graded shot to remove any rust, scale or other impurities pined into the surface by the shot. Shot blasting material is limited to iron, steel or synthetic shot which is applied by compressed air nozzles or centrifugal wheels. Shot blasting material shall be checked at least two times a week for replacement of abraded material.

6.6.3 Alternative methods of Surface Preparation

a) For Blast Cleaning

Power Tool Cleaning shall be confined to minor areas.

Unless otherwise specified it shall be carried out in accordance with the requirement of ISO 5404-3. Power Tool Cleaning (grinding) to bare metal shall be done in accordance with SSPC SP-11.

If Power Tool Cleaning is not feasible the surface cleanliness shall as a minimum, meet visual standard PSt3 in accordance with ISO 8501-2 at the time of coating. Care shall be taken to ensure that Power Tool Cleaning does not polish the steel surface. Hand tool cleaning is permitted prior to Power Tool Cleaning. If the surface being prepared lies adjacent to a coated surface the Power Tool Cleaning shall overlap the coated surface at least by 25 mm and the coated surface shall be feathered.

b) Centrifugal Abrasive Blasting: [Applicable for all local / field splice for structural & Piping Items]

A portable blasting machine using recyclable steel abrasive may be used to prepare steel decks and tank floors.

c) Vacuum Blasting

Vacuum Blasting may be used for spot repair of damaged or corroded area. It may be used in locations where open abrasive blasting is not permitted or desirable.

d) Wet abrasive blasting

Wet abrasive blast cleaning techniques may be used to avoid dust or in cases where fire and/or explosion risks are present. The cleaned surface should be washed off immediately after blast cleaning using fresh water. Corrosion inhibitors should not be used or only after written approval of the Principal. Corrosion inhibitor such as mass fraction 0.3 % sodium nitrite with mass fraction 1.2 % ammonium phosphate may be used to prevent flush rust, but Chromate inhibitors shall not be used. Corrosion inhibitor shall not be discarded into the sea. The paint Manufacturer shall approve the use of any inhibitor and the method for removing inhibitor residues before painting. The cleaned surface shall be dry at the time of painting. Particular care shall be taken to dry areas which are not self-draining, so that water collected is removed.

e) Water Jetting

High-pressure and ultrahigh-pressure water jetting may be used where abrasive blasting is not permitted because of the risk of damage to process equipment. Water jetting is restricted to maintenance painting. It is a hazardous operation and requires the use of well-trained, experienced operators. Where black spots occur in pits or other surface defects, the surface shall be cleaned again at higher pressure to remove the residual salts. Corrosion inhibitors should not be used or only after written approval of the Principal. Corrosion inhibitor such as mass fraction 0.3 % sodium nitrite with mass fraction 1.2 %ammonium phosphate may be used to prevent rust flash, but Chromate inhibitors shall not be used. Corrosion inhibitor shall not be discarded into the sea. The paint Manufacturer shall approve the use of any inhibitor and the method for removing inhibitor residues before painting.

The cleaned surface shall be thoroughly dry at the time of painting. Particular care shall be taken to dry areas which are not self-draining, so that water collected is removed.

7.0 COATING APPLICATION

7.1 General Application

All application, inspection and safety procedures shall be carried out in accordance with SSPC Painting Manuals, Vol. 1 Chapter 14.2 and Vol. 2 Chapter 5) and as set out below.

7.1.1 Supply and Storage

All coatings shall be furnished, mixed and applied in accordance with manufacturer's recommendations and as specified herein. Mixing of different manufacturer's coating applications on the same surface is not permitted.

All coating materials and thinners shall be in original, unopened containers being the manufacturers label batch numbers and instructions. For materials having a limited shelf life, the date of manufacture and the length of life shall be shown. Materials older than their stated shelf life shall not be used.

Materials shall be stored in accordance with the manufacturer's recommendations.

Coating materials that have gelled, other than thixotropic materials or materials that have deteriorated during storage shall not be used.

7.1.2 Pot Life

If the coating requires the addition of a catalyst, the manufacturer's recommended pot life for the application conditions shall not be exceeded. When the pot life is reached, the spray pot shall be emptied, cleaned and a new material catalyzed. Manufacturer's recommendations to be followed

7.1.3 Mixing

Mixing and thinning directions as furnished by the manufacturer shall be followed. Only thinners specified by the manufacturer shall be used.

All coating materials shall be stirred with a power mixer use, until the pigments, vehicles and catalysts are thoroughly mixed and then strained while being poured into the spray pot. During application the materials shall be agitated according to the manufacturer's recommendations.

Different brands or types of paints shall not be intermixed.

7.1.4 Unblasted Surfaces

Coating shall not be applied within 75 mm of an unblasted surface.

A 300 mm wide strip of uncoated, blasted surface shall be left between primed and unblasted surfaces, so as to prevent damage to the newly dried coating when additional blasting is done.

7.1.5 Application Requirements

A Coating procedure test (CPT) shall be used to qualify all coating procedures. A suitable location on the component to be coated may be selected on which the CPT shall be carried out. Alternatively, a test panel (minimum 1 m x 1 m) containing at least 1 pipe-end (1500mm long& 50 mm Ø., 1 angle and 1 flat bar (1500 mm in length)may be used for other coating systems. The coating procedures shall be qualified under realistic conditions likely to be present during coating. Selection of coating systems and application procedures shall be made with due consideration to conditions during fabrication, installation and service of the installation.

7.2 Cleanliness

All Surfaces shall be clean free from dust and dry. Any blast cleaning dust or grit remaining on the surfaces shall be removed by means of compressed air before priming or application of any coating. Any surface with a rust bloom shall be re-blasted as per ISO: 8504.

7.2.1 Temperature

Coating shall only be applied when the temperature of the steel is at least 3°C above the dew point, ambient air temperature must be within the limits specified by the manufacturer.

7.2.2 Weather Conditions

No coatings shall be applied during fog, mist or rain or when humidity is greater than 85% or on to wet surfaces. In case the minimum temperature at the fabrication yard is below 5^0C, the contractor shall propose alternate coating procedure for Company's approval at the bidding stage itself.

The company has the right to suspend application of coating when damage to the coating may result from actual or impeding weather condition.

When Relative Humidity (RH) is less than 60%, Zinc rich Epoxy Primer shall be used in instead of Inorganic Zinc Silicate as per recommendation of manufacturer and with prior approval of the company.

7.2.3 Coats

Each coat shall be applied uniformly and completely over the entire surface. Each coat shall be allowed to dry for the time specified by the manufacturer before the application or a succeeding coat. To reduce the possibility of intercoat contamination and to assure proper adhesion between successive coats, all coats shall be applied as soon as possible after the minimum specified drying time of the preceding coat.

7.3 Brush Application

7.3.1 General Requirements

A **stripe coat** shall always be applied by brush. The stripe coat shall be applied for each coat to all edges, corners, welding seams, bolt holes, back side of piping, stiffeners, vent and drain holes, notches and any other area that is difficult to reach by spray gun & where spraying may not be effective.

The colour of the stripe coat shall be different from the previous or subsequent coat.

Inorganic zinc primer coatings shall not be applied by brushing, not even for touch – up repairs.

7.3.2 Equipment for Brush Application

Brushes shall be of a style and quality that will permit proper application of coating. Round or oval brushes are most suitable for rivets, bolts, irregular surfaces and rough or pitted steel. Wide flat brushes are suitable for large flat areas. Brush width shall not be greater than 100 mm. No extension handles shall be used on brushes.

7.3.3 Procedure for Brush Application

Rounding of Edges and surface preparation shall be done as per Clause No. 6.3.1 of this Specification prior to brush application.

(a) Brushing shall be done so that a smooth coat, uniform in thickness, is obtained. There shall be no deep or detrimental brush marks.

(b) Paint shall be worked into all crevices and corners.

(c) All runs and sags shall be brushed out to prevent air pockets, solvent bubbles or voids.

(d) When applying solvent type, coatings, care shall be taken to prevent lifting of previous coats.

7.3.4 Finish Coat

An additional layer of finish coat shall be hand brushed at edges, corners, welds and hard-to spray areas to eliminate holidays in the final coats.

7.4 Spray Application

7.4.1 Equipment

(a) All equipment to be used for spray applications shall be inspected and tested before application begins.

(b) All equipment shall be maintained in good working order and shall be equal to that described in the manufacturer's instructions.

(c) All equipment shall be thoroughly cleaned before and after each use and before adding new material.

(d) An adequate moisture trap shall be installed between the air supply and each pressure pot. The trap shall be of the type that will continuously bleed off any water or oil from the air supply.

(e) Suitable pressure regulators and gauges shall be provided for both the air supply to the pressure pot and the air supply to the spray gun. Spray equipment and operating pressures shall comply with the recommendations of the manufacturer.

(f) The length of hose between the pressure pot and spray gun shall not exceed 15 m.

7.4.2 Procedures

(a) Pressure pot, material hose and spray gun shall be kept at the same elevation where possible. When spraying inorganic zinc, the elevation difference shall not exceed 3m.

(b) The spray gun shall be held at right angles to the surface.

(c) Each pass with the spray gun shall overlap the previous pass by 50%.

(d) The spray width shall not exceed 300 mm.

(e) All runs and sags shall be immediately brushed out or the surface re-coated.

(f) Large surfaces shall receive two passes (except when applying inorganic zinc) at right angles to each other (crosshatched).

(g) The coated surface shall be checked for chloride contamination before application of subsequent coating. When surface are encountered with chloride contamination soluble salt removers shall be used before application of coatings.

7.4.3 Airless Spray Equipment

(a) Airless spray equipment may be used for applying epoxy or aliphatic polyurethane coatings.

(b) The manufacturer's recommendations in selection and use of airless spray equipment shall be followed

7.4.4 Field Welds

No coating shall be applied within 150 mm of edges prepared neither for field welds nor to surfaces waiting non-destructive testing.

7.4.5 Tie/Sealer Coat

After application of Inorganic zinc Silicate primer, a tie/sealer coat of 25-35 µm DFT of polyamide epoxy on top of Primer coat shall be applied in order to seal porous surface of the zinc primer as per recommendation of the Manufacturer.

7.5 Roller Application

Roller application is permitted for paint materials where this is the manufacture's recommended method of application, such as for deck paints containing non-skid material. The manufacturer's recommended procedures shall be used.

7.6 Over spray and Drip Protection

Appropriate protection of buildings, structures and equipment from drips and spray and shall be provided to all equipment and facilities.

7.7 Safety Equipment

Appropriate safety equipment shall be provided for blasters, painters and other workers involved in the preparation and application of coating systems as per recommendation of paint manufacturer. Work areas shall be adequately ventilated.

7.8 Handling and shipping of coated items

Coated items shall be carefully handled to avoid damage to coated surfaces. No handling shall be performed before the coating system is cured to an acceptable level. Packing, handling and storage facilities shall be of non-metallic type.

8.0 REPAIR OF DAMAGED AREAS

All areas of paintwork that are locally damaged during transportation, handling or erection shall be repaired as specified below:

Prior to the application of any re-coat, damaged coatings shall be removed, preparing the surface and reapplying the protective coat(s).

Following steps of repair shall be followed for damaged painted / coated areas.

8.1 Repair Procedure for Damaged coating

Surfaces where coating is damaged after application of the finish coat shall be repaired as follows;

8.1.1 Top Coat

The top coat damaged, but base coat undamaged and the metal substrate is not exposed:

(a) Damaged coating shall be removed with a hand file and abraded back to the sound coating using emery paper or a fine grinder.

(b) The damaged area shall be wiped with a suitable solvent to remove debris. The periphery of repair area shall be feathered back for a minimum distance of 25 mm into the adjacent undamaged coating by light abrasion or grinding to produce a smooth chamfered surface profile.

(c) Apply a new topcoat as specified.

8.1.2 Base Coat

Coating damaged to base metal

(a) The damaged area greater than 0.2m^2 in area, the surface of exposed metal shall be prepared to the original specified standard prior to repairing by power tool cleaning as per SSPC-SP3 or spot blasting to SSPC-SP5 and applying primer, intermediate coat and final coat as specified. Alternatively, high solid surface tolerant epoxy coating may be used in place of primer & intermediate coats, followed by specified topcoat

(b) The damaged areas less than 0.2 m^2 in area may be repaired as per manufacturer's recommendation or by preparing the surface of exposed metal by power tool cleaning as per SSPC-SP11 to the original specified standard.

A primer, intermediate and final coat shall be applied as specified. Alternatively, high solid surface tolerant epoxy coating may be used in place of primer & intermediate coats, followed by specified topcoat. Brush application is acceptable. Even appearance and smooth feathering into surrounding coating in addition to correct dry film thickness and holidays must be achieved. Coating and surrounding repaired areas shall not be damaged and complete tie-in of the coating with surrounding areas shall be obtained. Zinc based products shall not be applied without Blast Cleaning to Sa 2 ½, instead Surface tolerant epoxy @ 100 microns shall be used as a primer in case blast cleaning is not possible or practical.

9.0 GALVANIZING

9.1 Galvanizing Standard

All steel grating, stair treads, handrails, ladders and other items specified as being galvanized shall be hot-dipped after fabrication in accordance with ASTM A123. Iron and steel hardware materials shall be hot-dip galvanized in accordance with ASTM A153.

9.2 Surface Preparation

All welding slag and burrs shall be removed. Surface contaminants and coatings, which cannot be removed by the normal chemical cleaning process in the galvanizing operation, shall be removed by abrasive blast cleaning.

Steelwork shall be prepared in accordance with the requirements of ASTM A123 and/or ASTM A153.

9.3 Zinc Coating Weight

The weight of zinc coating on structural items shall be **minimum 0.705 kg/m²** surface area. The composition of the zinc used in the galvanizing bath shall not be less than 98% zinc.

9.4 Surface Finish

The galvanized coating shall be continuous, adherent, as smooth and as evenly distributed as possible and free from any defect that is detrimental to the end use of the coated component. On 'silicon killed' steels, the coating may be dully gray, provided the coating is sound and continuous.

9.5 Welding

Galvanized members, that are to be permanently fixed by welding, shall be attached after the supporting members are primed, but before topcoats are applied. The heat affected area of the supporting structure shall be cleaned of all welding flux, the surface prepared and the coating reinstated. Grinding of edges prior to welding shall be permitted to reduce zinc oxide fumes formed during welding and eliminate the potential for weld porosity to occur. Adequate ventilation shall be provided and in confined spaces a respirator shall be used.

9.6 Damaged Items

All damage to galvanized items caused by fabrication, welding, handling and loading out of installation shall be reinstated.

9.7 Reinstatement of Damaged Surfaces

Repaired surfaces shall present the same appearance as adjacent galvanized areas and have approximately the same weathering characteristics as the galvanized surface.

9.8 Painting Galvanized Surfaces

All galvanized structural members shall be painted with coating system **13.7**. All galvanized surfaces shall be degreased, fresh water washed and treated as per SSPC SP1. It will then be sweep blasted and coated with surface tolerant epoxy – F8 as primer or compatible Epoxy primer followed by respective coating systems.

9.9 Painting Aluminium Helideck

a) For Blast Cleaning

All areas to be thoroughly cleaned with strong detergents and high pressure water to remove all traces of oil or grease. Aluminum oxide to be used as blasting media, abrasive size to be 60-80 mesh, surface profile 25-50 micron. Nozzle pressure to be 4-5 bars. During blasting nozzle movement should be continuous to avoid positional high density as aluminum is a soft metal.

For spot repairs after painting no spot blasting is recommended, positional rectification by mechanical tool cleaning shall be done. Part by part blasting and priming is preferable than one time area blasting and priming as it is difficult to identify oxidation over aluminum surface.

b) For Power Tool Cleaning

All areas to be thoroughly cleaned with strong detergents and high pressure water to remove all traces of oil or grease. 3M abrasive flap of 35or 60 meshes for electrical or pneumatic rotor grinding machine to be used for roughening the surface. It is preferable to prime the surface within 4-5 hours after cleaning to avoid oxidation over cleaned surface. Part by part blasting and priming is preferable than one time area blasting and priming as it is difficult to identify oxidation over aluminum surface.

9.10 Painting stainless steel (wherever applicable):

Stainless steel Handrails and ladders in the splash zone as specified in bid shall be coated as under:

i. Surface preparation: Sweep blasting with non-metallic and chloride free grit to obtain anchor profile of approximately 25 µm to 50 µm.

ii. Top coat: Two component epoxy- minimum 350µm DFT.

Also as per recommendation of paint manufacturer.

10.0 INSPECTION AND TESTING PLAN (ITP)

10.1 Quality Control

Procedures for testing and documenting quality control shall be prepared prior to the initial start up of any work covered by this specification & submitted to company for approval. The procedures shall include methods to assure the specification requirements are complied. Further, environmental conditions, surface temperature, coating applicator, surface(s) being coated and coating applied and status of required examinations and tests shall be documented.

Testing and inspection shall be carried out in accordance with Table-10.2. Surfaces shall be accessible until final inspection is carried out.

ITP shall include breakup of various activities, estimated time, duration for each activity. Time duration for painting activity wise is to be included in the in the planning package of the fabrication yard and adhere to the painting schedule during fabrication of structure in the yard.

Following inspection activities shall be included in the Inspection and Test Plan (ITP) furnished in Table 10.2.

10.2 Inspection and Testing requirements

Requirement for Inspection & Test Plan for Equipment, Pressure Vessels and Package items shall be followed as specified in respective Discipline Design Criteria or Specification

Table 10.2.1 - Inspection and testing requirement

Test type	Test Method	Test Frequency	Acceptance criteria	Consequence
Environmental conditions	Ambient and steel Temperature. Relative Humidity. Dew point.	Before start of each shift + minimum twice per Shift.	In accordance with specified requirements	No blasting or coating
Visual examination	Visual for sharp edges weld spatter slivers, rust grade, etc.	100 % of all surfaces	No defects, see specified requirements	Defects to be repaired
Cleanliness	a) ISO 8501-1 b) ISO 8502-3	a) 100 % visual of all surfaces b) Spot checks	a) In accordance with specified Requirements b) Maximum quantity and size rating 2	a) Re-blasting b) Re-cleaning and retesting until acceptable
Salt test	ISO 8502-6 and	Spot checks	Maximum	Repeated

	ISO 8502-9		conductivity Corresponding to 20 mg/m^2 NaCl for External surface and 50mg/m^2 for internal surface of Vessel.	washing with potable water and retesting until acceptable
Roughness	Comparator or stylus instrument (see ISO 8503)	Each component or once per 200 m^2	As specified	Re-blasting
Curing test (for Zn silicate).	ASTM D4752	Each component or once per 100 m^2	Rating 4-5	Allow to cure
Visual examination of coating	Visual to determine curing, contamination, solvent retention, pinholes / popping, sagging and surface defects	100 % of surface after each coat	According to specified requirements	Repair of defects
Holiday detection	NACE RP0188 and as per note -1 below	As per coating system specification	No holidays	Repair and retesting.
Film thickness	ISO 19840. Calibration on a smooth surface	ISO 19840	ISO 19840, and coating system data sheet	Repair, additional coats or recoating as appropriate
Adhesion	ISO 4624 using equipment with an automatic centred pulling force, and carried out when coating system are fully cured	Each component or once per 200 m^2	*See note-2 below	Coating to be rejected
Paint Material Inspection	Company appointed Inspector / representative shall physically verify paint material with respect to Test Certificate, data sheet, Batch Number, Date of Manufacture, date of expire and accept the Paint material.			
Surface preparation & Priming Coat application	Company appointed Inspector/ representative shall physically verify & record the surface preparation anchor pattern, record of surface salt acceptability and clear for priming coat application. Record of Temperature, Humidity and Dew point Temperature shall be recorded at the time of Start, at time interval Duration of Priming coating application and at completion time.			
Application intermediate coating	Company appointed Inspector/ representative shall carryout physical verification of coating application at random.			
Final coating application	Company appointed Inspector/ representative shall carryout physical verification of coating application at random.			
Final coating	Company appointed Inspector/ representative shall physically verify & record			

acceptance	coating thickness, other field test and accept the coating.

*Note:

1. Holiday Testing (Detection of pinholes)

   Holiday testing shall be conducted in accordance with NACE SP0188. For immersion and splash zone services, 100% of the coated areas shall be inspected for holidays. For atmospheric services, 10% of the coated areas, which include weld seams, corners, and edges, shall be holiday detected. Any holiday is unacceptable and shall be marked and repaired according to spot repair procedures. Lined vessel shall be marked clearly in black letters on vessel surface: "LINED VESSEL - HOT WORK PROHIBITED''.

   Pinhole testing for electrically non-conductive coatings shall be carried out in accordance with NACE RP0188 for splash zone coatings, internal tank coatings and the external coatings of buried tanks, vessels and piping.The high voltage technique shall be used; nominally set at 5 V/µm based on NDFT, or as agreed with the paint Manufacturer but not exceeding 25 kV in total. For coatings that are < 500 µm DFT, the wet sponge technique may be used if approved by the company.

2. Adhesion test

   Adhesion test shall be carried out on separate test plates, adhesion values in accordance with ISO 4624 shall be minimum 5.0MPa when using automatically centered test equipment.

10.3 **Role of Paint Vendor Representative**

The Contractor shall have to ensure that the Authorized technical–representative of the paint Manufacturer shall be qualified in accordance with NS 476 Inspector level II or certified as NACE coating inspector level II and physically inspects the painting / coating applications including surface preparation.

The Authorized technical representative of the paint Manufacturer shall have relevant knowledge of health and safety hazard, use of protection equipment, coating materials, mixing and thinning of coatings, coating pot-life, surface requirements, etc.

Paint manufacturers inspector /supervisor present in the fabrication yard shall be qualified according to NS 476 Inspector level-I, or certified as NACE coating inspector level-I and may carry out the inspection work under the guidance of an inspector level-II.

10.4 **Qualification of Contractor's QC personnel at Fabrication Yard (Inspectors, supervisors, foremen)**

The contractor's quality control inspector shall be qualified as a coating inspector in accordance with NS 476 Inspector level II or NACE level-II.

The contractor's quality control inspector shall qualify the tradesman level as blast-cleaner, painter, and applicator etc. for painting application.

The personnel shall have relevant knowledge of health and safety hazard, use of protection equipment, coating materials, mixing and thinning of coatings, coating pot-life, surface requirements etc.

Contractor shall carry out tests in accordance with the painting Specification, relevant Codes & standards for all coating systems applied as per clause 13.0 that are planned to be used for this project before commencement of painting work.

Contractor shall generate Painting application document & inspection report item wise specifying date & time of application for each step of painting with test record.

The test shall be carried out on a test panel in compliance to ISO 4624.

The acceptance criteria are the requirement to the visual and non – destructive inspection of the coating system described in this specification # section 10.2. Operators failing to meet the requirement shall not be allowed to carry out the work on this project.

10.5 **Qualification of passive fire protection operators**

Operators including pump machine operator, shall be qualified, trained and certified according to the manufacturer's procedures. Before any stud welding, the welders and the procedures to be used shall be qualified in accordance with the coating manufacturer's procedures.

If the operators or stud welders have not been working with the type of application or material within a period of 12 months, the applicator shall document that necessary supplementary training have been given before start of any work.

10.6 **Equipment and Material**

Materials, tools or equipment used in the surface preparation and coating applications, shall be inspected regularly and rejected if they do not comply with the Specification.

10.7 **Inspection of Instruments**

The following items shall be inspected using the inspection instruments listed below:

Instrument Item	Inspection Instrument
Surface Profile	Keane-tator Surface Profile Comparator or Testex Press-o-Film Elcometer 124 with 122 testex tape
Holidays	Tinker – Rasor Model M – 1

Surface Cleanliness	SSPC – Vis – 1
Viscosity	Zahn Viscometer or Ford Cup
Wet Film Thickness(WFT)	Nordson Wet Film Thickness Gauge Sheen WFT Gauge
Temperature & Humidity	Gardner Certified Hydrometer and Temperature Indicator
Surface Temperature	Pandux Surface Temperature Thermometer Elcometer Surface temperature gauge
Compressed Air Quality	Dry white cloth

10.7.1 **Calibration of Equipment**

Each test instrument shall be maintained and calibrated as prescribed by the manufacturer.

10.7.2 **Dry Film Thickness**

The dry film thickness of the coating system shall be determined in accordance with SSPC-PA2 or by a Microtest thickness gauge (ISO 2178) or comparable instrument in accordance with the following procedure.

(a) Ten readings shall be taken for every $10m^2$ of painted areas.

(b) 90% of all readings shall be within the specified dry film thickness.

(c) Where thickness accordance with the above procedure fall below the specified minimum an additional coat of the intermediate or finish coat shall be applied.

10.8 **Repair**

All work not done in accordance with this specification shall be redone or repaired as per coating system 13.10.

10.9 **Maintenance**

For maintenance of previously coated surfaces, the condition of the existing coating and the surfaces shall be checked using suitable methods, e.g. ISO 4628, to determine whether partial or complete repainting should be carried out. The type of surface preparation and protective paint system shall then be specified. The paint manufacturer should be consulted for recommendations. Test areas may be prepared to check the manufacturer's recommendations. Coating system 13.11 shall be followed for maintenance work.

10.10 **Role of Company Representative / Company appointed TPI /CA**

Refer Table 10.2

10.11 **Surface preparation, Coating application & field Test Report**

- Contractor shall generate Painting application & inspection report item wise (to be painted) for each activity of painting (surface preparation, priming, tie coat, 1st coat & final coat as applicable) specifying date & time of application. A daily record shall be maintained with following as minimum:

  (a) Air Temperature at the start, finish of the work and at intermediate stage.

  (b) Relative Humidity & dew point during application period (at interval of time).

  (c) surface roughness/ anchor profile,

  (d) Coating Material details & batch number (for traceability),

  (e) Salt test, chloride contamination

  (f) Dry film thickness applied & Identification of areas coated

  (g) Results of required field tests carried out.

  (h) Paint progress/ time lapse between coats.

These reports generated shall be documented in Construction Dossier in original and submitted to Company with PDF Copy.

11.0 PAINT MATERIALS:

The coating manufacturer shall provide a Coating System Data Sheet (CSDS) for each coating system to be used, containing at least the following information for each of the primer, intermediate coat (s) and topcoat product:

- Product name
- Colour
- Material type
- Batch numbers, Part A & B
- Manufacturing Date
- Shelf life
- Volatile organic compounds (VOCs) (g/l)

- Cleaning Solvent
- Thinner Type
- Maximum percentage of thinner content by volume
- Mixing ratio by volume or weight
- Application method
- Application temperature range (ºC)
- Application RH range (%)
- Induction time (minutes)
- Minimum recoat and dry to touch time (hours@ temperature)
- Maximum recoat time (days @temperature)
- Solids content by weight and/or volume
- WFT Range (µm)
- DFT Range (µm)
- Pot life (hours@ temperature)

Paint manufacturer shall furnish all the characteristics of paint materials on printed literature, along with the **test Certificate with actual test values of supplied batch for all the specified characteristics given in the specifications**.

Paint Material inward & traceability (location used) shall be maintained. These records shall be documented in Construction Dossier and submitted to Company.

All the paint materials shall be of first quality and conform to the following general characteristics are furnished below:

Table -11.1: Required **Characteristic of Paint Materials**

Description	Technical name	Type and composition	Volume of solids (approx.)	DFT (Dry film thickness) per coat (approx.)	**Theoretical covering capacity in M /coat/litre (approx.)**	Weight per litre in kgs/ litre (approx)	Touch dry at 25° C (approx.)	Hard dry at 25° C (approx.)	Over coating Interval (approx.)	Pot life (approx.) 25° C	Resistance to dry temperature
1	2	3	4	5	6	7	8	9	10	11	12
F-1	Epoxy MIO High build finish paint	Two pack polyamide cured epoxy, pigmented with lamellar micaceous iron oxide	80 ± 5%	100-125u	As per suppliers data	1.4	3 hours	Overnight	Min. : Overnight Max. : as recommended by manufacturer	2 hrs.	90°-120° C
F-2	Self -priming surface tolerant High build Epoxy coating	Two pack epoxy resin based suitably pigmented	80 ± 5%	100-125u	As per suppliers data	1.4	3 hours	24 hours	Min, :10 hours Max: as recommended by manufacturer	90 mints	90°-120° C
F-3	Inorganic zinc-silicate coating SSPC SP-20, Level-II, containing 80±3% zinc in dry film by weight & zinc as per ASTM D520 Type-II	A two pack air drying self - curing solvent based Inorganic Zinc silicate coating.	65 ± 5%	65-75u	8-9	2.3	30 mts	less than 6 hours	Min. 16-24 hours Max. as recommended by manufacturer	2 to 4 hours	400° C
F-4	Organic zinc rich epoxy coating SSPC SP-20, Level-II, containing 80±3% zinc in dry film by weight & Zinc as per ASTM D520 Type-II	Two pack epoxy resin based zinc primer	65 ± 5%	65-75u	As recommended by manufacturer	2.5	75 mts	less than 6 hours	Min. ; 6 hours Max. ; as recommended by manufacturer	5 hours	90-120° C

F-5	Heat resistant Aluminum paint based on Moisture Curing Silicone Binder	Single pack moisture cured silicon resin based medium with aluminum flakes	40±5 %	20-25u	As recommended by manufacturer	1.1	30 mts	1-2 hours	Min. ; 6-12 hours Max. ; as recommended by manufacturer	Not applicable	Upto 600° C
F-6	Epoxy Phenolic based Tank Lining (see note-1)	Two Pack Epoxy Phenolic Based	60±5 %	125 u-150µ	As per suppliers data	Min 1.4	5 hours	8 hours	Min. ; 8 hours Max. ; as recommended by manufacturer	As recommended by manufacturer	Upto 200° C
*F-7	Epoxy Passive fire protection (PFP)		100%	As per risk analysis Study report & as recommended by manufacturer							
F-8	Extra High Build Epoxy	Two pack polyamide/Polyamine cured epoxy resin	100± 5%	2500 µm	As per suppliers data	Min 1.9	8 hours	Overnight	**Min. 4 hrs Max. As recommended By manufacturer**	30 minutes	140° C
F-9	Glossy Aliphatic Acrylic Polyurethane Topcoat	Two Components Aliphatic Acrylic Polyurethane	60±5 %	65-80 µm	As per suppliers data	--	1.5 hours	6 hours	Min. 6 hours Max. ; As per suppliers data	1 hour	90° -120° C

F-10	Solvent free epoxy for Drinking / potable water tanks (see note-2)	A Two pack solvent free epoxy tank coating for drinking water	100%	250-300µm	As per supplier s data	--	6 hrs	12 hrs	Min. ; 10 hours Max. ; 40 hours	1 hour	Upto 60°C
F-11	Ultra High Build glass flake Epoxy	A Two component glass flake Epoxy coating	96%± 2%	600-1000 µm	As per supplier s data	--	Refer Product Data Sheet	Refer Product Data Sheet	Refer Product Data Sheet	Refer Product Data Sheet	90-100 °C
F-12	Pure epoxy pigmented with aluminium (see note-3)	A Two component Pure epoxy coating pigmented with aluminium	60%± 2%	100-200 µm	As per supplier s data	---	3 hours	6 hours	7 hours -14 days	2.5 hours	---
F-13	Low volume solid epoxy primer As recommended by paint manufacturer	A Two pack epoxy solvent free epoxy tank coating for drinking water	47%± 2%	40 µm	As per supplier s data	---	30 minutes	8 hours	As recommended by supplier	8 hours	--
F-14	Sealer coat TSAC As recommended by pain manufacturer	Used as Sealer for TSAC Operating temperature >120˚C	30%	15-25 µm	--	---	20 minutes	2 hours	As recommended by supplier	--	--
F-15	One coat of fast curing solvent less UV -resistant polyester resin (Isophthalic based reinforced with glass flakes.	Glass flake reinforced unsaturated Polyester coating	96%± 2%	600-1500 µm	As per supplier s data	--	2 hours	2 hours	As recommended by supplier	45 minutes	90-100 °C

Note:

1. IMO PSPC COT Resolutions MSC. 288 (87)

2. APAS Approved to specification 0213, 2974F & 2974P / UL Certified in accordance to NSF/ANSI 61. [In case of water tank constructed using stainless steel SS316L, tank internal coating is not required.]

3. Approved for PSPC for water ballast tank according to IMO Res. MSC 215 (82)

12.0 APPROVED VENDOR'S RECOMMENDED LIST OF PAINT MATERIALS

Table-12.1: **Approved Vendor's** Paint products

Description	Area of application	Berger Paints Ltd., India	Shalimar Paints, India	Carboline, USA (CDC) Carboline, India	**Sigma Coatings (PPG Coaings), USA**	Hempel, Denmark	Amercoat (PPG Coatings), USA	Jotun Coatings, Norway / Jotun India Pvt Ltd	KCC Corporation, Korea	International Brand (AKZO NOBEL), UK	Chogoku paints, Singapore	Excel India Paints, India
1	2	3	4	5	6	7	8	9	10	11	12	13
F-1 Epoxy MIO High Build finish	Exterior surface of steel <120° C	Epilux 455 HB MIO	epiguard XL	Carboguard 893 MIO	sigma cover 410 (MIO)	Hempadur Mastic 45880/1/W / 47550	Amercote 383 HS / 410 (MIO)	Penguard Midcoat MIO	Korepox EH2350	Interseal 547/ Intergard 475 HS	Univan MIO	EXLGUARD 495
F-2 Self-priming surface tolerant High build Epoxy	Maintainance Painting	Barger Protectomastic RPL	epiplus 56	Carbomastic 615	Sigma Cover 630 /620	Hempadur Mastic 45880/1/W / 47550	amerlock 400 / 400C	Jotamastic 80/Jotaprime mastic 80	Korepox Primer EP 1760	Interseal 547/ Interseal 670 HS	Umeguard HS	EXLGUARD 680
F-3 Inorganic zinc silicate coating SSPC SP-20, Level-II, containing minimum 80% zinc in dry film by weight & zinc as per ASTM D520 Type-II	Primer coat Relative Humidity> 60%	Zinc anode 304 Coating	tuffcote zilicate -IZS	cabozinc 11	Sigma Zinc 158	Hempel's Galvosil 15780	dimetcote 9	Resist 78/Resist 804A	Galvany IZ 180 (N)	Interzinc 2280	Galbon S-HB	EXLZINC 113

Description	Area of application	Berger Paints Ltd., India	Shalimar Paints, India	Carboline, USA (CDC) Carboline, India	**Sigma Coatings (PPG Coaings), USA**	Hempel, Denmark	Amercoat (PPG Coatings), USA	Jotun Coatings, Norway / Jotun India Pvt Ltd	KCC Corporation, Korea	International Brand (AKZO NOBEL), UK	Chogoku paints, Singapore	Excel India Paints, India
1	2	3	4	5	6	7	8	9	10	11	12	13
F-4 Organic zinc rich epoxy coating SSPC SP-20, Level-II,containing minimum 80% zinc in dry film by weight & Zinc as per ASTM D520 Type-II	Primer coat When Relative Humidity< 60% & Primer for repair work	Epilux ZR HB Primer	Epigard 4	Carbozinc 858	Sigma zinc 109 HS	Avantguard 1736 G	Amercoat 68 HS	Barrier 80	Korepox Zinc Rich Primer EZ 175	Interzinc 52	-	EXLZINC 108
F-5 Heat resistant Aluminum paint based on Silicone Binder	Temperature >400° C For equipments	Lumerous HR143	Iustotherm 600	Thermaline 4700 aluminium	Sigmatherm540	Hempel's Silicon aluminium 56914	Amercoat 878 HS	Solvalitt	Yeolcoat QT 606	Intertherm 50	-	EXLTHERM HR600
F-6 Epoxy Phenolic based Tank Lining	Internal lining of Diesel & crude oil tank	Epilux 9 HB Phenolic Coating	HB finish	Phenoline 187 VOC	sigma guard 730	hempadur 85671	Amercoat 90 HS	Tankguard storage	Kophenol EH2630	Interline 850	-	EXLLINI 965
*F-7 Epoxy (PFP) (See note-2 below)	Protection of platform from pool and jet fire	-	-	Pyroclad X1	PITT-CHAR XP	Hempafire Pool 200	PITT-CHAR XP	Jotachar JF 750/Jotachar 1709	Firemask 3200	Chartek-7/ Chartek-7E/ Chartek-1709	-	EXLINTUMESCENT 447

Description	Area of application	Berger Paints Ltd., India	Shalimar Paints, India	Carboline, USA (CDC) Carboline, India	**Sigma Coatings (PPG Coaings), USA**	Hempel, Denmark	Amercoat (PPG Coatings), USA	Jotun Coatings, Norway / Jotun India Pvt Ltd	KCC Corporation, Korea	International Brand (AKZO NOBEL), UK	Chogoku paints, Singapore	Excel India Paints, India
1	2	3	4	5	6	7	8	9	10	11	12	13
F-8 Extra High Build Epoxy	Steel plated helidec k,walk way,es cape route & laydow n area	-	-	Carboguard 1207	Sigmashield 1090	Hempel's Spray Guard 35490/3	Tideguard 171	Jotacote UHB	Korepox H.S.EH3200	Interzone 485	-	EXLGUARD-991UHB
F-9 Glossy Aliphatic Acrylic Polyuretha ne Topcoat	Finish coat for UV protect ion & require d colour	Bergerthane Acrylic PU Coating	Shalithane HB rich	Carbothane 134HG	Sigmadur 550 / 585	Hempathane topcoat 55910/55610/55210	Amercoat 450 Series	Hardtop XP/Futura OS	Korethan Topcoat UT6581	Interthane 990/ Interthane 990 SG	-	EXLTHANE 570
F-10 Solvent free Epoxy	Interna l lining of potabl e water tank	Epilux 155 SF	-	Carboguard 703	Sigma Guard CSF 650 / 585	Hempadur 35530/35531	Amercoat 391 PC	Tankguard 412	Korepox H.B.EH3100	Interline 925	-	EXLLINE-989SF
F-11 Ultra High Build glass flake Epoxy (Splash Zone)	Protect ion of structu re in splash zone	Steel shield 1100	Tuffkote Epishield ST glass flake epoxy	Carboguard 1209	Sigmashield 905 / 880GF	Hempadur Multistrength 35870	Amerlock 400GF	Marathon XHB	Korepox H.B.EH3200 (GF)	Interzone 1016	Permax no.3000 S	EXLGUARD- 890GF

Description	Area of application	Berger Paints Ltd., India	Shalimar Paints, India	Carboline, USA (CDC) Carboline, India	**Sigma Coatings (PPG Coaings), USA**	Hempel, Denmark	Amercoat (PPG Coatings), USA	Jotun Coatings, Norway / Jotun India Pvt Ltd	KCC Corporation, Korea	International Brand (AKZO NOBEL), UK	Chogoku paints, Singapore	Excel India Paints, India
1	2	3	4	5	6	7	8	9	10	11	12	13
F-12 Pure epoxy pigmented with aluminium	Internal lining of ballast water/ salt water tank	-	-	Carbomastic 615A	Sigmacover 280	Hempadur uniq 47741/ 4774 D	Amercoat182ZP	Jotacote universal alu./ Jotacote universal N10	Korepox EH2030	Intershield 300	-	EXLGUARD- 343
F-13 Low volume solid epoxy primer As recommended by paint manufacturer	Used as Sealer/ Tie coat for TSAC & Over Zinc silicate primer Operating temperature <120˚C	-	-	Carboguard 893ZP	Sigmacover 522/246	Hempadur 05990/15553/15570	Amercoat 182 ZP/71 TC	Penguard tie coat 100 /Penguard primer	Korepox Holding Primer EP 1700	Intergard 269	-	EXLGUARD- 247
F-14 Aluminum silicone sealer/tie coat As recommended by paint manufacturer	Used as Sealer for TSAC Operating temperature >120˚C	-	-	Thermaline 1248 E	Sigmatherm 540	Hempel's Silicon aluminium 56914	Amercoat 878	-	-	Intertherm 50/ Intertherm 972	-	EXLTHERM-530

Description	Area of application	Berger Paints Ltd., India	Shalimar Paints, India	Carboline, USA (CDC) Carboline, India	**Sigma Coatings (PPG Coaings), USA**	Hempel, Denmark	Amercoat (PPG Coatings), USA	Jotun Coatings, Norway / Jotun India Pvt Ltd	KCC Corporation, Korea	International Brand (AKZO NOBEL), UK	Chogoku paints, Singapore	Excel India Paints, India
1	2	3	4	5	6	7	8	9	10	11	12	13
F-15 Glass flake Polyester resin	Splash zone to bottom of Cellar deck (Maintenance Painting)	---	---	---	-	Hempel's Polyster GF -35920	Amercoat -4800	Baltoflake	------	Interzone-762	----	EXLGUARD 891GF

Note:

1. The paint materials indicated in the table are indicative only and may be updated. Bidder is to select & procure latest paint material from the list of the recommended list of paint manufacturers meeting the DFT requirement of paint.

2. *Epoxy (PFP) installed on the platform shall be Tough, durable and resistant to impact and vibration damage. Thickness of coating shall be based on risk analysis study report and fire safety measures to be taken to mitigate hydrocarbon and jet fire as per ISO 22899 jet fire certification up to 2 hours and NORSOK M-501. FIRE RATING shall implement based on safety study carried out in compliance to API 14J & API RP 2A.

3. Contractor to verify the latest status of availability & applicability of products before consideration and report their intention of change within the above listed vendors with proper reasons.

13.0 PROTECTIVE COATING SYSTEMS

13.1 SCOPE

Manufacturer's recommendations and latest product of paint shall be used to suit the environment conditions and for the intended purpose shall be followed. Care must be taken to achieve the specified dry film thickness indicated in the bid. **Items to be painted covered by these specifications are as follows:**

13.2 **Section A:** New Construction of offshore structure.

13.2.1 **<u>Coating System No.1</u>**: Atmospheric Zone-I: Thermally sprayed aluminium coating shall be applied for items having service life of 10 years or longer which are :

13.2.1.1 exposed to seawater & not easily accessible for inspection or maintenance:-

i) structural Steel items, piping, piping supports etc**. above splash zone** upto & including **Bottom plating of cellar deck**,

ii) Conductors, Pump & Sump Caisson.

iii) bottom framing beams & plating of Main Deck

iv) bottom chord of Bridge including bottom of plating , piping .piping supports etc

v) Boom of pedestal crane

vi) Steel framing supporting helideck above main deck and upto and including bottom of steel helideck.

TSAC is not required for bottom of aluminium pancake.

13.2.1.2 Exposed to operating temperature > 120 °C as heat resistance paint.

i) Flare Stack/vent boom

13.2.1.3 :Table for TSAC System

Application	Surface Preparation	Coating System	DFT
<u>Coating System No.1A</u> All carbon steel surfaces exposed to operating temperature > 120 °C. • Flare Stack/vent boom.	Cleanliness: SSPC –SP5/ Sa 3 Surface Profile: Minimum 75 µm	Thermally sprayed Aluminium or alloys of aluminium with sealer coat as top coat. *Sealer: (F-14)	225 µm (Min value 200 µm & Max value 250 µm) For sealer, see Note- 1
<u>Coating System No.1B</u> All carbon steel surfaces exposed to atmosphere & operating temperature < 120 °C. • Jacket above splash zone up to and including Cellar Deck bottom surface including bottom of deck plating, primary & secondary beams etc. • Main deck bottom surface including bottom of deck plating, primary & secondary beams etc. • Bottom chord of bridge connecting offshore platforms including	Cleanliness: SSPC-SP5/ Sa 3 Surface Profile: Minimum 75 µm	Thermally sprayed aluminium or alloys of aluminium with sealer, intermediate coat and final coat as top coat *Sealer: (F-13) *Intermediate Coat (F-1) *Final Coat(F-9)	225 µm (Min value 200 µm & Max value 250 µm) See Note- 1 : Sealer Coat Note- 2: Intermediate Coat Note- 3: Final coat

bottom of deck plating • Steel framing supporting helideck above main deck and upto bottom of helideck below aluminium pancake & bottom of steel helideck plating • boom of pedestal crane			

*Note-1: All metallized surfaces shall be sealed in accordance with the following requirements: The sealer shall fill the metal pores. It shall be applied until absorption is complete. There should not be a measurable overlay of sealer on the metallic coating after application.

The materials for sealing the metal coating shall be:

- Two-component epoxy for operating temperatures below 120 °C (F-13).
- Aluminium silicone above 120 °C (F-14).

Volume solids in the sealers when applied shall be 15 % nominal. The overlay of sealer should be less than 38 µm (1.5 mil) on the TSA coating after application.

The sealer should have a contrasting color to the TSA to aid visual inspection.

*Note-2: Intermediate coat over TSAC shall be applied as per coating system no.2

*Note -3: Final coat TSAC shall be applied as per coating system no.2

Application of Thermally Spread Aluminium Coating (TSAC) is specified in Clause No. 14.0.

- Repair, field coating of pipes and coating of in-fill steel

All requirements, including adhesion, applicable to metal spraying, shall apply.

The treating and handling of the substrate shall be done in such a manner that the product in its final condition will have a continuous and uniform coating. Before the metal spraying operation starts, the area 30 cm to 40 cm in distance from the weld zone shall be sweep-blasted to ensure that all contamination is removed. The uncoated welding zone shall be blast Cleaned as specified for coating system no. 1.

13.2.2 <u>Coating System No.2</u>: Atmospheric zone - II except for zone specified under coating system no. 1

Application Area	Surface Preparation	Coating System	Coating DFT Range
Carbon steel with	Cleanliness:	Primer Coat: Inorganic zinc silicate	65 - 75 µm

operating temperature < 120 °C – Structural steel, Exteriors of equipment, vessels, piping and valves (not insulated) in Atmospheric Zone **except for zone specified under coating system no. 1**	Near white metal shot/grit blasting as Per SSPC-SP-10 or Sa 2 ½ of SIS-05-5900 /ISO 8501-1 surface profile : 40-60 µm Max.	primer (F-3)	
		Sealer Coat/tie coat: Low volume solid epoxy (F-13) as recommended by paint manufacturer	25 - 30 µm
		Intermediate coats: Two coat of high build epoxy (F-1) with MIO pigments	100 – 125 µm (each coat)
		Finish Coat: One Topcoat of Gloss Aliphatic Acrylic Polyurethane (F- 9)	65 - 80 µm
		Total DFT µm	355-435 µm

Notes:

1. This system is applicable for carbon steel surface with operating temperature less than 120^{o} C.

2. Colour for final coat shall be as per colour code.

13.2.3 Coating System No.3: Protection of Carbon Steel in Submerged & Splash Zone

Application	Surface Preparation	Coating System	Coating DFT Range
Coating System No.3A: Submerged Zone • PLEM, SSIV and Riser clamps, etc.	Cleanliness: Near white metal shot/grit blasting as Per SSPC-SP-10 or Sa 2 ½ of SIS-05-5900/ISO 8501-1 Surface profile: 40-60 µm Max.	two coat epoxy based primer with aluminium pigment (F-12)	100-125µm (each coat) Total 200 µm min.
Coating System No.3B: Splash zone: Jacket, Boat landing, Barge Bumper, conductors, Riser & Conductor Protector, clamps, etc.	Cleanliness: Near white metal shot/grit blasting as Per SSPC-SP-10 or Sa 2 ½ of SIS-05-5900/ISO 8501-1 Surface profile: 75-100 µm Max.	Two coats of Ultra High Build Glass Flake Epoxy Coating (F- 11)	750 µm (each coat) Total 1500 µm min.

13.2.4 Coating System No.4: Galvanizing

All grating, handrails, ladders, safety cages shall be hot dip galvanized as per ASTM 123 & misc. hardware shall be hot dip galvanized ASTM A153. Galvanizing shall preferably be done after cutting shaping and welding has been done. The weight of zinc coating for grating, handrails, ladders, safety cages shall be minimum **705 g**m per square meter **(coating thickness of 100** µm).
The weight/thickness of zinc coating for iron and steel hardware shall be as per ASTM A153.

The galvanized surface shall be free from bare spots, dross pimples, Lumpiness and runs, flux inclusions and slag inclusions.

13.2.5 Coating System No.5: Painting of All Galvanized items (as per 13.2.4) except gratings & Aluminium Helideck shall coated as below:

Application	Surface Preparation	Coating System	DFT in µm
All Galvanised items and Aluminium Helideck painting for marking, lettering, etc.	SSPC SP1 & etched chemically with etch primer or physically by sweep blast cleaning with non-metallic and chloride free grit to SSPC SP7/Sa1. Cleaning with alkaline solution followed by hosing with fresh water as per SSPC SP1. surface profile : 25-50 µm	Low volume solid epoxy primer suitable for aluminium & galvanized surface. (F-13)	Minimum 50 µm
		Intermediate coat: Two coat of high build epoxy with MIO pigments. (F- 1)	100 µm (each coat)
		Finish Coat: One coat of aliphatic polyurethane. (F-9)	Minimum 75 µm
			Minimum DFT 325 µm

13.3 SECTION B: New Construction: Process vessels and piping, valves, manifolds, pumps, heat exchangers and related equipment

13.3.1 All submersible type pumps protective coating system shall be as per manufacturer's standard which shall be suitable for Marine environment. Holiday test shall be carried out on protective coatings of all equipment as per NACE SP 0188.

13.3.2 Piping & other equipment

For Piping & other equipment made up of Carbon Steel Material, Protective coating system shall be followed as per Clause No. 13.2.2.

13.3.3 Flanges

Flanges on equipment to be installed on offshore platforms shall be coated with the protective coating system given in 13.2.2 except for the seal surface. Care shall be taken to prevent damage of ring and gasket seal surface by covering during the blasting and coating. After the coating has cured, an anti-rust grease coating shall be applied to protect the seal surface. Apply a minimum dry film of an anti-rust grease coating as per MIL-C-161730 Amendment, 2, Grade 1 Qualified Products List.

A corrosion control material (such as inhibitor grease) should be installed correctly initially after the flange has been assembled on the offshore structure. The bolt holes and flange crevices shall be sealed off completely from the offshore corrosive environment.

Commercially available viscous sealants, such as soft and low-molecular-weight polyolefin sealants or corrosion inhibitor greases, should be injected into the flange crevice. The sealants are very hydrophobic. A polymeric tape or SS tape should then be wrapped around the flange to enclose the flange crevice completely.

Petrolatum or wax tapes may be used for ambient to moderate temperature service applications in accordance with the product manufacturer's specification.

For maintenance, the entire flange should be overwrapped with a commercial petrolatum or wax tape, which is made of polyester felt impregnated with very hydrophobic petrolatum or wax to provide encapsulation against water. Visual inspection cannot be conducted without unwrapping the tape. However, the flange shall be wrapped again after the inspection. Tape shall be replaced if the encapsulation is broken.

All Painted Nut bolts used for flanged/bolted connection shall be protected from corrosion by providing Protective caps with inhibited grease as per recommendation of the vendor.

13.4 **SECTION C: COMPRESSORS, GAS TURBINES, GENERATORS, ENGINES, MOTORS, INSTRUMENTS AND PANEL BOARDS**

Equipment in covered/enclosed areas shall be coated as per Equipment manufacturer's standards coating suitable for **offshore environment** as specified in respective Specification wherever provided in Bid, otherwise Protective System given in 13.2.2 shall be followed.

All skid surfaces which can be cleaned by blasting shall be coated with one of the protective system given in 13.2.2. Protect critical moving parts, bearing surfaces, machined surfaces, nameplates, glass next to machinery, etc. from entrance of blasting dust and damage from blasting by masking with polyethylene sheeting and masking tape. Equipment in enclosed areas shall be protected with the same paint systems as for exterior surface. Panel boards shall be disassembled before cleaning and coating.

13.4.1 When equipment manufacturer's coating is unsatisfactory and blast cleaning is not permissible. Clean and protect as follows:

a. Remove coating by use of a paint and varnish remover.

b. Surface preparation; solvent cleaning (SSPC-SPI) followed by power tool cleaning (SSPC-SP3 or St3) or of SIS-05-5900.

c. One coat of (F-2) Self priming surface tolerant epoxy coating @ 125 Microns DFT/Coat minimum (maximum DFT-150 microns)

d. Top Coat – 2 coats (F-1) High build epoxy MIO coating cured with Polyamide/polyamine hardener @ 125 microns DFT/Coat minimum (maximum DFT – 150 microns/coat) = 2 X 125 =250 microns.

e. Finish Coat- One Topcoat of (F- 9) Gloss Aliphatic Acrylic Polyurethane 60-80 micron dry film thickness

Total DFT: Minimum: 435 microns, Maximum: 530 microns.

13.4.2 When equipment manufacturer's shop primer is sound and blast cleaning is not permissible, clean and top coat as follows (where finish coat has to be done at the yard):

a. Surface preparation: Solvent cleaning (SSPC-SPI), followed by power tool cleaning (SSPC-SP) or St 3 of SIS-05-5900.

b. Apply one coat of (F-2) Self priming surface tolerant epoxy coating @ 125 Microns DFT/Coat minimum (maximum DFT-150 microns)

c. Top coat: as per 13.4.1 (d) above.

13.4.3 Painting System applicable for Equipment having surface temperature between 110°C (230°F) and 400° C (752°F), clean and protect as follows.

a. Surface Preparation: Near white metal shot/grit blasting as per SSPC-SP-10 or Sa 2 ½ of SIS-05-5900/ISO 8501 –1:2001 with a 40 – 60 microns anchor pattern. Items with metal to metal joints shall be seal welded prior to blasting and coating when possible or dismantled following all fittings and welding to obtain blasting and coating of the joint surfaces

b. Apply 1 coat inorganic zinc silicate primer coating (F-3) @ 65-75 Microns DFT minimum (maximum – 75 microns).

c. Apply 2 coats (F- 5) of Heat resistant silicon aluminum finish paint @ minimum 20 microns DFT per coat = 2 X 20 = 40 microns (maximum 2 X 25 = 50 microns).

Total DFT: 105 Microns minimum: 125 microns maximum.

Note : coating having operating temperature less than 120 degree shall be coated as specified in respective Discipline Spec.

13.4.4 Equipment with surface temperature above 400°C (752°F),

a. Surface Preparation: Near white metal shot/grit blasting as per SSPC-SP-10 or Sa 2 ½ of SIS-05-5900/ISO 8501 –1:2001 with a 40 – 60 microns anchor pattern. Items with metal to metal joints shall be seal welded prior to blasting and coating when possible or

dismantled following all fittings and welding to obtain blasting and coating of the joint surfaces

b. Apply 2-3 coats of Heat resistant silicon aluminium finish paint (F-5) (maximum 3 X 20 µm – 25 µm).

Total DFT: 60 µm minimum: 75 µm maximum.

13.5 SECTION D: Internal Surfaces of Tanks and Process Vessels/ Piping/SPM

13.5.1 Coating System No.6: Coating of Internal Surfaces of Tanks and Process Vessels/Piping/SPM

Application	Surface Preparation	Coating System	DFT in µm
System No.6A: Salt Water Service (ballast tanks/internal sea water filled compartments)	Cleanliness: Near white metal shot/grit blasting as Per SSPC-SP-10 or Sa 2 ½ of SIS-05-5900/ISO 8501-1 Surface profile: 60-85µm Max.	*Two coats of Epoxy pigmented with aluminium (F-12) (See Note-3)	150 µm (each coat) Total Min DFT 300 µm.
System No.6B: Hydrocarbon Services: all internal surfaces of Welded tanks handling hydrocarbons/diesel.	Cleanliness: Near white metal shot/grit blasting as Per SSPC-SP-10 or Sa 2 ½ of SIS-05-5900/ISO 8501-1 Surface profile: 50-85µm Max.	*Two coats of Phanolic Epoxy (F-6) (See Note-1)	125 µm – 150 µm (each coat) Total Min. DFT 250 µm.
System No.6C: Fresh Water Services: all internal surfaces of Welded tanks handling portable water having NSF Certificate	Cleanliness: Near white metal shot/grit blasting as Per SSPC-SP-10 or Sa 2 ½ of SIS-05-5900/ISO 8501-1 Surface profile: 50-85µm Max	Two coats of Solvent free Epoxy. (F-10) (See Note-2)	250 µm – 300 µm (each coat) Total Min DFT 500 µm

*Note-1: Certified to NSF / ANSI Standard.

*Note-2: Paint shall be tested and certified according to AS/NZS4020:2005 or UL classification in accordance with ANSI/NSF Standard for potable water.

Note-3: Paint shall be for PSPC for water ballast tank according to IMO Res .MSC 215.

Note-4: Paint System 6C: Potable Water Tank made up of SS316, internal surface painting is not required.

Note-5: Paint System 6B: Pressure vessel internal painting shall be followed as per respective Spec of the item provided in the bid. In case, respective specification do not have specific requirement then Paint System 6B is to be followed.

Note-6: Internal Surfaces of Tanks shall be coated as per intended use specified in this specification and respective Discipline Specification provided in the Bid.

13.6 **SECTION- E**: **Carbon steel Metal buildings**

13.6.1 **Equipment Skid**

Structural Steel: Method of surface preparation and acceptable coatings shall be followed as per Coating system No.2 Section – A (13.2.2).

Piping/Mechanical & Process: As per Section – B (13.3)

13.6.2 Structural Framing/Sheeting/Plating: Method of surface preparation and acceptable coatings shall be followed as per Coating System No.2

13.6.2.1 Exposed portions: Method of surface preparation and acceptable coatings shall be followed as per Coating system No.2 except Walkways, escape routes and lay down areas.

13.6.2.2 Unexposed portions covered by insulation, wall Panels, false roofing/ceiling and false flooring:

Coating System No.7: **Painting of Unexposed** portions

Application	Surface Preparation	Coating System	DFT in
Portions covered by insulation, wall Panels, false roofing and false flooring)	Cleanliness: Near white metal shot/grit blasting as Per SSPC-SP-10 or Sa 2 ½ of SIS-05-5900/ISO 8501-1 surface profile: 40-60 µm Max.	Primer Coat: Inorganic zinc silicate primer (F-3)	65 -85 µm
		Sealer Coat/tie coat: One coat of low volume of epoxy Primer (F- 13) as recommended by manufacturer.	25-30 µm
		Finish coat: One coat of high build epoxy with MIO pigments. (F- 1)	125 – 150 µm.
		Total DFT (Minimum)	Total Min DFT 215 µm

13.7 **Section F - Antiskid Painting**

Coating system is applicable on **Steel plated Helideck**, Plated **Laydown Area**, Plated **Walkway**, and applicable area on **plated floor**.

Coating System No.8: **Antiskid Painting**.

Application	Surface Preparation	Coating System	DFT in µm
Topcoat of **Steel plated Helideck, Laydown Area, Walkway (Plated)** & **Plated flooring**	Cleanliness: SSPC –SP5/ Sa 3 Roughness: 75 µm to 100µm	**Extra high** build epoxy coating. Application as recommended by paint Manufacturer. (F-8) Top coat: one coat of Aliphatic acrylic polyurethane (F-9) (Refer General Note 1 to 4 below)	2500-3000 µm 50-60 µm
		Total DFT	2550-3060 µm

General notes:

1. Light colour non-skid aggregates shall be used.
2. Quantity and Particle size of non-skid aggregate shall be as per recommendation of paint manufacture.
3. Aggregates shall have a uniform spread.
4. Coating system No.8 shall have adequate properties related to water absorption, impact resistance, coefficient of friction (**at least 0.65 in wetted condition**), hardness and flexibility.

13.8 Section G : Passive Fire Protection

The recommended use for this product is for fireproofing of steel beams below cellar deck, Deck Legs, Jacket legs & braces, pipes, vessel skirts, bulkheads etc. for fire rating as per Fire Safety Study Report.

The epoxy intumescent coating shall provide pool and jet fire protection minimum one hour for well platform & two hours for Process platform.

Coating thickness shall be as per recommended by paint Manufacturer to obtain desired Fire rating based on safety study.

Base coat & Top coating on top of the passive fire protection shall be compatible with the passive fire protection coating. No hot work shall be done after fireproofing material is installed. Procedure for application shall be as per manufacturers' quality and application manual.

13.8.1 ***Coating System No.9**: Epoxy Based Passive Fire Protection system (Epoxy Intumescent Coating)

Application	Surface Preparation	Coating System	DFT in µm
Carbon steel – Structural steel	Cleanliness: Near white metal shot/grit blasting as Per SSPC-SP-10 or Sa 2 ½ of SIS-05-5900/ISO 8501-1 surface profile : 40-60 µm Max.	Primer Coat: Inorganic zinc silicate primer (F-3)	65 -85 µm
		Sealer Coat/tie coat: Low volume epoxy primer as recommended by paint manufacturer (F-13)	25-30 µm
		Top Coat: Thickness of Fire proofing coating shall be provided by Manufacturer for the desired Fire	

		Rating (as per safety study). Coating material F-7.	

*Note: Epoxy Intumescent Coating shall be applied over a suitably primed substrate as per Recommendation of paint manufacturer. Procedure for application shall be as per Manufacture's Quality & application manual.

Epoxy Intumescent Coating shall be top coated to meet colour schemes & finish and as per recommendation of paint manufacturer.

The fire ratings of Epoxy Intumescent Coating shall be obtained from Certification Agencies like UL, LRS, DNV, BV, etc. and submitted.

No hot work shall be carried out after completion of Fire proofing coating.

13.9 **Deleted**

13.10 **Section H**: **Miscellaneous items**

13.10.1 Pipe Support: I-Rod Pipe Support System

Corrosion Control in Piping Support & anchoring Steel U- Bend shall be carried out as per NACE SP 0108 (Standard Practice – Corrosion Control of Offshore Structure by Protective Coating).

Pipe Support shall also be so designed that there is no water accumulation beneath the pipe to avoid corrosion and deterioration of paint. Design of pipe support shall be such that there shall be enough space for recoating even at the bottom of pipe. One of method is to use semi sphere-shaped **Thermoplastic Pipe Support** Rod under the pipe to facilitate water drainage for future corrosion inspection and access to recoat. Alternative option is to overwrap the pipe wherever the pipe is being supported with a layer of fiber glass reinforced composite tape with adhesive backing. Alternatively contractor may also propose other alternative accordingly.

The U-bend shall be protected with a carbon black pigmented polychloroprene rubber sheath. The rubber sheath shall also provide abrasion resistance. Corrosion inhibitor grease should be applied onto the steel U-bend prior to inserting in the rubber sheath.

Contractor shall address all such aspect in Piping support standard and put up to Company for approval.

13.10.2 **Coating System No.10**: Painting of carbon steel valves.

Application	Surface Preparation	Coating System	DFT in µm

Painting of carbon steel valves with operating temperature up to 150°C.	Cleanliness: Near white metal shot/grit blasting as Per SSPC-SP-10 or Sa 2 ½ of SIS-05-5900/ISO 8501-1 Surface profile: 40-60 µm Max.	Two coats of Phenolic Epoxy (F- 6)	150 – 175 µm (each coat) Total Min DFT 300 µm

13.11 <u>Section I</u>: Extension & Modification on Old Platform and **Repair & Maintenance painting of offshore platform**

i. Extension & Modification on Existing Platform Section A to H shall be followed with following change: Thermally Spread Aluminum Coating (TSAC) System is not applicable.

ii. Repair & Maintenance Painting.

13.11.1 Evaluation of condition monitoring

The Principal will review the condition monitoring reports. Facility components found to be Inaccessible for coating condition inspection should be reported to the responsible Technical Authorities and Management. The barrier coating should be maintained within Two (2) years after the condition of the barrier is found to be not in accordance with the Minimum acceptance criteria. If repair work has not been carried out in time or if severe (Local) corrosion has been observed during condition monitoring, the responsible disciplines should be consulted on further actions to ensure safety and technical integrity.

The condition of all the components of an area/block, including insulation and passive Fireproofing, should be reviewed when drafting scopes of work for maintenance. Consideration should be given to whether it is justified to repair the coating barrier of a Single component only, or whether the whole area/block should be maintained. Specialist Knowledge, including insight into costs and techniques, may be required to make the choice between repair and large-scale maintenance. Specific consideration should be given to and thorough analysis made of areas and/or parts that cannot be inspected or maintained. Appropriate actions, such as risk analyses etc., to ensure long-term integrity should be instigated.

VISIBLE CORROSION DEGREES

The pictures given in ASTM D 610 can be used for the visual determination of the percentage of surface Rusting. These pictures are based on standards such as the European Scale of Degree of Rusting and ASTM D 610.

13.11.2 **:** Maintenance Painting system (applicable for protective coating of wellhead/ process/ LQ platform projects) shall be as under:

Maintenance Protective Coating System	Area of application	Surface Preparation	Surface Profile	Product	DFT
Maintenance Protective Coating System no.1	From 1.0 meter below MSL to bottom of cellar deck framing (excluding underside of cellar deck framing, piping, pipe support, equipment) Including Jacket, Boat landing, Barge Bumper, conductors, Conductor Protector, riser clamps, conductors, conductor guides, sump/pump caissons and all other structural members. Design life for this system shall be 10 years.	SSPC-SP-10 / Sa 2 ½	50-75µm	One coat of fast curing solvent less UV -resistant polyester resin (lsophthalic based reinforced with glass flakes.(F-15)	750 µm
Maintenance Protective Coating System no.2	a) All items from bottom of cellar deck framing (including underside of cellar deck framing & plating (both sides) to top of topside including conductors & conductor guides , external surface of Pressure vessel, X-mas' tree and steel cable trays & structural items including existing handrails except helideck top, walkways, gratings, new handrails, Vent boom, barrier/fire wall facing well head area, crane exhaust b) All piping , equipment & piping supports above & below cellar deck framing c) All other items below cellar deck which are not included in Maintenance Protective Coating System no. 1. Design life for this system shall be 5 years.	SSPC-SP1 to be followed by hand/power tool cleaning SSPC-SP2/ SSPC-SP3.	-	a) 1 coat of 2 component surface tolerant polyamine cured self-priming epoxy @ 100-125µm/coat. (F-2) b) 1 coat of 2 component polyamide cured high build epoxy coating @ 100-125µm/coat.(F-1) c) 1 coat of Acrylic Aliphatic Polyurethane finish paint @ 50-75 µm/coat.(F-9) **Total DFT** **Note:** Carbon Steel Flange and external side of valves shall be painted by Brush applied stripe coat of 2 component surface tolerant polyamine cured self-priming epoxy coating at all corners, edges, pit, welds followed by application of (b) & (c) above .	100-125 µm 100-125 µm 50-75µm 250-325 µm
Maintenance Protective Coating system no.3	Carbon steel floorings For Walkways & Steel Helideck (NACE SP 0108) including markings and lettering.	SSPC-SP-10 / Sa 2 ½	50-75 µm	1 coat of Glass Flake epoxy with anti-skid system as crushed flint / anti slip beads @ 600-650 µm.(F-11)	600-650 µm

				1 coat of Acrylic Aliphatic PU @ 50-75 µ.(F-9) **Total DFT:** **Colour scheme shall be conforming to CAP 437 offshore helicopter landing areas**.	50-75 µm **650-725 µm.**
Maintenance Protective Coating system no.4	Top coat of fire/barrier walls	SSPC-SP1 to be followed by hand/power tool cleaning SSPC-SP2/ SSPC-SP3.	-	Under epoxy based fire protection system 1 coat of Surface Tolerant Epoxy @ 100 µm.(F-2)	100 µm
				Intumescent paint as per UL 1709. (F-7) Top Coat: Thickness of Fire proofing coating shall be provided by Manufacturer for the desired Fire Rating.	As recommended by Manufacturer Please see Note-3 below.
Maintenance Protective Coating system no.5	Heat resistant paint for flare boom, pedestal crane exhaust 120°C to 400 °C.	SSPC-SP-10/ Sa 2 ½	40-60 µm	Apply 3 coats of Heat resistant silicon aluminium finish paint. (F-5) **Total DFT:**	20-25 µm (each coat) **60-75 µm**
Maintenance Protective Coating system no.6	Carbon Steel Threaded Joints, Flange Joints, Nuts & Bolts.	The surfaces, crevices and corners should be cleaned of rust and dust, loose paint etc. by manual cleaning and compressed air. After cleaning the flange joints, nuts and bolts shall be further protected by **anticorrosive polytetrafluoroethylene (PTFE)/Synthetic tapes** of size suitable for the flanges and studs of the platform as per recommendation of manufacturer.			Not Applicable

Note: 1. Paint manufacturers to be asked to provide their best suitable paint as per the scheme made for the environment and as per warranty and paint life required by ONGC.

2. For any coating scheme, all the coats will be from same manufacturers to avoid any compatibility issues.
3. Epoxy Intumescent Coating shall be applied over a suitably primed substrate as per Recommendation of paint manufacturer. Procedure for application shall be as per Manufacture's Quality & application manual. Epoxy Intumescent Coating shall be top coated to meet colour schemes & finish and as per recommendation of paint manufacturer. The fire ratings of Epoxy Intumescent Coating shall be obtained from Certification Agencies like UL, LRS, DNV, BV, etc. and submitted. No hot work shall be carried out after completion of Fire proofing coating.

13.12 **Miscellaneous Accessories**

Three identification boards, with name of the platforms shall be provided on North and South faces of the platforms and on the top of the helideck respectively. The details to be written on the boards shall be approved by the Company. The letters on the boards shall be at least 900 mm in size.

13.13 **Handling and shipping of coated items**

Coated items shall be carefully handled to avoid damage to coated surfaces. No handling shall be performed before the coating system is cured to an acceptable level. Packing, handling and storage facilities shall be of non-metallic type.

14.0 APPLICATION OF THERMALLY SPRAYED ALUMINIUM COATING (TSAC)

14.1 TSAC REQUIREMENTS

14.1.1 Surface Preparation

All the parts to be sprayed shall be degreased according to SSPC-SP 1. The absence of oil and grease after degreasing shall be tested by method given elsewhere in the specification. Thereafter the surface to be abrasive blasted to white metal finish as per NACE 1/SSPC-SP 5 for marine and immersion service. Using SSPC VIS 1, it is to be visually assessed that the blast cleaned surface meets requirement of SSPC-SP 5. Thereafter clear cellophane tape test as per ISO 8502-3shall be used to confirm absence of dust on the blasted surface. Finally blasted surface shall be tested for presence of soluble salts as per method ISO8502-9. Maximum allowable salt content shall be considered 50mg/M^2. (5 micrograms/cm^2) .In case salt content exceeds specified limit, the contaminated surface shall be cleaned by method as per Annex –C of IS 12944-4 (Water Cleaning.) After cleaning the surface shall be retested for salt content after drying.

The blasting media shall be either chilled iron or angular steel grit as per SSPC-AB-3 of mesh size G-16 to G-40. Copper or Nickel slag or Garret as abrasive will also be suitable having mesh size in the range of G16 to G24, conforming to SSPC-AB-1. Mesh size shall be required as appropriate to the anchor tooth depth profile requirement and blasting equipment used. The blasted surface should be having angular profile depth not less than 75 microns with sharp angular shape. The profile depth shall be measured according to NACE standard RP 0287 (Replica Tape) or ASTM D 4417 method B (Profile depth gauge).

For manual blasting one profile depth measurement shall be taken every 10-20 M^2 of blasted surface.

Surface preparation shall be completed in one abrasive blast cleaning operation wherever possible.

If rust bloom (visual appearance of rust) appears on the blast cleaned surface before thermal spraying, the affected area shall be re-blasted to achieve specified degree of cleanliness after which only thermal spraying shall commence. Air blasting pressure at nozzle shall be normally maintained at 100 psi. Air pressure and media size should be reduced and adjusted to preclude damage/distortion to thin gauge materials. Blasting time on work piece should be adjusted to only clean the surface and cut required anchor tooth with minimum loss of metal. Blast angle should be as close to perpendicular as possible but in no case greater than 30° from perpendicular to work surface. Blasting media must be free of debris, excessive fines, contaminants such as NaCl and sulphur salts (Ref. Clause 14.4.6.1 of this spec.).

14.1.2 Blasting Equipment

The TSAC applicator shall use mechanical (centrifugal wheel) or pressure pot blast cleaning equipment and procedures. Suction blasting equipment shall not be used. Clause No. 14.4.6.2 shall be used to validate clean and dry air.

14.1.3 Feed Stock

The feed stock shall be in the form of wire. The feed stock shall be 99.5% aluminium of commercial purity grade, its composition shall be in accordance with requirement of BS 1475 or ASTM B833 (wrought aluminium and aluminium alloys, wire).

14.1.4 Thickness Requirement

The nominal thickness of finished TSAC shall be 225 microns having minimum value of 200 microns at low thickness areas (valleys) and not more than 250 microns at peak areas.

The finished thickness shall be measured using SSPC-PA 2 type 2 fixed probe gauge (Magnetic Gauge).

14.1.5 Coating Bond Strength Requirement

During CPT coating bond strength requirement shall be as indicated in table 14.2.1.3.

The Minimum Tensile Bond Requirements during production shall be 7 Mpa (1000 psi) for any single measurement.

The TSC tensile bond shall be measured according to ASTM D 4541 using a self-aligning adhesion tester.

For non-destructive measurement: Tensile force shall be measured to the 7Mpa (1000 psi). The tensile force shall then be reduced and the tensile fixture Removed Without damaging the TSAC.

14.1.6 Porosity

All thermally sprayed metallic coatings will have porosity. For thermally sprayed aluminium coatings porosity shall not exceed 15% of total surface area for flame sprayed coating and 8% for arc spray coating.

14.2 THERMAL SPRAY APPLICATION PROCEDURE

Thermally spread Aluminium coating shall be applied by either Flame spray or Arc spray method and shall refer to Clause No. 13.2.1 for application area/zone.

14.2.1 Equipment set up

14.2.1.1 Thermal spray equipment shall be set up calibrated, operated (1) according to manufacturer instructions/technical manuals and also TSAC applicators refinement thereto and (2) as validated by Procedure Qualification.

14.2.1.2 Spray parameters

Spray parameters and thickness of each crossing pass shall be set and shall be validated with bend test.

14.2.1.3 Table:

Spray Parameters	Method of Application	
	Arc wire Spray	Flame Wire Spray
Arc voltage	27 V	-
Air pressure	80 psi	80 psi
Steel surface cleanliness	NACE-1 white metal	NACE-1 white metal / or Near white metal
Steel surface profile	75 microns (minimum)	75 microns (min.)
Arc current	250-280A	-
Coating thickness	225 microns (nominal)	225 microns(Nominal)
Coating adhesion	> 1500 psi (Total coating),see 3.1.4	>1000 psi
Coating porosity	Less than 8%	Less than 15%.
Spray distance (spray Gun work piece)	6-8”	5-7”
Spray Pass width	40mm	20 mm

The above parameters to be validated with a bend test by the contractor before start of work (for detail of Bend Test see clause No. 14.4.5 of this Spec.)

14.2.2 **Post Blasting Substrate condition and thermal spraying period.**

14.2.2.1 The steel surface temperature shall be at least 5°C above dew point of ambient air temperature. Steel substrate surface temperature shall be recorded by with a contact pyrometer.

Thermal spraying should commence within 15 minutes from the time of completion of blasting

14.2.2.2 Holding Period

Time between the completion of final anchor tooth blasting and completion of thermal spraying of blasted surface should be no more than four hours.

14.2.3 Pre-Heating

For flame spraying, the initial starting area of 1-2 square feet to be preheated to approx.120° C to prevent condensation of moisture in the flame on the substrate. For arc spraying the preheating is not required.

14.2.4 Thermal spraying

Spraying should commence only after validation of equipment set up by procedure qualification test and spray parameter validation tests described in Clause No. 14.5 and 14.4.5 respectively. **Thermal spraying must commence within 15 minutes from the time of completion of blast cleaning.**

The specified coating thickness shall be applied in several crossing passes. The coating bond strength is greater when the spray passes are kept thin. Laying down an excessively thick spray pass increases the internal stresses in TSAC and decreases the bond strength of total TSAC. The suitable thickness for crossing passes shall be determined by procedure qualification test.

For manual spraying, spraying to be done in perpendicular crossing passes to minimize thin spots in coating. Approx. 75-100 microns of TSAC shall be laid down in each pass.

The surface geometry of the item or area to be sprayed should be inspected before application. The spraying pass and sequence shall be planned according to following.

- Maintain Gun as close to perpendicular as possible and within ± 30°C from perpendicular to the substrate.

- Maintain nominal stand-off distance and spray pass width as given below:

Table 14.2.4.1:

Spray method	Standoff (Inches)	Spray pass width (Inches)

Arc Wire	6 – 8	1 ½ inch (40mm)
Flame wire	5 – 7	¾ inch (20mm)

14.2.4.1 Rust Bloom (Visual appearance of rust or Discoloration)

If Rust bloom appears on the blasted surface before thermal spraying, the affected area shall be re-blasted to achieve the specified level of cleanliness.

If Rust bloom in form of discoloration, or any blistering or a degraded coating appears at any time during application of TSAC, then spraying shall be stopped and acceptable sprayed area shall be marked off. The unsatisfactory areas shall be repaired to the required degree of surface cleanliness and profile.

Blast the edges of the TSAC to provide for 2-3" feathered area overlap of the new work into existing TSAC.

Then apply TSAC to the newly prepared surfaces and overlap the existing TSAC to the extent of feathered edge so that overlap is a consistent thickness.

14.2.4.2 Masking

Masking all those parts and surfaces which are not required to be thermally sprayed as instructed by purchaser should be inspected by applicator to ensure that they are properly marked and covered by purchaser.

Complex geometries (flanges, valve manifolds, intersections) shall be masked by applicator to minimize overspray i.e. TSAC applied outside spray parameters (primarily gun to substrate distance and spray angle).

14.2.4.3 TSAC Finish

The deposited TSAC shall be uniform without blisters, cracks, loose particles, or exposed steel as examined with 10 X magnification.

14.3 SEALER

Sealant shall be applied after satisfactory application of TSAC and completion of all testing and measurements of the finished TSAC as per clause No.14.4 of this specification.

For shop work Sealer shall be applied immediately after thermal spraying and for field work sealer shall be applied within 8 hours. The sealcoat shall be thin enough to penetrate into the body of TSAC.

The materials for sealing the metal coating shall be:

- Two-component epoxy for operating temperatures below 120 °C (F-13).
- Aluminium silicone above 120 °C (F-14).

Volume solids in the sealers when applied shall be 15 % nominal. The overlay of sealer should be less than 38 µm (1.5 mil) on the TSA coating after application.

The sealer should have a contrasting colour to the TSA to aid visual inspection.

During application of seal coat, complete coverage shall be ensured by visual observation.

14.4 TESTING AND MEASUREMENT SCHEDULE

14.4.1 Surface Finish

That the blasted cleaned surface meets the required criteria (NACE 1/SSPC-SP 5) shall be visually inspected using SSPC-VIS 1. The clear cellophane-tape test shall be used to confirm absence of dust or foreign debris on the cleaned surface.

14.4.2 Blast Profile Measurement: (In-Process testing during actual production before application of TSA coating)

The angular profile depth measurement shall be done by profile tape as per method NACE Standard RP 0287 or ASTM D 4417 method B (Profile depth gauge micrometre). Spot measurement shall be carried out every 15m^2 of blasted surface. At each spot three measurements shall be taken over an area of 10 cm^2 and average of measurements to be recorded and reported.

If profile is <75 microns blasting shall continue till greater than 75 microns depth profile is achieved.

14.4.3 TSAC Thickness (In-Process Testing For finished coating during regular production)

14.4.3.1 TSAC finished thickness shall be measured using SSPC-PA 2 type 2 fixed probe gauge.

14.4.3.2 For flat surfaces, measurements shall be taken along a straight measurement line; one measurement line for every 15 m^2 of applied TSAC shall be selected along which 5 measurements to be taken at 25 mm internal and average to be reported.

14.4.3.3 For curved surface or complex geometry, 5 measurements shall be taken at a spot measuring 10 cm^2 in area. One spot to be taken for every 15 m^2 of applied TSAC area.

14.4.3.4 The TSAC thickness in surface changes or contour changes, welds and attachments shall be also measured and reported.

14.4.3.5 If TSAC is less than specified minimum thickness, apply additional TSAC until specified thickness range is achieved.

14.4.3.6 All locations and values of TSAC thickness measurements shall be recorded in Job Record (JR).

14.4.4 Tensile Bond Testing (In-Process testing for finished coating during regular production)

- Tensile Bond strength of the TSAC finish coat shall be determined according to ASTM D 4541 using a self-aligning adhesion tester.
- One measurement shall be made every 50 m2. If tensile bond at any individual spot is less than 1000 psi for flame sprayed coating and 1500 psi for arc sprayed coating the degraded TSAC shall completely remove and reapplied.
- The tensile bond portable test instrument to be calibrated according to ASTM C 633.

14.4.5 Bend Tests

Bend test shall be carried out at beginning of each work shift. Bend tests shall also be conducted on sample coupons before start of thermal spraying work to qualify the following as mentioned earlier in this specification.

- To qualify spray parameters and thickness of each crossing pass.

14.4.5.1 Test Procedure

a) Five corrosion control steel coupons each of dimension 50 mm x 150 mm x 1.25 mm thick are to be prepared.

b) Surface shall be prepared by dry abrasive blast cleaning as per this specification.

c) TSAC shall be applied as per specified thickness range. TSAC should be sprayed in crossing passes laying down approx. 75-100 microns in each pass.

d) TSAC applied coupons shall be bent 180° around a 13 mm diameter mandrel.

e) Bend test shall be considered passed if, on bend radius there is

- No cracking or spalling or lifting by a knife blade from the substrate.
- Only minor cracking that cannot be lifted from substrate with a knife blade.

f) Bend test fails if coating cracks with lifting from substrate.

14.4.6 Tests for blasting media, blasting air

14.4.6.1 Blasting Media (For every fresh batch of media and one random test during blasting)

a) Blasting Media shall be visually inspected for absence of contamination and debris using 10 X magnification.

b) Inspection for the absence of oil contamination shall be conducted using following procedure:

- Fill a small clean 200 ml bottle half full of abrasive.
- Fill the bottle with potable water, cap and shake the bottle.
- Inspect water for oil film/slick. If present, the blasting media is not to be used.

c) Soluble salt contamination if suspected shall be verified by method ASTM D 4940 if present media is to be replaced.

d) Clean blasting equipment, especially pot and hoses and then replace blasting media and retest.

14.4.6.2 Test for Blasting Air (Once Daily before start of blasting and once at random during blasting)

The air for blasting shall be free from moisture and oil. The compressor air shall be checked for oil and water contamination per ASTM D 4285.

14.4.7 Test for presence of oil/grease and contamination

The steel substrate after degreasing as per SSPC-SP 1 shall be tested as per following procedure to validate absence of oil and grease contamination.

a) Visual inspection - Continue degreasing until all visible signs of contamination are removed.

b) Conduct a solvent evaporation test by applying several drops or a small splash of residue- free trichloromethane on the suspect area especially pitting, crevice corrosion areas or depressed areas. An evaporation ring formation is indicative of oil and grease contamination.

Continue degreasing and inspection till test is passed.

14.5 TSAC APPLICATOR QUALIFICATION

Following tests to be carried out as part of procedure qualification test for the applicator.

- Thickness measurement
- Coating bond strength
- Porosity test
- Bend strength

TSAC applicator's surface finishing and application process and equipment set up, calibration and operation shall be qualified by application of TSA on a reference sample which shall be used as a comparator to evaluate the suitability of application process. Only that applicator will be permitted to carry out the work when test specimens coated by the applicator meets the desired requirements as cited below.

The sample shall be made of a steel plate measuring approx. 18"x 18" x ¼" thick. If the actual work is less than ¼" thick then the sample to be made from material of representative thickness.

The surface preparation, TSAC application shall be made with actual field equipments and process/spray parameters and procedures as per the specification. The depth profile of blasted surface, TSAC coating thickness for each cross pass and total thickness range shall be as per specification.

The surface preparation and thermal spraying shall be carried out in representative environmental conditions spraying with makeshift enclosure.

14.5.1 After application of coating, thickness and tensile bond measurements shall be made in following manner.

- Divide the sample piece into four quadrants.
- Measure thickness along the diagonal line of each quadrant.
- Four each quadrant five in-line thickness measurements at 1" interval shall be done using SSPC-PA 2 type 2 fixed probe gauge. Thus a total of four "five in line" thickness measurements are to be done for the whole sample.
- One tensile bond measurement using ASTM D 4541 type III or IV portable self-aligning test instrument to be done at centre of each quadrant. Total of 4 measurements for the sample.

- One porosity evaluation test by Metallographic examination shall be conducted to demonstrate the achievement of porosity within the limits specified. Sample shall be prepared for Metallographic examination as per ASTME – 3.

- The procedure shall be considered qualified if thickness and tensile bond strength and porosity values meets the specification requirement.

14.5.2 Bend test**:** Bend test shall be carried out as detailed at Clause No. 14.4.5 of this specification.

- Applicators thermal spray equipment set-up, operation and procedure of application including in-process QC checkpoints adopted during procedure qualification as described above should be always subsequently followed during entire duration of work.

14.6 Documentation

The following information shall be provided by TSAC applicator before award of work.

- TSAC application process consisting of equipment capabilities and their technical parameters, feed stock material and source of procurement.

- Detailed application procedure and in-process quality control check points for (a) surface preparation (b) thermal spraying (c) seal coats.

- Type and specification of instruments to be deployed for measurement of blast profile depth, TSAC thickness and tensile bond.

- Paint manufacturer data sheet for the selected sealing coat to be applied.

14.7 **RECORDS**

- The TSAC applicator shall maintain job record to record production and QC information. All the results of the tests and quality control checks shall be entered in the record for each component/part thermally sprayed. All the result of tests (thickness, tensile bond, bend tests) and other validation tests (e.g. procedure qualification test, test for surface cleanliness after abrasive blasting, test for cleanliness of abrasives and air) shall also be recorded and duly signed by owner.

- All the information mentioned in Clause no.14.6 above should also form part of the Job record.

- Any modification affected after procedure qualification in the procedure, QC, spray parameter, equipment spec to the original information (submitted before award of the work) must also form part of Job record.

14.8 **WARRANTY**

The TSAC applicator shall warrant the quality of material used by providing the purchaser with a certificate of materials used to include

- Spray feed stock: Alloy type/designation, Lot Number, wire diameter, chemical analysis, name of supplier, manufacturer.

- Sealant: Name of manufacturer, application data sheet.

14.9 **SAFETY**

The TSAC applicator shall follow all safety procedures required by the purchaser/owner. Owner shall also give compliance requirement to be followed by applicator. The applicator shall follow all appropriate regulatory requirements.

14.10 **CODES AND STANDARDS**

This specification shall apply in case of conflict between specification and following applicable Codes & Standard:

AWS C.2.17	Recommended Practice for Electric arc Spray.
ASTM C 633	Test Method for Adhesive/Cohesive Strength of Flame Sprayed Coatings.
ASTM D 4285	Method for indicating Oil or Water in Compressed Air.
ASTM D 4417	Test Method for Field Measurement of Surface Profile of Blasted Steel.
BS 2569	Specification for Sprayed Metal Coating.
NACE Standard RP 0287	Field Measurement of Surface Profile of Abrasive Blast Cleaned Steel Surfaces Using a Replica Tape.
ASTM D 4541	Test method for Pull-Off Strength of Coating Using Portable Adhesion Testers.
ANSI/AWS C2.18	Guide for the Protection of Steel with Thermal Spray Coatings of Aluminium, Zinc and Their Alloys and Composites.
NACE No. 12/AWS C2.23M/SSPC-CS	Specification for the application of thermal spray coatings (Metallizing) of aluminium, zinc, and their alloys and composites for the corrosion protection of steel.
SSPC Publication	The inspection of coatings and linings: A Handbook of Basic practice for Inspectors, Owners, and Specifiers.
SSPC-AB 1	Mineral and Slag Abrasives.
SSPC-AB 3	Ferrous Metallic Abrasives.
SSPC-PA 1	Shop, Field, and Maintenance Painting of Steel.
SSPC-PA 2	Measurement of Dry Coating Thickness with Magnetic Gages.

NACE No. 1/SSPC-SP 5	White Metal Blast Cleaning.
NACE No. 2/SSPC-SP 10	Near –White Metal Blast Cleaning.
SSPC-VIS 1	Guide and Reference Photographs for Steel Surfaces Prepared by Dry Abrasive Blast Cleaning.

14.11 Suggested List of TSA applicators

1) *Metallizing Equipment Corporation, Jodhpur, India*
2) *Larsen & Toubro Ltd (Eutectic division), New Delhi, India*
3) *Chemtreat India Ltd, Navi Mumbai, India*
4) *Deepwater MPE, Malaysia*
5) *Associated Thermal Spray, Ahmedabad, India*
6) *M/S PS Coatings Pvt. Ltd, New Mumbai, India*
7) *M/S NPCC, Abu Dhabi*
8) *M/S McDermott, Jebal Ali, Dubai*
9) *M/S PTSC, Vietnam*

TSAC coating shall be applied to bottom of bridge including bottom of deck plating of bridge & piping.

Other applicators will also be acceptable if they qualify the applicator qualification procedure given in the specification. If any one of the above suggested applicator is selected, even then he has to perform the qualification test before commencing of the work.

15.0 COLOUR SCHEDULE

15.1 Table No.

Sr. No.	Item to be Painted	Finish Coat Colour	Colour Code No.
A	**SPLASH ZONE**		
1	All areas coated with XHB Glass flake reinforced Epoxy	Golden Yellow	RAL 1004
2	Handrails & Ladders		
B	**ATMOSPHERIC ZONE-I & II**	Golden Yellow	RAL 1004
1	All areas coated with System 2	Golden Yellow	RAL 1004

	(Structural steelwork)		
2	Handrails and ladders	Golden Yellow	RAL 1004
3	Buildings (outside surfaces)	Spanish Tile Red	RAL 3000
4	Overhead Obstructions	Golden Yellow with Black Strip	RAL 1004 & RAL 9012
5	Helideck top surface (marking colour shall be as per CAP437 ,(7th Edition) guideline)	Dark Green	RAL 6020
6	Walkway of Helideck	Dark Green with White side strips	RAL 6020+ RAL NO.9010
C	**EQUIPMENT, PIPING AND TUBING**	Natural Grey	RAL 7038
1	Pumps and Drivers	Dark Grey or Factory Finish	RAL 7039
2	Compressors and Drivers	Dark Grey or Factory Finish	RAL 7039
3	Generators and Drivers	Dark Grey or Factory Finish	RAL 7039
4	Water Makers	Light Blue or Factory Finish	RAL 5012
5	Tanks and Vessels	White	RAL 9003
6	Electrical Equipment incl. Cable Trays	Aluminium Grey	RAL 9007
7	Electrical Danger Points	Orange	RAL 2004
8	Instrumentation	Factory Finish	--------
9	Fire Fighting Equipment	Red	RAL 3000
10	First Aid Equipment	Signal Green	RAL 6032
11	Risers and J Tubes	Aluminium Grey	RAL 9007
12	Process Piping Systems	Refer Table 15.2 of this specification	

The name tag no. and service of all equipment should be stencilled in capital letters 40 – 50 mm in height on the equipment. The Colour shall be black or white and in contrast to the colour on which the lettering is superimposed.

15.2 Pipe Colour Coding

All piping shall be colour coded to identify service in accordance with the following criteria. The colour coding shall comply with the schedule in Table 15.

TABLE No. 15.2: PIPE COLOUR CODING (PROCESS) COLOUR SCHEME FOR PRODUCT BEING PROCESSED

Sr. No.	Description	Colour code Identification	
		Base/Ground Colour (Colour code no.)	Colour Bands 1)1st Band 2)2nd Band
GASES			
1	Sour Gas	Canary Yellow (RAL	1)Grey(RAL NO.7000)

		NO.1012)	2)Dark Violet (RAL NO.5022)
2	Sweet Gas	Canary Yellow (RAL NO.1012)	Grey (RAL NO.7000)
3	Fuel Gas	Canary Yellow (RAL NO.1012)	1)Light Brown(RAL NO.8025) 2)White(RAL NO.9010)
4	Residue	Canary Yellow (RAL NO.1012)	1)Oxide Red(RAL NO.3009) 2)White(RAL NO.9010)
5	Flare Gas	Canary Yellow (RAL NO.1012)	Aluminum(RAL NO.9006)
6	Regenerated Gas	Canary Yellow (RAL NO.1012)	1)White(RAL NO.9010) 2)Dark Violet(RAL NO.5022)
LIQUIDS			
1	Crude Oil	Light Brown (RAL NO.8025)	No Band
2	High Speed Diesel	Light Brown (RAL NO.8025)	1)Brilliant Green(RAL NO.~~6038~~, 6017) 2)White(RAL NO.9010)
3	Hydrocarbon Condensate	Dark Admiralty Grey(RAL NO.7031)	1)Signal Red (RAL NO.3020) 2)Black(RAL NO.9005)
4	Chlorine	Dark Violet(RAL NO.5022)	Signal Red(RAL NO.3020)
5	Triethyline Glycol (TEG)	Dark Admiralty Grey (RAL NO.7031)	1)Sea Green (RAL NO.6018) 2)Yellow (RAL NO.1003)
6	Oily Water	Black (RAL NO.9005)	No Band
7	Dosing Chemical	Dark Admiralty Grey (RAL NO.7031)	No Band
8	ATF	Dark Admiralty Grey(RAL NO.7031)	Black(RAL NO.9005)
UTILITY ITEMS: WATER			
1	Drinking Water	Sea Green (RAL NO.6018)	1)French Blue (RAL NO.5002) 2)Signal Red(RAL NO.3020)
2	Domestic Hot Water	Sea Green (RAL NO.6018)	Light Green(RAL NO.6021)
3	Cold Water from Storage Tank	Sea Green (RAL NO.6018)	1)French Blue (RAL NO.5002) 2)Canary Yellow(RAL NO.1012)
4	Demineralized Water	Sea Green (RAL NO.6018)	Gulf Red(RAL NO.8019)
5	Cooling Water	Sea Green (RAL NO.6018)	French Blue (RAL NO.5002)
6	Condensate Water	Sea Green (RAL NO.6018)	1)Light Brown (RAL NO.3009) 2)Signal Red(RAL NO.3020)
7	Wash Water	Sea Green(RAL NO.6018)	Canary Yellow(RAL NO.1012)

8	Fire water	Fire Red(RAL NO.3000)	Crimson Red(RAL NO.3007)
9	Sea Untreated water	Sea Green(RAL NO.6018)	White(RAL NO.9010)
10	Waste Water	Sea Green (RAL NO.6018)	1)Canary Yellow(RAL NO.1012) 2)Signal Red(RAL NO.3020)
11	Treated Water	Sea Green (RAL NO.6018)	Light Orange(RAL NO.2003)
UTILITY ITEMS: COMPRESSOR			
1	Compressed Air (upto 15kg/sq.cm)	Sky Blue (RAL NO.5015)	-----
2	Compressed Air (above 15kg/sq.cm)	Sky Blue(RAL NO.5015)	Signal Red(RAL NO.3020)
3	Plant Air	Sky Blue(RAL NO.5015)	Silver Grey(RAL NO.7001)
4	Instrument Air	Sky Blue(RAL NO.5015)	French Blue(RAL NO.5002)
5	Drainage	Black(RAL NO.9005)	------
UTILITY ITEMS: OILS			
1	Lubricating Oil	Light Brown(RAL NO.8025)	Light Grey(RAL NO.7004)
2	Hydraulic Power	Light Brown(RAL NO.8025)	Dark Violet(RAL NO.5022)
UTILITY ITEMS: GASES			
1	Acetylene	Canary Yellow (RAL NO.1012)	Service Brown(RAL NO.8022)
2	Nitrogen	Canary Yellow(RAL NO.1012)	Black(RAL NO.9005)
3	Oxygen	Canary Yellow(RAL NO.1012)	White(RAL NO.9010)
4	Carbon di Oxide	Canary Yellow(RAL NO.1012)	Light Grey(RAL NO.7004)

Note: Any product which is not covered will be governed as per colour code standard IS: 2379/other standard under reference.

15.3 Colour Bands for Identification & Size of pipe

15.3.1 Identification & Size

The Width of colour band shall confirm to the following (As per IS 2379)

Sr. No.	Nominal Pipe Size	Width : L (mm)
1	80NB and below	25
2	Over 100 NB upto 150 NB	50
3	Over 200 NB upto 300 NB	75
4	Over 350 NB	100
The relative proportional width of first colour band to be subsequent bands shall be 4:1. Maximum stripe spacing shall be 150mm		

15.3.2 Application

Color Bands shall be applied at the following location as per # 7.3 of IS: 2379

- Battery Limit points
- Intersection point and change of direction points in piping ways
- Other point such as midway of each piping way, near valves, junction joints of service appliances, walls on either side of pipe culverts
- For long stretch yard piping at 50 m interval and at start and terminating points

Color bands shall be arranged in the sequence shown in table below and sequence follows the direction of flow.

For Insulation pipes, nominal pipe size means the outside diameter of insulation.

15.4 **Colour of Valves**

The Valves shall be painted as per As per IS: 2379

Table No. 15.4

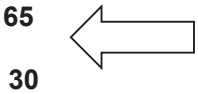
Sr. No.	Condition	Colour of Valves
1	Valves on Fire fighting	Red (RAL NO.3000)
2	Valves on Pipe carrying Hazardous Material	Golden Yellow with black strips (RAL NO.1004) & (RAL NO.9005)
3	Valves on Water carrying piping (Potable or Non-Potable)	French Blue(RAL NO.5002)
4	All Other Piping	Same colour as the main piping

15.5 SIZE OF FLOW DIRECTION ARROWS

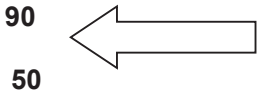
Pipes shall be marked with arrow to indicate flow direction as per the isometrics. Arrow markers shall be placed at starting / termination point of piping above each floor generation at a height of about 1.5m from floors. The service or the origin place shall also be marked in the

same colour, stenciled in capital letters 40 - 50 mm in height. The colour or arrow shall be black or white and in contrast to the colour on which they are superimposed. The arrow indicating flow directions as per piping isometrics shall be marked on both sides of the pipe 180 deg. apart at an interval of 3 to 6 meters or less depending on pipe length.

Size (in mm) of the arrow shall be either of the following:-



FOR PIPES 8" NB & BELOW



FOR PIPES 10''NB AND ABOVE

15.6 Colour Schedule for Fire Fighting Equipment:

Fire Equipment/ Fire Fighting Equipment colour shall be in accordance with following criteria as detailed in table 15.6

Table No. 15.6

Sr. No.	Item/Application	Standard Painting Color	RAL No.
1	A. Extinguishers: Water based,foam,DCP,CO2	Fire red Paint	(RAL NO.3000)
	B. all fire tenders /jeep, fire engines(except foam tank, foam piping & nitrogen cylinder)		
	C. Fire tender, water piping/wet risers. Fire hose reel box		
	D. Fire tender water pump & engine/water pipeline in engine & pump		
	E. DCP tank & DCP piping		
2	Foam tank & foam piping/nitrogen cylinder	Dark Admiralty Grey	(RAL NO.7031)
3	Driver compartment & inside of the lockers	Pale Cream	(RAL NO.9001)
4	Chassis & wheel arches	Black	(RAL NO.9005)
5	Fire services insignia & identity nomenclatures	Canary yellow with black border on sides of sides of the vehicle/equipment	(RAL NO. 1012) & (RAL NO. 9005)

16.0 RECOMMENDED LIST OF PAINT MANUFACTURERS

Refer updated Vendor List in ONGC Website.

17.0 PROCEDURE FOR APPROVAL OF NEW COATING MATERIAL

Following procedure recommended is to be followed for approval materials Manufactured by new manufactures (indigenous and foreign):

1. The manufacturer should arrange testing of the coating materials as per the List of tests given in para 5 below from one of the reputed Government Laboratories.

2. Samples of coating materials should be submitted to the Govt. Laboratory in sealed containers with batch number and test certificate on regular format of manufacturer's testing laboratory.

3. All test panels should be prepared by the Govt. testing agency-colored photographs of test panels should be taken before and after the test and should be enclosed along with test report.

4. Sample batch number and manufacturers test certificate should be enclosed along with the report. Test report must contain detail of observations and rating if any as per the testing code. Suggested Govt. laboratories are:

   RRL, Hyderabad

   HBTI, Kanpur

   DMSRDE, Kanur

   IIT, Bombay

   BIS, Laboratories

   UDCT, Mumbai

   Or Other International Test houses of Repute to be approved by the Company.

5. Manufacturers should intimate the company, details of sample submitted for testing, name of Govt. testing agency, date. Contact personnel of the Govt. testing agency. At the end of the test the manufacturer should submit the test reports to the Company for approval. The manufacturer(s) shall be qualified based on the results of these tests and the Company's decision in this regard -shall be Final and binding on the manufacturer.

All tests required for evaluation of acceptance coating materials for offshore application shall be as per the relevant ASTM standards.

18.0 PAINT PERFORMANCE GUARANTEE OF PROTECTIVE COATING:

Contractor shall provide performance Guarantee for protective coating for the period as specified in Volume – I of Bid Document.

-----END----

PIPING DESIGN CRITERIA

Prepared / Revised By	Reviewed By	Approved By	Total No. of Pages	Date	Rev. No.
PS	GJ	CBS	15	23.08.11	4

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1. SCOPE OF THIS DOCUMENT

1.1 INTRODUCTION

This outlines to the minimum and mandatory requirements of designs activities for piping, piping components, piping specialties including piping within battery limits of various skids/ package/ modules etc. These criteria shall be read in conjunction with project specifications (as per clause 2. of this specification) attached elsewhere in the bid package.

All piping assemblies, specialties & materials supplied or installed under these specifications shall be in accordance with sound engineering principles. Any omission from this specification shall not relieve the contractor from his responsibility of furnishing equipment or materials to meet the specific process parameters, environmental parameters, safety parameters and any other applicable statutory laws or relevant codes & standards.

1.2 APPLICATION OF THIS DOCUMENT

This document applies to this project for the Company. This document shall not be used for any other purpose.

2. STANDARDS & SPECIFICATIONS

2.1 PROJECT SPECIFICATIONS

1. Piping Design (spec. 2004-A)
2. Piping Fabrication (spec. 2004-B)
3 Piping Material Specification (Spec. 2008)
4. Piping Specialties (spec. 2004-D)
5. Protective Coatings (spec. 2005)
6. Piping & Equipment Insulation (spec. 2006)
7. Welding & NDT (spec. 2009F)

2.2 CODES & STANDARDS

The latest codes shall be used**.**

1. ASME B1.20.1 Pipe threads

2.	ASME B16.5	Pipe flanges and flanged fittings
3.	ASME B16.9	Factory made wrought steel butt-welding fittings
4.	ASME B16.10	Face-to-face & end-to-end dimensions of valves
5.	ASME B16.11	Forged fittings (socket welding and threaded)
6.	ASME B16.20	Metallic Gaskets for pipe flanges: Ring joint spiral wound and jacketed.
7.	ASME B16.21	Non-metallic flat gaskets for pipe flanges
8.	ASME B16.25	Butt welding ends
9.	ASME B16.34	Valves- flanged, threaded & welding ends
10.	ASME B31.3	Process piping
11.	ASME B31.4	Pipe line transportation systems for liquid hydro carbons and other liquids.
12.	ASME B31.8	Gas transmission and distribution piping system
13.	ASME B36.10 M	Welded and seamless wrought steel pipe
14.	ASME B36.19 M	Stainless steel pipe
15.	ASME SEC.VIII	Pressure vessel code
16.	ASME SEC.IX	Welding and brazing qualifications
17.	ASTM A105	Specification for forgings. Carbon steel, for piping components
18.	ASTM A106	Specification for seamless carbon steel pipe for high temperature service
19.	ASTM A 153	Zinc coating (hot dip) on iron & steel hardware
20.	ASTM A 182	Specification for forged or rolled alloy steel pipe flanges, forged fittings & valves and parts for high temperature service.

21.	ASTM A193	Specification for alloy steel & stainless steel bolting materials for high temperature service.
22.	ASTM A 194	Specification for carbon & alloy steel nuts for bolts for high pressure & high temperature service
23.	ASTM A216	Specification for carbon steel casting suitable for fusion welding for high temperature service.
24.	ASTM A234	Specification for piping fittings of wrought carbon steel & alloy steel for moderate & elevated temperature
25.	ASTM A262	Recommended practice for detecting susceptibility to inter granular corrosion attack in stainless steels
26.	ASTM A 312	Specification for seamless & welded austenitic stainless steel pipe
27.	ASTM A370	Test methods and definitions for mechanical testing of steel products
28.	ASTM A403	Specification for wrought, austenitic stainless steel, piping fittings
29.	ASTM A453	Specification for bolting materials high temperature 50 to 120 ksi with expansion coefficient comparable to austenitic steels
30.	ASTM A578	Straight beam ultrasonic examination of plain & clad steel for special applications.
31.	ASTM A 694	Std specs. For forgings, carbon and alloy steel for pipe flanges, fittings, valves & parts for high-pressure transmission service
32.	ASTM A 790	Seamless and welded ferritic/austenitic stainless steel pipe
33.	ASTM A 799	Std. practice for steel castings, stainless instrument calibration for estimating ferrite content.
34.	ASTM B42	Std. spec. For seamless copper pipe
35.	ASTM B 124	Std. spec. For copper and copper alloy forging rod, bar and shapes.

36.	ASTM B 165	Std. spec. For nickel copper alloy (UNS4400) seamless pipe and tube
37.	ASTM B 337	Std. spec. For seamless and welded titanium and titanium alloy pipe
38.	ASTM B 363	Std. spec. for unalloyed titanium and titanium alloy welding fittings
39.	ASTM B 366	Std. spec. for factory made wrought nickel and nickel alloy welding fittings.
40.	ASTM B 423	Std. specs. for nickel-iron-chromium-molybdenum copper alloy (UNS no.8825&8221) seamless pipe and tube
41.	ASTM B 425	Std. spec. for NI-FE-CR-MO- CU alloy (UNS no. 8825 & 8221) rod & bar.
42.	ASTM B466	Std spec. for seamless copper nickel pipe and tube
43.	ASTM D 1785	Spec. for poly vinyl chloride (PVC) plastic pipe
44.	ASTM D 2665	Spec. for poly vinyl chloride (PVC) plastic drain, waste and vent pipe fittings
45.	ASTM E18	Rockwell hardness testing of metallic materials
46.	ASTM E 45	Determining inclusion content of steel
47.	ASTM E 92	Vickers hardness of metallic materials
48.	ASTM B 142	Controlling quality of radiographic testing
49.	ASTM E 165	Liquid penetrant inspection method
50.	ASTM E 709	Recommended practice for magnetic particle examination.
51.	API 5 L	Line pipe Specification
52.	API 6A	Wellhead and Chrismas tree Equipment
53.	API 6D / ISO 14313	Petroleum and natural gas industries-Pipeline Transportation

		system –Pipeline valves
54.	API 6FA	Fire test for valves
55.	API RP 14C	Analysis, design, installation testing for basic surface safety system
56.	API RP 14E	Design and installation of offshore production platform piping system
57.	API RP 14G	Fire prevention and control on open type offshore production platform.
58.	API 598	Valve inspection and testing
59.	API 600 / ISO 10434	Steel gate valves, flanged and butt-welded ends (nps 1 through nps 24)
60.	API 607	Fire test for soft-seated Quarter turned valves
61.	BS 1868	Spec. for steel check valves (flanged &butt welding ends) for the petroleum petrochemical & allied industries
62.	BS 1873	Spec. for steel globe and globe stop and check valves (flanged & butt welding ends) for the petroleum, petrochemical &allied industries.
63.	BS EN 1092-3	Flanges and their joints. Circular flanges for pipes, valves, fittings and accessories, PN designated Copper Alloy Flanges.
64.	BS EN ISO 17292	Metal Ball Valve For The Petroleum, Petrochemical And Allied Industries
65.	BS EN ISO 5761	Steel Gate, Globe & Check valves For DN 100 and Smaller for Petroleum and Natural Gas Industries
66.	BS 5353	Specification For Steel Plug Valve
67.	BS 6755-2	Testing Of Valves. Specification for Fire Type-Testing requirements.
68.	MSS SP 44	MSS Steel Pipe Line Flanges
69.	MSS SP 75	Spec. For High Test Wrought Butt Welding Fittings

70.	NACE MR 01-75/ ISO 15156-1/2/3	Material for use in H2S containing environments in Oil and Gas Production. Part-1: General principles for Selection of Cracking resistant Materials. Part-2: Cracking resistant Carbon and Low Carbon Steels, and the use of Cast irons. Part-3: Cracking resistant CRAs (Corrosion Resistant alloys) and other alloys.
71.	NACE TM-01-77	Laboratory-Testing Of Metals For Resistance To Sulfide Stress Cracking and stress corrosion cracking in H2S Environments.
72.	NACE-TM-02-84	Evaluation Of Pipeline and pressure vessel steels for Resistance To Hydrogen induced Cracking
73.	NFPA-Volume-6	National Fire Code For Sprinklers, Fire Pumps And Water Tanks
74.	NFPA Volume 8	National Fire Code For Portable And Manual Fire Control Equipment.
75.	NFPA 15	Standard for Water spray fixed systems for fire protection.
76	ASTM D 2996	Specifications For Filament Wound Reinforced Thermosetting Pipes & Fittings.
77	ASTM D 2992	Method For Obtaining Hydrostatic Design Basis For Reinforced Thermosetting Resin Pipes & Fittings.
78	ASTM A 815	Specification for wrought ferritic, ferritic/austenitic and martensitic stainless steel piping fittings.
79	ASTM G 36	Performing stress corrosion-cracking tests in a boiler magnesium chloride
80	ASTM G 48A/B	Standard Test Method For Pitting & Crevice Corrosion Resistance of Stainless Steel & Related Alloy By Use Of Ferritic Chloride Solution
81	ASME B16.24	Cast Copper Alloy Pipe Flanges and flanged fittings – Classes 150,

300, 400, 600, 900, 1500 and 2500.

82	ASME B 16.18	Cast Copper Alloy Solder Joint Pressure Fittings
83	MSS-SP-80	Bronze Gate, Globe, Angle and Check Valve
84	MSS-SP-97	Integrally reinforced forged branch outlet Fittings-Socket Welding, Threaded and Butt Welding Ends

2.3 ORDER OF PRECEDENCE

In case of conflict the order of precedence shall be as follows:
1. P&ID
2. Data Sheets
3. This Specification
4. Other referenced project specifications
5. Codes and Standards

3. ENVIRONMENTAL DESIGN CRITERIA AND UTILITIES

3.1 BASIC CLIMATIC CONDITIONS

Refer Structural design criteria Section 3.4, Vol- II, attached in bid package.

3.2 SEISMIC AND TRANSPORTATION LOADS

All equipment supports and braces, pipe supports and other support steel work, including temporary braces, shall be designed to withstand seismic loads applicable to the present location. Refer to the Structural Basis of Design for seismic design considerations.

All equipment supports and braces, pipe supports and other support steel work, including temporary braces, shall be designed to withstand the operating, lifting, transport (by road and by sea) and hydro-test loads specified in Project Specification.

3.3 DESIGN LIFE

The process facilities design life requirement is 25 years.

3.4 DIMENSIONS

SI units shall be used. Dimensions shall be in mm and be related to the Platform datum's or reference lines.

4. DESIGN REQUIREMENTS

All materials shall conform to project Specification and the identified API, ASME, ASTM, BS and NACE codes and Standards.

Design and fabrication shall conform to this Specification and ASME B31.3, API RP14E. In case, any other applicable codes are proposed/ referred the same shall be complied with sufficient information/justification after approval by company.

Ingress of sand particle is envisaged in well fluid and injection water services. Therefore, the valves in well fluid and water injection systems shall be metal seated. Trim of the metal seated valves as a minimum shall be Tungsten Carbide coated.

From piping of pig-barrel of Launcher/Receiver up to piping-pipeline interface shall be designed as a minimum to ASME B 31.4 and ASME B 31.8.

For smooth pigging operation, attempts shall be made to keep the ID of the piping from reducer of pig-Barrel to pipeline –piping interface as close as possible to the ID of the riser in splash zone.

In case, ID of the topside is more than the the ID of the riser, thickness of the topside piping shall be increased to match the ID's

If the ID of the topside piping is less than that of riser in splash zone, thickness of the topside piping may be adjusted to match the ID's by:

- Considering the actual design conditions of the line in place of class conditions for pipe wall thickness calculations.
- Reducing the corrosion allowance for topside piping but not less than the internal corrosion allowance in corresponding pipeline.

In case, the difference in ID's still exists, ID of the topside piping shall be checked for the passage of gauge plate with out interference. The gauge plate
Diameter shall be calculated by the formula given in the 'specification of submarine pipelines'.

All cupro-nickel piping as a minimum shall be supplied in 16-bar system.

Velocity in Cu-Ni piping shall not exceed 1.6 m/sec for 2" NB and below and 3.3 m/sec. for 3" NB and above.

Fluid velocities in copper piping shall not exceed 1.5 m/sec.

Design of Cu-Ni piping system shall be such as to avoid excessive turbulence in the system.

Monel shall be used at locations such as bends, reducers, downstream of restriction orifices, downstream of control valves, downstream of check valves, etc. wherever there may be a possibility of Flow velocities exceeding the limits given above, Impingement of flow stream on piping, or Excessive turbulence.

In all cases where Monel is used it shall be in the form of spool pieces with electrical isolation with Cu-Ni material for minimizing galvanic corrosion of Cu-Ni piping.

Wherever dissimilar metals are in contact, insulating gasket or sacrificial spool piece of 600 mm shall be provided to avoid galvanic corrosion.

Piping general arrangement drawings/isometrics/support drawings etc. shall be prepared using good engineering practices and as per guidelines furnished in project Specifications.

As good engineering practices, piping carrying hydrocarbon and toxic/hazardous chemicals passing through safe area shall be of continuous lengths with welded joints such that valves, regulators, flanges etc. are not located in the safe area.

Thermal Insulation shall be provided wherever required during detail Engineering, and General Specification 2006. Also, wherever necessary, acoustic insulation shall be provided to limit the piping emitted noise to the permissible values as per API standards.

Piping schedule/thickness shall be calculated for each size, service & piping class including corrosion allowance indicated in material of construction of piping specification as per ASME B-31.3 for various services based on piping class conditions upto class 900 & actual design conditions for 1500 class of the system. The same shall be submitted to the Company for review & approval.

The piping connected to equipment shall be analyzed by the Contractor for flexibility and maximum stresses developed, along with nozzle reaction on equipment, and these shall not exceed the permissible limits/values as specified in relevant codes & standards. Vendor data for maximum permissible nozzle loadings shall be obtained while analyzing piping for flexibility analysis. The above permissible limits shall not be exceeded in any case. The analysis shall be furnished to the Company for review & approval.

Piping shall be suitably supported, as necessary, to prevent sagging, mechanical stresses and vibrations. In general, piping shall be fastened to pipe racks with appropriate sizes hot dip galvanized or cadmium plated U bolts {3/8" (9.5 mm) minimum} and shall be double nutted.

The layout of equipment and piping shall be based on following principles:-

a) To locate all equipments identified on equipment list.

b) To comply with codes & standards, regulations, P&ID's, project specifications and sound engineering industrial practices.

c) To maximize safety of personnel, equipment and facilities.

d) To ensure operability & maintainability of equipment with a provision of adequate clearances all around in horizontal & vertical direction.

e) To provide means of escape & access for fire fighting.

f) To satisfy all requirements indicated in process documents.

g) To minimize shutdown duration.

h) To provide neat and economical layout, allowing for easy supporting and adequate flexibility to meet equipment allowable nozzle load.

i) To comply with the requirements of Area classification.

5. FLEXIBILITY ANALYSIS CRITERIA & GENERAL GUIDELINES

Flexibility analysis shall be carried out as per requirements of ASME B 31.3. In addition to the requirements stated in ASME B31.3, following considerations shall be taken care during flexibility analysis:-

5.1 Stress analysis is carried out to prevent the following:–

a) Failure of pipe work or support from over stress or fatigue.

b) Detrimental stresses or distortion in pipe work or in connected equipment resulting from excessive thrusts and moments in the pipe work.

c) Excessive vibration in pipe work.

d) Leakage at joints.

5.2 The following guidelines shall be used to determine the type and extent of analysis to be adopted. The guidelines are not exhaustive and do not replace good judgment and experience particularly with regard to safety aspects of the design.

a) Process pipe work 3" diameter and larger connected to centrifugal compressors or turbines.

b) Pipe work connected to other strain sensitive equipment.

c) 4" diameter and larger at 200ºC design temperature and above.

d) 2" diameter and larger at 300ºC design temperature and above.

e) 8" diameter and larger at 150ºC design temperature and above.

f) 16" diameter and larger at 80ºC design temperature and above.

g) 6" diameter and larger at 65ºC design temperature and above connected to equipment like

h) air coolers, fin fan coolers, fired heaters / furnaces or any other sensitive equipment.

i) Lines subject to vibration, e.g. Reciprocating compressors and turbines.

j) Pipe work larger than 3" diameter at -50ºC design temperature and below.

k) Large diameter piping between closely spaced equipment.

l) All suction and discharge lines connected to pump.

m) All lines connected to wellhead.

n) All GRE, PVC, CPVC piping.

o) If applicable, all high pressure flange joints (i.e. 900# and above) shall be checked for leakage at flange joints due to combined effect of induced bending moments and internal pressure. This check shall also be performed for all lines ≥ 24" NB irrespective of rating.

p) All lines shall be analyzed at design / operating temperature range.

5.3 All the piping on the bridge between two platforms shall be designed taking into account the differential movements of the two structures under extreme storm conditions (Refer table – 5 of section 3.4 structural design criteria, Volume -II). Flexibility analysis of the piping system shall be carried out wherever required by the design conditions / platform movements and provide necessary loops / supporting arrangement.

6. FIRE SUPPRESSION SYSTEM

Fire Suppression system shall consists of a deluge type fire water spray network, fire water and foam hose reel, portable dry chemical/CO_2 fire extinguishers and dry chemical skid with hose reels.

A deluge type water spray network shall be designed in accordance with NFPA-15.

For quantity of fire water and foam hose reel refer Mechanical design criteria. The location and length of the hose reel shall be suitable to cover all points on the deck.

Offshore Suggested Vendor List of Engineering and Projects Excellence Centre, ONGC as on 27.03.2026		
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Important Notes		
1	Vendors with no remark in this list are confirmed vendors.	
2	The '**On Hold**' status for vendors has been categorised into following groups:	
	B1	On Hold due to Non-submission of Name change / Other documents by the vendor to ONGC
	B2	On Hold due to No response from the vendor
	B3	On Hold due to Poor / Negative Performance Feedback received against the vendor from ONGC Asset.
	Banned Firms	Banned Firm under Corporate MM Portal
	The reasons for which a vendor has been put 'On Hold' have been marked against the vendor as **B1, B2 & B3.**	
3	Vendors, who are traders, have been identified	
4	The contractor can continue to get quotation from vendors who are 'On Hold' under group **'B1**', provided that the reasons for which a vendor has been put 'on hold' are resolved by the vendor at the time of submission of Purchase Specification (PS)	
	Vendors in Group '**B2'** will be delisted automatically in case No-response is received from Vendor.However,the LSTK contractors are free to contact these vendors, if they respond to them.	
	The 'On Hold' status of a vendor under group '**B3**' in this Suggested Vendor List means that the contractor shall not place order(s) on the vendor for the product for which the vendor has been put 'On Hold' by ONGC.	
	The '**Banned Firms**' status of a vendor in this Suggested Vendor List means that the contractor shall not place order(s) on the vendor for the product for which the vendor has been put 'Banned Firms' by ONGC.	
5	**a)** Manufacturing locations of MNC vendors other than the enlisted location in the Suggested Vendor List, can also be accepted for supply of product after obtaining an Undertaking in the prescribed proforma from Principal/ enlisted vendor. Such manufacturing locations shall possess all accreditations required for the product and comply with the Functional specification.The OEM vendor who is enlisted already in ONGCs SVL and wants his other Manufacturing Locations / API certified fabrication yards etc., also to be accepted by ONGC, shall intimate all such locations where the material will be sourced to ONGC.In this case, the SVL shall be updated to incorporate all such locations wherefrom the items are being sourced. Any changes from these locations shall be intimated by the already enlisted OEM vendor.	
	b) For any vendor, where the Purchase Orders are to be placed on agencies authorized by the enlisted vendor, the enlisted vendor shall furnish Authorization letter to the agency for a particular product with an undertaking for agreeing to the following conditions:- 1) Purchase specification (For PS items) shall be submitted & signed by enlisted vendor only. 2) Warrantee/Guarantee shall be provided by the enlisted vendor 3) Enlisted Vendor shall ensure maintenance support for the product	

	c) Manufacturing locations established in India by an Indian Company can also be considered for supply/enlistment provided such Indian Company forms a Joint venture registered in India with: 1. an enlisted foreign vendor in the suggested vendor list. 2. any foreign vendor who is not enlisted but meets the qualification criteria for vendor enlistment. Above arrangement shall be considered subject to meeting the following requirements: (i) Undertaking in the prescribed proforma from both the partners shall be submitted. (ii) The manufacturing location established in India shall possess all the accreditations required for the product and also comply with the functional specifications. (iii) Factory inspection may be carried out to ascertain the capability of the Joint Venture Company. (iv) In the event of collapse of such joint venture arrangement, supply from the JV company shall not be accepted.
	d) Manufacturing locations established in India by an Indian Company can also be considered for supply/ enlistment for particular product with an undertaking in the prescribed Proforma from the enlisted foreign vendor, provided such Indian company has technical License agreement with an enlisted foreign vendor in the suggested vendor list. Above arrangement shall be considered subject to meeting the following requirements:- I. The Technical License agreement shall be valid for minimum 3 years on the date of submission of PQD. After expiry of Technical License agreement , renewed agreement shall be required to be submitted. II. In case of cancellation/non renewal of Technical License agreement, the name of the vendor shall be delisted. III. Undertaking in the prescribed proforma from enlisted foreign Vendor. IV. Undertaking in the prescribed proforma from Indian company. V. The manufacturing location established in India shall possess all the accreditations required for the product and also comply with the functional specifications. VI. Factory Inspection may be carried out to ascertain the capability of the Company.
6	For enlistment of a Packager as vendor following conditions must be satisfied: a) Packager is the Process Licensor OR the OEM of the main equipment(Main equipment is listed in the respective Functional specification/ Design Criteria. b) If the packager is not Process Licensor or OEM of main equipment, the packager shall submit the authorization letter from the ONGC enlisted Vendor of main equipment for carrying out packaging activity. c) The Packager shall select the major items from the ONGC Suggested Vendor List as mentioned in the respective Functional Specification/Design Criteria. d) Any registered Packager who is the Process Licensor OR the OEM of the main equipment wants to relinquish his Over-all single-point responsibility of the Packaged Item and transfer the same to Other Vendor(s) of the Packaged Items, he may do so provided all such major sub-vendors of such packaged Items mutually agree to Provide the similar Services to ONGC under the leadership of the new sub-vendor who accepts to take single Point responsibility.
7	Vendor should agree / guarantee supply of spares for the entire life cycle of the item / equipment (i.e. 7 years in case of electronic equipment/items and 10 years in case of mechanical equipment/items)
8	Vendor should agree to comply with ONGC Terms & Conditions for Supply of Spares
9	# indicates change over previous Suggested Vendor List

Piping			
Product	**1**	**Pipe-(Duplex SS)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Sumitomo Corporation	Japan	**Trader**
2	JFE Steel Corporation	Japan	
3	NSC	Japan	**On Hold-B1**
4	Sandvik	Sweden	
5	Mannesmann-werke AG'	Germany	**On Hold-B1**
6	NKK	Japan	**On Hold-B1**
7	Coprosider Srl	Italy	
8	Schultz Export GMBH	Germany	
9	T.T.I.Tubacex Inoxidables SA	Spain	
10	Inoxtech SPA	Italy	
11	Gieminox (Welded Pipes)	Italy	
12	H. Butting GmbH & Co. KG (Welded Pipes)	Germany	
13	Rivit SpA (Welded Pipes)	Italy	
14	Adinox Overseas	India	
15	Maxim Tubes Company Pvt. Ltd	India	
Product	**2**	**Fittings -Duplex SS (DSS)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Sumitomo Corporation	Japan	**Trader**
2	Sandvik	Sweden	
3	Shimoda Iron Works Co.Ltd	Japan	
4	Coprosider Srl	Italy	
5	Mannesmann--werke AG	Germany	**On Hold-B1**
6	NKK	Japan	**On Hold-B1**
7	Schulz Export GMBH	Germany	
8	ALLIED INTERNATIONAL S.r.l	Italy	

9	M.E.G.A S.p.a (Forged Fittings ASTM A182 Grade F60/F51 with UNS 31803 / UNS 32205)	Italy	
10	Fittinox SRL	Italy	
11	Raccortubi SPA (Butt Welded)	Italy	
12	Promat BD Middle East (Forged)	UAE	
13	Viar S.P.A.	Italy	
14	Sawan Engineers Pvt. Ltd.	India	
15	TK Corporation	South Korea	
16	Anggerik Laksana India Private Limited	India	
Product	**3**	**Flanges-Duplex SS (DSS)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Sumitomo Corporation	Japan	**Trader**
2	Sandvik	Sweden	
3	Coprisider Srl	Italy	**On Hold-B1**
4	Mannesmann-werke AG	Germany	**On Hold-B1**
5	Officine Ambrogio Melesi & C Srl	Italy	
6	MGI	France	
7	Officine Nicola Galperti Figlio S.p.A	Italy	
8	Officine Satafede SRL	Italy	
9	Promat BD Middle East	UAE	
10	Viar S.P.A.	Italy	
11	Sawan Engineers Pvt. Ltd.	India	
12	Felix Technology Co. Ltd.	South Korea	
13	Ulma Forja S.Coop	Spain	
14	Anggerik Laksana India Private Limited	India	
Product	**4**	**Ball Valves-Duplex SS (DSS)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	KTM	Japan	**On Hold-B1**
2	KITZ	Japan	
3	T.K. Valves Ltd, Abudhabi	UAE	**On Hold-B1**

4	Petrol Valves S.r.l	Italy	
5	Dresser Spa	Italy	**On Hold-B1**
6	Flowserve Flow Control GmbH,	Germany	
7	Argus	Germany	**On Hold-B1**
8	Grove Italia Spa	Italy	**On Hold-B1**
9	O.M.S. SALERI	Italy	
10	L&T Valves Ltd	India	
11	Emerson Process Management (India) Private Limited, India	India	
12	LCM Italia SPA	Italy	
13	ZA.VE.RO. S.r.l	Italy	
14	Emerson Process Management Virgo Valves, Srl	Italy	
15	M/s VALVINOX s.r.l. (I) PVT. LTD	India	
16	Hawa Valves (India) Pvt Ltd	India	
17	BAC Valves S.A.	Spain	
18	M/s NOVEL VALVES INDIA PVT. LTD.	India	**# NEW VENDOR**
Product	**5**	**Other Valves (Gate Globe, Check, Needle:Duplex SS)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	KITZ	Japan	
2	KTM	Japan	**On Hold-B1**
3	Petrol Valves S.r.l	Italy	
4	TK Valve Ltd, Abu Dhabi	UAE	**On Hold-B1**
5	Truflo Rona S.A.	Belgium	
6	LVF Spa (upto 2")	Italy	
7	ALCO Italia (Gate Valve) (150 - 2500 Lbs / 0.25 - 2 ")	Italy	**On Hold-B1**
8	ALCO Italia (Globe Valve) (800 - 2500 Lbs, 0.25 - 2 ")	Italy	**On Hold-B1**
9	L&T Valves Ltd	India	
10	Douglas Chero Spa (2" and below)	Italy	

11	Valvinox S.R.L.,Milan	Italy	
12	Firsa S.p.a	Italy	
13	M/s VALVINOX s.r.l. (I) PVT. LTD (For Check Valves only)	India	
14	M/s NOVEL VALVES INDIA PVT. LTD.	India	**# NEW VENDOR**
Product	**6**	**Shutdown Valves (Duplex SS)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	KTM	Japan	**On Hold-B1**
2	KITZ	Japan	
3	Flowserve Flow Control GmbH	Germany	
4	TK Valves Ltd, Abu Dhabi	UAE	**On Hold-B1**
5	Dresser Spa	Italy	**On Hold-B1**
6	Petrol valves S.r.l	Italy	
7	Argus	Germany	**On Hold-B1**
8	Grove Italia Spa	Italy	**On Hold-B1**
9	Cameron Italia S.R.L (Upto 60" / Up to 300 Lbs)	Italy	**On Hold-B1**
10	Cooper Cameron, Singapore (Up to 24" / Upto 1500 Lbs)	Singapore	**On Hold-B1**
11	EL-O-MATIC India (P) Ltd., (Up to 8" / Upto 300 Lbs)	India	
12	Flowserve PTE Ltd. (Up to 12" / Upto 300 Lbs)	India	**On Hold-B1**
13	HINDLE Cockburns Ltd. (Upto 16" / Upto 300 Lbs & Upto 12" / 600-2500 Pounds)	India	**On Hold-B1**
14	Metso Automation PTE Ltd. (Upto 30" / Upto 1500 Lbs)	India	**On Hold-B1**
15	Perrin Gmbh, Germany (8" / Upto 600 Lbs & 16" / Upto 300 Lbs)	Germany	
16	Pibiviesse SPA (Upto 16" / 1500 Lbs)	Italy	**On Hold-B1**
17	Weir Valves & Controls UK Ltd. (Upto 42" / Upto 1500 Lbs)	UK	
18	Pentair Valves and control India Pvt. Limited (Upto 16" / Upto 300 Lbs & Upto 12" / 600 to 2500 Lbs)	India	
19	Emerson Process Management (India) Private Limited, India	India	
20	L&T Valves Ltd	India	

21	Emerson Process Management Virgo Valves, Srl	Italy	
22	MIR Valve Sdn. Bhd	Malaysia	

Product	**7**	**Pipe (C.S NACE)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Sumitomo Corporation	Japan	**Trader**
2	NKK	Japan	**On Hold-B1**
3	Nippon Steel Corporation	Japan	**On Hold-B1**
4	Mannesmann-werke AG	Germany	**On Hold-B1**
5	Mitsubishi	Japan	**Trader**
6	Tenaris Dalmine	Italy	
7	Raccordi Forgiati	Italy	
8	JFE Steel Corporation	Japan	
9	OMR Officine Meccaniche	Italy	
10	Schulz Export GMBH	Germany	
11	Vallourec & Mannesmann	Germany/France	**On Hold-B1**
12	Maharashtra Seamless Ltd.	India	
Product	**8**	**Fittings (C.S. NACE)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Sumitomo Corporation	Japan	**Trader**
2	Mega Spa	Italy	
3	Gam Recordie	Italy	
4	Raccordi Forgiati	Italy	
5	Fittinox Srl	Italy	
6	OMR Officine Meccaniche	Italy	
7	Schulz Export GMBH	Germany	
8	Coprosider Srl	Italy	
9	Igwara Industrial Service and Trading Pte Ltd.	Singapore	**Trader**
10	Tecnoforge	Italy	
11	ALLIED INTERNATIONAL S.r.l	Italy	
12	ERNE [source of raw material Tenaries (Dalamine) Italy] Seamless fittings	Austria	
13	Tube Products Incorporate	India	
14	Sawan Engineers Pvt. Ltd.	India	

15	T K Corporation	Korea	
16	Sungkwang Bend Co., Ltd.	Korea	
17	Promat BD Middle East (Forged)	UAE	
18	Anggerik Laksana India Private Limited	India	
Product	**9**	**Flanges (CS NACE)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Sumitomo Corporation	Japan	**Trader**
2	Coprosider Srl	Italy	
3	Officine Nicola Galperti Figlio S.p.A	Italy	
4	Trouvay & Cauvin	France	**Trader**
5	Officine Ambrogio Melesi & C Srl	Italy	
6	MGI	France	
7	Schulz Export GMBH	Germany	
8	Special Flanges Srl	Italy	
9	Tube Products Incorporate	India	
10	CHW Forge Private Ltd.	India	
11	Officine Santafede SRL.	Italy	
12	Sawan Engineers Pvt.Ltd.	India	
13	Promat BD Middle East	UAE	
14	Felix Technology Co. Ltd.	South Korea	
15	Dae-Heung Ind. Co., Ltd	Korea	
16	Anggerik Laksana India Private Limited (Size upto 16" and Rating upto 1500#)	India	
17	Good Luck Engineering Co	India	
Product	**10**	**Ball Valves (CS NACE)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Cooper Cameron Valves	USA	**On Hold-B1**
2	Dresser Spa	Italy	**On Hold-B1**
3	KTM	Japan	**On Hold-B1**
4	Flowserve Flow Control GmbH	Germany	
5	KITZ	Japan.	

6	Serck Audco Valves International	U.K.	**On Hold-B1**
7	Flow Control Technology	France	**On Hold-B1**
8	Crane	USA	**On Hold-B1**
9	Orbit Valve	U.K.	**On Hold-B1**
10	T.K. Valve, Abu Dhabi	UAE	**On Hold-B1**
11	OMB Spa (Upto 1.5")	Spain	
12	Petrol Valves Srl	Italy	
13	LCM, Italia. SRL	Italy	
14	L&T Valves Ltd	India	
15	ZA.VE.RO. S.r.l	Italy	
16	Multi Valve Technology	UK	
17	Raimondi	Italy	**On Hold-B1**
18	Yoneda Kogyo Co	Japan	**On Hold-B1**
19	Truflo Rona S.A	Belgium	
20	Emerson Process Management (India) Private Limited, India	India	
21	Argus	Germany	**On Hold-B1**
22	Cameron Iron Works	USA	**On Hold-B1**
23	O.M.S. SALERI	Italy	
24	Valbart SRL	Italy	
25	JCF Fluid Flow FZE	UAE	**On Hold-B1**
26	Valvinox s.r.l- Milan	Italy	
27	Emerson Process Management Virgo Valves, Srl	Italy	
28	Oswal Industries Limited	India	
29	NSSL Italia S.R.L	Italy	
30	VALVINOX s.r.l. (I) PVT. LTD	India	
31	Anand Teknow Aids Engineering India Ltd.	India	
32	PHBB Valves Private Limited	India	
33	J C Valvulas India Pvt. Ltd.	India	
34	DelVal Flow Controls Pvt. Ltd.	India	
35	Novel Valves India Pvt. Ltd.	India	

Product	**11**	**Gate, Globe, Check, Needle Valves (CS NACE)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Cooper Cameron Valves	USA	**On Hold-B1**
2	Dresser Spa	Italy	**On Hold-B1**
3	KTM	Japan	**On Hold-B1**
4	KITZ	Japan	
5	Serck, Audco Valves International	U.K	**On Hold-B1**
6	Flow Control Technology	France	**On Hold-B1**
7	Crane	USA	**On Hold-B1**
8	Orbit Valve	UK	**On Hold-B1**
9	T.K.Valves, Abu Dhabi	UAE	**On Hold-B1**
10	Petrol Valves Srl	Italy	
11	LVF Spa	Italy	
12	BFE Spa	Italy	
13	SACCAP	France	
14	Flow serve Flow Control, GmbH	Germany	**On Hold-B1**
15	L&T Valves Ltd	India	
16	Goodwin International Limited (wafer Check valves Only)	England	
17	Argus	Germany	**On Hold-B1**
18	Cameron Iron Works	USA	**On Hold-B1**
19	Douglas Chero S.p.A (for 2" and below)	Italy	
20	JCF Fluid Flow FZE	UAE	**On Hold-B1**
21	Firsa Spa (For Gate, Globe & Check Valves)	Italy	
22	Niton Valves Industries (P) Ltd	India	
23	Valvinox s.r.l- Milan	Italy	
24	J C Valvulas India Pvt. Ltd. (For Gate, Globe & Check Valves)	India	
25	Novel Valves India Pvt. Ltd. (For Gate, Globe & Check Valves)	India	

Product	**12**	**Shutdown Valves- CS NACE**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Cooper Cameron Valves	USA	**On Hold-B1**
2	KITZ	Japan	
3	Flow Control Technology	France	**On Hold-B1**
4	Crane	USA	**On Hold-B1**
5	Orbit Valves	UK	**On Hold-B1**
6	T.K.Valves, Abu Dhabi	UAE	**On Hold-B1**
7	Tyco	Italy	**On Hold-B1**
8	Petrol valves S.r.l	Italy	
9	Dresser Italia Srl.(Grove Divisions)	Italy	**On Hold-B1**
10	Flowserve Flow control GmbH	Germany	
11	L&T Valves Ltd	India	
12	Emerson Process Management (India) Private Limited, India	India	
13	Argus	Germany	**On Hold-B1**
14	Grove Italia Spa	Italy	**On Hold-B1**
15	Cameron Iron Works	USA	**On Hold-B1**
16	Valbart Srl	Italy	
17	Valvinox S.R.L.,Milan	Italy	
18	Emerson Process Management Virgo Valves, Srl	Italy	
19	OMS SALERI S.p.A	Italy	
20	DelVal Flow Controls Pvt. Ltd.	India	
21	JC Valvulas India Pvt Ltd	India	
Product	**13**	**Pipe (CS Non-NACE)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	BHEL, Trichy	India	
2	Indian Seamless, Ahmed Nagar	India	**On Hold-B1**
3	Jindal Pipes, New Delhi	India	
4	Jindal Saw ltd, Delhi	India	
5	NSC	Japan	**On Hold-B1**
6	Mitsubishi	Japan	**Trader**
7	MGI	France	

8	Maharashtra Seamless	India	
Product	**14**	**Pipe Fittings in CS Non-NACE**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Commercial Supplying Agencies, Mumbai	India	**On Hold-B1**
2	Pipe Fit Engineers ,Vadodara (For CS,NON-NACE Butt welded fittings, BW Elbows Reducers/TEE/CAP (Seamless/Welded)	India	
3	Tube Products Incorporate	India	
4	Sawan Engineers Pvt Ltd	India	
5	T K Corporation	Korea	
6	Forged Industrial Corporation	India	
7	Tube Turn (India) Pvt Ltd	India	
8	Montex Forge Industries	India	
9	DEE Development Engineers Ltd.	India	
Product	**15**	**Forged Flanges in CS Non-NACE**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Echjay Industries, Mumbai /Rajkot	India	
2	Paramount Forge, Mumbai	India	
3	Tube Products Incorporate, Vadodara	India	
4	Sawan Engineers Pvt Ltd	India	
5	Forged Industrial Corporation	India	
6	Montex Forge Industries	India	
Product	**16**	**Ball Valves-(Fire Safe)-CS Non-NACE**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	L&T Valves Ltd	India	
2	Petrol Valves S.r.l	Italy	
3	Emerson Process Management (India) Private Limited, India	India	
4	Microfinish, Hubli	India	
5	MIR VALVE SDN BHD	Malaysia	
6	Camtech Manufacturing FZCO	UAE	
7	Novel Valves India Pvt. Ltd. (Upto 8 ")	India	
Product	**17**	**Ball Valves-CS Non-NACE**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	L&T Valves Ltd	India	

2	Forward Alloys & Castings, Mumbai (Upto 8"X900#)	India	
3	Emerson Process Management (India) Private Limited, India	India	
4	Microfinish, Hubli	India	
5	O.M.S. SALERI	Italy	
6	Friulco S.p.A	Italy	
Product	**18**	**Gate, Globe, Check , Needle Valves (CS Non-NACE)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	L&T Valves Ltd	India	
2	Forward Alloys & Castings, Mumbai (For Check valves CS upto size 6" x 900#)	India	
3	Niton Valve Industries Pvt.Ltd.,Mumbai	India	
4	Douglus Chero S.p.a (Upto 8" size)	Italy	
5	BHEL, Trichy	India	
6	Oswal Industries Ltd.(for Globe & Check Valves, size up to 6" and rating upto 600#)	India	
7	Friulco S.p.A.	Italy	
8	Valvinox s.r.l- Milan (For Needle Valves- CS Non NACE)	Italy	
9	Valve Tech Industries	India	
Product	**19**	**Pipe (Stainless Steel)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Choksi Tubes, Ahmedabad	India	
2	Productos Tubulares S.A.U	Spain	
3	Ratnamani Metals and Tubes	India	
4	TTI Tubacex Tubos Inoxidable SA	Spain	
5	Suraj Limited	India	
6	Maxim Tubes Company Pvt. Ltd	India	
7	Adinox Overseas {For Pipe(Stainless, Seamless Upto 16")}	India	
8	Venus Pipes and Tubes Limited {(For Pipes(Stainless Steel Non-NACE)}	India	
Product	**20**	**Pipes (Stainless Steel, Welded)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Ratnamani Metals & Tubes, Ahmedabad	India	
2	SOSTA BV (NACE, ASTM A 358)	The Netherlands	
3	Adinox Overseas	India	

Product	**21**	**Ball Valves (SS NACE and Non NACE)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	L&T Valves Ltd	India	
2	Emerson Process Management (India) Private Limited, India	India	
3	Friulco S.p.A (Only for SS Non NACE)	Italy	
4	Valvinox S.R.L.,Milan	Italy	
5	FG Valvole Srl	Italy	
Product	**22**	**Fittings (SS NACE & Non NACE)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Gujarat Infra Pipes, Mumbai	India	
2	Tube Products Incorporate	India	
3	Sawan Engineers Pvt Ltd	India	
4	ALLIED INTERNATIONAL S.r.l	Italy	
5	EBY Fasteners (Socket welded fittings for sizes less than 2")	India	
6	DEE Development Engineers Ltd.	India	
Product	**23**	**Flanges (SS NACE & Non NACE)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Echjay Industries, Mumbai	India	
2	CHW Forge Private Ltd.	India	
3	Officine Ambrogio Melesi & C.Srl	Italy	
4	Tube Products Incorporate	India	
5	Officine Nicola Galperti Figlio S.p.A	Itlay	
Product	**24**	**Globe , Check & Needle Valve (SS NACE)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	L&T Valves Ltd	India	
2	Niton Valve Industries Pvt.Ltd.,Mumbai,	India	
3	Douglus Chero S.p.a, (Upto 8" size)	Italy	
4	Petrol Valves S.r.l (for check valves only)	Italy	
5	M/s VALVINOX s.r.l. (I) PVT. LTD (For Globe Valves only)	India	
Product	**25**	**Gate Valve (SS NACE)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Niton Valve Industries Pvt.Ltd.,Mumbai	India	
2	Douglus Chero S.p.a, (Upto 8" size)	Italy	

Product	**26**	**Gate/Globe/Check/Needle valve (SS Non-NACE)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	L&T Valves Ltd	India	
2	Niton Valve Industries Pvt.Ltd.,Mumbai	India	
3	Friulco S.p.A	Italy	
4	Valvinox S.R.L.,Milan	Italy	
Product	**27**	**Pipes (Cu-Ni)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Alcabex Metals, Jodhpur,	India	**On Hold-B1**
2	Cubex Tubing, Hyderabad	India	**B3 Hold removed for Seamless pipes upto 14”**
3	Lebronze Industrial	France	
4	Haewon Industries Co.Ltd.	Korea	
Product	**28**	**Pipes & Fittings(Cu-Ni)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Le Brone Industrial	France	
2	TTS Ind.Co., Ltd	Korea	
3	V.D.M.	Germany	**On Hold-B1**
4	Sam Yang Metal Ind.Co.Ltd	Korea	
5	Haewon Industries Co.Ltd.	Korea	
6	KM Europa Metal Aktlengesell Schaft, (Pipes, Fittings & Flanges)	Germany	
7	Sitindustrie	Italy	
8	Bo Myung Metal Co Ltd.(Pipes, Fittings & Flanges)	Korea	
9	Cubex Tubing Limited (Seamless upto 12")	India	
Product	**29**	**Ball Valve (Aluminium Bronze)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Brooksbank Valves Ltd	U.K	
2	Orseal Ltd, Bolton	UK	
3	Hawa Valves (India) Pvt Ltd	India	
4	Valvinox S.R.L.,Milan	Italy	
5	DelVal Flow Controls Pvt. Ltd.	India	
Product	**30**	**Ball/Gate/Globe/Check/Needle valve (Al-Bronze, Leaded-Tin Bronze)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Shipham Valves	U.K	

2	Douglus Chero S.p.a,(Globe Valve-Al Bronze Only)	Italy	
3	Forward Alloys & Castings (Nickel Aluminium Bronze)	India	
4	Brooksbank Valves Ltd (For Gate, Globe, Check Valves) (Ni-Al-Bronze)	UK	
Product	**31** **Titanium valves**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Shipham Valves	UK	
2	Douglus Chero S.p.a,(Check Valves upto 6" size)	Italy	
Product	**32** **Flanges(SDSS/ Incoloy/Titanium/CS)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Officine Ambrogio Melesi & C Srl	Italy	
2	Officine Nicola Galperti Figlio S.p.A	Italy	
Product	**33** **Ball Valves (Incolloy)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Petrol Valves S.r.l	Italy	
2	BAC VALVES S.A	Spain	
Product	**34** **GRE-Pipes- Fittings / Flanges**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Future Pipe Industries B.V.	Netherlands	
2	Fiber Glass System	UAE	**On Hold-B1**
3	NOV FGS Singapore (Pte) Limited	Singapore	
4	Ameron	Netherlands	**On Hold-B1**
5	Future Pipe Industries (LLC)	Kingdom of Saudi Arabia	
6	Chemical Process Piping Pvt. Ltd.	India	
Product	**35** **Ball Valve (CPVC/PVC)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Georg Fisher Piping Sytems Ltd	Switzerland	
Product	**36** **Shutdown Valves (CPVC)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Georg Fisher Piping Sytems Ltd	Switzerland	
Product	**37** **Scrapper Tees (ASME-B31.4/31.8)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	TD Williamson	USA	
2	Pipeline Engineering Supply	UK	
3	Schulz Export GMBH	Germany	

4	Gam Raccordi, (For CS-Non NACE, CS-NACE)	Italy	
5	ALLIED INTERNATIONAL S.r.l	Italy	
6	FAI Office Di Carvico S.p.A (For CS-Non NACE, CS-NACE)	Italy	
7	Sawan Engineers Pvt. Ltd. (For CS-NACE, DSS)	India	
Product	**38** **Hinged Closures (ASME SEC VIII Div 1)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Perry Equipment Corp	USA	**On Hold-B1**
2	TD Williamson	USA	
3	Pipeline Engineering Supply	UK	
4	GD Engineering, (For CS-Non NACE & CS-NACE)	UK	
5	FAI Officine di Carvico S.p.a.(For CS-NON NACE & CS-NACE)	Italy	
6	Vee Kay Vikram & Co. LLP (CS NACE & Non-NACE)	India	
Product	**39** **Corrosion Probes**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Casasco Division	USA	**On Hold-B1**
2	Rohrback Cosasco	USA	
3	Caproco	Canada	
4	Emerson SRL	Romania	
5	Korosi Specindo	Indonesia	
Product	**40** **Pig Detector**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Pipeline Engineering & Supply Co.	UK	
2	TD Williamson	USA	
3	GD Engineering (For CS-Non NACE, CS-NACE)	UK	
4	FAI Office Di Carvico S.p.A (For CS-Non NACE, CS-NACE)	Italy	
5	Online Electronics Ltd (for Non-Intrusive only)	UK	
6	Vee Kay Vikram & Co. LLP (for CS NACE (Intrusive type))	India	
Product	**41** **FW/FM Reels & Utility Hose Reels-(UL/USCG/FM Approval)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Marine Hydraulic	India	
2	M/s Gaytri Industrial Corporation Pvt. Ltd	India	

3	Royal India Corporation, Mumbai	India	
4	Dooley Tackaberry	USA	
Product	**42**	**Strainer-(Basket type as per ASME SEC VIII Div 1 and others)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Armstrong International SA	Belgium	
2	Multitex Filtration Engineers Limited, New Delhi	India	
3	Filtration Engineers (India) Pvt Ltd, Mumbai,	India	
4	Vermeer Process Technology, BV	Netherlands	
Product	**43**	**Spray Nozzles-(NFPA 15 and appv. By UL,USA/VJTI,INDIA)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Marine Hydraulics	India	
2	Wormald Fire systems	UK	**On Hold-B1**
Product	**44**	**Continuous Drainers**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Armstrong International SA	Belgium	
Product	**45**	**5D Bends-(ASME B 31.4 / 31.8)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Fabricom	Belgium	
2	Igawara Industrial Service and trading Pte Ltd.	Singapore	**Trader**
3	PSL, Kandla	India	
4	Tenaris Dalmine	Italy	
5	Dai iChi High Frequency Co Ltd	Japan	
6	Pipeline Engineering	UK	
7	Salzgitter Mannesmann Grobblech GmbH (5D Bends - Cladded)	Germany	
8	Jindal Saw Ltd.	India	
9	Sawan Engineers Pvt Ltd	India	
10	FTV Proclad LLC,	UAE	
Product	**46**	**Chemical & Utility Hoses & Hose Connection.(Fire Resistant)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Marine Hydraulics, Mumbai	India	
2	Gaytri Industrial Corporation Pvt. Ltd	India	

3	Royal India Corp., Mumbai	India	
4	Gutteling B.V. (ABS approved)	The Netherlands	
5	Alfagomma Hydraulics SpA (ABS approved)	Italy	
Product	**47** **Choke valves**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Mokveld	USA	**On Hold-B1**
2	Petrol Valves S.r.l.	Italy	
3	WEIR Valves & Controls UK Ltd	UK	
4	Cooper Cameron	Singapore	**On Hold-B1**
5	CCI	USA	
6	JVS Engineers	India	
7	IMI Critical Engineering Korea (CS NACE)	Korea	
8	KOSO India Private Limited (CS NACE)	India	
Product	**48** **2" Adjustable choke valve 900# class NACE**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	BHEL	India	
Product	**49** **Sample Bomb-(ASME Sec VIII Div1)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Harsh Engg. Components Co., Mumbai	India	
Product	**50** **Launcher /Receiver-(ASME-B31.4/31.8) -(ASME Sec VIII Div1)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Pipeline Engineering	UK	
2	TD Williamson	USA	
3	L&T	India	
4	FAI Officine di Carvico S.p.a.,(CS-NACE & CS-NON NACE)	Italy	
5	Multitex Filtration Engineers Ltd. (CS-NACE)	India	
6	Vee Kay Vikram & Co. LLP	India	
Note	Fabrication of barrel of Launcher / Receiver can be carried out by Fabrication agency approved for LSTK Contracts with speciality items as per ONGC's Offshore Suggested Vendor List.		
Product	**51** **Gaskets**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Madras Industrial Product	India	
2	IGP	India	
3	Goodrich Gasket Pvt. Ltd.	India	

Product	**52**	**Flanges-Super DSS**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Officine Nicola Galperti Figlio S.p.A	Italy	
Product	**53**	**Shutdown Valve - CS**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Cameron Italia S.R.L (Upto 60" / Up to 300 Lbs)	Italy	**On Hold-B1**
2	Cooper Cameron, Singapore (Up to 24" / Upto 1500 Lbs)	Singapore	**On Hold-B1**
3	EL-O-MATIC India (P) Ltd., (Up to 8" / Upto 300 Lbs)	India	
4	Flowserve PTE Ltd. (Up to 12" / Upto 300 Lbs)	India	**On Hold-B1**
5	HINDLE Cockburns Ltd. (Upto 16" / Upto 300 Lbs & Upto 12" / 600-2500 Pounds)	India	**On Hold-B1**
6	Metso Automation PTE Ltd. (Upto 30" / Upto 1500 Lbs)	India	**On Hold-B1**
7	Per rin Gmbh, Germany (8" / Upto 600 Lbs & 16" / Upto 300 Lbs)	Germany	
8	Pibiviesse SPA (Upto 16" / 1500 Lbs)	Italy	**On Hold-B1**
9	Weir Valves & Controls UK Ltd. (Upto 42" / Upto 1500 Lbs)	UK	
10	Pentair Valves and control India Pvt. Limited (Upto 16" / Upto 300 Lbs & Upto 12" / 600 to 2500 Lbs)	India	
Product	**54**	**Shutdown Valve - SS - Non NACE**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Emerson Process Management (India) Private Limited, India	India	
Product	**55**	**Multi Port Valves**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Framo Engineering AS	Norway	
2	Bettis Emerson Process Management (Valve Automation)	Canada	
Product	**56**	**Shutdown Valve - Ni-Al-Br (NAB)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Brooksbank Valves Ltd. (Up to #150)	UK	
2	DelVal Flow Controls Private Limited	India	**(Up to 150#)**

Product	**57**	**Double Block & Bleed (DBB) Arrangement Valves (CS-NACE)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Oliver Valves Limited (Up to 2" and #1500)	UK	
2	Galperti Engineering and Flow Control Spa	Italy	
3	Novel Valves India Pvt. Ltd.	India	
Product	**58**	**Double Block & Bleed (DBB) Valves (DSS)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Galperti Engineering and Flow Control Spa	Italy	
2	Panam Engineers Limited	India	
3	Novel Valves India Pvt. Ltd.	India	
Product	**59**	**Gate, Globe, Check, Needle Valves (Incolloy/Inconel)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Petrol Valves SRL (Check Valves Incolloy only)	Italy	
2	Douglas Chero S.p.a (for 2" and below)	Italy	
3	Firsa S.p.a (for Cladded only)	Italy	
Product	**60**	**Flame Arrestors**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Petrol Service	France	
2	Groth Corporation	USA	
3	GPE Controls	USA	
4	Braunschweiger Flammenfilter GmbH	Germany	
5	Whersoc, S.A.	France	
6	Shand & Jurs	USA	
7	Safety Systems UK Ltd.	UK	
8	Protego India Pvt.Ltd	India	
Product	**61**	**In-Line Separator (De-Liquidiser)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	FMC Separation Systems BV	The Netherlands	
2	Sulzer Netherlands Holding BV (Sulzer AG)	The Netherlands	
Product	**62**	**Fittings (Incoloy 825/ Inconel 625)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Promat BD Ltd. Middle East	UAE	

Product	**63**	**Flanges (Incoloy 825/ Inconel 625)**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Promat BD Ltd. Middle East		UAE	
Product	**64**	**Incoloy 825/ Inconel 625 Clad Fittings, Pipes & Flanges**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	PT Cladtek Bi-Metal Manufacturing		Indonesia	
2	FTV Proclad L.L.C		UAE	
Product	**65**	**Flexible Hoses for Jumpers. API17K**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Contitech Rubber Industrial KFT		Hungary	
Product	**66**	**Butterfly Valves - NAB**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Forward Alloys & Castings		India	
Note 1	Vendors qualified for CS NACE shall also be considered qualified for CS Non NACE items. The list given above for CS Non NACE is additional list for vendors supplying CS Non NACE items only.			
Note 2	Vendors qualified for DSS material shall also be considered qualified for SS NACE / SS-Non NACE materials.			
Note 3	Vendors qualified for Nickel-Aluminium-Bronze (NAB) shall be considered qualified for Lead-Tin-Bronze (LTB) also and vice versa.			
Note 4	Vendors qualified for SS NACE shall also be considered qualified for SS Non NACE items. The list given above for SS Non NACE is additional list for vendors supplying SS Non NACE items only.			
Note 5	Vendors qualified for items under Pipeline Discipline can be considered acceptable for similar items under Piping Discipline provided the applicable specifications are similar and meet the specification requirements.			
Note 6	Vendors listed in the category of Ball Valves for all types of MOC can be considered for metal seated ball valves also in the respective caregories. However, Vendor is required to submit track record of supply of metal seated valves at the time of submission of PS.			
Note 7	Listed vendors for Globe valves can also be considered as qualified for Needle valves provided they manufacture the same.			

Mechanical and Rotary Equipment				
Product	**1**	**Deck Crane-API-2C monogram**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Favelle Favco Cranes		Australia / Malaysia	
2	Raina Engineers, Mumbai		India	
3	Oil States Skagit SMATCO, LLC		USA	
4	ECI, Energy Cranes International		USA	**On Hold-B1**
5	National Oil Well		USA	
6	ITALGRU SRL		Italy	
7	Oil States Industries (India) Private Limited		India	
8	M.E.P-Pellegrini Marine Equipments S.R.L		Italy	
Product	**2**	**Pneumatic (Air/HC Gas) driven pumps**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Haskel Energy		USA/UK	
2	CP Pumps & Systems FZE (For both Air and Gas)		UAE	
3	McFarland TriTan LLC		USA	
Product	**3**	**Fabricator of Pressure Vessels- ASME "U" code stamped (Air Receiver, Nitrogen Receiver, Glycol Seal Pot, Crude Condensate Drum, Well Activation Vessel, Hot Oil Expansion Tank, Oxygen Scavenger Reaction Vessel & Skimmer vessels)**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	L&T Hydrocarbon Engineering Ltd.		India.	
2	R.D.Engineers, Mumbai		India	
3	Kilburn Engg.Ltd, Mumbai		India	
4	BHEL		India	
5	GMMOS		Dubai	
6	FAI Officine di Carvico S.p.A.,		Italy	
7	Indcon Projects & Equipment Ltd., New Delhi		India	
8	Deutche Babcock		Abu dhabi	**On Hold-B1**
9	ISGEC		India	
10	Multitex Filtration Engineers Ltd.		India	
11	Perfect International FZC		UAE	
12	NPCC, Abu Dhabi		UAE	

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13	Sparklet Engineers Pvt Ltd	India	**Banned Firms** (20.04.2022 to 19.04.2024)
14	Petra Fabricators Sdn Bhd	Malaysia	
15	NASH Engineering FZCO	UAE	
16	Unique Chemo Plant Equipments	India	
17	Grand Prix Engineering Pvt.Ltd	India	
18	Steelcon Heat Transfer Equipment SDN BHD	Malaysia	
Product	**4** **Inst./Utility Gas Systems-Packaged Item (Process Licensor)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Kilburn Engg Ltd, Mumbai	India	
2	INDCON Projects & Equipments Ltd.	India	
Note	All the vendors under the Fuel Gas Conditioning Skid Category in this list can be considered acceptable for IUG Systems also.		
	Fabricator for the above Vessels to be selected from one of those Vendors listed in the Suggested Vendor List under item "Fabricator of Pressure Vessels"(Product Sl No.3). Any fabricator other than as stipulated in Sl. No. 3, selected shall be ASME 'U' code stamp authorized fabricator with proven PTR (Past Track Record) of minimum two years of offshore experience and shall be fabricating in the last ten years. Necessary documents are to be submitted / incorporated in Purchase Specification (PS) for review of company.		
Product	**5** **Chain Pulley Block**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Ingersoll Rand	USA	
2	Beebe International Inc	USA	**On Hold-B1**
3	V.G.Ajmera & Brothers	India	**Trader**
4	Kito	Japan	
5	Air Dyne	UK	**Trader**
6	J Barnsley	UK	
7	Ansel Jones	UK	
8	Hercules Hoist	India	
9	STAHL Crane Systems GmbH	Germany	
Product	**6** **Heat Exchangers-Shell-Tube TEMA code**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Bharat Heavy Plates & Vessels	India	
2	FBM – Hudson	Italy	
3	Alfa Laval Olmi S.p.A	Italy	
4	HHI	Korea	
5	L&T Ltd.	India	
6	GEI Industrial Systems Ltd. (For Gas Coolers)	India	**On Hold-B1**
7	FAI Officine di Carvico S.p.A	Italy	

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8	Bukit Fraser Thermal Technology Sdn Bhd	Malaysia	**On Hold-B3**
9	Godrej & Boyce Mfg. Co. Ltd	India	
10	GR Engineering Works	India	
11	ISGEC	India	
12	TEMA India Ltd.	India	
13	Indcon Projects & Equipments Ltd	India	
14	Unique Chemo Plant Equipments	India	
15	Tinita Engineering Private Limited	India	
Product	**7**	**Process Licensors-Separators (HP/LP / Cyclone / Test / Surge Tank / Knock Out Drums - ASME 'U' code stamped)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Frames Process Systems, B.V	Netherland	
2	Global Process System	Dubai / Malaysia	
3	GasTech Engineering LLC	USA	
4	Laksel EPS Technologies Pte Ltd	Singapore	
5	Cameron Manufacturing (India) Pvt Ltd	India	
6	Multitex Filtration Engineers Ltd	India	
7	Koene Consultancy Holland BV (KCH Separation BV)	Netherlands	
8	FMC Separation System B.V.	Netherlands	
9	Zeta-pdm Ltd	United Kingdom	
10	Jord International Pte Ltd	Singapore	
11	Innovative Synergy Solutions Sdn Bhd	Malaysia	
12	Bliss Anand Private Limited (In technical License agreement with M/s Gastech Engineering LLC, USA)	India	Enlisted under technical license agreement with M/s GasTech Engineering LLC USA, valid upto 12.02.2029 as a process licensee, subject to endorsement of all design documents by licensor Gastech Engineering LLC, USA
Note	Fabricator for the above Separators to be selected from one of those Vendors listed in the Suggested Vendor List under item "Fabricator of Pressure Vessels". Any other fabricator selected shall be ASME 'U' code stamp authorized fabricator with PTR (Proven Past Track Record) of minimum two years of offshore experience and shall be fabricating in the last ten years. Necessary documents are to be submitted / incorporated in Purchase Specification (PS) for review of company.		

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Product	**8**	**Fuel Gas Conditioning Skid-Packaged Item (Process Licensor)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Oil & Gas Systems Ltd	UK	
2	GasTech Engineering LLC	USA	
3	Texas Systems & Controls	USA	**On Hold-B1**
4	Global Process System	Dubai/Malaysia	
5	Premabergo Italiano SRL	Italy	**On Hold-B1**
6	Multitex Filtration Engineers Ltd.	India	
7	GlobalTech Systems Engineering Pte Ltd	Singapore	
8	Innovative Synergy Solutions Sdn Bhd	Malaysia	
9	BGR Energy Systems Limited	India	
10	Bliss Anand Private Limited (In Technical License Agreement with M/s GasTech Engineering LLC, USA)	India	Enlisted under technical license agreement with M/s GasTech Engineering LLC USA, valid upto 12.02.2029 as a process licensee, subject to endorsement of all design documents by licensor Gastech Engineering LLC, USA.
Note	Fabricator for the above Vessels to be selected from one of those Vendors listed in the Suggested Vendor List under item "Fabricator of Pressure Vessels"(Product Sl No.3). Any fabricator other than as stipulated in Sl. No. 3, selected shall be ASME 'U' code stamp authorized fabricator with proven PTR (Past Track Record) of minimum two years of offshore experience and shall be fabricating in the last ten years. Necessary documents are to be submitted / incorporated in Purchase Specification (PS) for review of company.		

Product	**9**	**HVAC-Packaged Item (Process Licensor/Compressor OEM or Authorised by Compressor OEM)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Specialist Services	UAE	
2	Mpower Projects Pty Ltd.	Australia	
3	ABB	Switzerland	**On Hold-B1**
4	HVAC & Refregeration Engineering Ltd., ABERDEEN	Scotland	
5	Hi Air Korea Co. Ltd.	Korea	
6	Harve Engineering Pte. Ltd. (Major equipments of the package should be from M/s Carrier)	Singapore	
7	JL Marine & Engineering Pte Ltd.	Singapore	
8	Hertal Offshore	The Netherlands	
9	Seven Seas Services LLC	Dubai	
10	Universal Voltas LLC (For Pressurization & Ventilation System only)	Dubai	

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11	FlaktGroup Singapore Pte Ltd	Singapore	
12	AAE Engineering (Thailand) Company Limited	Thailand	
Product	**10** **Sewage Treatment Unit**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Red Fox, Lousiana (Biological type)	USA	
2	Sasakura Engg. Co. Ltd.	Japan	
3	De Nora Water Technologies Texas LLC (Electro-catalytic type)	USA	
4	EVAC OY (Biological Type)	Finland	
Product	**11** **Utility Water Pump**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Peerless	Australia	
2	DMW Corporation	Japan/India*	
3	ITT Corporation	USA	
4	WPIL	India	
5	Flowserve Hamburg GmbH	Germany	
Note	**** Vendor is required to fulfil requirement of Note 5(a) of SVL for future supplies***		
Product	**12** **Rotary Gear Pumps**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Albany	UK	
2	Viking	UK	**On Hold-B1**
3	Hayward Tayler	UK	
4	MAAG	Italy	
5	Viking Pump, Inc.	USA	
Product	**13** **Fire Water Pump-Packaged item (Pump/Enginer OEM or Authorised by Pump/Engine OEM)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Fluid Power	USA	
2	Peerless Pumps/Peerless Pty Australia Ltd	USA/Australia	
3	Flowserve Corporation	USA	
4	Flowserve (Austria) GmbH	Austria	**On Hold-B1**
5	KSB AG.	Germany	**On Hold-B1**
6	SPP Pumps Ltd	UK	
7	Nuovo Pignone International S.r.l	Italy	
8	Gruppo Aturia (Pump OEM)	Italy	
Product	**14** **Instrument & Utility Air Compressors-Packaged Item (Compressor OEM or Authorised by Compressor OEM)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Tide Air	USA	**On Hold-B1**
2	Atlas Copco	USA/India*	
3	Norwalk Compressor Co.	USA	

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4	Comp air	UK	
5	Ingersoll Rand	USA/India*	
6	Airpack	Netherland	
7	Bellies & Morcam	UK	
8	Kobe Steel Ltd	Japan	
9	Neuman & Essar	USA	
10	Indcon Projects & Equipment Ltd., New Delhi (only for Instrument Air Dryers/Compressed Air Dryer)	India	
11	Elgi Sauer Compressor Ltd.	India	
12	Bellelli Engineering S.p.A	Italy	
13	Sinergia Srl	Italy	
14	Firscomp Co., Ltd	Korea	
Note	** Vendors are required to fulfil requirement of Note 5(a) of SVL for future supplies*		
Product	**15**	**Sea Water Lift Pumps**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Clyde Union Ltd.	UK	
2	KSB AG.	Germany	**On Hold- B1**
3	Ebara Corporation	Japan	
4	DMW Corporation	Japan	
5	SPP Pumps Ltd	UK	
6	BPCL	India	
7	Flowserve (Austria) GmbH	Austria	**On Hold- B1**
8	Hayward Tyler Engineered Products Ltd	UK	**On Hold- B1**
Product	**16**	**Produced Water Conditioner System (PWCS)-Packaged Item (Process Licensor)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Peerless Europe Limited	UK	
2	Global Process System	Dubai/Malaysia	
3	Siemens Pte.Ltd.	Singapore	
4	FLSmidth gMAX Systems	USA	
5	Process Group Pty Ltd	Australia	
6	Cetco Oilfield Services	Malaysia	
7	Enhydra Ltd	UK	
8	Innovative Synergy Solutions Sdn Bhd	Malaysia	

# Indicates changes in Rev No. : 12.3

Product	**17**	**Engine Supplier and Packager**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Cooper Engergy Services		USA	**On Hold-B1**
2	Man Energy Solution SE		Germany	
3	Wartsila		Finland	
4	Caterpillar		USA	
5	Ruston Diesel		USA	
6	Cummins		USA	
7	MTU Detroit Inc.		USA	
NOTE	Vendors who are Engine Manufacturer from the above list are also acceptable as Engine Supplier for other products of suggested vendor list. All the vendors listed above can be considered acceptable for portable DG Sets also			
Product	**18**	**Chlorinators**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Engel Hard		UK	**On Hold-B1**
2	Titanium Tantalum Products Ltd., Chennai		India	
3	De Nora Water Technologies Texas LLC		USA	
4	PREMABERGO ITALIANA SRL		Italy	
5	Sespi Srl		Italy	
6	Cumberland International LLC		UAE	
7	Kalf Engineering Pte Ltd.		Singapore	
8	AZIENDA CHIMICA GENOVESE S.R.L. (ACG)		Italy	
9	Cumberland Electrochemical Limited		UK	
Product	**19**	**Fine Filters**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	GPS (Global Process Systems Sdn Bhd)		Malaysia	
2	Eta Process Plant		UK	
3	Innovative Synergy Solutions Sdn Bhd		Malaysia	

Product	**20**	**Water Deoxygenation Tower**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Eta Process Plant Ltd.		UK	
2	Edwards Limited		UK	
3	Innovative Synergy Solutions Sdn Bhd		Malaysia	
Product	**21**	**Main Injection / Main Oil Line Pumps**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Clyde Union Ltd.		UK	
2	Ebara Corporation		Japan	
3	Clyde Union SAS		France	
4	Clyde Union DB Ltd		UK	
5	Sulzer Pump		UK/Switzerland/India*	
6	DMW Corporation		Japan	
7	Flowserve S.A.		Argentina	
8	Arai Pump Manufacturing Co. Ltd.		Japan	
9	KSB Limited (JV with KSB SE & Co. KGaA Frankenthal, Germany)		India	**Banned Firms** {15.03.2024 to 14.03.2026)
Note	**** Vendor is required to fulfil requirement of Note 5(a) of SVL for future supplies***			
Product	**22**	**Sump Caisson**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	ESI		USA	
2	ENVIRO-TECH SYSTEMS L.L.C		USA	
Product	**23**	**Chemical Storage & Dosing System-Packaged Item (Chemical Dosing Pump OEM or Authorised by Chemical Dosing Pump OEM)**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Accudyne Industries India Pvt Ltd.		India	
2	Nikkiso		Japan	
3	Bran & Luebbe		UK	
4	Petronash FZE		UAE	
5	CP Pumps & Systems FZE		UAE	
6	GlobalTech Systems Engineering Pte Ltd		Singapore	
7	Innovative Synergy Solutions Sdn Bhd		Malaysia	
8	Enpro Industries Pvt Ltd		India	
9	Controls & Electrics Pte Ltd.		Singapore	
9 (a)	Controls And Electrics System Manufacturing Ltd, UAE		UAE	**Additional Manufacturing Location of vendor "M/s Control & Electrics Pte. Ltd., Singapore**
10	Positive Metering Pupms India Pvt. Ltd		India	
Product	**24**	**Sump Pumps**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Sulzer Pumps		India	
2	DMW Corporation		Japan	
3	KirloskarEbara Pumps Ltd		India	
4	Ebara Corporation		Japan	
5	Weir Gabbioneta Srl		Italy	

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Product	**25**	**Controlled Volume Pumps**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Union		USA	**On Hold-B1**
2	Nikkiso		Japan	
3	Milton Roy		USA	
4	Bran & Luebbe		Germany	
5	Lewa		Germay	
6	Wallace & Tieman		UK	**On Hold-B1**
7	Check Point Fluidic Systems International Ltd		USA	
8	Officine Meccaniche Gallaratesi S.P.A.		Italy	
9	Pulsafeeder Inc		USA	
10	SEKO S.p.A		Italy	
11	Positive Metering Pupms India Pvt. Ltd.		India	
Product	**26**	**Rotary Positive Displacement Pumps**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Bomeman		Germany	**On Hold-B1**
2	Viking Pump, Inc.		USA	
3	Ingersoll / Dresser Intl.		USA	**On Hold-B1**
4	Seepex Germany		Germany	
5	Netzsch Pumpen & Systeme Gmbh		Germany	
Product	**27**	**Reciprocating Plunger Pumps**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	DMW Corporation		Japan	
2	WEPUKO Hydraulic GMBH		Germany	
3	RAM PUMPS LTD.		UK	
4	Clyde Union Canada Limited		Canada	
5	Peroni Pompe SpA		Italy	
Product	**28**	**Vacuum Pumps**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Nash		USA	**On Hold-B1**
2	Busch GVT Ltd		UK	
3	Edwards Ltd		United Kingdom	
Product	**29**	**Process Gas Compressor (PGC)-API 617**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	KHI		Japan	
2	Dresser Rand		USA	
3	Cooper Energy Services		USA	**On Hold-B1**
4	Man Energy Solution SE		Germany	
5	Ebara Corporation		Japan	

6	Solar Turbines Inc.	USA	
7	Demag-De Laval (Siemens AG, Germany)	Germany	
8	Nuovo Pignone S.p.A (GE Group of Company)	Italy	
Product	**30 Gas Turbine- API 616**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	General Electric	USA	
2	Rolls-Royce	UK	
3	Alstom, UK (M/s Siemens AG, Germany)	Germany	1) CT1 and CT2 blade MOC for SGT 400 Gas Turbine shall be INCO 6203 DS. 2) Whenever H2S level is 600 ppm or more in fuel gas composition, SGT 400 Gas Turbine shall be supplied with Gold plated C-seal burners.
4	Solar Turbines	USA	
5	Nuovo Pignone International S.r.l.	Italy	
Product	**31 CO2 Snuffing System**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	New Age Industries, Mumbai	India	
2	Newage Fire Protection Engineers Pvt Ltd	India	

Product	**32**	**Air Cooled Heat Exchanger**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	GEA Energy Systems India Ltd.		India	**On Hold-B1**
2	GEI Industrial Systems Ltd		India	
3	L&T Ltd		India	
4	Paharpur Cooling Towers		India	
5	Bharat Heavy Plates & Vessels		India	
6	Bukit Fraser Thermal Technology Sdn Bhd		Malaysia	
7	BGR Energy Systems		India	
8	Jord Malaysia Sdn Bhd		Malaysia	
9	Patels Airtemp (India) Limited		India	
10	Enginemates Heat Transfer Pvt. Ltd		India	
Product	**33**	**Waste Heat Exchangers / Waste Heat Recovery Unit-WHRU**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Bharat Heavy Plates & Vessels		India	**On Hold-B1**
2	Thermax Babcock & Wilcox Energy Solutions Limited		India	
3	Citech Energy Recovery System Malaysia Sdn Bhd		Malaysia	
Product	**34**	**Demisters**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Evergreen Technologies Pvt. Ltd		India	
2	Continental Profiles Ltd		India	
3	Pace Engg Industries, Mumbai		India	
Product	**35**	**Nitrogen Generator-Packaged Item (Membrane/Air Compressor OEM or Authorised by Membrane/Air Compressor OEM)**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Atlas Copco Air & Gas Purification (Membrane & PSA Type)		Netherlands	
2	Air Pack (Membrane)		Netherlands	
3	Parkar Hannifin Corporation (Membrane & PSA Type)		USA	
4	Air Liquide (Membrane)		France	
5	Bellelli Engineering S.p.A (Membrane & PSA Type)		Italy	
6	Sinergia Srl (Membrane Type)		Italy	
7	Sinergia Srl (PSA type)		Italy	
8	Firscomp Co., Ltd (Membrane & PSA Type)		Korea	

Product	**36**	**Blow Case for Sump Caisson**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	ESI		USA	
2	ENVIRO-TECH SYSTEMS L.L.C		USA	
Product	**37**	**Water Maker**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	MEMBRANE SRL, (RO Type)		Italy	
2	AQUA - CHEM (RO & Vapour Compression Type)		USA	
3	MECO (Vapour Compression Type)		USA	
4	Rochem Separator System (India) Pvt. Ltd.		India	
5	Alfa Laval Copenhagen A/S (Vapour Compression type)		Denmark	
6	ITT C'treat LLC (RO Type)		USA	
7	Hatenboer-Water BV (RO Type)		Netherlands	
8	QUANTUM INGENIERIA EUROPEA, S.L., (Reverse Osmosis Type)		Spain	
Product	**38**	**ATF Handling Pump**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Blackmer (A Dover Resources Company)		USA	
Product	**39**	**GDU (Gas De-hydration Unit)-Packaged Item (Process Licensor)**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Cameron Manufacturing (India) Pvt Ltd		India	
2	Global Process Systems Sdn Bhd		Malaysia	
3	Frames Process Systems, BV		Netherlands	
4	GasTech Engineering LLC		USA	
5	Process Group		Australia	
6	Innovative Synergy Solutions Sdn Bhd		Malaysia	
7	Bliss Anand Private Limited (In Technical License Agreement with M/s GasTech Engineering LLC, USA)		India	Enlisted under technical license agreement with M/s GasTech Engineering LLC USA, valid upto 12.02.2029 as a process licensee, subject to endorsement of all design documents by licensor Gastech Engineering LLC, USA.

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Note	Fabricator for the above Vessels to be selected from one of those Vendors listed in the Suggested Vendor List under item "Fabricator of Pressure Vessels"(Product Sl No.3). Any fabricator other than as stipulated in Sl. No. 3, selected shall be ASME 'U' code stamp authorized fabricator with proven PTR (Past Track Record) of minimum two years of offshore experience and shall be fabricating in the last ten years. Necessary documents are to be submitted / incorporated in Purchase Specification (PS) for review of company.			
Product	**40**	**GSU (Gas Sweetening Unit)-Packaged Item (Process Licensor)**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Cameron Manufacturing (India) Pvt Ltd		India	
Note	Fabricator for the above Vessels to be selected from one of those Vendors listed in the Suggested Vendor List under item "Fabricator of Pressure Vessels"(Product Sl No.3). Any fabricator other than as stipulated in Sl. No. 3, selected shall be ASME 'U' code stamp authorized fabricator with proven PTR (Past Track Record) of minimum two years of offshore experience and shall be fabricating in the last ten years. Necessary documents are to be submitted / incorporated in Purchase Specification (PS) for review of company.			
Product	**41**	**HP & LP Flare System**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Zeeco Inc.		USA	
2	GBA Engineering & Construction Ltd.		UK	
3	Callidus Technologies		USA	
4	Argo Flare Service Ltd		UK	
5	Flareon Inc		Malaysia	
6	Flare Industries		USA	
Product	**42**	**Travelling Bridge Crane**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	ICON Engineering LLP		India	
2	STAHL Crane Systems GmbH		Germany	
3	Safex Energy Pvt Ltd		India	
Product	**43**	**Jet Fuel System**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Wilmax Engineering Pte. Ltd.		Singapore	
2	IMENCO AS		Norway	

Product	**44**	**Diesel Centrifuge**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Alfa Lavel Mid Europe GmbH		Germany / India*	
2	GEA Westfalia Separator Systems GmbH		Germany	
Note	** Vendor is required to fulfil requirement of Note 5(a) of SVL for future supplies*			
Product	**45**	**Monorail Hoist (Manually Operated Geared Trolley with Electric Hoist)**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	ICON Engineering LLP		India	
2	STAHL Crane Systems GmbH		Germany	
3	Safex Engineers Pvt. Ltd.		India	
Product	**46**	**Filters & Filter Coalescers**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Multitex Filtration Engineers Limited		India	
2	Bea Technologies Spa		Italy	
3	Sulzer Chemtech AG		Switzerland	
4	Petromar Engineered Solutions Pvt Ltd		India	
Product	**47**	**Air/Natural Gas Operated Double Diaphragm Pumps**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	WARREN RUPP INC		USA	
Product	**48**	**Natural Gas Compressors (For Low Pressure Application) - API 618 / API 619**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Howden Compressors Ltd. (Screw Compressor only)		UK	
2	IHI		Japan	
Product	**49**	**Hot Oil Pump**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	DMW Corporation		Japan	
2	Flowserve India Controls Pvt. Ltd.		India	
3	Arai Pump Manufacturing Co. Ltd.		Japan	
4	Weir Gabbioneta Srl		Italy	
5	KSB Limited		India	**Banned Firms** (15.03.2024 to 14.03.2026)

Safety			
Product	**1**	**Dry Chemical Powder Skid**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Tyco Fire Products LP (Tyco Fire Protection Products)	USA	
2	Wormald	USA	**On Hold- B1**
3	Vijay Fire Vehicles & Pumps Ltd	India	
Product	**2**	**Portable Fire Extinguishers**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Doopley Fire System Inc.	USA	**On Hold- B1**
2	Zeneth Fire Services, Mumbai	India	
3	SHM Products Private Limited, Mumbai	India	**Trader**
4	Nitin Fire Protection Ind.Ltd	India	
Product	**3**	**Life Preservers (Life Jackets, Life Ring Buoys) Inflatable Life Rafts,First Aid Kits, , Fire Blanket, Fireman's outfit, Breathing equipment,Stretcher, Personnel, Baskets, Scramble nets**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Billy Pugh Co. Inc. (Personnel Basket, Work Vest, Wind Socks, Swing Ropes, Stretcher, Ring Buoys, Scramble Net)	USA	
2	Galvanisers India	India	
3	SHM Products Private Limited, Mumbai	India	**Trader**
4	Unicare Emergency Eqpt Pvt Ltd , Mumbai (Life Buoy, Life Jacket)	India	
5	Eastern Shipchandlers	India	**Trader**
6	Besto Life Saving (Life Jackets)	The Netherlands	
7	Viking Life-Saving Equipment Pte. Ltd (Life Raft, Work Vest, Life Jacket)	Singapore	
8	Solas Marine Services Co., LLC	UAE	**Trader**
9	Vijay Sabre Safety Pvt. Ltd., (Only for Breathing Apparatus Set)	India	

Product	**4**	**Breathing Air Compressor**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Aerotecnica Coltri Spa		Italy	
Product	**5**	**Eye Wash & Safety Shower**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Unicare Emergency Equipment, Mumbai.		India	
2	Nippon Encon. MFG.Co.Ltd.		Japan	
Product	**6**	**Helicopter Rescue Kit**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Bristol Uniform Ltd., Bristol (Fire Protective clothing)		UK	
2	Doopley Fire Systems Inc.		USA	**On Hold- B1**
3	Joseph Leslie & Co.Mumbai (Helmet, Safety Lamps, Fire Protective Clothing, Gloves,Fireman outfit)		India	
4	Solas Marine Services Co., LLC		UAE	**Trader**
Product	**7**	**Personnel Protection Equipment from H2S Exposure**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Dooley Fire Systems Inc.		USA	**On Hold- B1**
2	Nohmi Fire Safety Equipt.Co.		USA	**On Hold- B1**
3	Drager Aktiengesell Schaft		Germany	
Product	**8**	**Survival Craft**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Palfinger Marine Safety AS		Norway	
2	Norsafe		Norway	
Product	**9**	**Clean Agent System**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Nitin Fire Protection India Ltd.(HFC 227ea-FM200)		India	
2	Tyco Fire Suppression & Building Products (Sapphire System-Novec 1230)		US	
3	Kidde Fire Protection (Novec 1230)		UK	
Product	**10**	**Water Mist & CAF Fire Extinguisher Trolley Mounted (Capacity-50L)**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Aska Equipments Ltd		India	

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Instrumentation				
Product	**1**	**Pressure Gauges**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Budenberg Gauge Co. Ltd.		UK	
2	Wika Alexander Wiegand SE & Co. KG, Germany & Wika Instruments India Pvt. Limited, India		Germany/India	
3	Aschcroft		USA	
4	Ametek		USA	
5	Manometer		India	
6	Gauges Bourdon (India) Pvt Ltd		India	
7	Baumer Technologies India Pvt Ltd		India	
8	ITT Barton		USA	
9	Thermal Instrument India Pvt. Ltd.		India	
10	Goa Instruments Industries Private Limited		India	
Product	**2**	**Diff. Pressure Gauges & Indicators**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Budenberg Gauge Co. Ltd		UK	
2	Wika Alexander Wiegand SE & Co. KG, Germany & Wika Instruments India Pvt. Limited, India		Germany/India	
3	Burdon		France	
4	Cameron Technology UK Ltd		UK	
5	Meriam Process Technologies		USA	
6	Aschcroft		USA	
7	Gauges Bourdon (India) Pvt Ltd		India	
8	Baumer Technologies India Pvt Ltd		India	
9	ITT Barton		USA	
10	Thermal Instrument India Pvt. Ltd.		India	
Product	**3**	**Telemetry Interface Cabinet**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Fabricons Technocrat Solutions Pvt. Ltd.		India	
2	Yokogawa		Japan/India	
3	e-production solution		USA	**On Hold-B1**
4	Elec. Mech. Corporation, Mumbai		India	

Product	**4**	**Well Fire Shutdown Panel & Test Separator Shutdown Panel-Packaged Item**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	e-production solution	USA	**On Hold-B1**
2	WT Petrotech	USA	
3	Controls and Electrics (Boustead) Pte Ltd	Singapore	
4	Wormald	UK	**On Hold-B3**
5	S.A.Technologies Pte.Ltd	Singapore	
6	Wilmax Controls System Pte.Ltd	Singapore	
7	Frames Energy Systems	Netherlands	
8	Sulaiman Petrotech ME FZE	UAE	
9	Haven Automation	Singapore	
10	Proserv Middle East	UAE	
11	Petronash FZE, Dubai	UAE	
12	Bliss Anand Pvt. Ltd	India	
13	Inrada Oil & Gas	Netherlands	
14	Trisystems Engineering Sdn Bhd - for limited purpose of **Smart Platforms.**	Malaysia	
15	Control and Electrics System Manufacturing Ltd	UAE	
Product	**5**	**Distributed Control System (DCS)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Foxboro (Invensys)	USA, India	
2	Toshiba	Japan	
3	Honeywell Automation India Ltd (HAIL)	India	
4	Yokogawa	Japan	
5	Emerson Process Management	USA/UK/ Singapore/ India	
6	Yokogawa India Ltd	India	
7	ABB	USA, India	
8	Siemens AG	Germany	
Product	**6**	**Pneumatic Pressure Switch/HI-LO Pilots (Indicating dial type)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	e-Productions solution	USA	**On Hold-B1**
2	WT Petrotech	USA/BATAM	

3	Halliburton Energy	USA	
4	Ruelco	USA	
Product	**7** **Pressure Switches (Explosion Proof)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	SOR	USA	
2	ITT Neo Dyn	USA	
3	Aschcroft	USA	
4	Yokogawa	Japan	
5	Delta Mobrey Limited	UK	
6	e-Productions solution	USA	**On Hold-B1**
7	Dag Process	India	**On Hold-B3**
8	United Electric Control	USA	
9	Pyropress Engineering Company Ltd.	UK	
Product	**8** **Pressure Relief Valves (PRV)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Sapag – Alshtom	France	**On Hold-B1**
2	Tyco Flow Control	UK/USA	**On Hold-B1**
3	Fukui	Japan	
4	Farris Engineering- A division of Curtiss-Wright Flow Control Corporation, USA)	USA	
5	Anderson Greenwood Crosby Sanmar Ltd. (All MOC's except Hastalloy and Titanium)	India	
6	Triangle (Tyco Flow Control)	UK	**On Hold-B1**
7	Ham-Let Advanced Control Technology	Israel	
8	Safety System	UK	**On Hold-B1**
9	Bliss Anand Pvt. Ltd	India	
10	LESER Gmbh & Co. KG	Germany	
11	Fainger Leser Valves Pvt.Ltd.	India	
Product	**9** **Pressure Indicating Controllers (PIC)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Yokogawa	Japan	
2	Emerson Process Management (Fisher)	UK/USA	
3	Foxboro	USA	

4	Bristol Babcock	USA	
5	ITT Barton	USA	**On Hold-B1**
Product	**10** **Large Case Field Recorders**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Foxboro	USA	
2	Cameron Technology UK Ltd.	UK	
3	Yokogawa	Japan	
4	Bristol Babcock	USA	
5	ITT Barton	USA	**On Hold-B1**
Product	**11** **Field Pneumatic Integrators**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Yokogawa	Japan	
Product	**12** **Gauge Glass & Cocks**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Jerguson Gauge & Valve Company	USA	
2	Penburty	USA	**On Hold-B1**
3	Technomatic	India	
4	Protolina, Mumbai	India	
5	Chemtrols Samil India Pvt Ltd	India	
6	Daniels	USA	**On Hold-B1**
7	Bliss Anand Private Limited	India	
8	Levcon	India	
9	Samil	South Korea	
10	Gauges Bourdon India Pvt Ltd	India	
Product	**13** **Level Instruments- Pneumatic (Level- controls & Level Switches)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Magnetrol International NV (Only Level Switches)	Belgium	
2	Dresser Maseneilan (Only Level Controls)	USA	
3	S.O.R. (Only Level Switches)	USA	
4	e-Productions solution (Weatherford Production)	USA	**On Hold-B1**
5	Chemtrols (Only Level Switches)	India	**On Hold-B3**
6	GE Oil & Gas India Pvt. Ltd (Displacer Type Level Transmitters)	India	
7	Dresser Maseneilan	France	

8	Eckardt	West Germany	**On Hold-B3**
Product	**14**	**Level Switches (Explosion Proof)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	S.O.R.	USA	
2	Tokyo Keiso	Japan	
3	Magnetrol	Belgium	
4	Chemtrol Samil India Pvt Ltd	India	
5	Dag Process	India	**On Hold-B3**
6	Pyropress Engineering Company Ltd.	UK	
Product	**15**	**Pressure Transmitters(PT) / P to I Converters**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Emerson Process Management / Emerson Process Management India Private Limited (Rosemount make)	USA/UK/Singapore/India	
2	ABB Limited	India	
3	Yokogawa / Yokogawa India Limited	Japan/India	
4	Honeywell / Tata Honeywell	USA/India	
5	ABB-Kent	UK	
6	Instrumentation Ltd. Kota (PT only)	India	
7	Pyropress Engineering Company Ltd.	UK	
Product	**16**	**Solenoid Valves**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	ASCO	USA	
2	Maxseal (IMI Norgren Herion Ltd.)	UK	
3	Skinner (Parker)	USA	
4	Blakeborough (Weir Valve and Control)	UK	
5	Herion (IMI Norgren Herion Ltd.)	Germany	
6	Bifold Fluid Power Limited	UK	
7	Versa Products Company, Inc	USA	
8	Rotex Automation Limited	India	
Product	**17**	**Turbine Flow Meters**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Daniel OpCo LLC (Daniel Measurement and Control)	USA	

2	Flow Tech.	USA	
3	FMC Smith Meters Inc	USA	
4	Cameron Flow Control Technology (UK) Ltd.	U.K	
5	Bopp & Reuther	Germany	
6	Faure Herman (For Liquid only)	France	
7	Rockwin Flowmeter India Pvt. Limited	India	
Product	**18** **Temperature Transmitters**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Emerson Process Management / Emerson Process Management India Private Limited (Rosemount make)	USA/UK/Singapore/India	
2	Yokogawa/ Yokogawa India Limited	Japan / India	
3	ABB-Kent	UK	
4	Honeywell Automation, India Ltd	India	
5	Fuji	Japan	
6	Pyropress Engineering Company Ltd.	UK	
Product	**19** **Temperature Gauges**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Aschcroft	USA	
2	Nagano Keiki	Japan	
3	A.N. Instruments	India	**On Hold-B3**
4	Gauges Bourdon (India) Pvt Ltd	India	
5	Wika Alexander Wiegand SE & Co. KG, Germany & Wika Instruments India Pvt. Limited, India	Germany/India	
6	Baumer Technologies India Pvt Ltd	India	
7	Goa Instruments Industries Pvt Ltd	India	
8	Thermal Instrument India Pvt. Ltd.	India	
Product	**20** **ESD & FSD Valves**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	e-Productions solution (Weatherford Production)	USA	**On Hold-B1**
2	Sigma	USA	
3	Versa Products Company Inc.	USA	

Product	**21**	**Temperature Switches**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Delta Mobrey Limited		UK	
2	NAGANO-KEIKI		Japan	
3	SOR		USA	
4	ITT, Snider		Germany	
5	Aschcroft		USA	
6	ASCO		USA	
7	United Electric Control		USA	
8	Pyropress Engineering Company Ltd.		UK	
Product	**22**	**Fusible Plugs**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	e-Productions solution		USA	**On Hold-B1**
2	Sigma		USA	
3	Ruelco		USA	
3	M/s Fusion Technologies		India	**# NEW VENDOR**
Product	**23**	**Orifice Plates & Flanges and Restriction Orifices**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Daniel OpCo LLC (Daniel Measurement and Control)		USA	
2	ABB Limited		India	
3	Micro Precision, Faridabad		India	
4	Minco (India) Pvt. Ltd		India	
5	Rank Controls Instruments Pvt. Ltd		India	
6	Flow Metering Company Ltd		UK	
7	Minco (India) Flow Elements Pvt. Ltd		India	
8	Dynafluid Valves and Flow Controls Pvt. Ltd		India	
Product	**24**	**Senior,Junior, Simplex Orifice Fittings**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Daniel OpCo LLC (Daniel Measurement and Control)		USA	
2	Flow Metering Company Ltd		UK	
3	Petrol Valves S.r.l		Italy	
4	Canalta Controls Ltd.		Canada	
Product	**25**	**Filter Regulators**		
S. No.	**Vendor**		**Country**	**Remark/Note**

1	Shavo Norgren (India) Private Limited	India	
2	Emerson Process Management/ Emerson Process Management India Private Limited (Make - ASCO Numatics (India) Pvt. Limited)	UK/Singapore/India	
3	Dresser Masoneilon	USA	
4	ABB Limited	India	
5	Versatile Controls Ltd	UK	
6	GE Oil & Gas India Pvt. Ltd	India	
7	Bifold Fluid Power Limited	UK	
8	Versa Products Company Inc.	USA	
Product	**26** **Flow Switches (Pneumatic & Electrical)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Magnetrol International NV	Belgium	
2	Tokyo Keiso	Japan	
3	Yokogawa India Ltd	India	
4	Cameron Technology Uk Ltd	UK	
5	Pyropress Engineering Company Ltd.	UK	
Product	**27** **Control Valves**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Fisher Controls	USA/India	
2	Dresser Masoneilon	USA	
3	ABB – Kent	UK	
4	Blakeborugh (Weir Valve and Control)	UK	
5	CCI	USA	
6	KOSO India Private Limited	India	
7	GE Oil & Gas India Pvt. Ltd	India	
8	KSB MIL Controls Limited	India	
9	Richards Industries	USA	
10	KOSO Controls	Singapore	
11	KOSO Fluid Control Pvt. Ltd.	India	
12	Tejas Research & Engineering LP	Texas, USA.	**On Hold-B1**
13	Severn Glocon Valves Private Limited	India	

14	Emerson Process Management Chennai Private Limited	India	
15	Flowserve Corporation, Flow Control Division	USA	
Product	**28** **Self Actuated Pressure Control Valves**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Emerson Process Management Chennai Private Limited	India	
2	Dresser Masoneilon	USA	
3	ESME (Electrical Steam & Mining Equiptment Ltd.)	UK	
4	WEIR Valves & Controls Ltd. (Hopkinsons Ltd)	UK	
5	GE Oil & Gas India Pvt. Ltd	India	
6	Jordan Valve	USA	
Product	**29** **Fire and Gas Detection System-Packaged Item**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	General Monitors Ireland Ltd.	Ireland	
2	Det-tronics	USA	
3	Haven Automation	Singapore	
4	Nohmi Bosai	Japan	
5	Safety Systems	USA	
6	ICS Triplex (Rockwell Automation)	UK	
7	Yokogawa Industrial Safety Systems	Malaysia	
8	Honeywell Automation India Ltd (HAIL)	India	
9	Control & Electric Pte. Ltd (Bousted Group)	Singapore	
10	Honeywell Analytic	India/UK	
11	Autronica Fire & Security (Addressable type for Living Quarter and Technical Buildings)	Norway	
12	Scame Sistemi, srl (Addressable Type - open and closed area)	Italy	
13	Detector Electronics Corporation (Det-Tronics) Inc. (Addressable type - open and closed area)	USA	
14	Trisystems Engineering Sdn Bhd	Malaysia	

15	Control and Electrics System Manufacturing Ltd	UAE	
14	Safe-Tronics Automation Pvt. Ltd.	India	
Product	**30** **Deluge Valve with Test facility**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Cla-Val	USA	
2	MIL Inbal Valve Ltd	Israel	
3	Wormald	Hongkong/Singapore	**On Hold-B1**
Product	**31** **Water Cut Meter**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Emerson Process Management (Microwave type)	USA/UK	
2	Weatherford Oil Tool Middle East Ltd. (Near Infrared type)	UAE	
Product	**32** **Rotameter**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	AL Flow Glass Equipment	India	
2	Tokyo Keiso Co. Ltd	Japan	
Product	**33** **Annubars / Pitot Tubes**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Emerson Process	USA	
2	Intra Automation GMBH	Germany	
Product	**34** **Instrumentation Fittings**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Swagelok	USA	
2	Circor (HOKE) Instrumentation Ltd	USA	
3	Parker	USA/Singapore	
4	Astec Valves & Fittings Private Limited	India	
5	DK Tech	South Korea	
6	Ham-Let Advanced Control Technology	Israel	
7	Precision Engineering Industries, Mumbai	India	
8	Fluid Controls Pvt. Ltd.	India	
9	Hy-Lok Corporation	Korea	
10	Panam Engineers Ltd., Mumbai	India	
11	Allied Engineering Pvt.Ltd (only for SS 316/316L)	India	

12	Havi Engineering India Private Limited	India	
13	MET-LOK Valves and Fittings (Only for SS316)	India	
Product	**35** **Flow computer (Liquid/Gas)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Emerson Process Management (Daniel Measurement & Control)	USA	
2	Emerson Process Management, Bristol	USA	
3	Cameron Flow Control Technology (UK) Ltd. (for Liquid / Gas)	UK	
Product	**36** **Flow totalizer**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Cameron Technology UK Ltd.	UK	
2	Emerson Process Management (Daniel Measurement & Control)	USA	
3	Yokogawa	Japan	
4	Contrec	Australia	
Product	**37** **Ultrasonic flow Meter**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Emerson Process Management (Daniel Measurement & Control)	USA/Singapore	
2	GE Sensing EMEA, Ireland / GE Oil & Gas India Pvt. Ltd.	Ireland / India	
3	Sick AG	Germany	
Product	**38** **Portable Calibrator/Hand Held Multifunctional Calibrato**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Fluke	USA	
2	Druck/GE Druck Ltd	UK	
3	Beamax	Finland	
Product	**39** **Chlorine Analyzer**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Emerson Process Management	USA	
Product	**40** **Dissolved Oxygen Analyzer**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Hach Company	USA	
Product	**41** **Programmable Logic Controller (PLC)**		
S. No.	**Vendor**	**Country**	**Remark/Note**

1	Yokogawa Industrial Safety Systems, (Prosafe) (only TUV approved applications) (Yokogawa)	Malaysia/Japan/ Netherlands	
2	ABB	USA/India	
3	Allen Bradley(Rockwell Automation)	UK	
4	Siemens	Germany	
5	ICS Triplex (Rockwell Automation)		
6	Honeywell Automation India Ltd (HAIL)	India	
7	Hima Paul Hidebrandt GmBH	Germany	
8	Bristol Babcock (Emerson Process Management)	Australia	
9	Rockwell, (Only non TUV approved applications)	India	
10	Triconix (only TUV approved applications) (INVENSYS)	USA	
11	Schneider Electric Industries Sdn Bhd	Malaysia	
12	GE Intelligent Platform inc (only TUV approved application)	USA/India	
13	Yokogawa India Ltd	India	
14	Emerson Process Management (For Safety PLC)	USA	
Product	**42** **Multiphase Flow Meter (MPFM)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Schlumberger	USA	
2	mPm, Norway	Norway	
3	Agar Corporation	USA	
4	Emerson SRL	Romania	
Product	**43** **Coriolis Meter**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Emerson process management / Emerson Process Management India Private Limited (Micromotion make)	USA/India	
2	Heinrichs Gmbh	Germany	
Product	**44** **High pressure Control Valves (HPCV)**		
S. No.	**Vendor**	**Country**	**Remark/Note**

1	Dresser Mansoneilan	USA	
2	Mokveld	Holland	
3	Fisher Controls	USA	
4	Control Component Inc.	USA	
5	Kent Process control Ltd	UK	**On Hold-B1**
6	WEIR Valves & Controls Ltd	UK	
7	GE Oil & Gas India Pvt. Ltd	India	
Product	**45 Level Transmitters**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	GE Oil & Gas India Pvt. Ltd. (Displacer Type)	India	
2	Chemtrols Industries Ltd (Displacer Type)	India	
3	Endress+Hauser Germany/India Pvt Ltd (Guided Wave Radar Type)	Germany/India	
4	Magnetrol International Inc (Guided Wave Radar Type)	USA	
5	Magnetrol International NV (Displacer type)	Belgium	
6	Emerson Process Management (Rosemount) (Guided Wave Radar Type)	Sweden	
7	Pyropress Engineering Company Ltd.(Differential Pressure Type)	UK	
8	Emerson Process Management Chennai Private Limited (Displacer Type)	India	
Product	**46 Isolators & Barriers (Active & Passive)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	MTL	India	
2	P&F	Germany	
Product	**47 Limit Switches**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Emerson Process Management Valve Automation Inc (BETTIS)	USA	
2	Honeywell Automation India Ltd (HAIL)	India	
Product	**48 HC/H2/H2S Gas Detectors**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Crowcon Detection Instruments	UK	

2	Det-tronics	USA	
3	Drager Safety AG	UK	
4	General Monitors Ireland Ltd.	Ireland	
5	Honeywell International Inc (Honeywell Life Safety),USA/ M/s Honeywell International India Private Limited ("HIIPL"),INDIA (Honeywell Analytics)	USA/INDIA	
6	Detcon Inc. (Addressable & Normal)	USA	
7	Detector Electronics Corporation (Det-Tronics) Inc. (For Addressable type)	USA	
8	JSC-Electronstandart-Pribor	Russia	
Product	**49** **Fire Detectors**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Crowcon Detection Instruments	UK	**On Hold-B3**
2	Det-tronics	USA	
3	General Monitors Ireland Ltd.	Ireland	
4	Honeywell International Inc (Honeywell Life Safety),USA/ M/s Honeywell International India Private Limited ("HIIPL"),INDIA (Fire Sentry)	USA/INDIA	
5	Emerson Process Management Net Safety Monitoring Inc (For IR/Triple IR/UV-IR)	Canada	
6	Detector Electronics Corporation (Det-Tronics) Inc. (For Addressable type)	USA	
7	JSC-Electronstandart-Pribor	Russia	
Product	**50** **Smoke / Thermal Detectors**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Apollo Fire Detectors Limited	UK	
2	Honeywell International Inc (Honeywell Life Safety),USA/ M/s Honeywell International India Private Limited ("HIIPL"),INDIA (System Sensor)	USA/INDIA	
3	Autronica Fire & Security (Addressable type for Living Quarter and Technical Buildings)	Norway	
4	Detector Electronics Corporation (Det-tronics) (for thermal detectors)	USA	
Product	**51** **Manual Call Point, Abandon Switch, Flash Light**		

S. No.	Vendor	Country	Remark/Note
1	MEDC	UK	
Product	**52 Hand Control Valves**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Mansoneilan	France	
2	Mokveld	USA	
3	Fisher Controls	USA	
4	Kent Process Control Ltd	UK	**On Hold-B1**
5	KOSO India Private Limited	India	
6	Control Component Inc.	USA	
7	Hopkinsons Blackborough	Dubai	
Product	**53 Moisture Analyzer System Integrator**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Chemtrol Engineering Pvt. Ltd., Mumbai (With Amtek make Mositure Analyser)	India	
Product	**54 Magnetic Flow Meter**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Emerson Process Management, Rosemount Inc.	USA	
2	KROHNE AG	Switzerland	
Product	**55 Actuators for Valves (Topside)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	AUMA RIESTER GMBH & CO.	Germany	
2	BIFFI ITALIA S.R.L	ITALY	
3	EIM CONTROLS LTD	UK	
4	FLO - TORK INC	USA	
5	FLOWSERVE LIMITORQUE	India	
6	Emerson Process Management Valve Automation Inc (EPM Valve Automation) (A) Shafer Make (B) Bettis Make	USA	
7	Rotork Controls (India) Ltd.	India	
8	Cameron Italy S.R.L	Italy	
9	ATC	The Netherlands	
10	Imtex Controls Limited	United Kingdom	

11	DVG AUTOMATION SPA	Italy	
12	DelVal Flow Controls Private Limited	India	
13	Emerson Process Management (India) Private Limited, Bettis Make	India	
Product	**56 Actuators for Valves (Subsea)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	ATC	The Netherlands	
2	Emerson Process Management Valve Automation Inc (EPM Valve Automation) (A) Shafer Make (B) Bettis Make	USA	
3	Cameron Ledeen Facility	Italy	
4	Perar S.P.A.	Italy	
Product	**57 Instrumentation Tubing**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	AB Sandvik Material Technology	Sweden	
2	Tubacex Tubos Inoxidables, S.A.	Spain	
3	Fine Tubes	UK	
4	Ratnamani Metals & Tubes Ltd	India	
5	Dong Nam Korea Co. Ltd.	South Korea	
6	Maxim Tubes Company Pvt. Ltd	India	
7	M/s TUBACEX TUBES AND PIPES PRIVATE LIMITED	India	**# NEW VENDOR**
Product	**58 Weather Monitoring System (WMS)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	MIROS AS	Norway	
Product	**59 Subsea Umbilical**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	JDR Cable Systems Ltd	UK	
2	Fibron BX Limited (Thermoplastic)	UK	
3	Mfx do brasil equipamentos de petroleo Ltda (Thermoplastic)	Brasil	
Product	**60 Wet Gas Meter**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Emerson process management (Roxar Flow measurement AS)	Norway	

Product	**61**	**Cables (Instrument)**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Brooks		India	
2	Draka		Spain	
3	Associated Flexible and Wires		India	
4	Associated Cables		India	
5	KEI Industries Ltd		India	
6	Nexans (Kukdong)		Korea	
7	Polycabs Wires Pvt. Ltd		India	
8	Teldor Cables & Systems Ltd		Israel	
9	Radiant - RSCC Specialty Cable Pvt Ltd		India	
10	Thermo Cables Ltd		India	
11	Cords Cable Industries Ltd		India	
Product	**62**	**Cone Meter**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	McCrometer Inc		USA	
2	Cameron Flow Control Technology (UK) Ltd.		UK	
Product	**63**	**HIPPS (High Integrity Pressure Protection System)- Packaged Item**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Sella Controls Limited., with components from following vendors: 1. M/s HIMA – Logic solver 2. M/s ATC – Actuator 3. M/s TYCO FCT – SDV subject to submission of corrosion test results before supply of valves 4. M/s ABB – Pressure Pilots 5. M/s Bifold - SOVs		UK	
2	Emerson Process Management India Private Limited., with components from following vendors: 1. Emerson Process management (I) Pvt. Ltd – Logic solver 2. Emerson Process Management (Bettis) – Actuators 3. Emerson Process management (I) Pvt. Ltd - Transmitters		India India Singapore India	

	4. Astava , Netherland – Manifold	Netherland	
	5. Emerson Process Management, Singapore - Digital Valve Controller	Singapore	
	6. ASCO - Solenoid Valves	India	
	7. Valbart - Ball Valve	Italy	
Instrumentation Note:			
1	For all instrumentation items that are to be located in the field (hazardous area), certification by FM / UL / BASEEFA / CSA / DGMS / CMRS or equal is essential, stating that the said Instrumentation item is suitable for use in the hazardous area. (This applies to all the items in the Suggested Vendor List)		
2	For all instruments, weather proofness of NEMA 4X / IP 65, as a minimum, is essential. (This applies to all the items in the Suggested Vendor List)		
3	For all electrical instruments, explosion proofness of NEMA 7, as a minimum, is essential (This applies to: Flow Totalizer, Electronic Flow Transmitter, MPFM, Pressure Switch (Electrical) and Pressure Transmitter (Elect.)		
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Electrical

Product	**1**	**Transformers (Cast Resin Type)**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	General Electric		USA	
2	Fuji Electric Co Ltd		Japan	
3	Kirloskar Electric Co Ltd		India	
4	ABB		India	
5	Schneider Electric Industries SA		France	
6	Trasfor, SA		Switzerland	
7	Unelectra (Only for Lighting Transformers)		Taiwan	
Product	**2**	**UPS System-Packaged Item**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	AEG Power solutions Pte		Singapore	
2	Chloride Industrial System		UK/France	
3	Fuji Electric Co Ltd		Japan	
4	GUTOR		Switzerland	
5	Vertiv Energy Private Limited		India	
6	Hitachi Hi-Rel Power Electronics Pvt Ltd		India	
7	BORRI SPA		Italy	
8	Ametek Solidstate Controls		USA	
Product	**3**	**LT Switch Gear**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Eaton		USA	
2	Fuji Electric Co Ltd		Japan	
3	IME Quadri		Italy	
4	Terasaki Electric Co Ltd		Japan	
5	L&T, Mumbai		India	
6	ABB		Korea / India	
7	Schneider Electric Industries SA		France	
8	COMECA EBT		France	

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9	Siemens	India	
10	GE Industrial	India / USA	
Product	**4 HT Switch Gear**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Eaton	USA	
2	Hitachi	Japan	
3	ABB	India/Norway	
4	Siemens	Germany /Indonesia/ India	
5	Fuji Electric Co Ltd	Japan	
6	Schneider Electric Infrastructure Ltd	India	
7	Schneider Electric Industries SA	France	
8	GE Industrial	India / USA	
9	Tamco Switchgear (Malaysia) Sdn Bhd	Malaysia	
10	L&T Ahmednagar (Technical License agreement with Tamco Malaysia)	India	
Product	**5 Battery & Battery Charger-Packaged Item**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Chloride Industrial System	France	
2	Yuasa	Japan	**On Hold-B1**
3	Gutor	Switzerland	
4	HBL Power Systems Ltd	India	
5	Chhabi Electricals Pvt Ltd (For Battery Chargers)	India	
6	AEG Power solutions Pte	Singapore	
7	BORRI SPA	Italy	
8	Hitachi Hi-Rel Power Electronics Pvt Ltd	India	
Product	**6 Cathodic Protection System**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Alico Industries Co. Ltd.	UAE	
2	Nippon Corrosion Engineering Co Ltd	Japan	
3	PSL, Mumbai	India	
4	Emirates Techno Casting LLC	UAE	
5	CP TECH	Singapore	

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6	SBN Industries SDN BHD	Malaysia	
7	Metec SARL	Tunisia	
Product	**7**	**Fluorescent / Incandescent Class-"B" LTG Fixtures**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Cooper Crouse-Hinds	USA	
2	Appleton/EGS electrical	USA	
3	Baliga Lighting Eqpt Ltd	India	
4	Pepperl and Fuchs manufacturing (India) Pvt. Ltd	India	
5	Chalmit Lighting	UK	
6	R STAHL	Germany	
Product	**8**	**Navigational Aid System-(NAVAID)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Pharos Marine Automatic Power Inc	USA	
2	Tata BP Solar India Ltd	India	
3	SOLAPAK	UK	
4	Tideland Signal Ltd	UK	
5	ORGA BV	Holland	
6	SICE s.r.l	Italy	
Product	**9**	**Lead Acid Batteries (Solar Power System)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Amara Raja Batteries Ltd	India	
2	Yuasa	Japan	**On Hold-B1**
3	HBL Power Systems Limited	India	
Product	**10**	**Multi Cable Transit (MCT)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Roxtac	Sweden	
2	Hawke Transit System	Spain	
3	MCT Brattberg AB	Sweden	
4	Wallmax India Enterprises Pvt Ltd.	India	
Product	**11**	**Div.2 Aviation Marker Lighting Fixtures**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Tranberg A/S	Norway	
2	Tideland Signal Ltd	UK	

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Product	**12**	**Lighting & Power Distribution Panel, AC & DC Distribution Board**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Legrand (India)	India	
2	Novateur Electrical & Digital Systems Private Limited	India	
3	BCH Electric Ltd	India	
4	Fabricons Technocrat Solutions Pvt. Ltd.	India	
5	Siemens	Germany/India	
6	HHI	Korea	
7	Marine Electricals	India	
8	Schneider Electric India Pvt. Ltd.	India	
9	Middle East Switch Gear Ltd	Sharjah	
10	Cooper Crouse-Hinds	USA	
11	R STAHL	Germany	
Product	**13**	**Solar Power System-Packaged Item**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Central Electronics Ltd	India	
2	BHEL	India	
3	Tata Power Solar System	India	
4	Solapak	UK	
5	SICE s.r.l	Italy	
6	Fabricons Technocrat Solutions Pvt. Ltd.	India	
7	TSS4U BV	Netherlands	
Product	**14**	**Fire Survival Cables**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Universal Cables, Satna, MP	India	
2	INCAB	India	
3	Cable Corporation of India	India	
4	Polycab	India	
5	Prysmian PowerLink Srl	Italy	
6	Associated Flexibile & Wires Pvt. Ltd	India	
7	KEI Industries Ltd	India	

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8	Nexans (Kukdong)	Korea	
9	Radiant - RSCC Specialty Cable Pvt Ltd	India	
10	Thermo Cables Ltd (Upto 1.1 KV)	India	
Product	**15 FRP Cable Trays**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Ercon, Jodhpur	India	
2	General Composites Pvt Ltd	India	
3	INDIANA, Mumbai	India	
4	Enduro	USA	
5	Sumip Composites Pvt.Ltd	India	
6	Sintex Industries Ltd	India	
7	Oglaend System	Malaysia	
8	Kemrock Industries & Exports Ltd	India	
9	Satyam Composites Pvt. Ltd.	India	
10	Bravo Cable Trays Pvt. Ltd.	India	
Product	**16 Batteries (Ni-Cd)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Saft India Private Limited	India	
2	HBL Power systems Ltd	India	
Product	**17 Hazardous Area Lighting Fixtures**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Baliga Lighting Equipment	India	
2	Flexpro Electrical,Mumbai	India	
3	FCG Flame Proof Control Gears Pvt. Ltd	India	
4	FCG Power Industries Pvt. Ltd	India	
5	Flameproof Equipments Pvt. Ltd		
6	Sudhir Switchgears	India	
7	EGS Electrical Group	USA	
8	Chalmit Lighting	UK	
9	Cooper Crouse-Hinds	USA	
10	R STAHL	Germany	
11	Raytec Ltd	UK	

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Product	**18**	**MV Motors (Flame Proof Exd, EX e, EX n)**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	CG Power & Industrial Solutions Limited		India	
2	Bharat Bijlee		India	
3	KEC		India	
4	Siemens		India	
5	WEG Equipamentos Electricos S.A.		Brazil	
6	Laxmi Hydraulics Pvt. Ltd		India	
Product	**19**	**Junction Boxes**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Baliga		India	
2	Flexpro Electricals Pvt Ltd		India	
3	FCG Flame Proof Control Gears Pvt. Ltd		India	
4	FCG Power Industries Pvt. Ltd		India	
5	Cooper Crouse Hinds		USA	
6	Weidmuller		Singapore/Germany	
7	Pepperl and Fuchs manufacturing (India) Pvt. Ltd		India	
8	Sudhir Switchgears		India	
9	Appleton/EGS Electricals		USA	
10	Flameproof Equipments Pvt.Ltd		India	
11	Rose Systemtechnik GMBH		Germany	
12	R STAHL		Germany	
13	Phoenix Mecano India Pvt. Ltd		India	**Technical Licence Agreement with M/s Rose Gmbh,Germany valid till 31.12.2022**
Product	**20**	**Cables (Power)**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Universal Cables		India	
2	Apar Industries Limited		India	
3	KEI Industries Ltd		India	

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4	Nexans (Kukdong)	Korea	
5	AmerCable Incorporated	USA	
6	Cable Corporation of India	India	
7	Polycab	India	
8	Prysmian PowerLink Srl	Italy	
9	Associated Flexibile & Wires Pvt. Ltd	India	
10	Radiant - RSCC Specialty Cable Pvt Ltd	India	
11	Thermo Cables Ltd (Upto 1.1 KV)	India	
12	Cords Cable Industries Ltd (Upto 1.1 KV)	India	
13	Nangalwala Industries Pvt. Ltd (Upto 1.1 KV)	India	
Product	**21 Cable Glands**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	HAWKE	UK	
2	Cooper Crouse-Hinds	USA	
3	Baliga	India	
4	Flexpro Electrical Co.Ltd	India	
5	Comet Brass Product	India	
6	FCG Flameproof Control Gears Pvt Ltd	India	
7	R STAHL	Germany	
8	CMP Products	UK	
Product	**22 Electrical Heat Tracing**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Thermopads Pvt Ltd	India	
2	Thermon		
3	Lloyds	India	
4	Xicon	India	
5	Pentair Thermal Management India Pvt. Ltd.	India	
6	Bartec GmbH	Germany	
Product	**23 i. Explosion Proof Control Stations**		
	ii. Industrial Type Plug and Sockets		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Pepperl and Fuchs manufacturing (India) Pvt. Ltd	India	

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2	Baliga	India	Vendors from 2 to 10 are for Explosion Proof Control Stations Only
3	FCG Flame Proof Control Gears Pvt. Ltd	India	
4	Ex-protecta	India	
5	Flexpro Electricals Pvt Ltd	India	
6	Sudhir Switchgears	India	
7	Flameproof Equipments Pvt Ltd	India	
8	Cooper Crouse-Hinds	USA	
9	R STAHL	Germany	
10	Phoenix Mecano India Private Limited	India	
Product	**24 Submarine Cable**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	EXSYM Corporation	Japan	
2	Nexans	France	
3	J Power Systems	Japan	
4	Prysmian PowerLink Srl	Italy	
5	JDR Cable System Ltd	UK	
6	J-Power Systems Saudi Co. Ltd	Saudi Arabia	
Product	**25 HV Motors-(Ex e/ Ex n)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	ATB Laurence Scott Ltd	UK	
2	Fuji Electric Co Ltd	Japan	
3	Parson Peebles	UK	
4	BHEL	India	
5	Siemens		
6	CG Power & Industrial Solutions Limited	India	
7	KEC	India	
8	Nidec ASI S.p.A	Italy	
9	WEG Equipamentos Electricos S.A.	Brazil	
10	Toshiba Mitsubishi Electrical Systems Corporation (TMEIC)	Japan / India*	

Note	***All HV Motors manufacturers are enlisted for MV Motors also.***			
	**** Vendor is required to fulfil requirement of Note 5(a) of SVL for future supplies***			
Product	**26**	**Hazardous Area Lighting Panel (Flameproof)**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Baliga Lighting Equipment Ltd		India	
2	FCG Flame Proof Control Gears		India	
3	Flexpro Electrical Pvt Ltd		India	
4	Flameproof Equipments Pvt. Ltd		India	
5	FCG Power Industries		India	
6	Sudhir Switchgears		India	
7	R STAHL		Germany	
Product	**27**	**Welding Receptacles & Convenience Outlets (Flameproof)**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	FCG Flame Proof Control Gears		India	
2	Flameproof Equipments Pvt Ltd		India	
3	Sudhir Switchgears		India	
4	Appleton/EGS Electricals		USA	
Product	**28**	**Marine Passenger Elevators (Electric)**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Kone Elevators Ltd		Finland	
2	Alimak Group Norway AS		Norway	
3	Maspero Elevatori S.P.A		Italy	
Product	**29**	**Utility Generator- Alternator for Diesel Generator Package**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	KATO		Japan	
2	LEROY SOMER			
Product	**30**	**Main Injection / Main Oil Line Pumps- Electric Motors**		**Product merged with HV Motors at Sl no 25**

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Product	**31**	**Turbine Generator Set- Alternators**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Brush Electric		UK	
2	General Electric		USA	
3	Siemens		Germany	
4	Fuji Electric		Japan	
5	Hyundai Ideal Electric Co.		USA	
6	Converteam UK Ltd		UK	
7	Kato Engineering Inc.		USA	
Product	**32**	**Diesel Generator Packager**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Fuji Electric Co. Ltd.		Japan	
2	Stewart and Stevenson		USA	
3	Pon Power BV		Netherlands	
4	KATO		USA	
5	McPEC Marine & Offshore Engineering Pte Ltd.		Singapore	
6	East Asia Power Pte Ltd.		Singapore	
7	Intergen S.p.a,		Italy	
Product	**33**	**Motorized Actuator for Valve**		**Product Deleted**
Product	**34**	**Helideck Lighting System**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Point Lighting Corporation		USA	
2	Orga B.V		The Netherlands	
3	Q-Aviation B.V.		India	**# NEW VENDOR**
Product	**35**	**Solar Charge Controller**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Fabricons Technocrat Solutions Pvt. Ltd.		India	
2	SICE S.R.L		Italy	
Product	**36**	**Solar Panels**		
1	Central Electronics Limited (A Govt. of India Enterprise)		India	
2	Tata Power Solar System		India	

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Pipeline			
Product	**1**	**Carbon Steel Line Pipe (Sour & Non-Sour Service)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Sumitomo Corporation	Japan	**Trader**
2	Tenaris Dalmine	Italy	
3	Mannesmannrohren-werke AG	Germany	**On Hold-B1**
4	NKK Corporation	Japan	**On Hold-B1**
5	Nippon Steel & Sumitomo Metal Corporation	Japan	
6	Marubeni	Japan	**Trader**
7	Mitsui	Japan	**Trader**
8	Itochu	Japan	**Trader**
9	Thyssen	Germany	**On Hold-B1**
10	Vallourec Deutschland GmbH	Germany	
11	Tenaris Tamsa	Mexico	
12	ArcelorMittal Tubular Products Roman S.A. (For Seamless Line Pipes only)	Romania	
13	Jindal Saw Ltd. (For LSAW Pipes only)	India	
14	Man Industries (India) Ltd.(For LSAW pipes only)	India	
15	Volzhsky Pipe Plant of TMK Group	Russia	**On Hold-B1**
16	EEW GmbH & Co (For LSAW Pipes only)	Germany	
17	EISENBAU KRAMER GmbH(For LSAW Pipes only)	Germany	
18	Welspun Corporation Ltd (For LSAW Pipes only)	India	
19	Jubail Energy Services Company (JESCO) (For Seamless Line Pipes only)	Saudi Arabia	
20	Jindal SAW Ltd, Malegaon MIDC, Sinnar, district Nashik, (For Seamless Line Pipes only) [Under DMI&SP Policy]	India	
21	Maharashtra Seamless Ltd. (MSL), Nagothane, Pipe Nagar (For Seamless Line Pipes only) [Under DMI&SP Policy]	India	
22	ISMT Limited, Baramati, Maharashtra (For Seamless Line Pipes only) [Under DMI&SP Policy] { for seamless	India	
Product	**2**	**Long Radius Bend**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Fabricom	Belgium	

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2	Dai-ichi High Frequency Co. Ltd.	Japan	
3	Tenaris Dalmine	Japan	
4	PSL	India	
5	Igawara Industrial service and Trading Pte. Ltd	Singapore	
6	Proclad Induction Bending	UK	
7	Jindal Saw Ltd.	India	
8	Welspun Corp. Limited	India	
9	Salzgitter Mannesmann Grobblech GmbH (CRA Clad)	Germany	
10	Fabricom (CRA Clad)	Belgium	
Product	**3** **Concrete & Anti-corrosion Coating**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Bredero Price	Sharjah	
2	NPCC	Abu Dhabi	
3	PSL	India	
4	Jindal Saw Ltd.	India	
5	Welspun Corp Ltd (External 3LPP Coating)	India	
6	Wasco Coatings Malaysia Sdn. Bhd.	Malaysia	
7	Welspun Wasco coatings Pvt Ltd (Only for Concrete Coating)	India	Technical Licence Agreement with Wasco Coatings Malaysia Sdn Bhd Malaysia, valid till 24.07.2028
8	M/s Man Industries (India) Limited	India	
Product	**4** **Nickel-Copper Alloy- UNS N04400 plates (Monel)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	VDM Metals GmbH	Germany	
2	Mitsubishi	Japan	**Trader**
3	Special Metals Wiggin Ltd.	UK	
4	Special Metals Corporation	USA	
5	Kanematsu Corporation, Tokyo {Trader of OEM M/s Nippon Yakin Kogyo Co. Ltd., Japan}	Japan	**Trader/Exporter**
6	Kirtanlal and Sons, Mumbai	India	**Trader**
Product	**5** **Flexible Linepipe**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Well Stream	UK	
2	National Oilwell Varco Denmark I/S	Denmark	

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3	Technip	France	
4	Asiaflex Products Sdn. Bhd	Malaysia	
Product	**6**	**Subsea Flanges – Carbon Steel (Sour & Non-Sour Service)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	CHW Forge Private Ltd	India	
2	Officine Nicola Galperti & Figlio SPA	Italy	
3	Officine Ambrogio Melesi & C, SRL	Italy	
4	Vilmar	Romania	
5	Vilmar (for CRA Clad)	Romania	
6	Viar S.p.A	Italy	
7	Vee Kay Vikram & Co. LLP	India	
Product	**7**	**Misalignment Flange**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Oil States Industries (OSI), USA	USA	
Product	**8**	**Subsea Carbon Steel Ball Valves (Sour & Non Sour Service)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Valbart Srl	Italy	
2	T. K. Valves	UK	**On Hold-B1**
3	CAMERON	USA	
4	GROVE Italia	Italy	**On Hold-B1**
5	PERAR	Italy	
6	BEL Valves	UK	
7	HIT Valve S.p.A	Italy	
8	Breda Energia SpA (Only non-sour)	Italy	
9	Advanced Technology Valve S.p.A	Italy	
10	MIR Valve Sdn. Bhd	Malaysia	
11	Ampo S.Coop - Poyam Valves	Spain	
12	Fluitek Orsenigo Valves Srl	Italy	
Product	**9**	**Subsea Carbon Steel Check Valves (Sour & Non Sour Service)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Tom Wheatley (Cameron)	USA	**On Hold-B1**
2	BEL Valves	UK	
3	Goodwin	UK	
4	Velan	Italy	**On Hold-B1**
5	Valvitalia	Italy	

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Product	**10**	**Subsea Carbon Steel Flow Tee (Sour & Non Sour Service)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Pipeline Engineering & Supply Co. Ltd.	UK	
2	GAM Raccordi SPA	Italy	
3	Sawan Engineers Pvt. Ltd.	India	
4	Vee Kay Vikram & Co. LLP	India	
Product	**11**	**Subsea Carbon Steel Hinged Closure (Quick Opening)(Sour & Non Sour Service)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Pipeline Engineering & Supply Co. Ltd.	UK	
Product	**12**	**Subsea Piggable Wye (Sour service)**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	M.E.G.A SpA	Italy	
Product	**13**	**CRA Clad Pipes**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	H. Butting GmbH & Co KG	Germany	
2	Eisenbau Kramer GmbH	Germany	
3	FTV Proclad L.L.C	UAE	
4	PT Cladtek Bi-Metal Manufacturing	Indonesia	
Product	**14**	**Nickel-Alloy UNS NO6625 Seamless line Pipes**	**Product Deleted**
*******			
Product	**15**	**Internal Field Joint Coating**	
S. No.	**Vendor**	**Country**	**Remark/Note**
1	M/s NOV -Tuboscope (Sleeves for WI Pipelines only)	Germany	
2	M/s Seal for Life-CCB (Sleeves for WI Pipelines only)	USA	
3	M/s Reliable Pipe & Tubes Limited (Sleeves for WI Pipelines only)	India	
4	M/s Aegion Coating Services (Robotic Arm Method for both WF as Well as WI Pipeline)	USA	
Pipeline Note for Product No. 15:-			
1	New Vendor shall be evaluated for compliance to the Pipeline Design Criteria (Annexure to Section 4A), Past Track Record, Performance feed-back.		

Structural				
Product	**1**	**Rubber Fenders**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	IRM Offshore and Marine Engineers		India	
2	Regal Marine Products, Offshore Kinematics Inc, Texas		USA	**On Hold-B1**
3	Trelleborg Hercules Pte.Ltd.		Singapore	
4	Hi-Tech Elastomers Ltd, Ahmedabad		India	
Product	**2**	**Grout Seal**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	IRM Offshore and Marine Engineers		India	
2	Regal Marine Products, Offshore Kinematics Inc, Texas		USA	**On Hold-B1**
3	Hi-Tech Elastomers Ltd, Ahmedabad		India	
4	Trelleborg Hercules Pte.Ltd.		Singapore	
Product	**3**	**Diaphragm Closure**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	IRM Offshore and Marine Engineers		India	
2	Regal Marine Products, Offshore Kinematics Inc, Texas		USA	**On Hold-B1**
3	Hi-Tech Elastomers Ltd, Ahmedabad		India	
4	Trelleborg Hercules Pte.Ltd.		Singapore	
Product	**4**	**Marine Growth Preventer (MGP)**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	IEV Manufacturing Sdn Bhd		Malaysia	
Product	**5**	**Marine Paints (Protective coating) for offshore structure**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Chugoku Marine Paints (S) Pte Ltd.		Singapore	
2	Ameron (PPG Coating)		USA	
3	Sigma Coating (PPG Coating)		USA	
4	Hempel A/s		Denmark	
5	Carboline Products		USA	
6	Berger Paints (I) Ltd		Kolkata	
7	Jotun Coatings		Norway	

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8	KCC Products	Korea	
9	Shalimar Paints Ltd	Kolkata	
10	International Paint	UK	
11	EXCEL India Protective Paints Pvt. Ltd	India	
Product	**6** **Fibre glass Reinforced Plastic Gratings & Handrails**		**Product Deleted**
Product	**7** **Aluminium Heli Deck**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Aluminium Offshore Pte Ltd	Singapore	
2	Bayards Aluminium Constructies, B.V	Netherlands	
3	Marine Aluminium, Norway (Through Walchand Industries, CBD Navi Mumbai)	Norway	
4	NCMP Ltd.	UK	
5	SEBOTECH Co, Ltd	Korea	
6	Helitecnica Aluminum Heliports, S.L.	Spain	
Product	**8** **Barge Bumper Shock Cell**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Hi-Tech Elastomers Ltd, Ahmedabad	India	
2	IRM Offshore and Marine Engineers	India	
3	Trelleborg Hercules Pte.Ltd.	Singapore	
Product	**9** **Rubber Strips**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Hi-Tech Elastomers Ltd, Ahmedabad	India	
2	IRM Offshore and Marine Engineers	India	
3	Trelleborg Hercules Pte.Ltd.	Singapore	
Product	**10** **Single Point CALM Buoy System and conventional boat mooring system (SALM)**		
S. No.	**Vendor**	**Country**	**Remark/Note**
1	Blue Water Energy Services	Netherlands	
2	Single Buoy Moorings Inc. (SBM Inc.)	Switzerland	

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Architectural				
Product	**1**	**External Fire Proofing H rated**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Specialist Services		Dubai	
2	Booth Engineering Ltd.		UK	
3	Van Dam		Netherlands	
4	Solent Composite Solutions (SCS) Ltd		United Kingdom	**(Composite Material only)**
Product	**2**	**External Fire Proofing A Rated**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Specialist Services		Dubai	
2	Booth Engineering Ltd		UK	
3	Van Dam		Netherlands	
4	Mech Tools Engineering		UK	
5	Solent Composite Solutions (SCS) Ltd		United Kingdom	**(Composite Material only)**
Product	**3**	**Internal Fire Wall A rated & B rated**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Specialist Services		Dubai	
2	Booth Engineering Ltd.		UK	
3	Van Dam		Netherlands	
4	Mech Tools Engineering		UK	
5	IMAC		Sweden	
6	BD System		UK	
7	Shinsung Engg.		South Korea	
8	InterDam BV (A rated only)		The Netherlands	
9	Sungmi Co Ltd		Korea	
10	Solent Composite Solutions (SCS) Ltd (A rated only)		United Kingdom	**(Composite Material only)**
11	STACO Co Ltd		Korea	
Product	**4**	**Pre Fabricated Toilet System**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Specialist Services		Dubai (UAE)	
2	Flush Metal partition Corporation		UK	
3	Norac AS		Norway	
4	Shinsung Engg.& Archi Co Ld.		Korea	
5	Sungmi Co Ltd		Korea	

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6	STACO Co Ltd		Korea	
Product	**5**	**B-15 Suspended Ceiling**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	TNF Ceiling		Denmark	
2	STACO (Shinsung Total Accommodation System)		South Korea	
3	BD System		UK	
4	IMAC		Sweden	
5	USG Ceiling, New Delhi		India	
6	Sungmi Co Ltd		Korea	
Product	**6**	**Floor Screed / Underlay**		**Product Deleted**
Product	**7**	**Vinyl Tiles / Vinyl Sheets**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Marley Floors			**On Hold-B1**
2	Armstrong		USA	
3	LG Chemicals		South Korea	
4	Tarkett Sweden/Tarkett Hong Kong Ltd.		Sweden/Hongkong	
Product	**8**	**Ceramic Tiles**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Pilkington		USA	
2	Woomi Ceramics		South Korea	**On Hold-B1**
3	Armstrong		USA	
4	Marley Floors			**On Hold-B1**
5	H & R Johnson Tiles Limited		UK	
6	Prism Johnson Limited - H & R Johnson (India) Division		India	
Product	**9**	**Access Floor System**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Tate Floor		USA	
2	Gamma Industries		France	
3	Van Dam		Netherlands	
4	MERO		Germany	

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Product	**10**	**External Fire Proofing Coating H Rated**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Liquid Plastic Ltd		UK	
2	Chartek IV, Ms Textron Speciality Material		USA	
3	Thermolag 2000, M/s Thermal Science Inc		UK	
Product	**11**	**H-60 & A-60 Fire Rated Door**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Specialist Services		Dubai	
2	Booth Engineering Ltd.		UK	
3	Van Dam		Netherlands	
4	Sung Mi Geun Cheul		Korea	
5	STACO (Shinsung Total Accommodation System)		Korea	
6	Rapp Bomek AS		Norway	
7	Solent Composite Solutions (SCS) Ltd (A rated only)		United Kingdom	**(Composite Material only)**
8	Sung Mi Co., Ltd (for A-60 fire rated door only)		Korea	
9	E.S TECH MARINE Co. (for A-60 fire rated door only)		Korea	
Product	**12**	**B-15 Fire Rated Door**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Specialist Services		Dubai	
2	Booth Engineering Ltd.		UK	
3	Van Dam		Netherlands	
4	Mech Tools Engineering		UK	
5	IMAC		Sweden	
6	BD System		UK	
7	STACO (Shinsung Total Accommodation System)		Korea	
8	Sung Mi Geun Cheul		Korea	
9	Sung Mi Co., Ltd		Korea	

Product	**13**	**A-60 Fire Rated Window**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Samgong Co Ltd		Korea	
2	Nordam A/S		Norway	
3	Jung Gong Co.		Korea	
4	JUNG GONG IND.CO.,LTD		Korea	
Product	**14**	**A-0 Fire Rated Rolling Shutter**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Atlas Door			
2	Bolton Gate co.Ltd		UK	
Product	**15**	**Insulation**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Lloyd Insulation (I) Ltd. N Delhi		India	
2	Rockwool A/S		Denmark	
3	KCC Corporation		Korea	
4	Thermal Ceramics		UK	
**********				

Infocom				
Product	**1**	**Paging and Intercom System**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	GAI-TRONICS S.r.l.		USA / Italy	
2	Federal Signal Corporation		USA / UK	
3	PSC - Vodec		UK	
4	Comtrol Pte Ltd.		Singapore	
Product	**2**	**Telephone System**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Progility Technologies Pvt Ltd		India	
2	Federal Signal		UK	
3	Tadiran Telecom Ltd		Israel	
Product	**3**	**Radio System**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	Southern Avionics Company		USA	
2	Park Electronics Ltd		UK	
3	Amplidan			**On Hold-B1**
4	Jotron		Norway	
5	Thrane & Thrane A/S		Denmark	
6	MOTOROLA (VHF Portable Radio Sets & Accessories)		USA	
7	Furuno Electric Co Ltd (only for VHF and HF-SSB system including antennae)		Japan	
Product	**4**	**CCTV System**		
S. No.	**Vendor**		**Country**	**Remark/Note**
1	BOSCH		Germany	
2	HERNIS SCAN SYSTEM		Norway	
3	PELCO		USA	
4	Vicon Industries Ltd		UK	
5	Technocrats Security Systems Pvt. Ltd (as a Packager of OEM, M/s PELCO,USA)		India	

# Indicates changes in Rev No. : 12.3

PROFORMA OF UNDERTAKING (Refer Important Notes Clause 5(a))
(from the enlisted vendor)

We <name of the Principal/ enlisted vendor with location> declare that the < name of the vendor with location to be accepted> is our subsidiary unit with our stake more than 50%

We guarantee that < name of the vendor to be accepted> shall follow the same design, manufacturing, quality processes and testing standards of the <name of product> as that of the < name of the Enlisted vendor in Suggested Vendor List>.

And < name of the vendor to be accepted > shall comply with the Functional Specification.

We also stand warranty for the item supplied for a period of ____ years.

We guarantee for supply of spares and provide maintenance support in India for the entire life cycle of the item / equipment (i.e. 7 years in case of electronic equipment/ items and 10 years in case of mechanical equipment/ items).

(Authorized Signatory of Principal/enlisted vendor)

PROFORMA OF UNDERTAKING (Refer Important Notes Clause 5(c))
(from enlisted vendor/ from foreign partner who meets the enlistment criteria)

We <name of the Principal/enlisted vendor/ foreign partner who meets the enlistment criteria with location> declare that the < name of the partner in India> is our Joint venture partner in India with our _____% stake.
We guarantee that < name of the joint venture to be accepted> shall follow the same design, manufacturing, quality processes and testing standards of the <name of product> as that of the < name of the Enlisted vendor in Suggested Vendor List/ foreign partner who meets the enlistment criteria >.
And < name of the joint venture to be accepted > shall comply with the Functional Specification.

We agree for the Factory inspection to ascertain the capability of <name of Joint Venture Company>.

We also stand warranty for the item supplied for a period of ____years.
We guarantee for supply of spares and provide maintenance support in India for the entire life cycle of the item / equipment (i.e. 7 years in case of electronic equipment/ items and 10 years in case of mechanical equipment/ items).

(Authorized Signatory of Principal/enlisted vendor/foreign partner who meets the enlistment criteria)

PROFORMA OF UNDERTAKING (Refer Important Notes Clause 5(c))
(from Indian partner)

We <name of the Indian partner> declare that the < name of the enlisted vendor/ foreign partner who meets the enlistment criteria > is our Joint venture partner in India with our _____% stake.
We guarantee that < name of the joint venture to be accepted > shall follow the same design, manufacturing, quality processes and testing standards of the <name of product> as that of the < name of the Enlisted vendor in Suggested Vendor List/ foreign partner who meets the enlistment criteria >.
And < name of the joint venture to be accepted > shall comply with the Functional Specification.

We agree for the Factory inspection to ascertain the capability of <name of Joint Venture Company>.

We also stand warranty for the item supplied for a period of _____years.
We guarantee for supply of spares and provide maintenance support in India for the entire life cycle of the item / equipment (i.e. 7 years in case of electronic equipment/ items and 10 years in case of mechanical equipment/ items).

(Authorized Signatory of Indian Partner)

Annexure-IV

PROFORMA OF UNDERTAKING
(Refer Important Notes Clause 5(d))

(From the enlisted foreign vendor having Technical License agreement with the Indian Company for manufacturing facilities in India)

We <name of the enlisted foreign vendor with location> guarantee that < name of the Indian company to be accepted with manufacturing location in India > shall follow the same design, manufacturing, quality processes and testing standards of the <name of product> as that of the < name of the Enlisted vendor in Suggested Vendor List> and shall comply with the Functional Specification.

We undertake the full responsibility of quality of product to be supplied by <name of the Indian company to be accepted with manufacturing location in India >

We further understand that if, the product manufactured by the Indian company fails to maintain the same design, manufacturing, quality processes and testing standards of the <name of product> as that of the < name of the enlisted foreign vendor in Suggested Vendor List>, the Company reserves the rights of delisting us <name of the enlisted vendor with location in Suggested Vendor List > for <name of product> in Suggested Vendor List.

In case of cancellation as well as renewal of technical license agreement, we shall inform ONGC well in advance.

(Authorized Signatory of enlisted foreign vendor)

PROFORMA OF UNDERTAKING (From Indian Company)
(Refer Important Notes Clause 5(d))

We <name of the Indian Company> declare that the < name of the enlisted foreign vendor > has ***Technical License agreement*** with us for <name of product>

We guarantee that <name of the Indian Company> shall follow the same design, manufacturing, quality processes and testing standards of the <name of product> as that of the < name of the Enlisted foreign vendor in Suggested Vendor List.

We <name of the Indian Company> shall possess all the accreditations required for the product and also comply with the functional specifications.

We agree for the Factory inspection to ascertain the capability of <name of the Indian Company>.

We also stand warranty for the item supplied for a period of _____years.

We guarantee for supply of spares and provide maintenance support in India for the entire life cycle of the item / equipment (i.e. 7 years in case of electronic equipment/ items and 10 years in case of mechanical equipment/ items).

In case of cancellation as well as renewal of technical license agreement, we shall inform ONGC well in advance.

(Authorized Signatory of Indian Company)